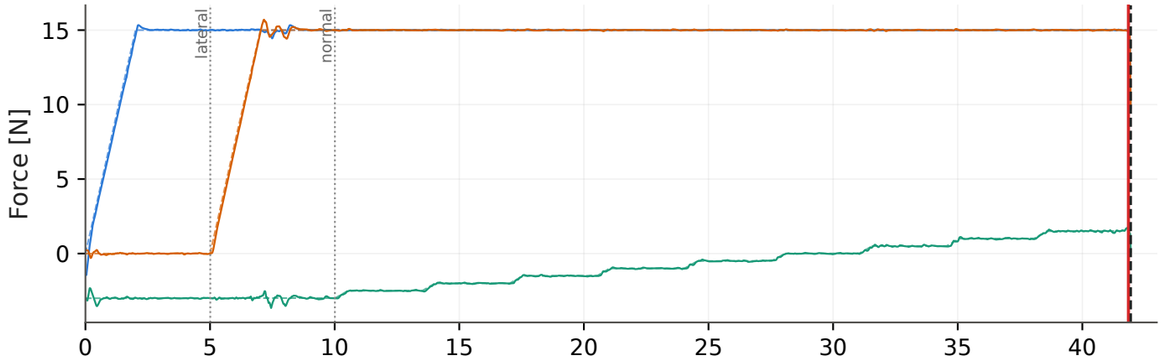
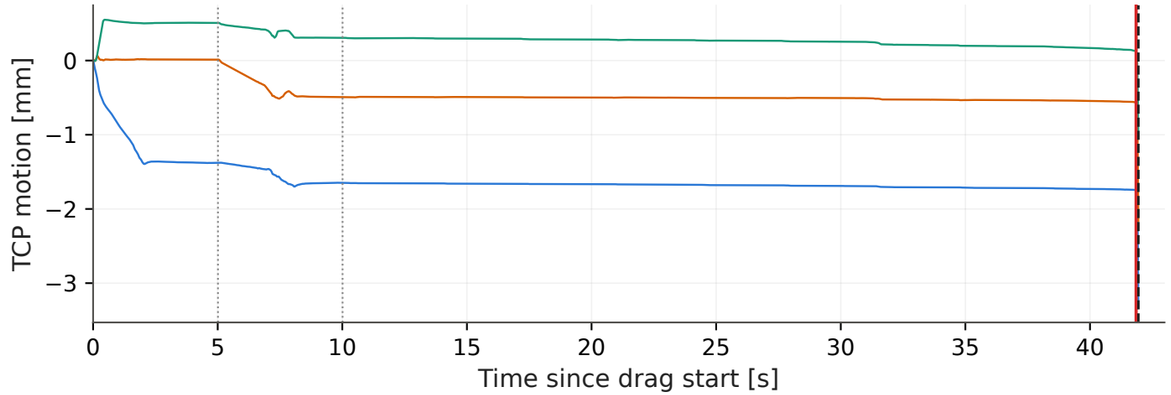
Trial audit | 1–3 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

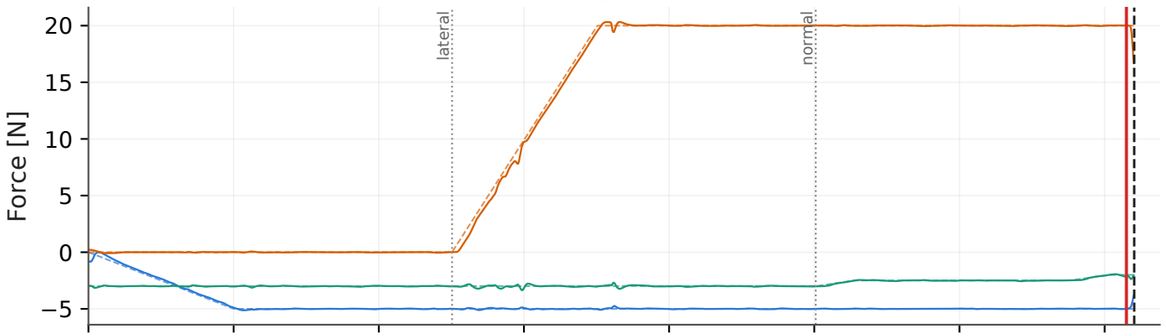
T1 R1 S=15, L=15 N | Normal tension (Fn > 0)



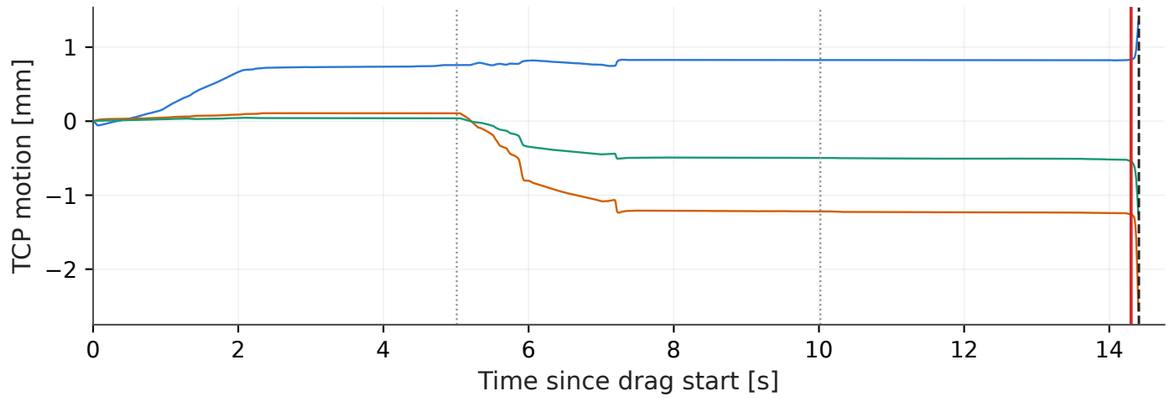
abrupt motion; re-engaged: False; review: False



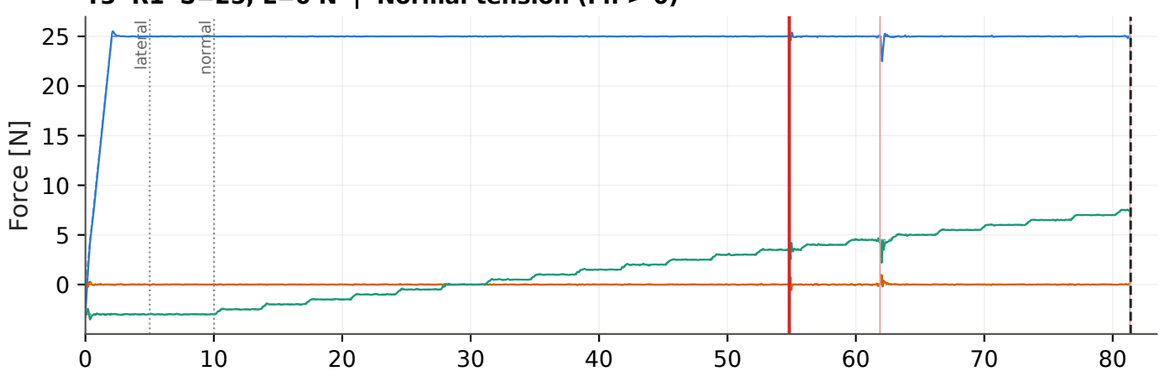
T2 R1 S=-5, L=20 N | Normal unloading (Fn <= 0)



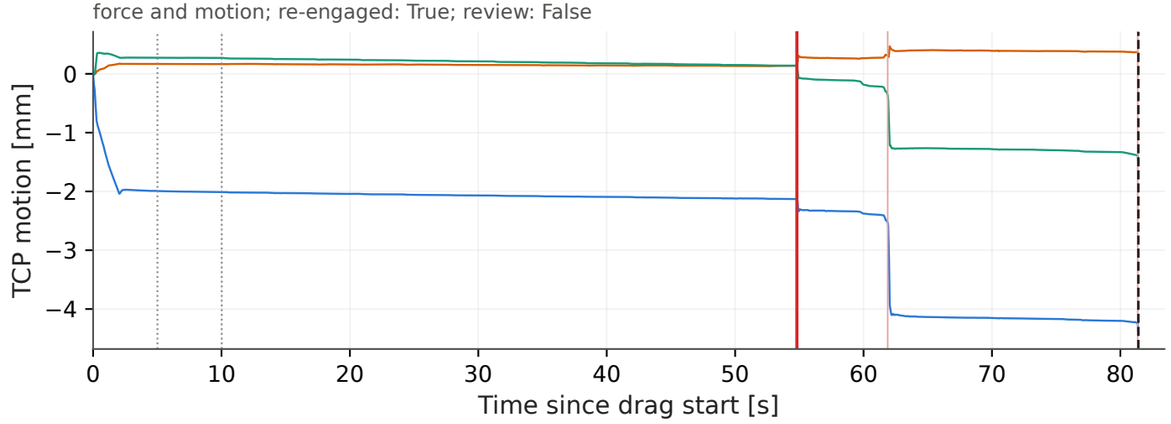
abrupt motion; re-engaged: False; review: False



T3 R1 S=25, L=0 N | Normal tension (Fn > 0)



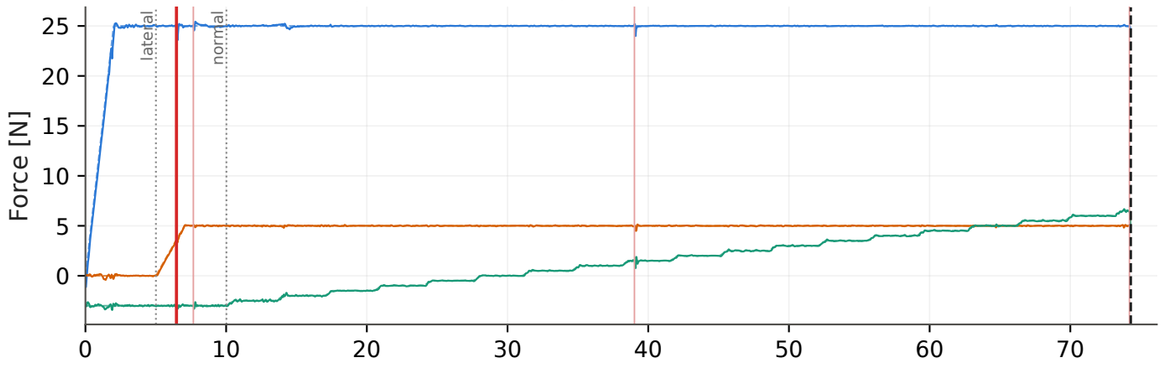
force and motion; re-engaged: True; review: False



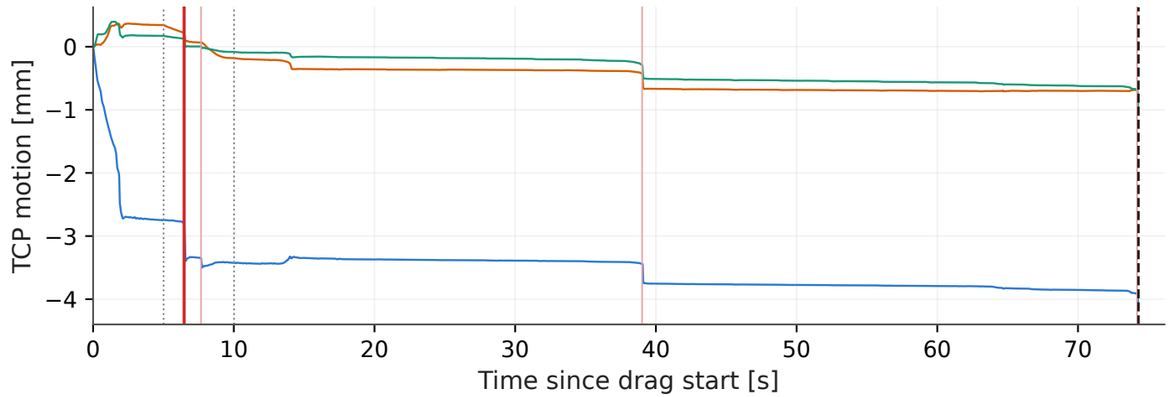
Trial audit | 4–6 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

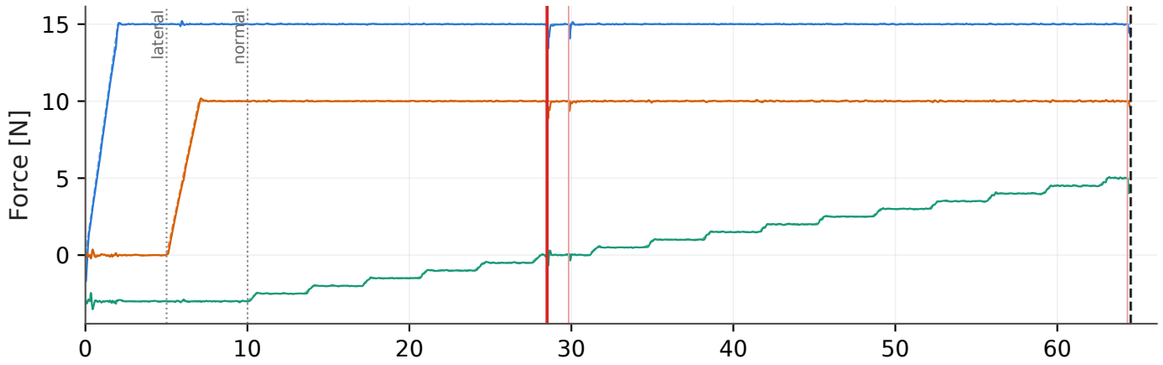
T4 R1 S=25, L=5 N | Lateral loading



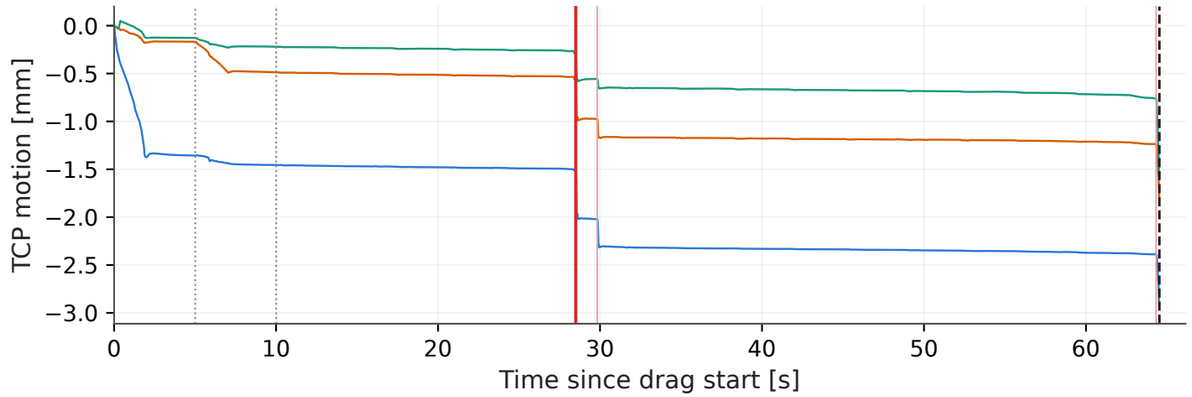
force and motion; re-engaged: True; review: False



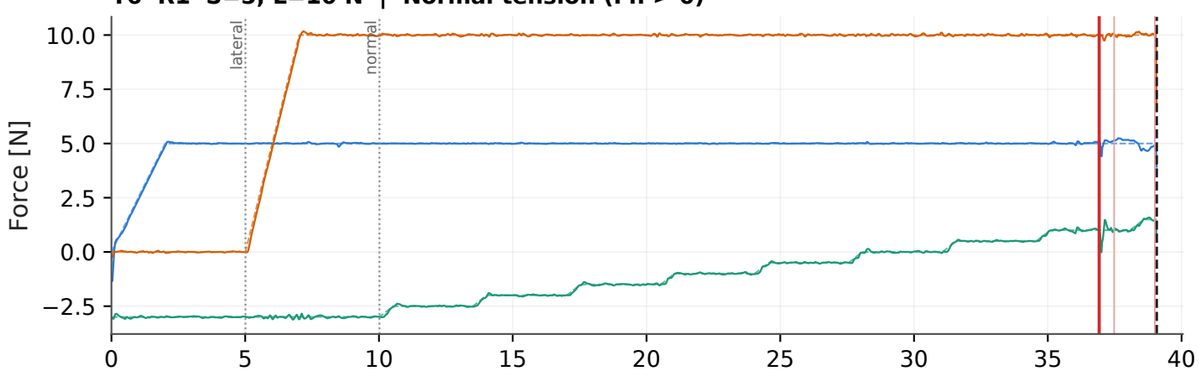
T5 R1 S=15, L=10 N | Normal unloading (Fn <= 0)



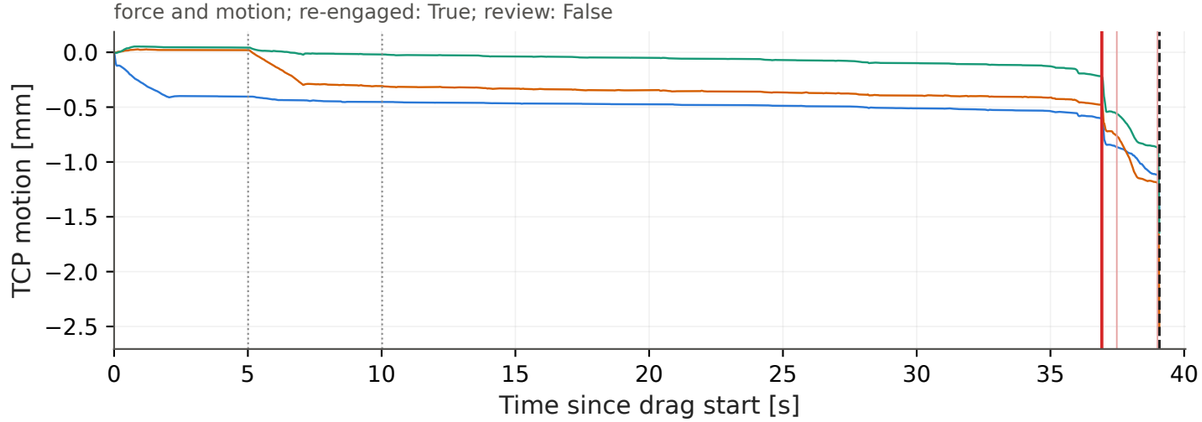
abrupt motion; re-engaged: True; review: False



T6 R1 S=5, L=10 N | Normal tension (Fn > 0)

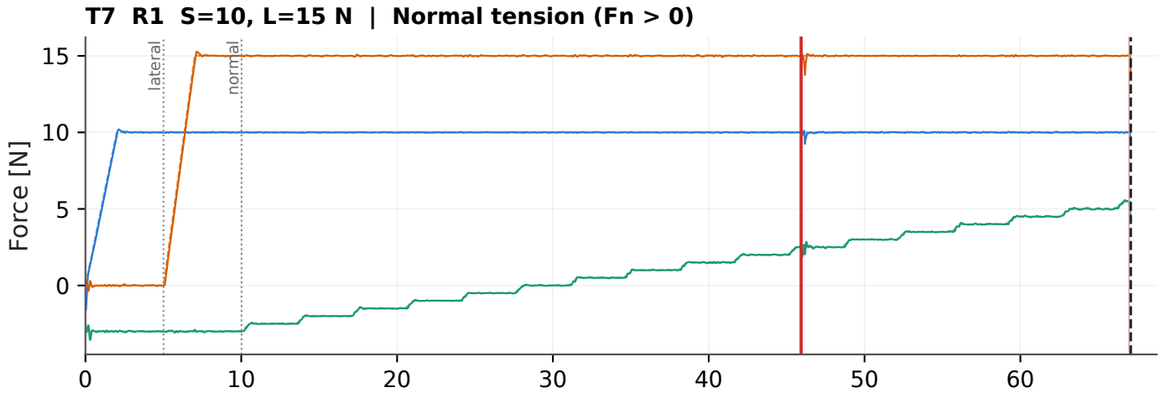


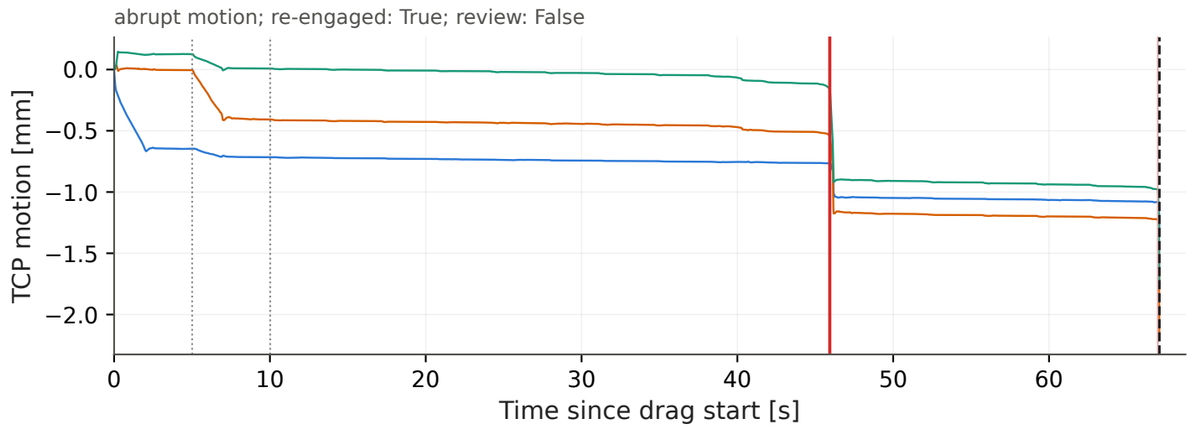
force and motion; re-engaged: True; review: False

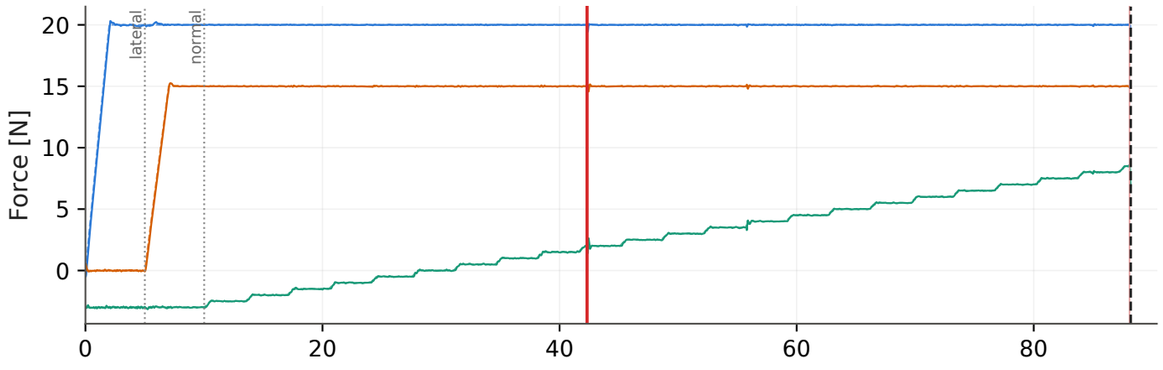


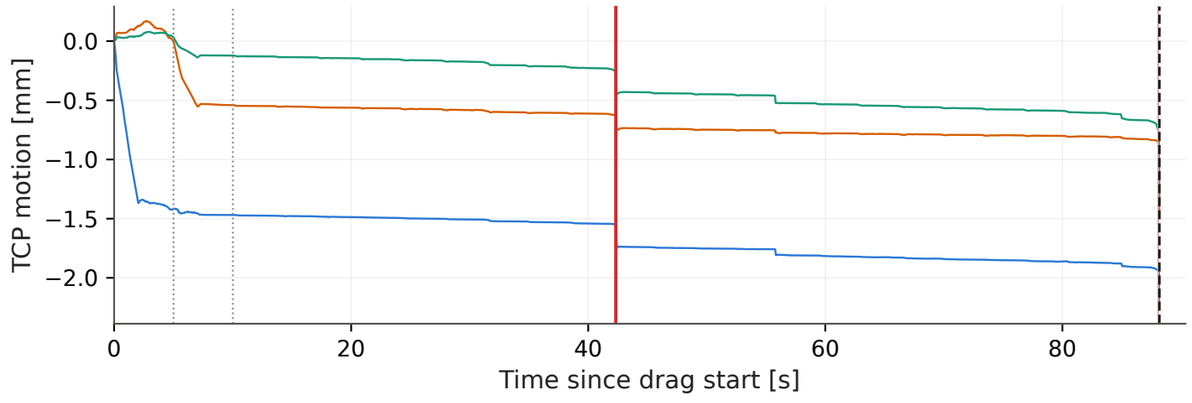
Trial audit | 7–9 / 231

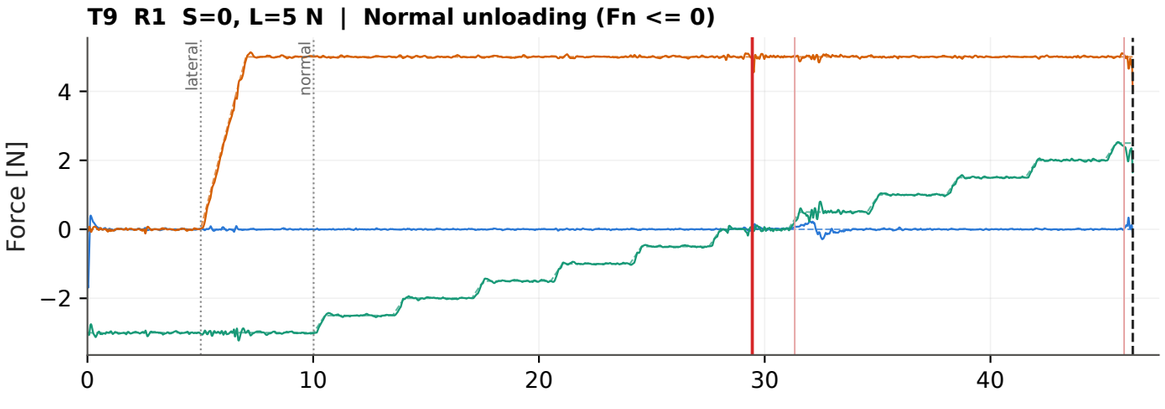
Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

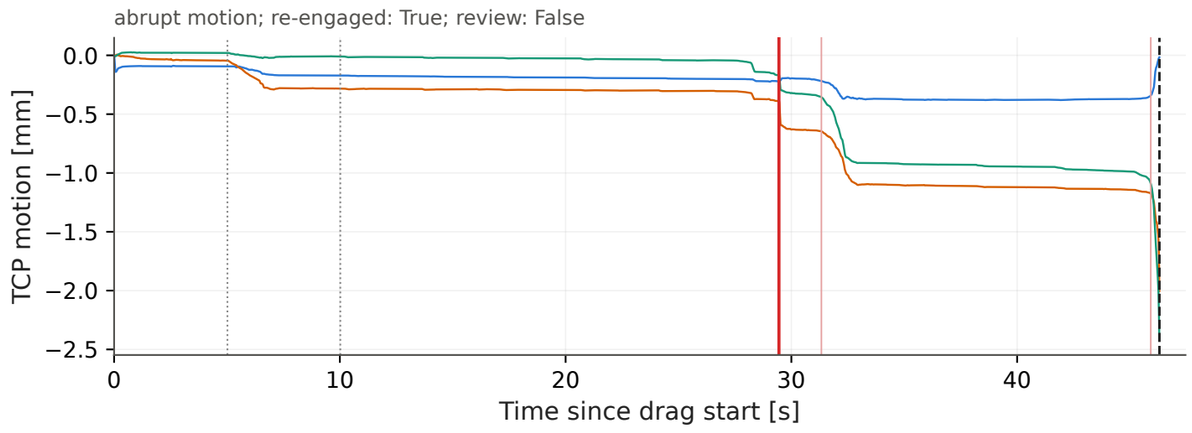








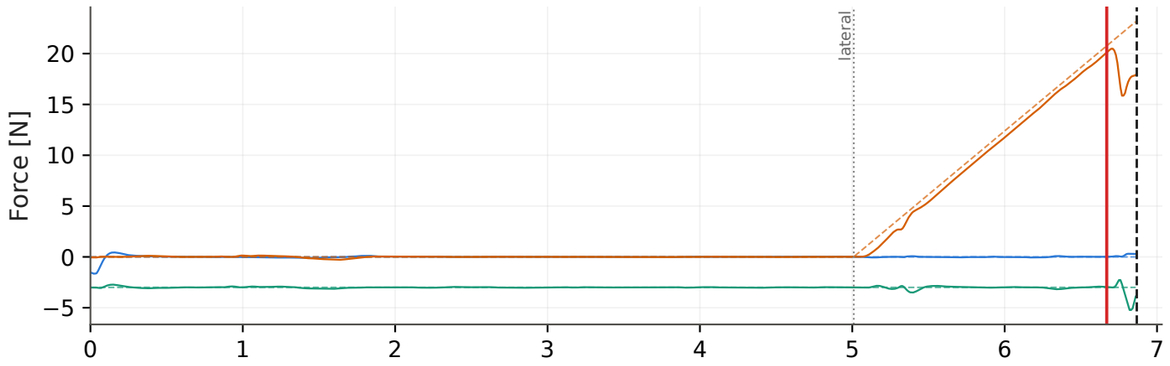




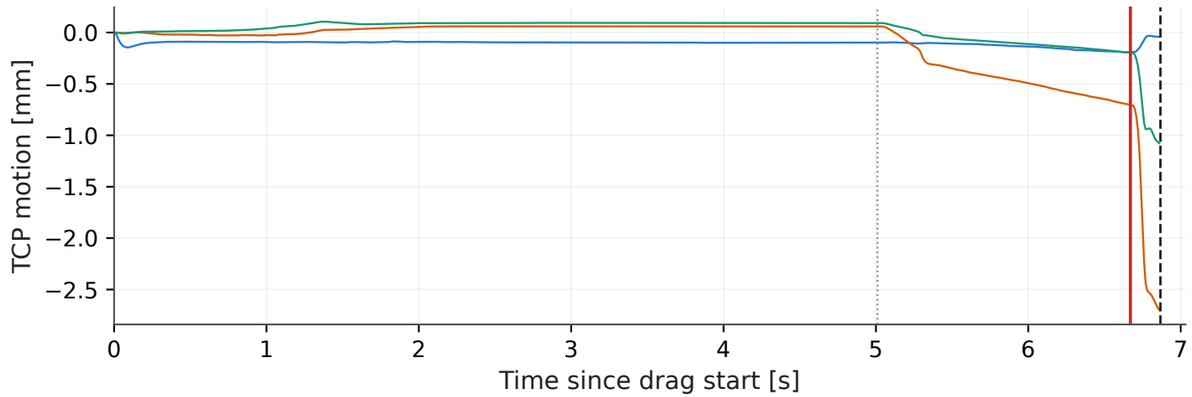
Trial audit | 10–12 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

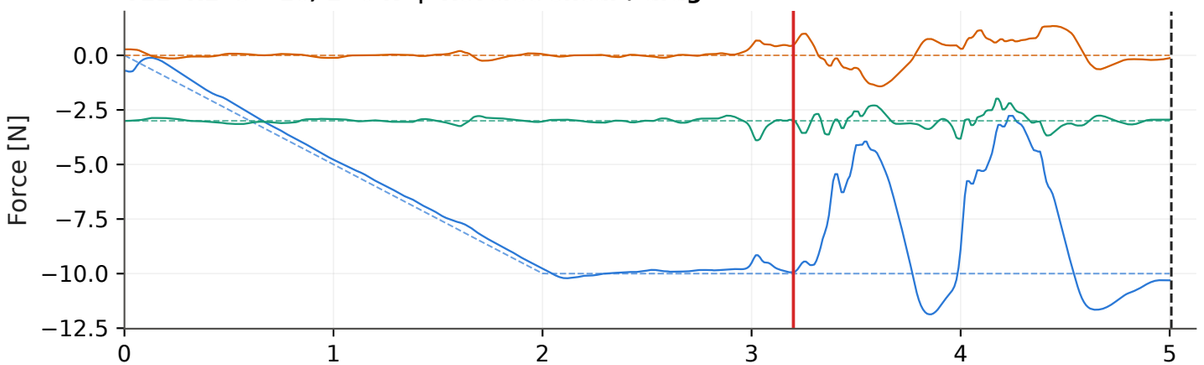
T10 R1 S=-0, L=25 N | Lateral loading



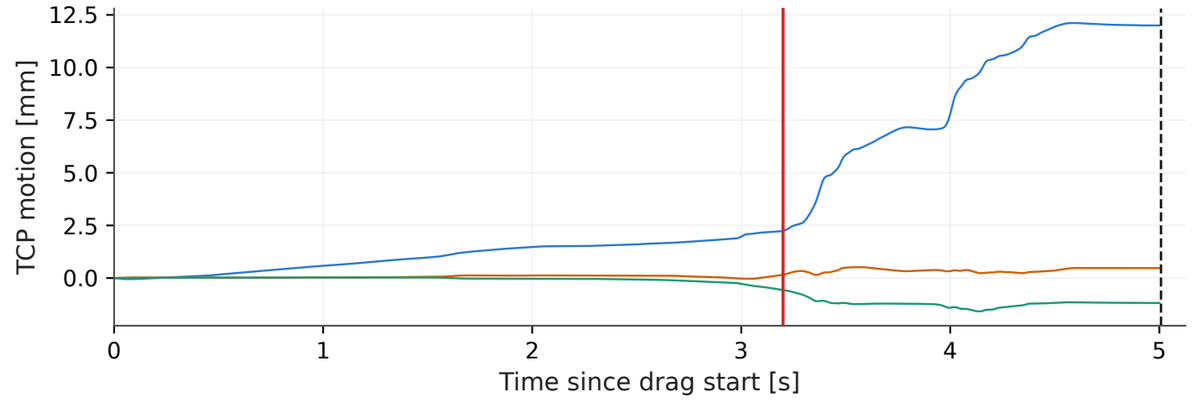
force and motion; re-engaged: False; review: False



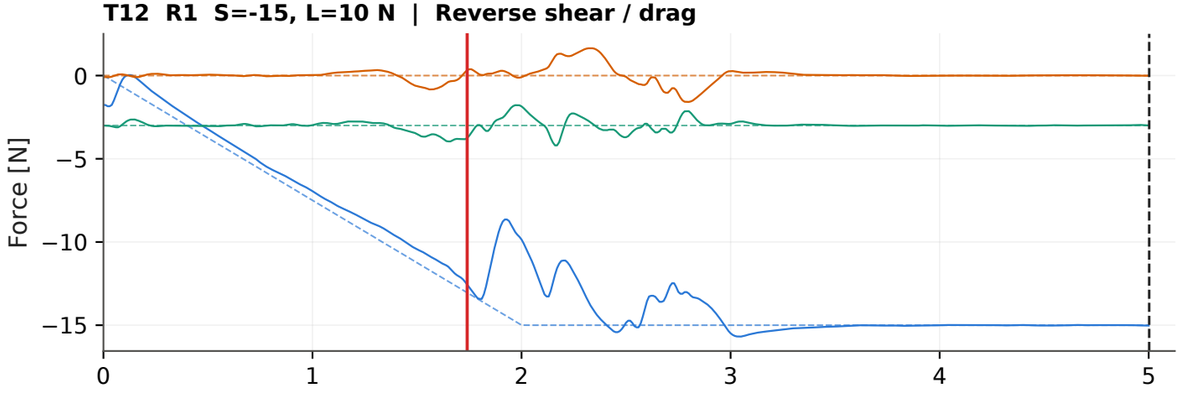
T11 R1 S=-10, L=0 N | Reverse shear / drag



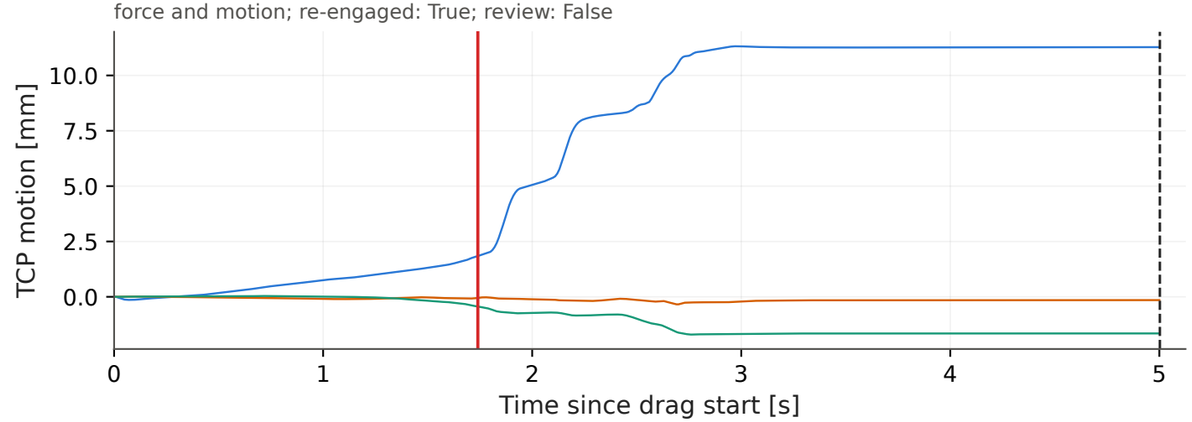
force and motion; re-engaged: False; review: False



T12 R1 S=-15, L=10 N | Reverse shear / drag



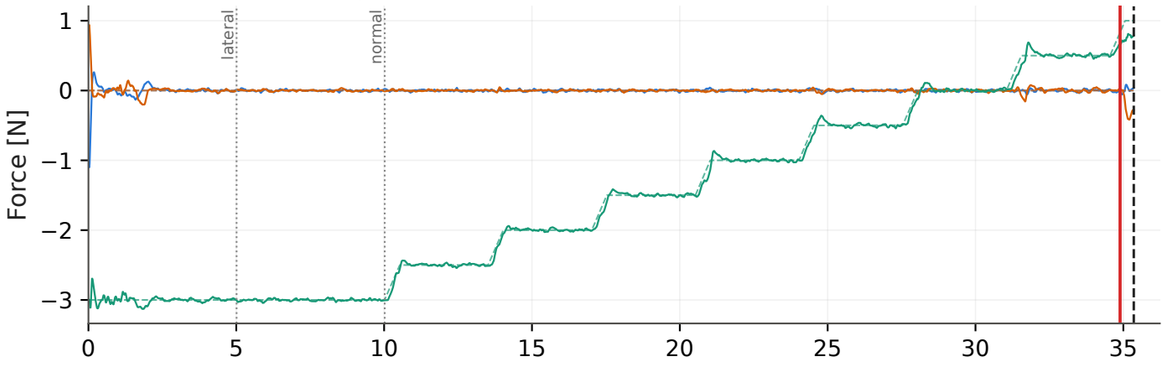
force and motion; re-engaged: True; review: False



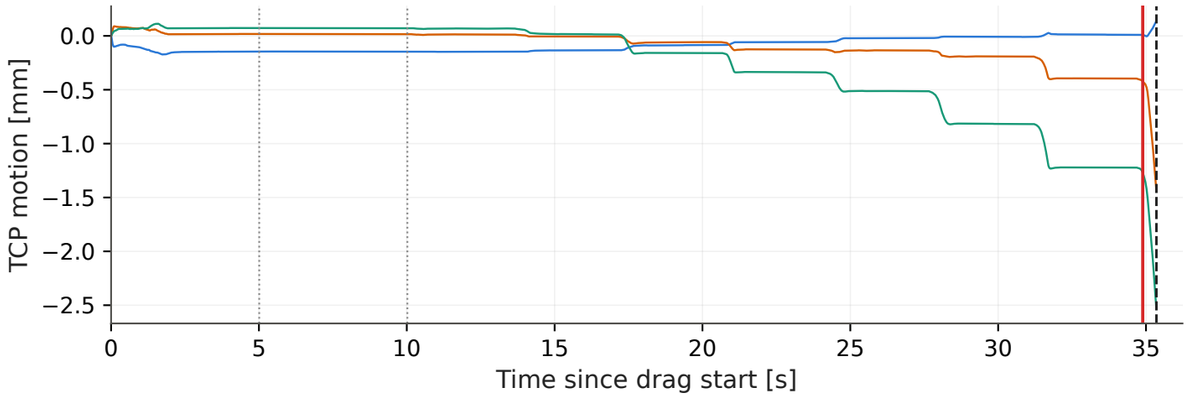
Trial audit | 13–15 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

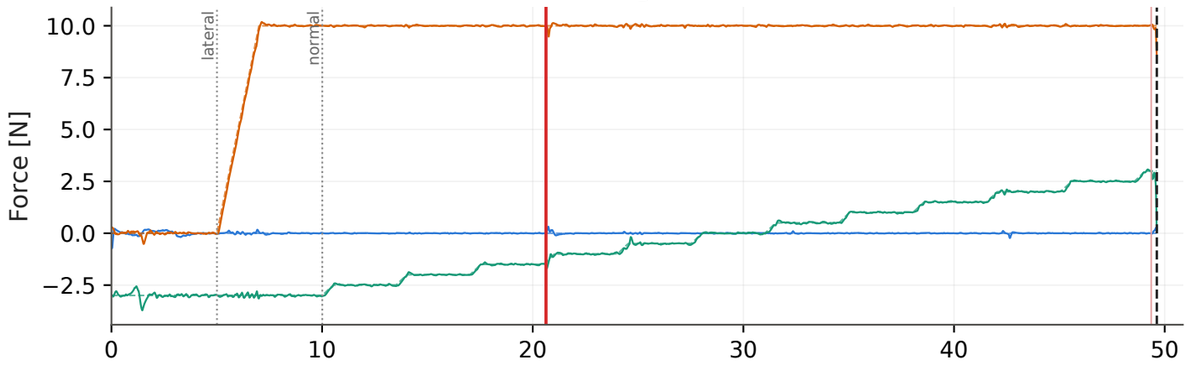
T13 R1 S=0, L=0 N | Normal tension (Fn > 0)



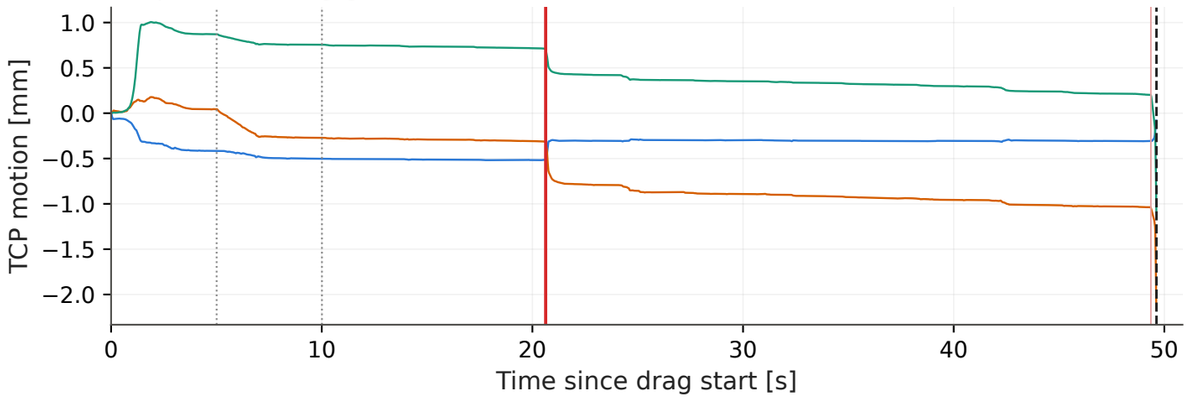
abrupt motion; re-engaged: False; review: True



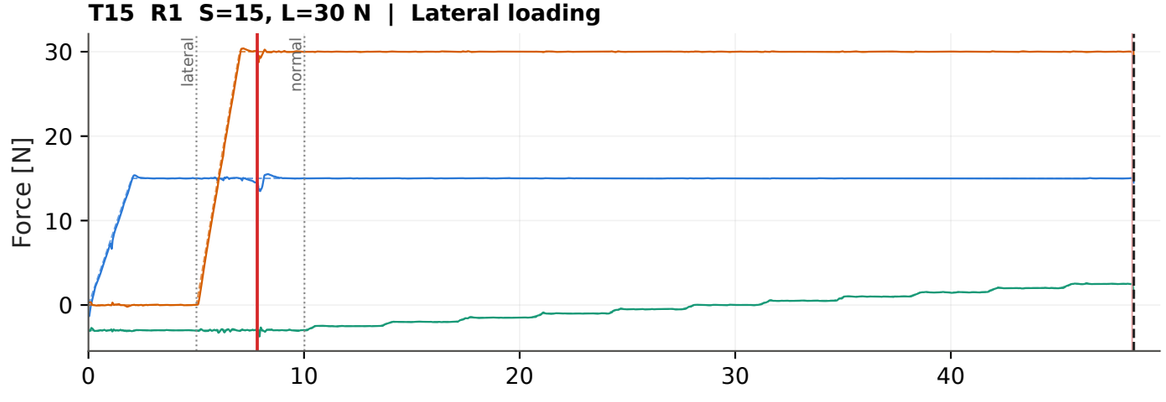
T14 R1 S=-0, L=10 N | Normal unloading (Fn <= 0)



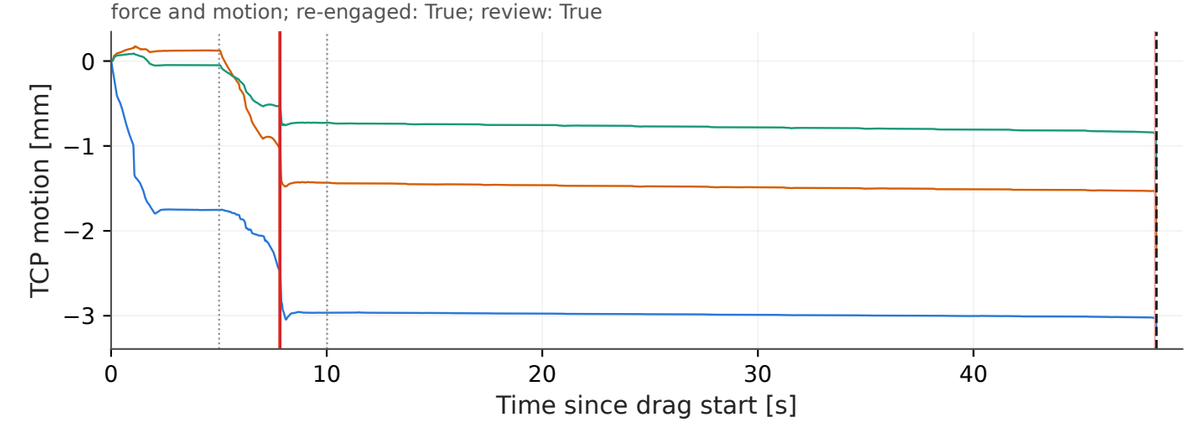
abrupt motion; re-engaged: True; review: False



T15 R1 S=15, L=30 N | Lateral loading



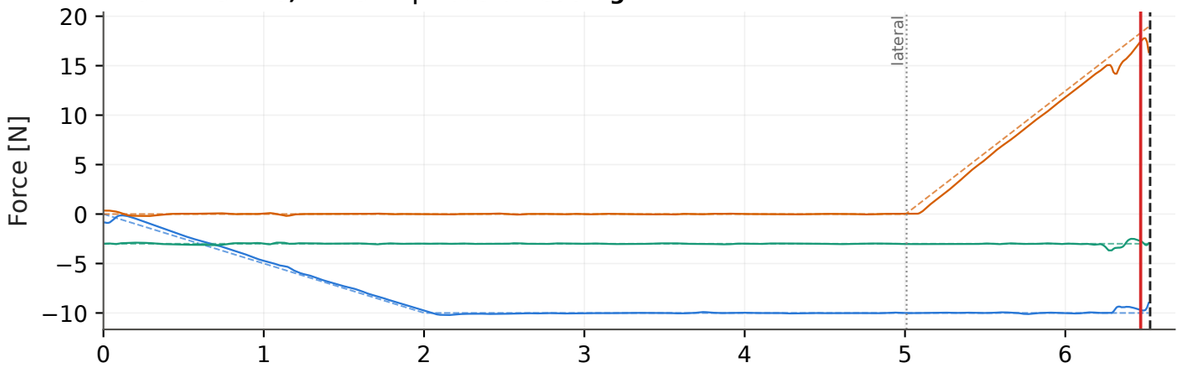
force and motion; re-engaged: True; review: True



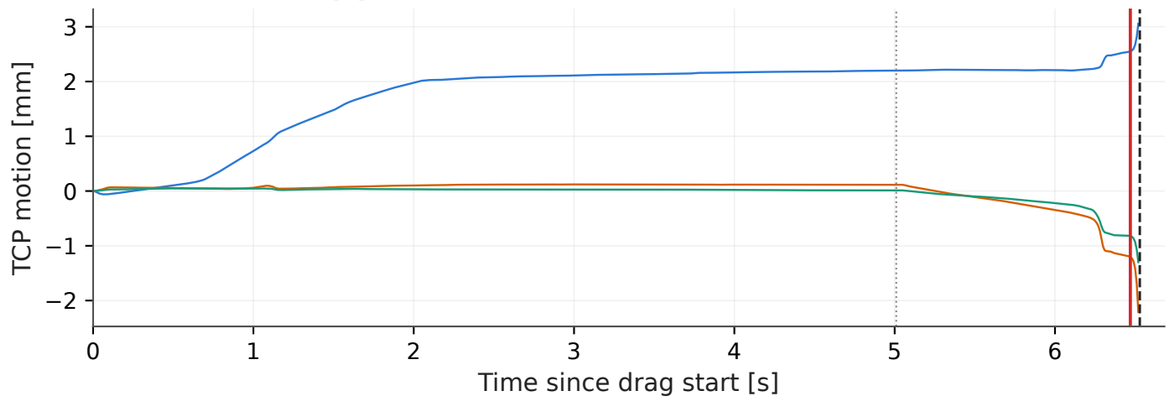
Trial audit | 16–18 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

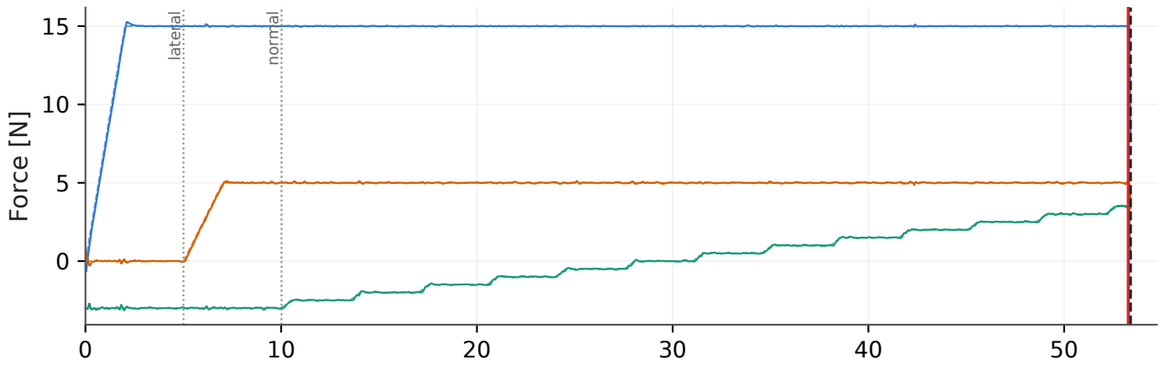
T16 R1 S=-10, L=25 N | Lateral loading



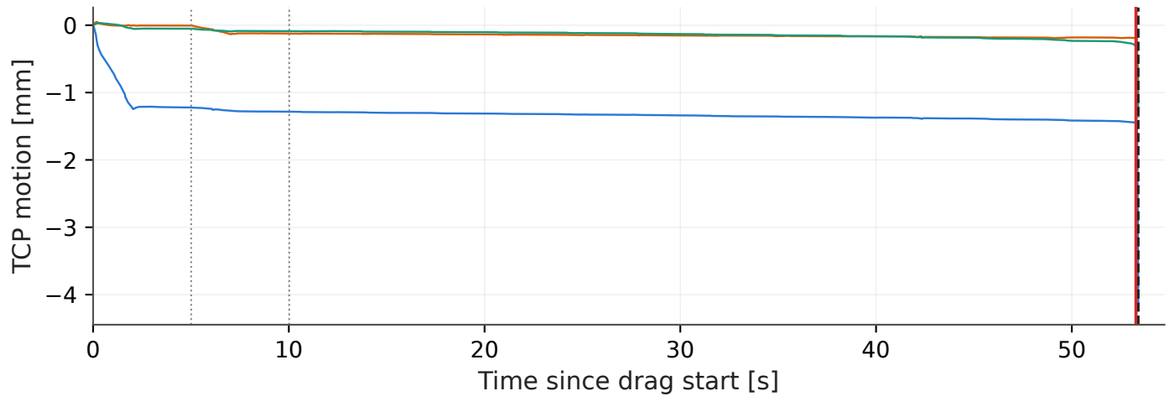
force and motion; re-engaged: False; review: False



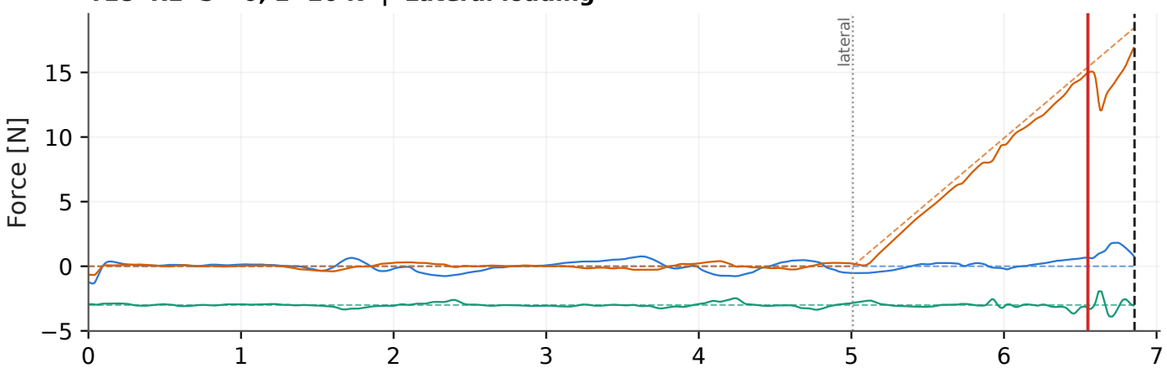
T17 R1 S=15, L=5 N | Normal tension (Fn > 0)



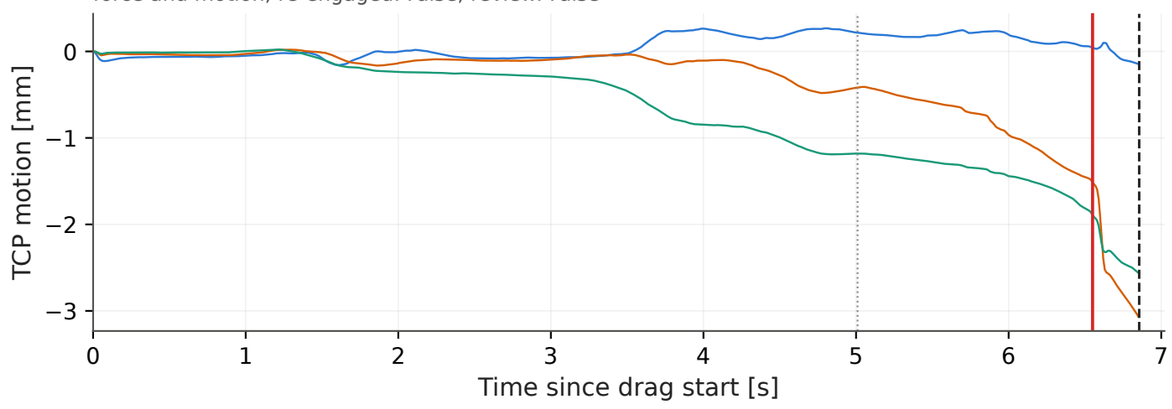
abrupt motion; re-engaged: False; review: False



T18 R1 S=-0, L=20 N | Lateral loading



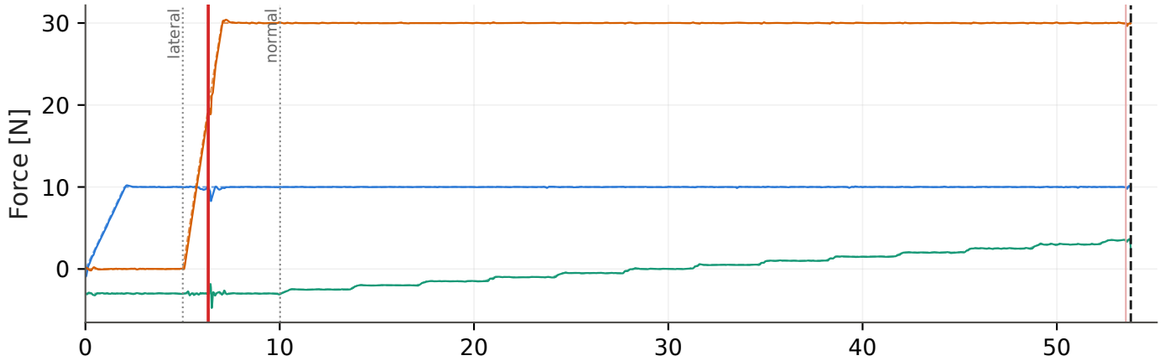
force and motion; re-engaged: False; review: False



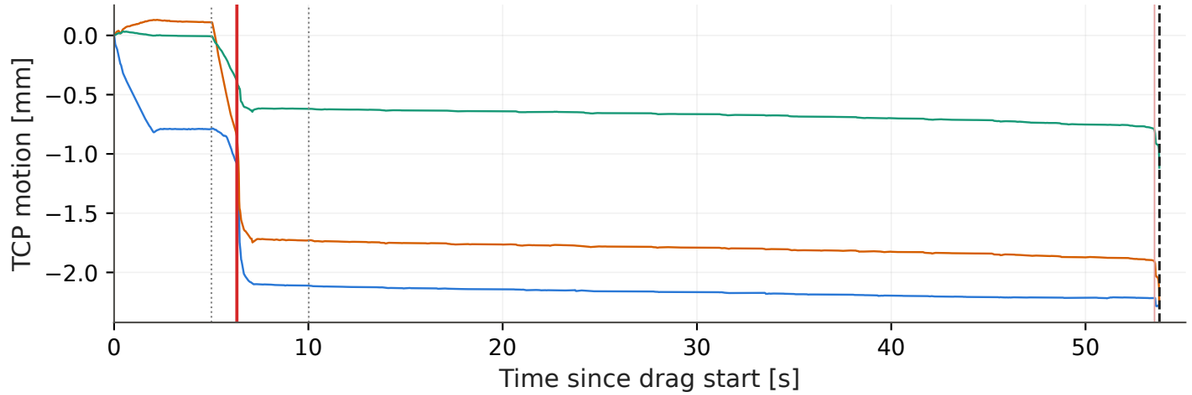
Trial audit | 19–21 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

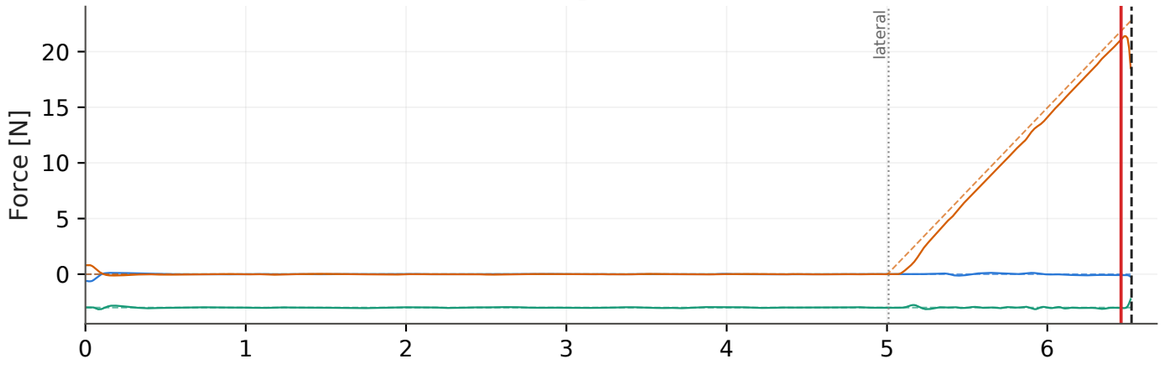
T19 R1 S=10, L=30 N | Lateral loading



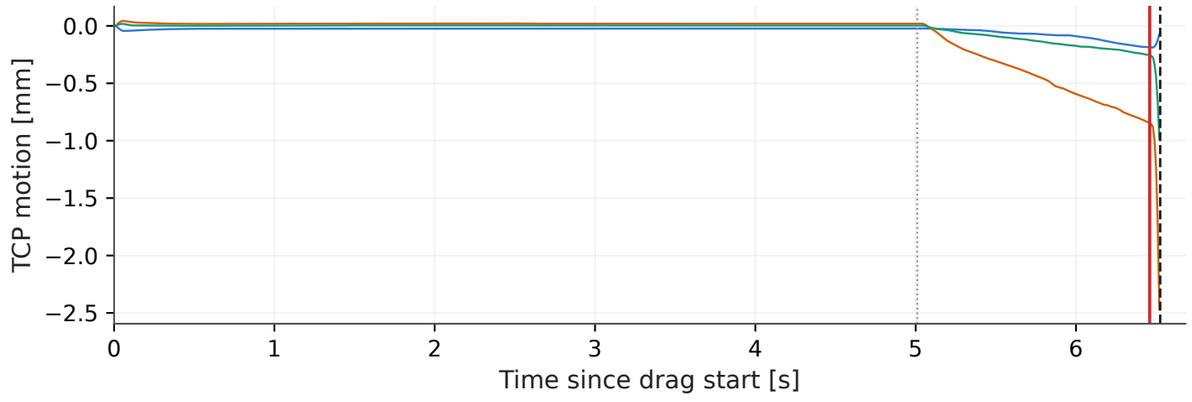
force and motion; re-engaged: True; review: False



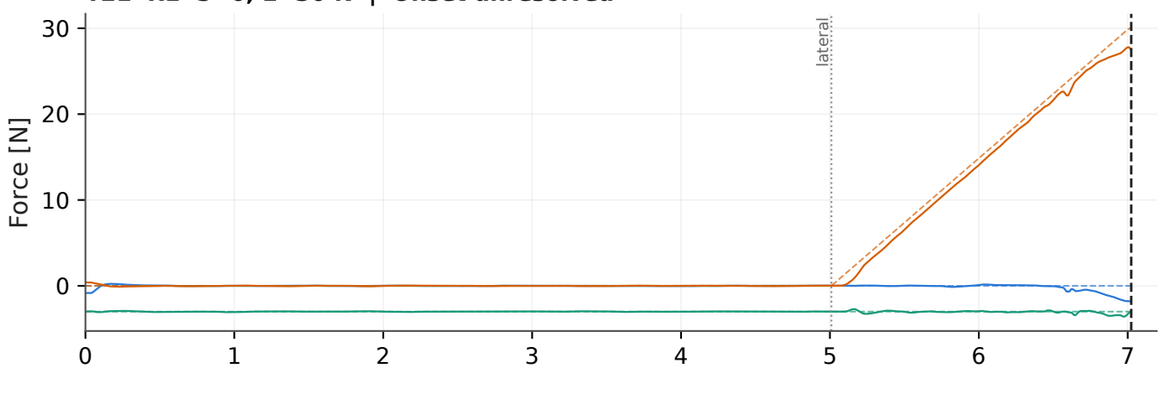
T20 R1 S=-0, L=30 N | Lateral loading



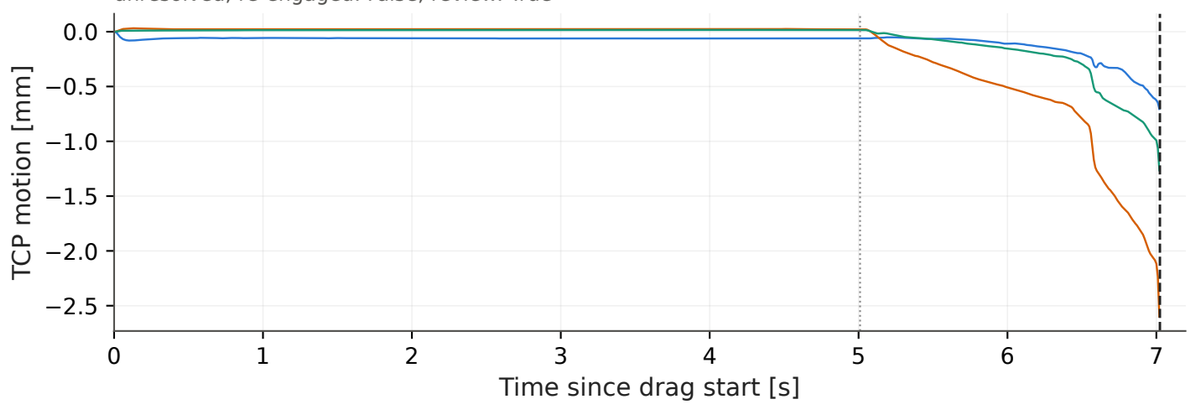
force and motion; re-engaged: False; review: False



T21 R1 S=0, L=30 N | Onset unresolved



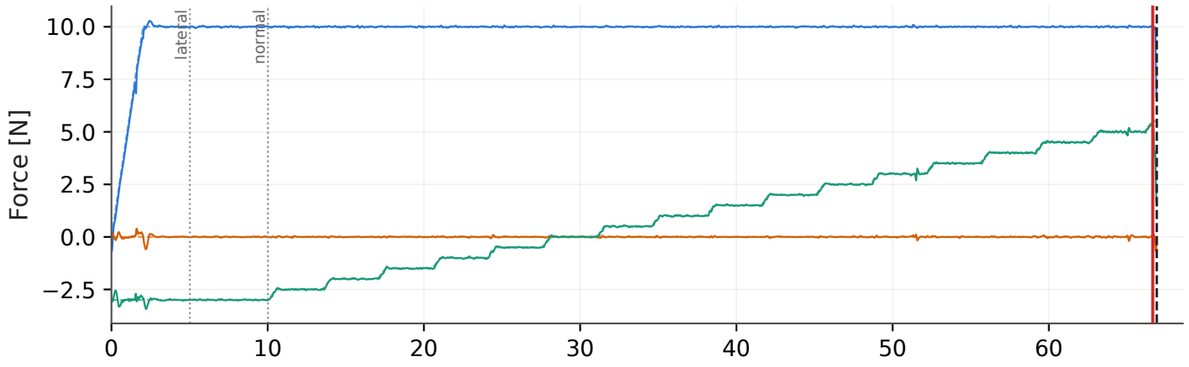
unresolved; re-engaged: False; review: True



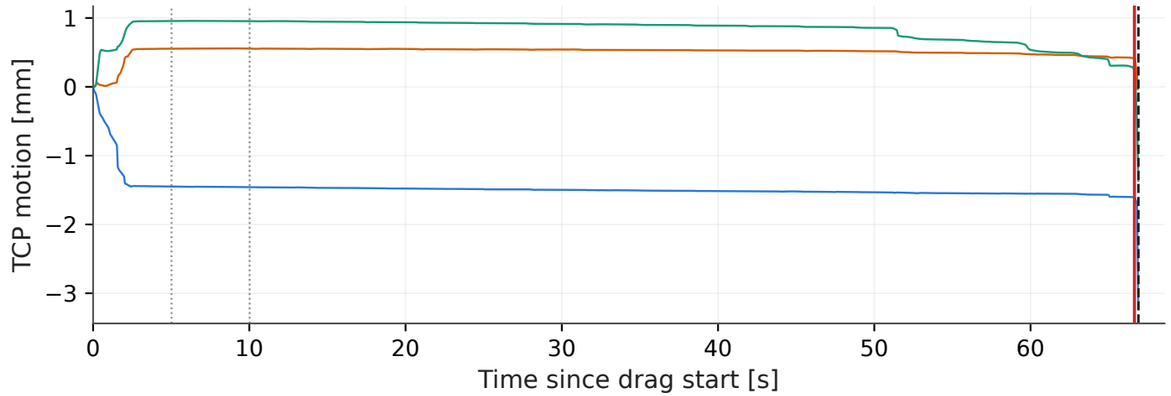
Trial audit | 22–24 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

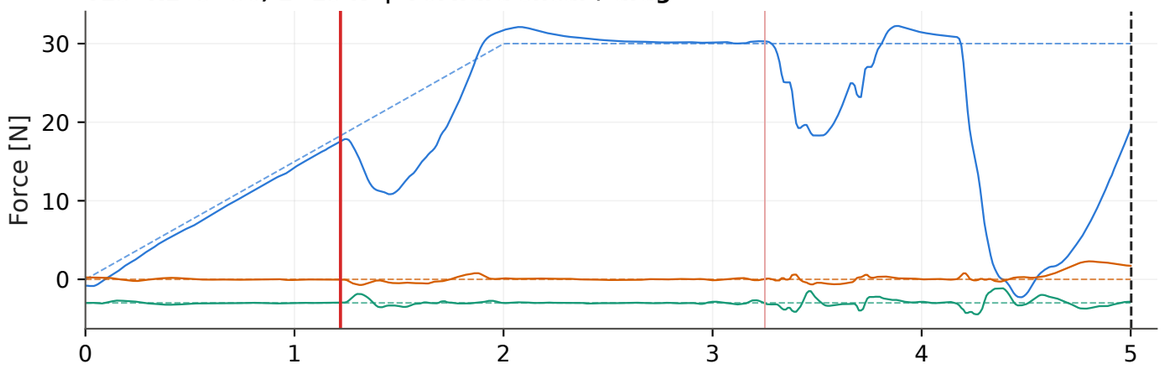
T22 R1 S=10, L=0 N | Normal tension (Fn > 0)



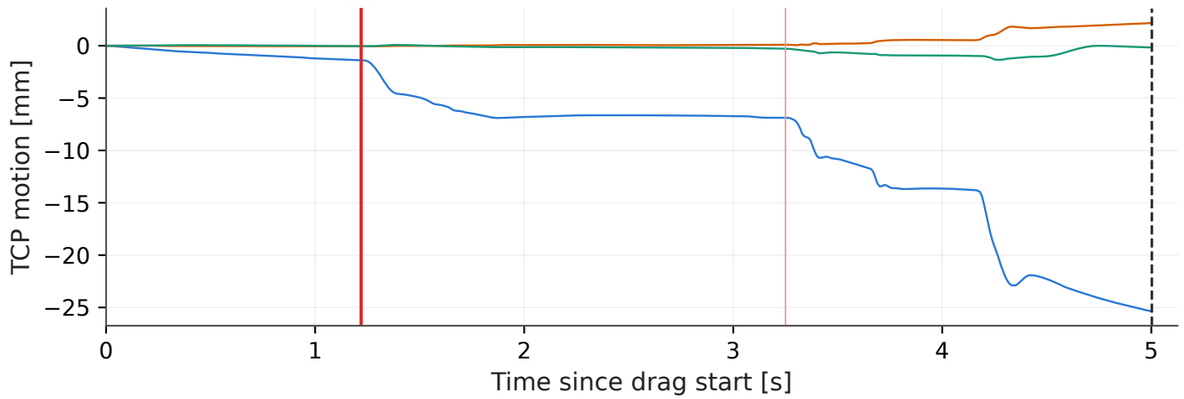
abrupt motion; re-engaged: False; review: False



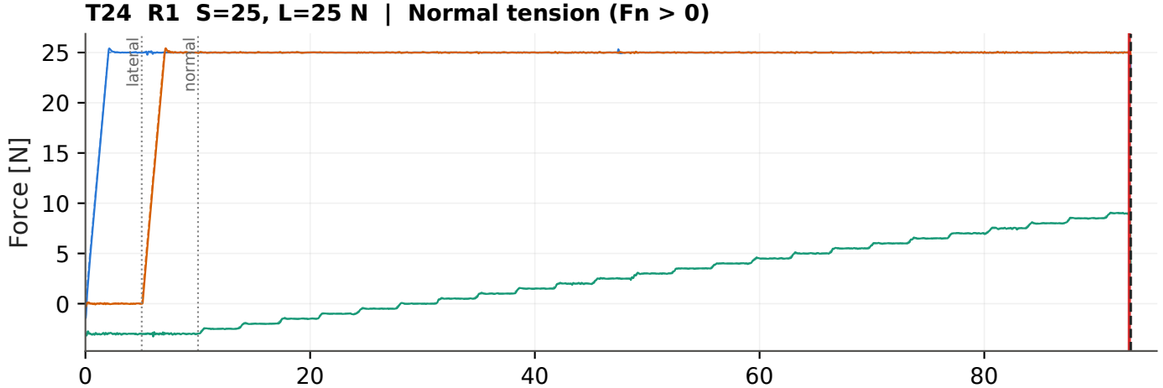
T23 R1 S=30, L=15 N | Forward shear / drag



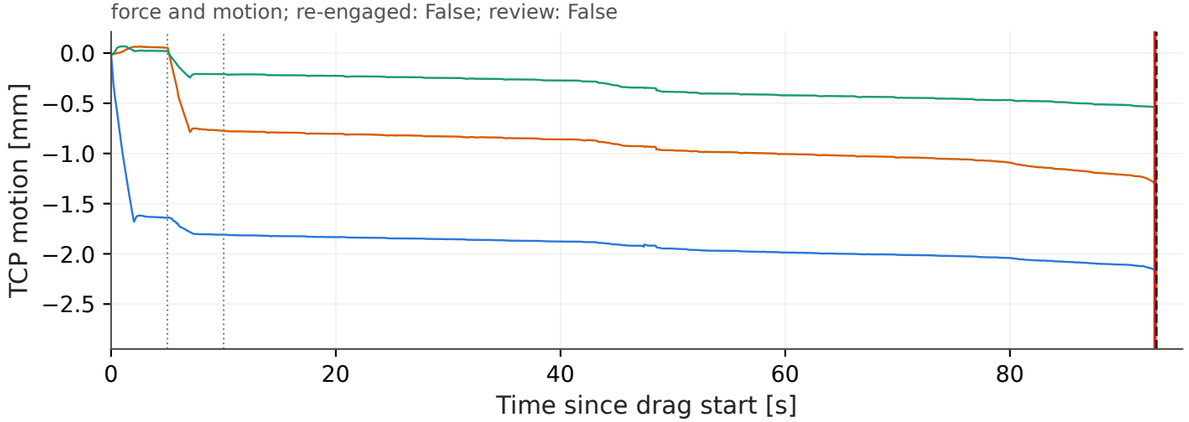
force and motion; re-engaged: True; review: False



T24 R1 S=25, L=25 N | Normal tension (Fn > 0)



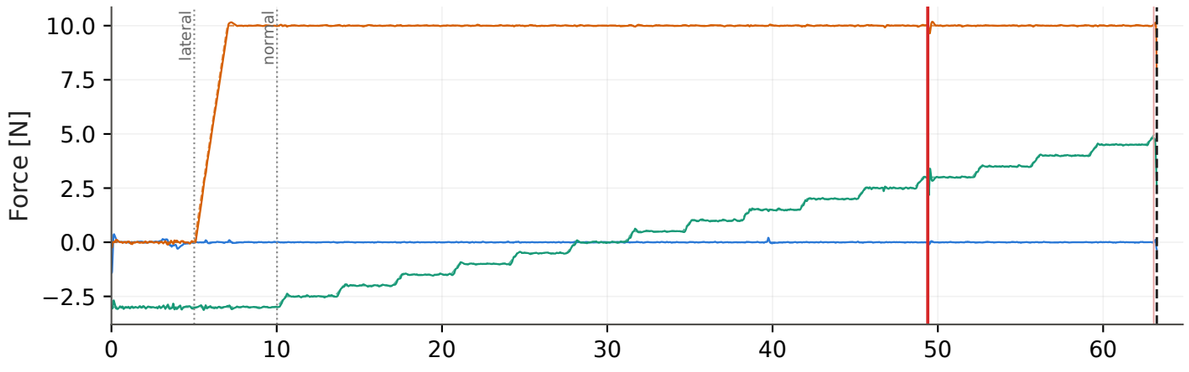
force and motion; re-engaged: False; review: False



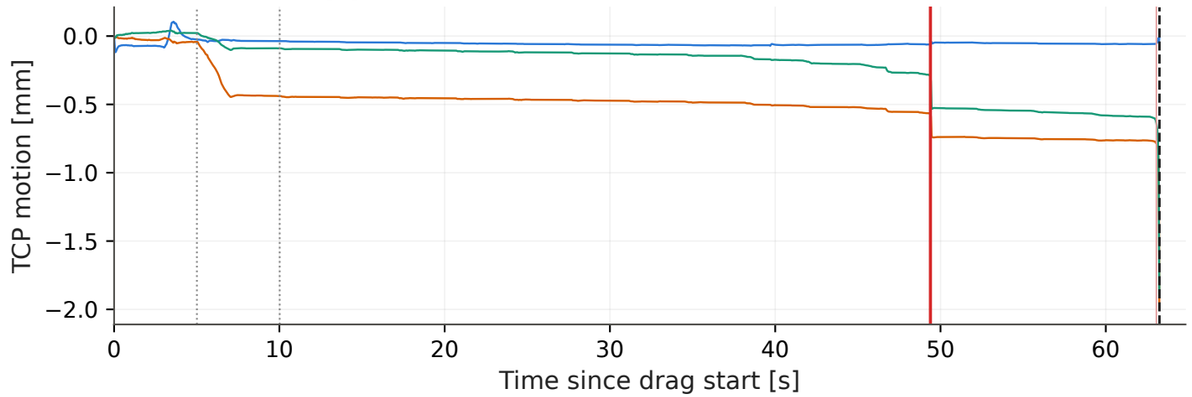
Trial audit | 25–27 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

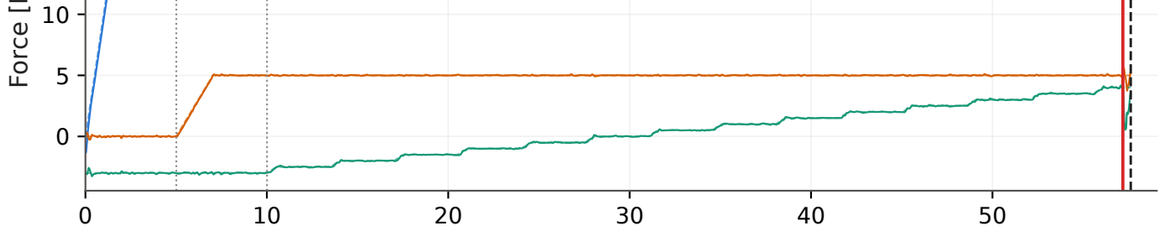
T25 R1 S=0, L=10 N | Normal tension (Fn > 0)



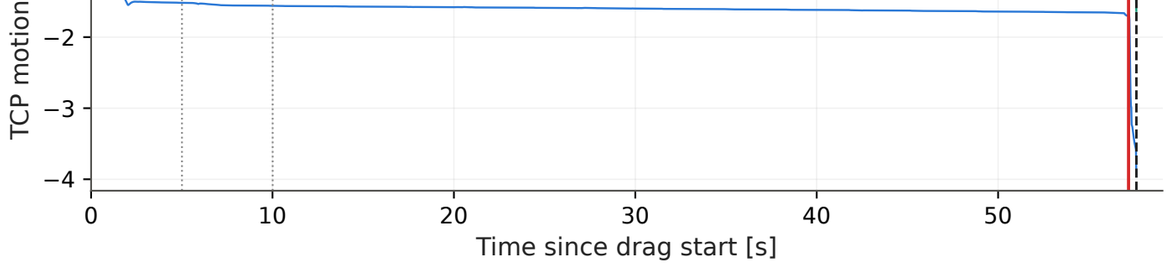
abrupt motion; re-engaged: True; review: False



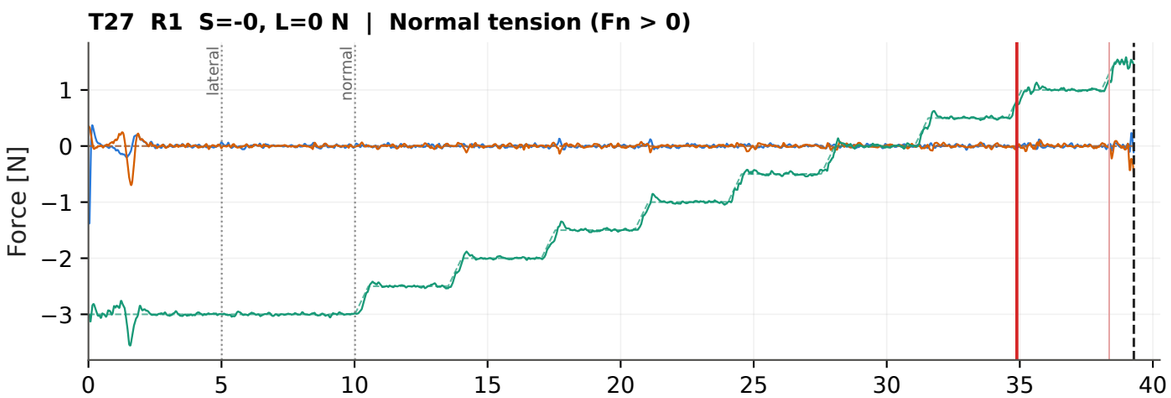
T26 R1 S=20, L=5 N | Normal tension (Fn > 0)



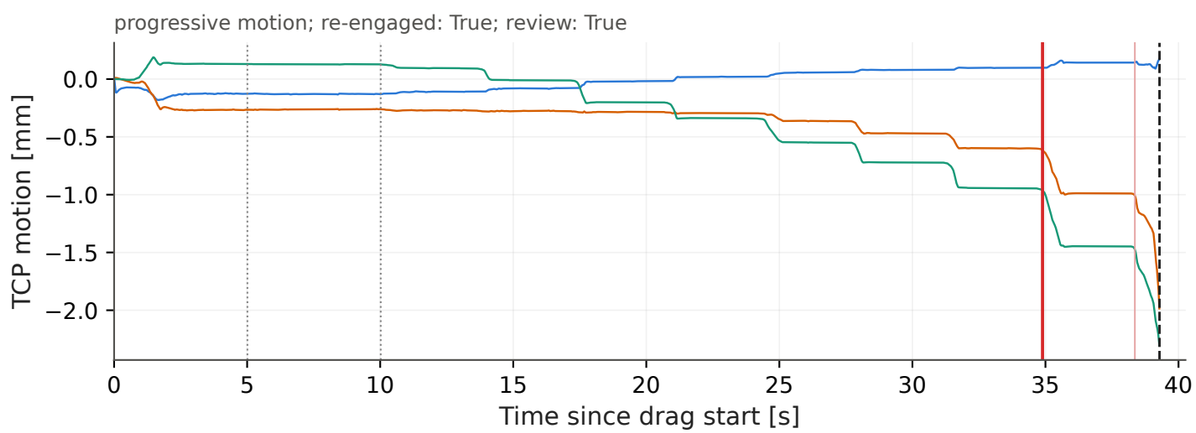
force and motion; re-engaged: False; review: False



T27 R1 S=-0, L=0 N | Normal tension (Fn > 0)



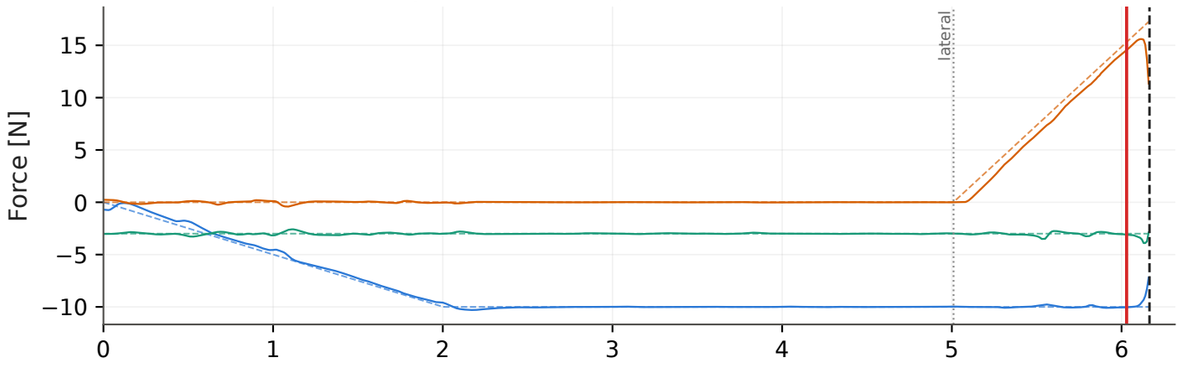
progressive motion; re-engaged: True; review: True



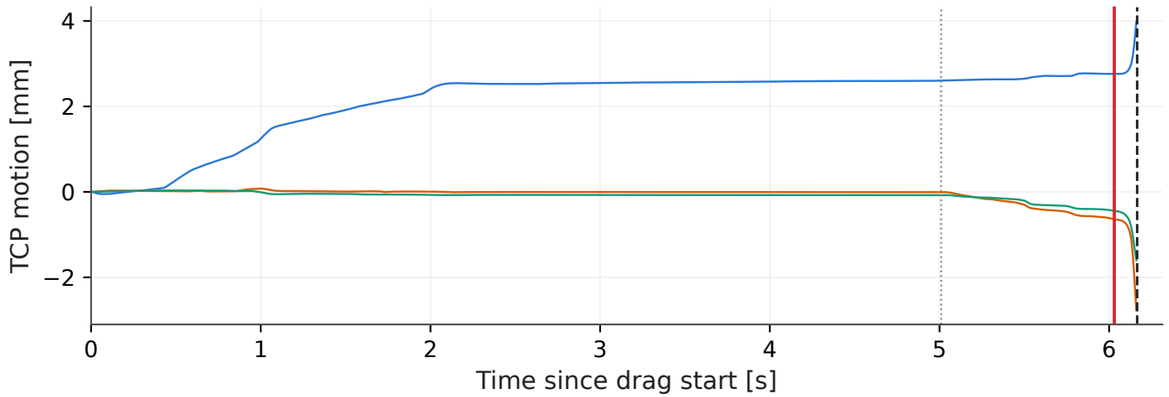
Trial audit | 28–30 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

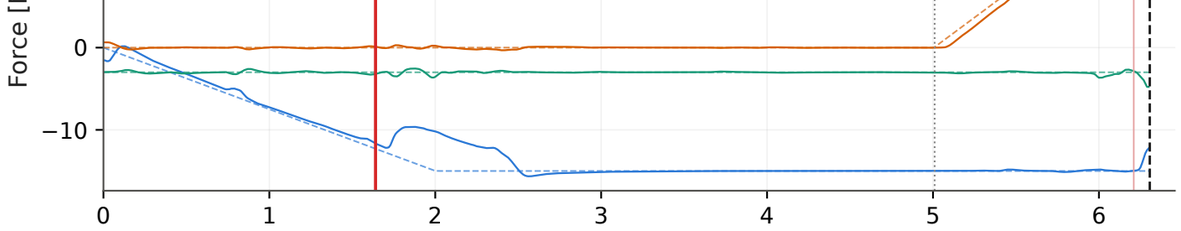
T28 R1 S=-10, L=30 N | Lateral loading



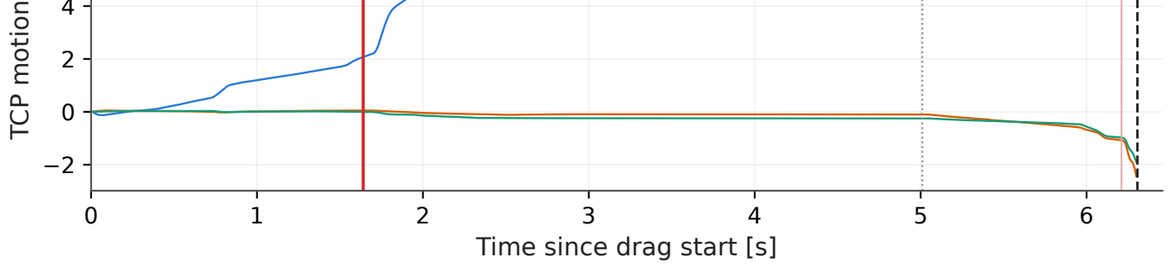
force and motion; re-engaged: False; review: False



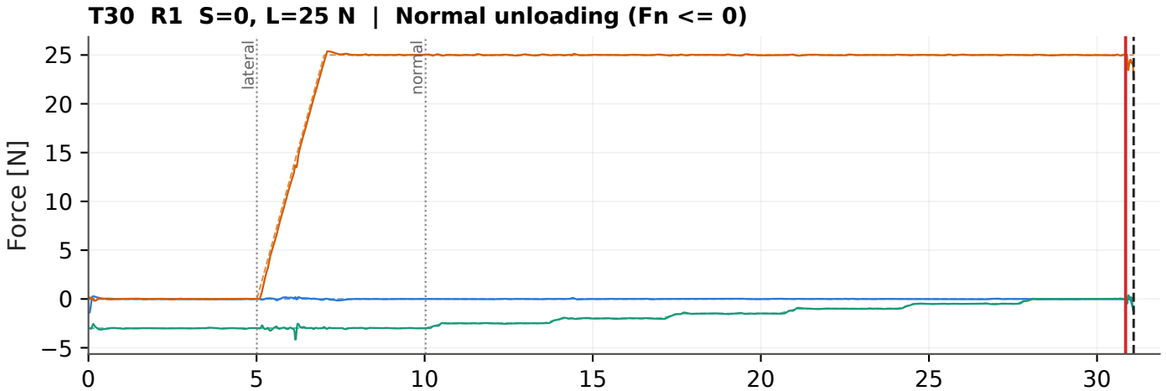
T29 R1 S=-15, L=30 N | Reverse shear / drag



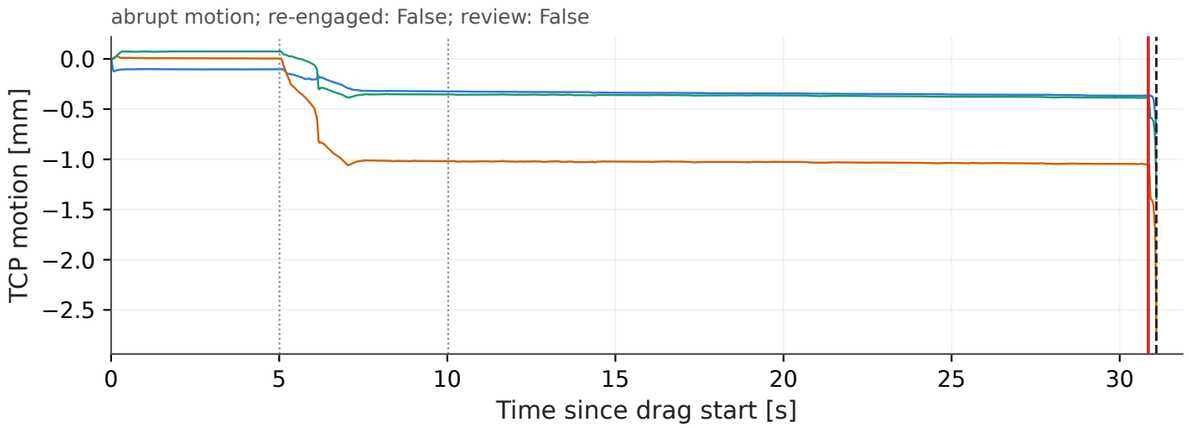
force and motion; re-engaged: True; review: False



T30 R1 S=0, L=25 N | Normal unloading (Fn <= 0)



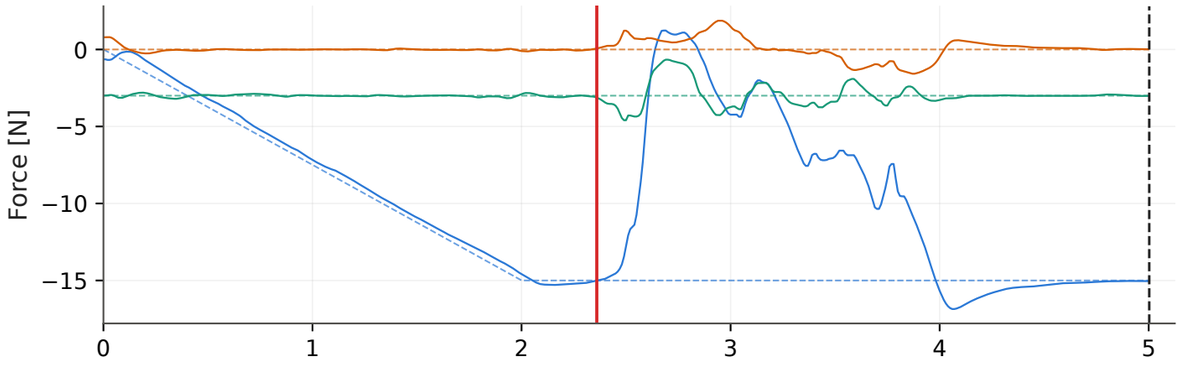
abrupt motion; re-engaged: False; review: False



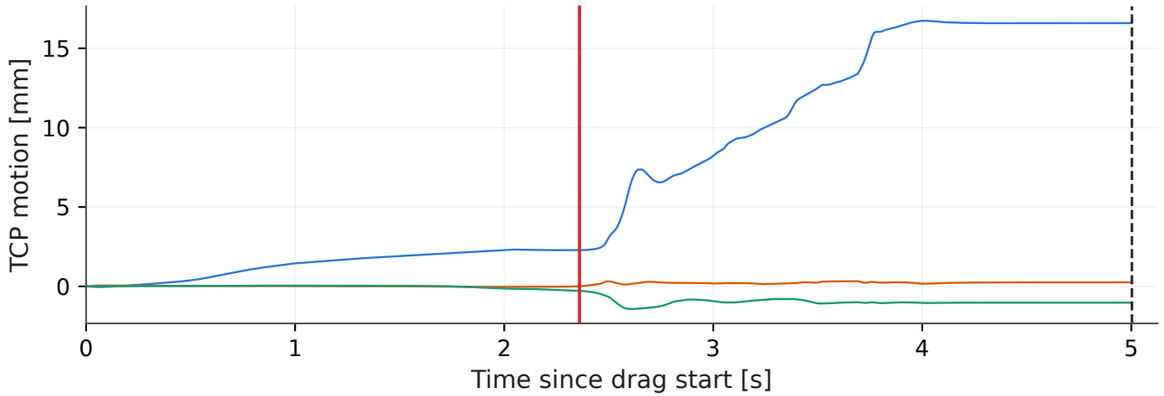
Trial audit | 31–33 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

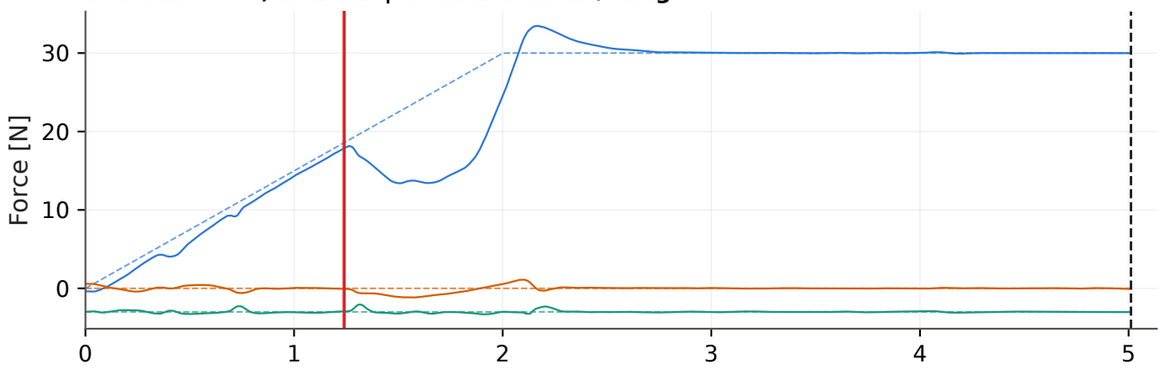
T31 R1 S=-15, L=5 N | Reverse shear / drag



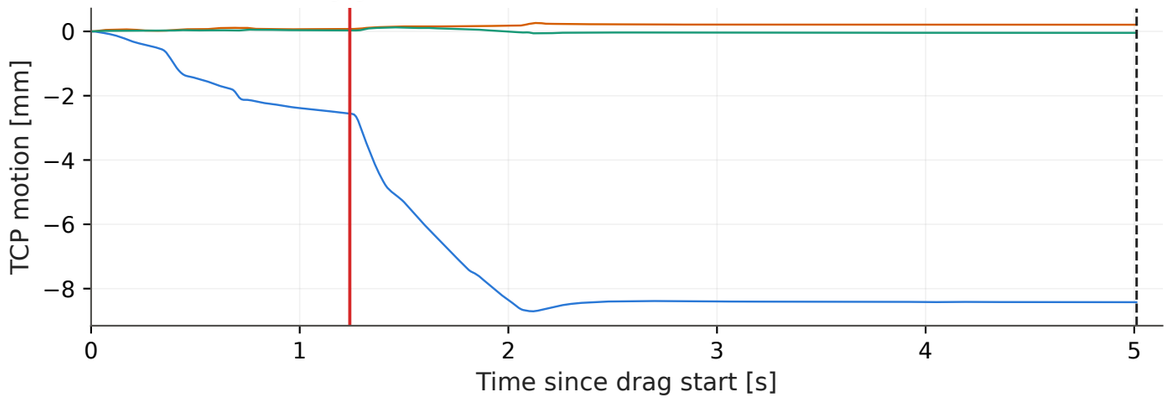
force and motion; re-engaged: True; review: False



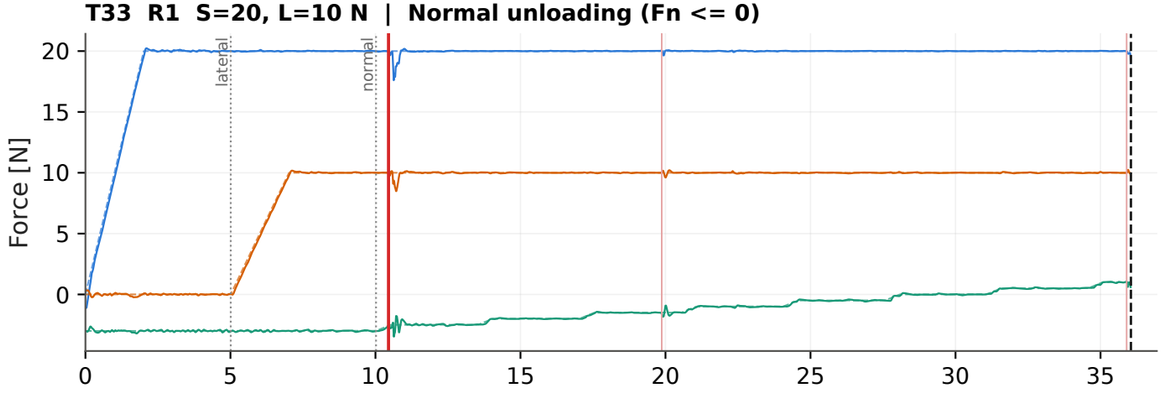
T32 R1 S=30, L=10 N | Forward shear / drag



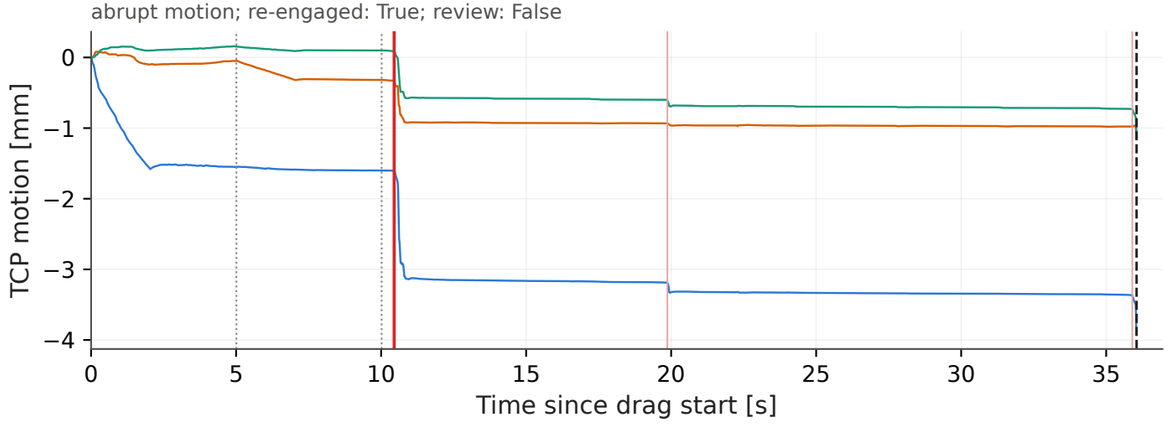
force and motion; re-engaged: True; review: False



T33 R1 S=20, L=10 N | Normal unloading (Fn <= 0)



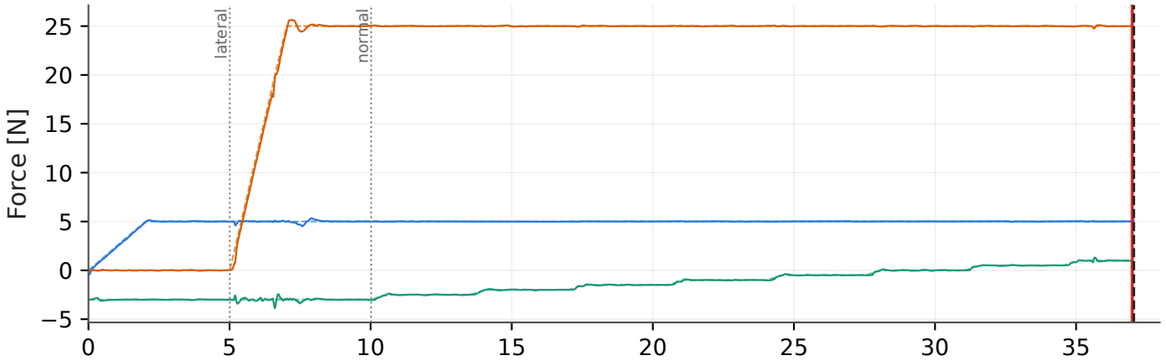
abrupt motion; re-engaged: True; review: False



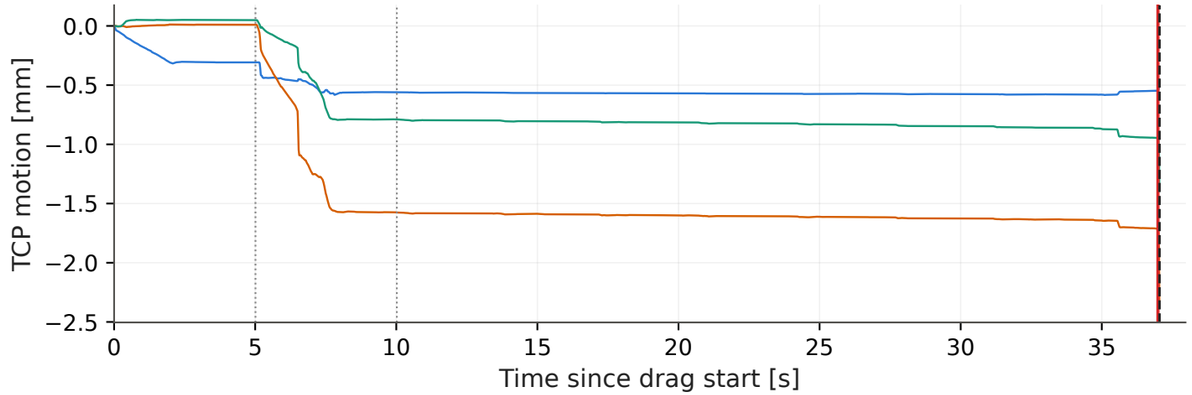
Trial audit | 34–36 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

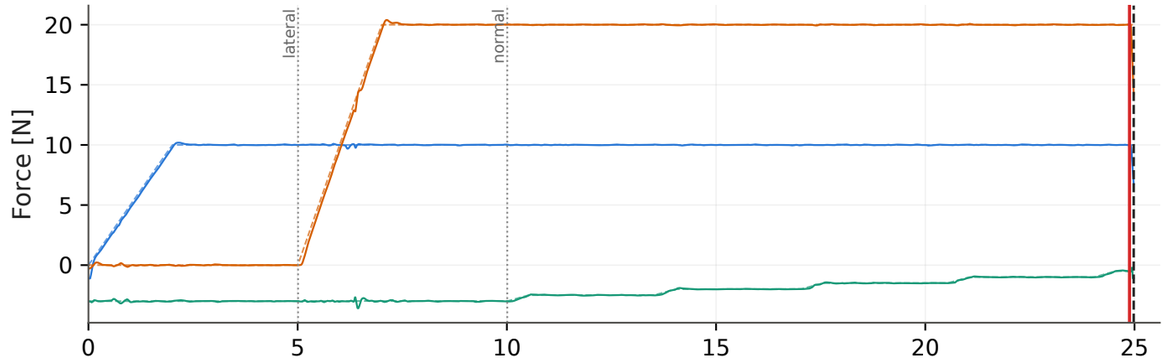
T34 R1 S=5, L=25 N | Normal tension (Fn > 0)



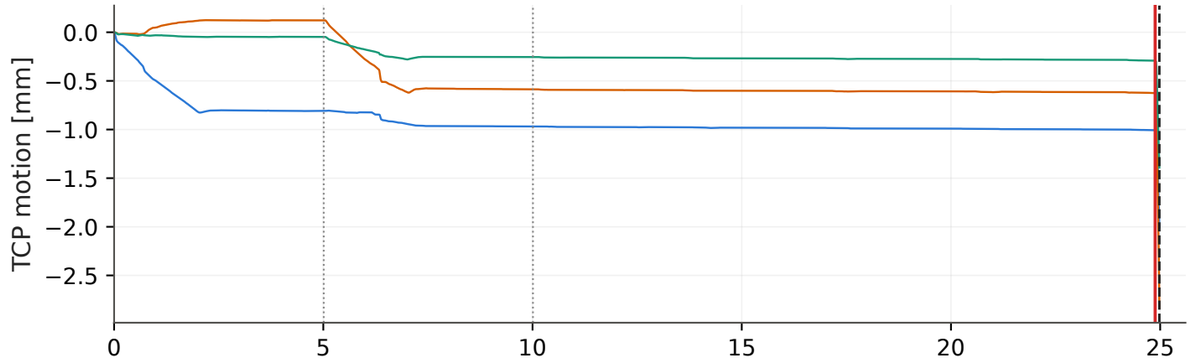
abrupt motion; re-engaged: False; review: False



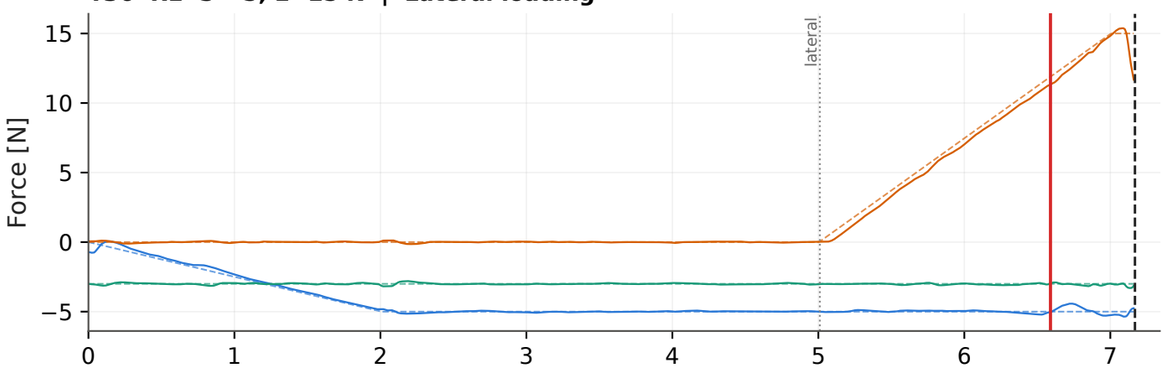
T35 R1 S=10, L=20 N | Normal unloading (Fn <= 0)



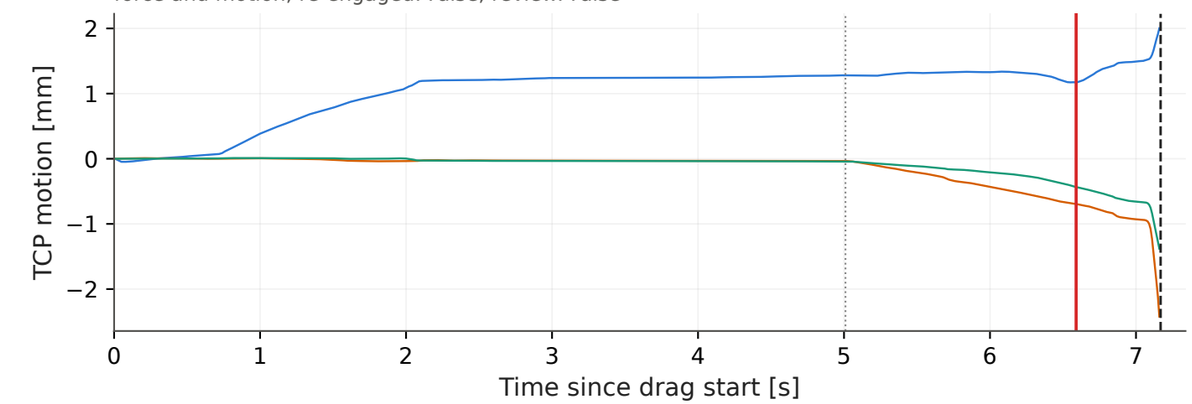
abrupt motion; re-engaged: False; review: False



T36 R1 S=-5, L=15 N | Lateral loading



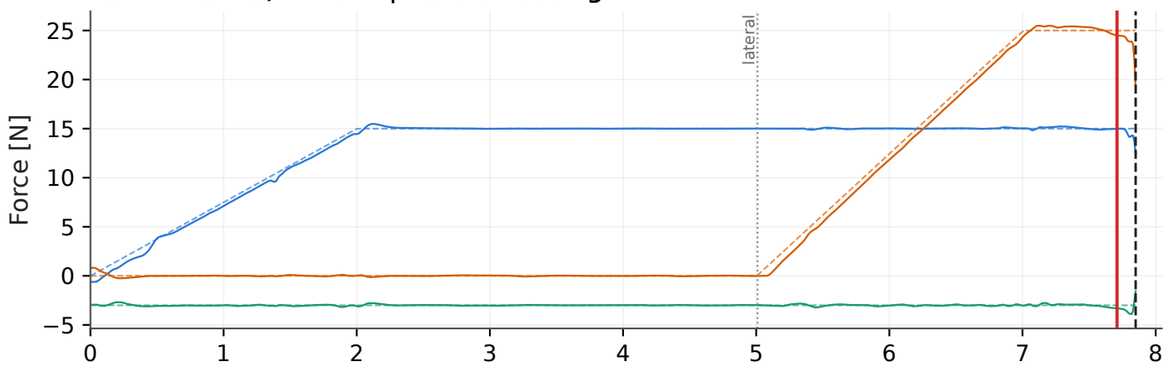
force and motion; re-engaged: False; review: False



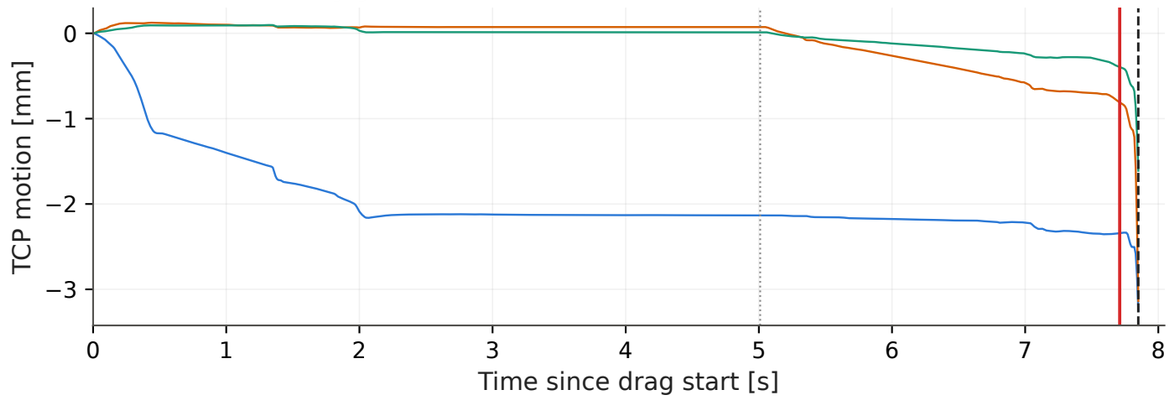
Trial audit | 37–39 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

T37 R1 S=15, L=25 N | Lateral loading



force and motion; re-engaged: False; review: False



T38 R1 S=-5, L=30 N | Lateral loading

force and motion; re-engaged: False; review: True

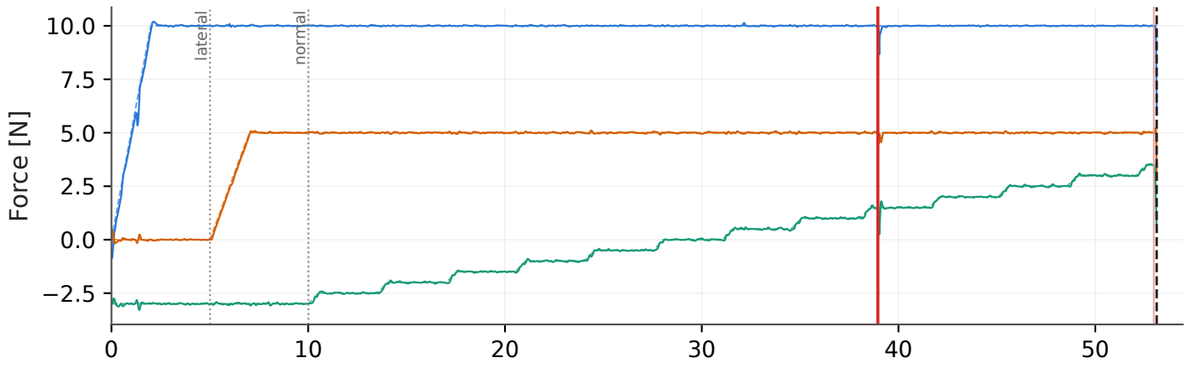
T39 R1 S=-10, L=20 N | Reverse shear / drag

abrupt motion; re-engaged: True; review: False

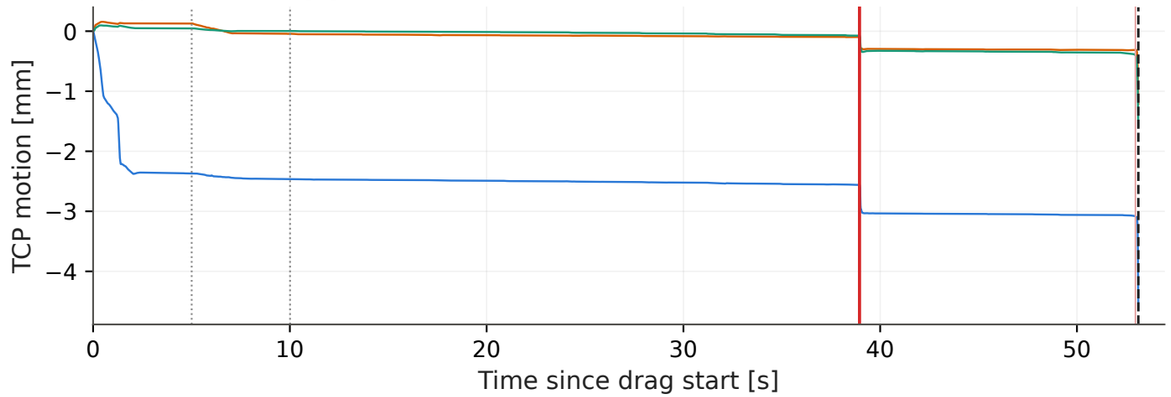
Trial audit | 40–42 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

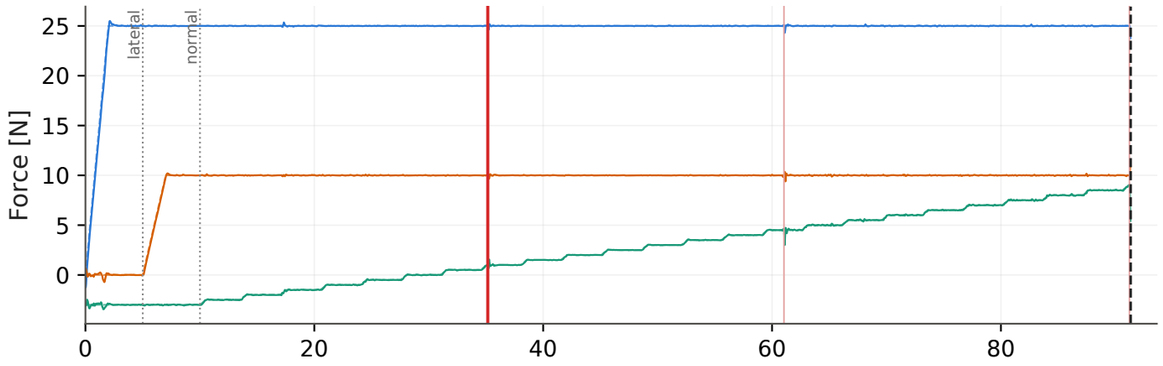
T40 R1 S=10, L=5 N | Normal tension (Fn > 0)



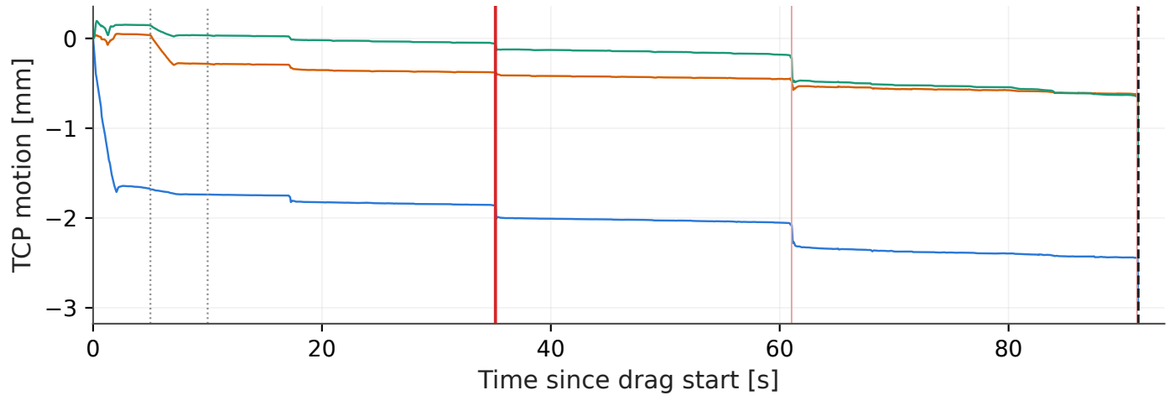
abrupt motion; re-engaged: True; review: True



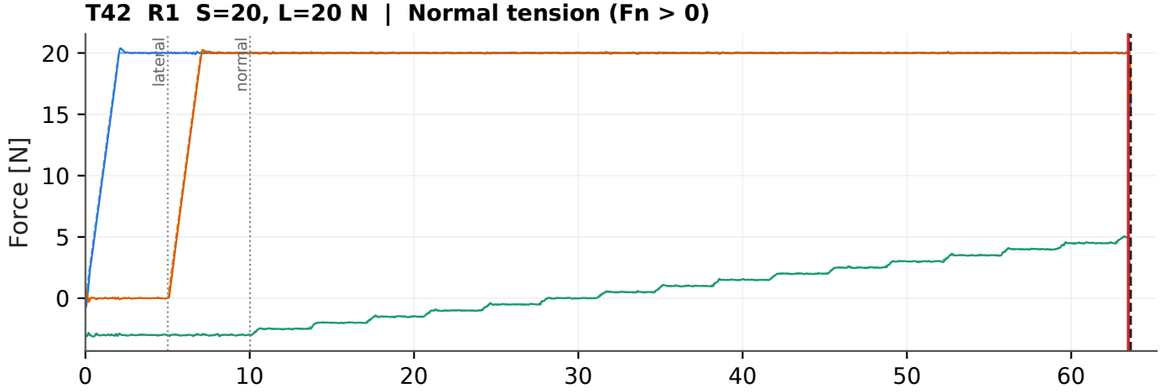
T41 R1 S=25, L=10 N | Normal tension (Fn > 0)



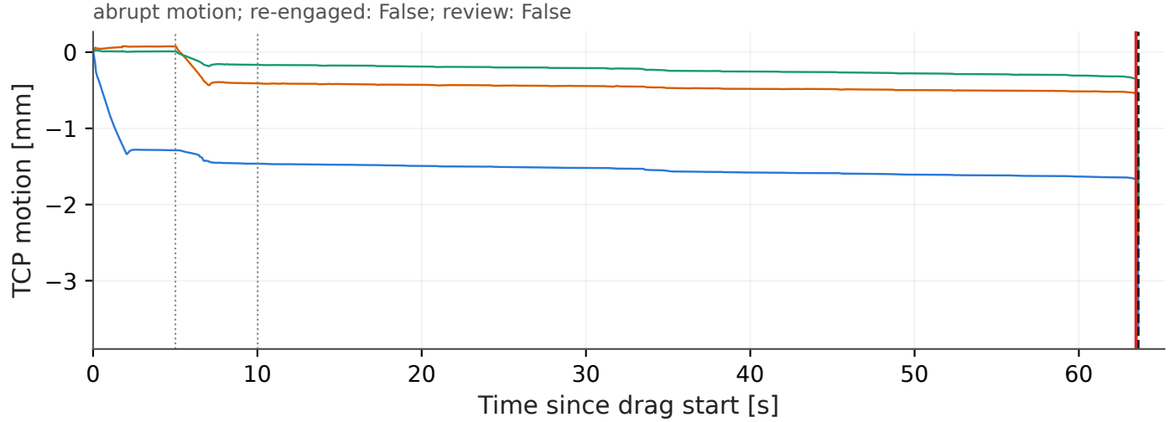
abrupt motion; re-engaged: True; review: True



T42 R1 S=20, L=20 N | Normal tension (Fn > 0)



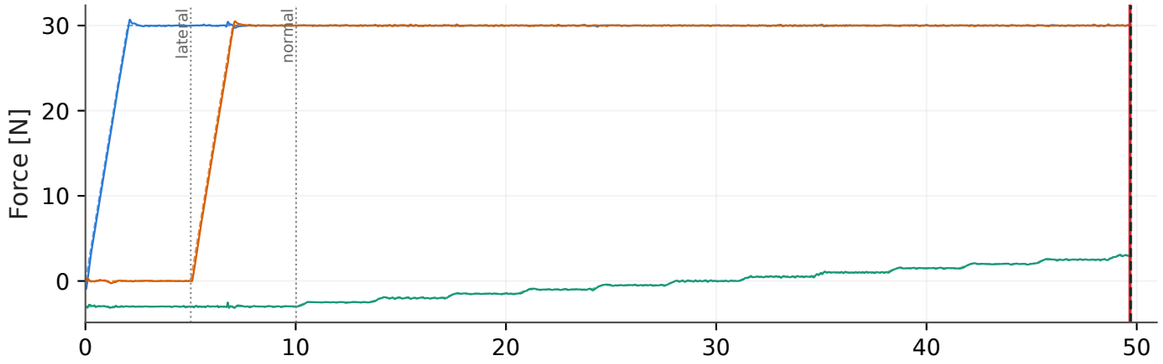
abrupt motion; re-engaged: False; review: False



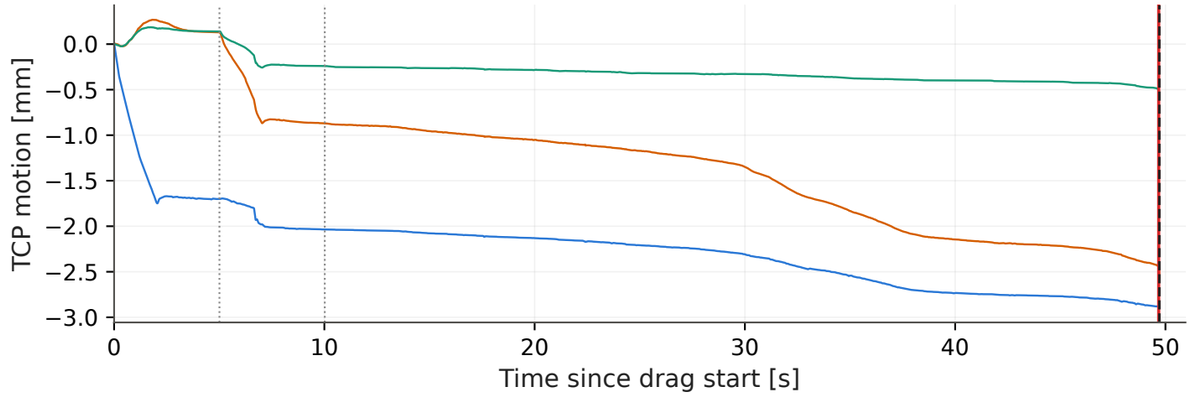
Trial audit | 43–45 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

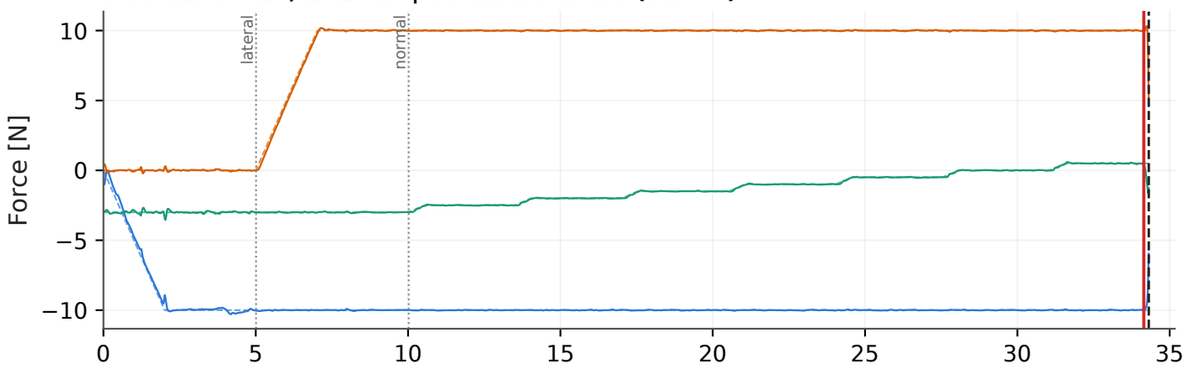
T43 R1 S=30, L=30 N | Normal tension (Fn > 0)



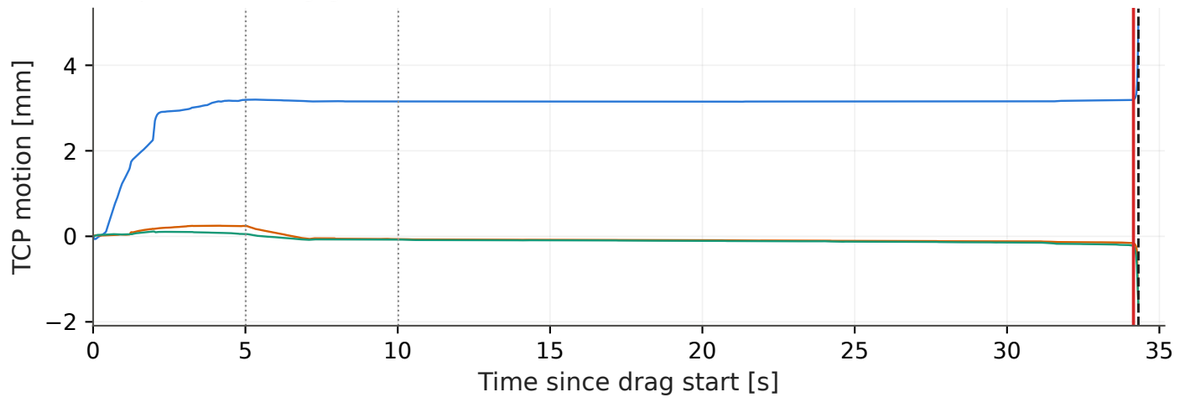
abrupt motion; re-engaged: False; review: True



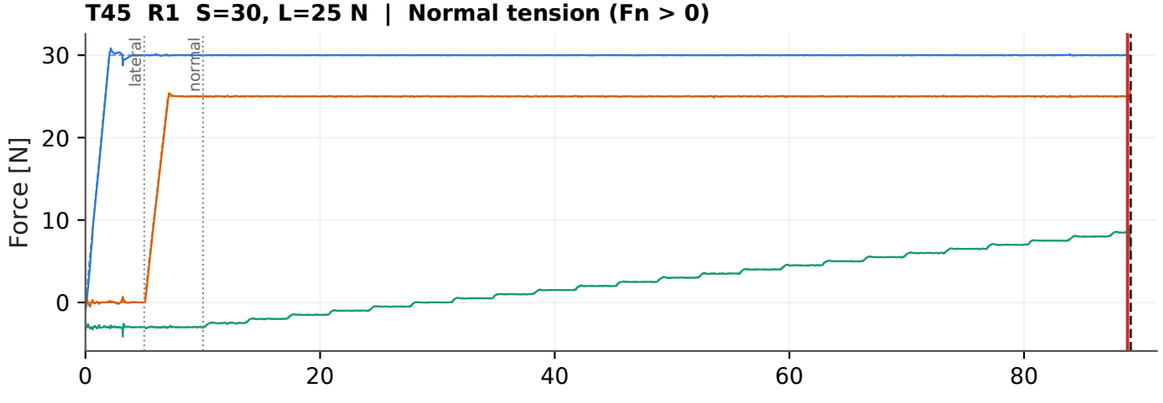
T44 R1 S=-10, L=10 N | Normal tension (Fn > 0)



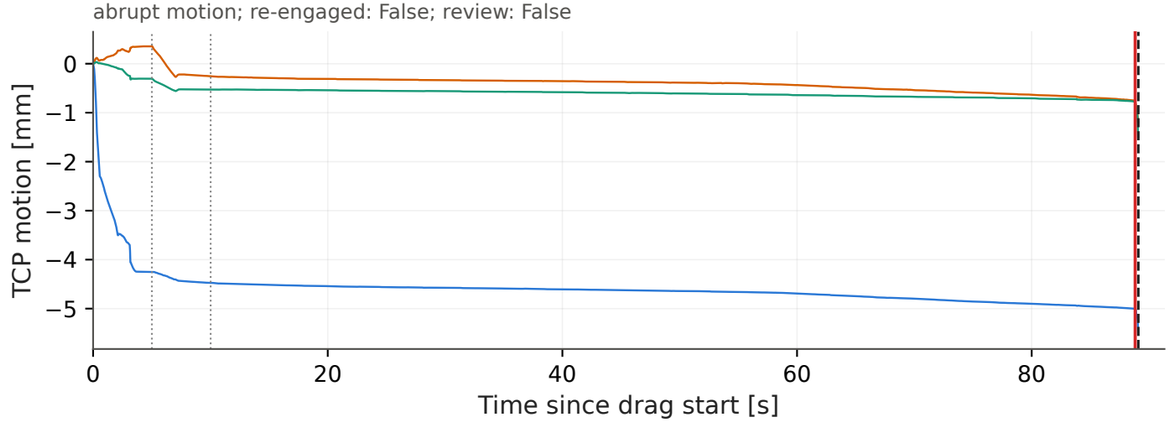
abrupt motion; re-engaged: False; review: True



T45 R1 S=30, L=25 N | Normal tension (Fn > 0)



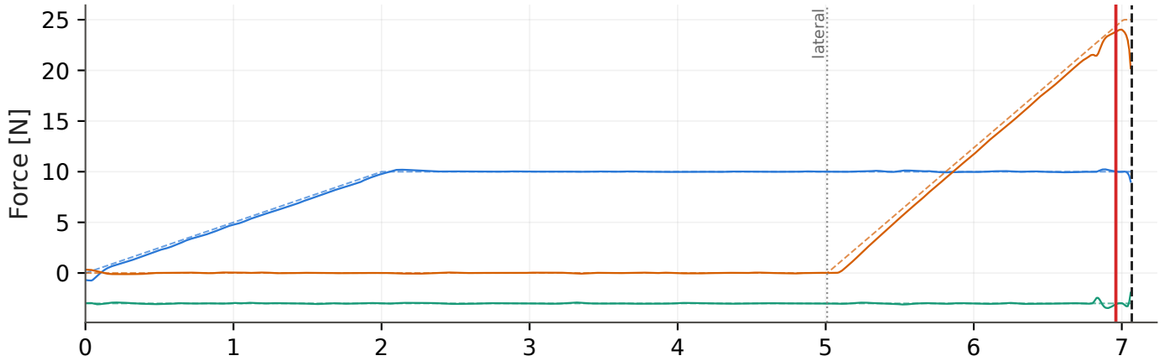
abrupt motion; re-engaged: False; review: False



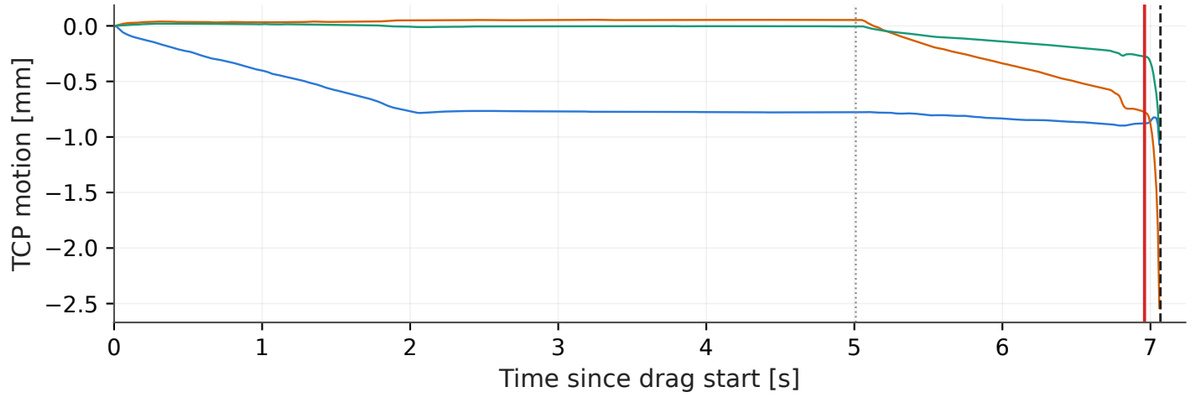
Trial audit | 46–48 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

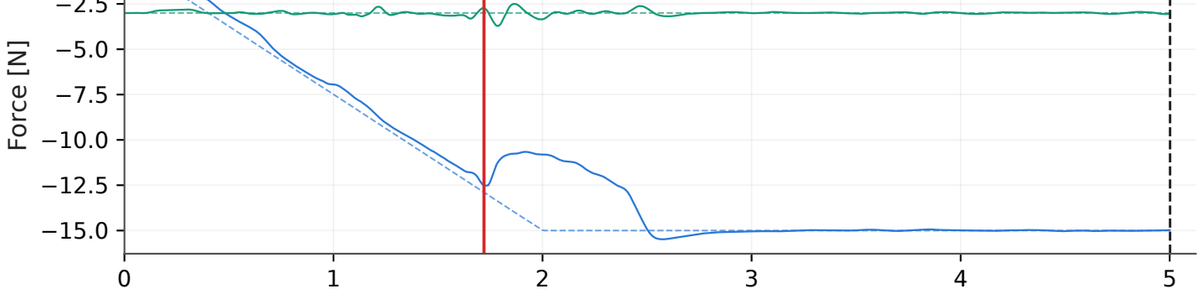
T46 R1 S=10, L=25 N | Lateral loading



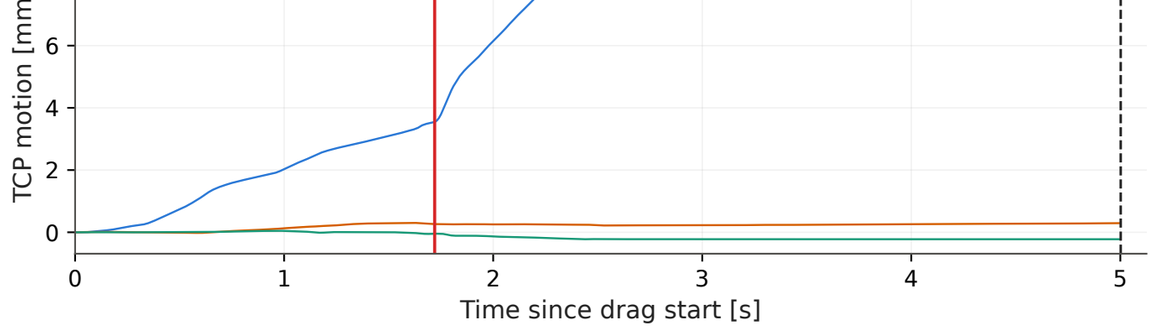
force and motion; re-engaged: False; review: False



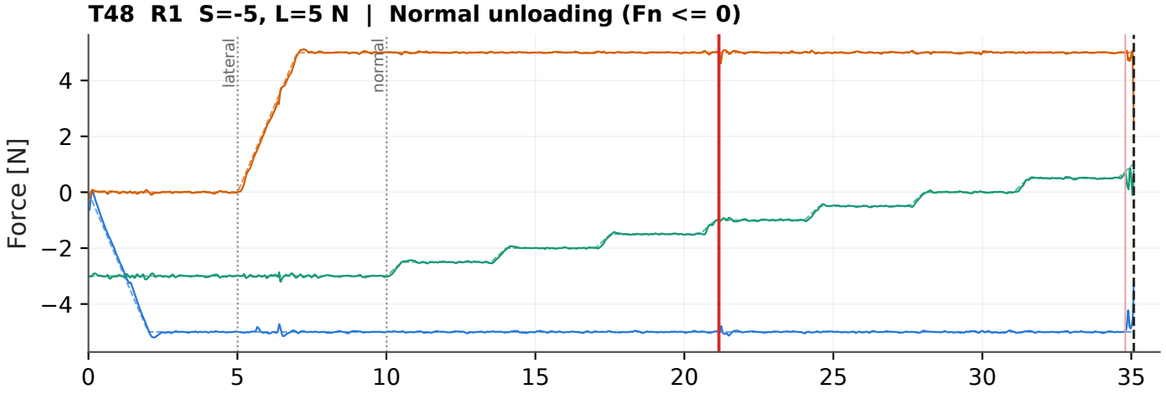
T47 R1 S=-15, L=20 N | Reverse shear / drag



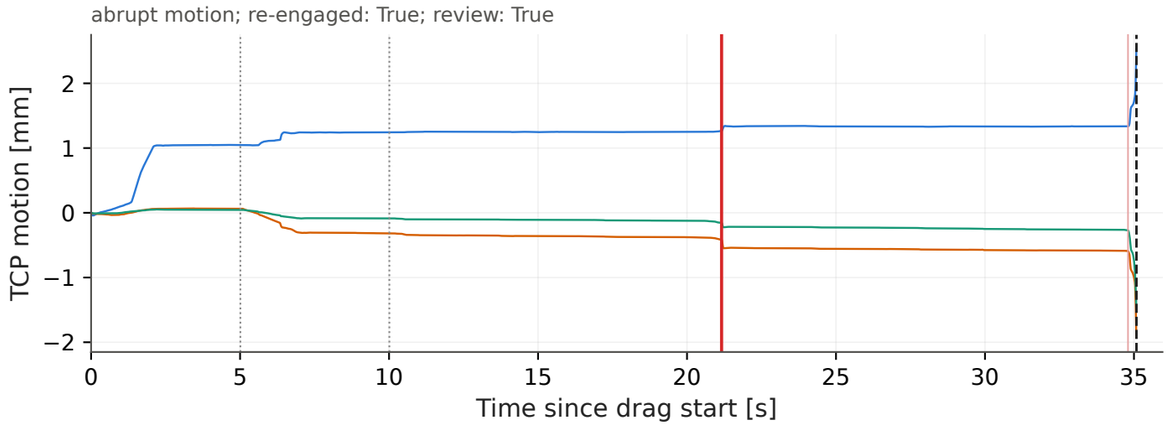
force and motion; re-engaged: True; review: False



T48 R1 S=-5, L=5 N | Normal unloading (Fn <= 0)



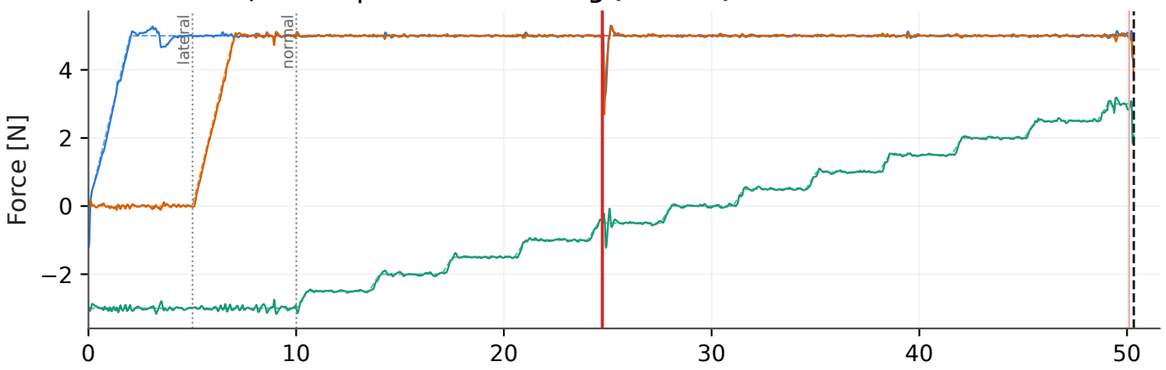
abrupt motion; re-engaged: True; review: True



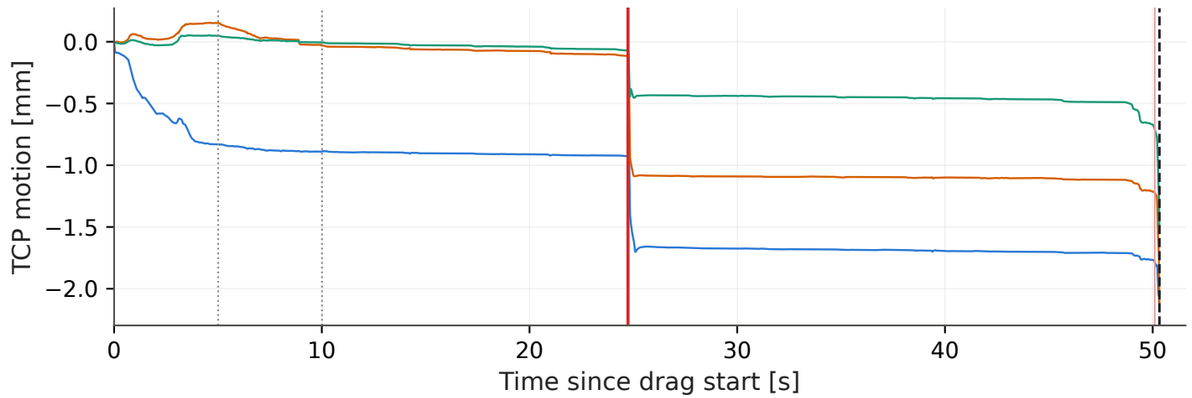
Trial audit | 49–51 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

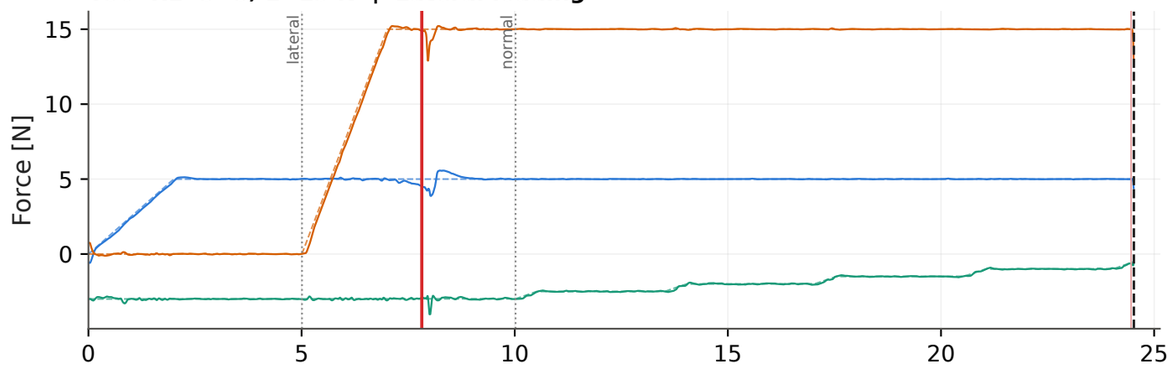
T49 R1 S=5, L=5 N | Normal unloading (Fn <= 0)



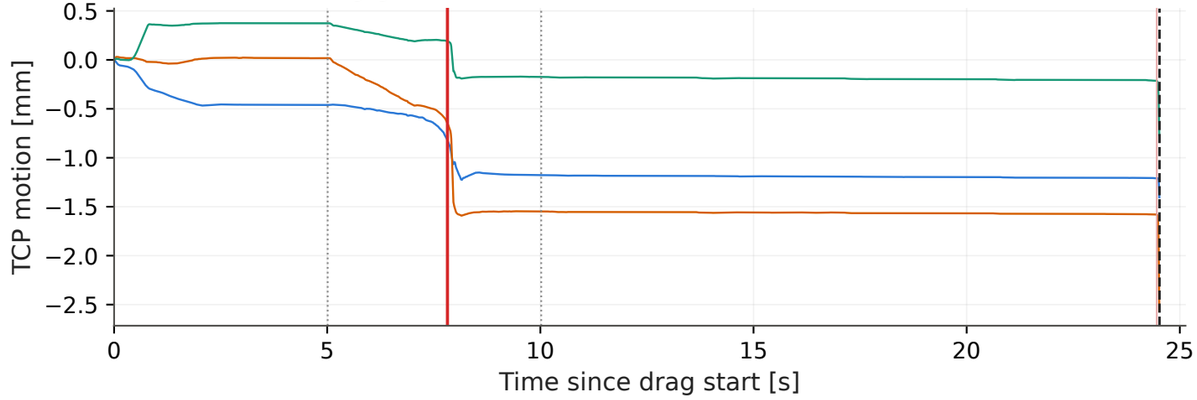
abrupt motion; re-engaged: True; review: False



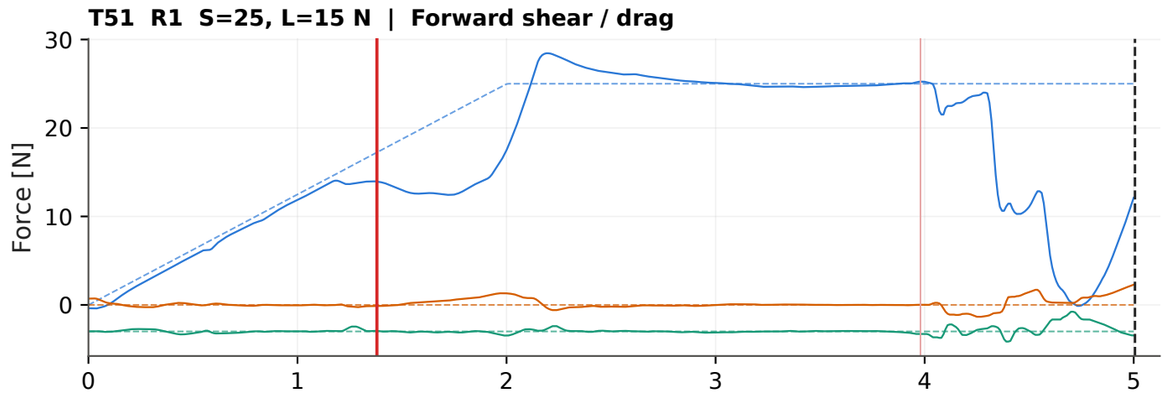
T50 R1 S=5, L=15 N | Lateral loading



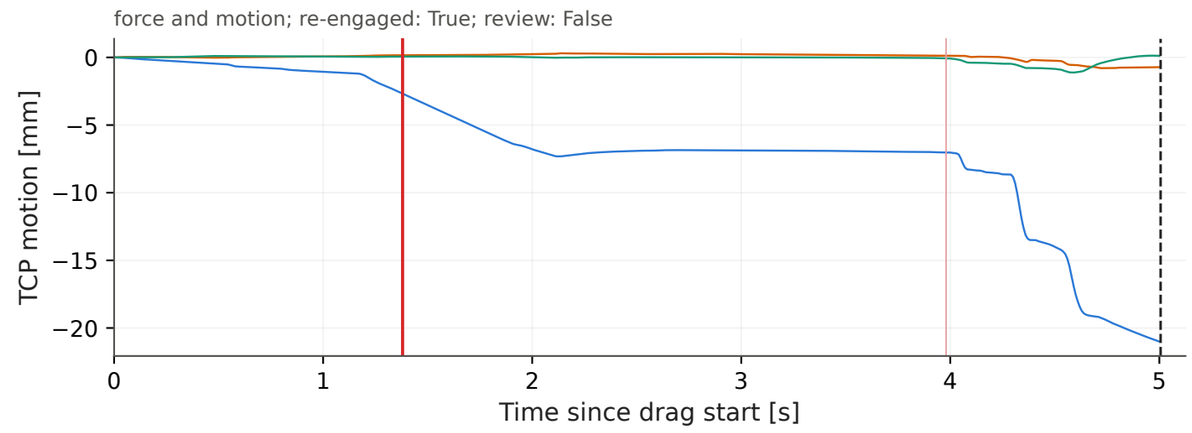
force and motion; re-engaged: True; review: False



T51 R1 S=25, L=15 N | Forward shear / drag



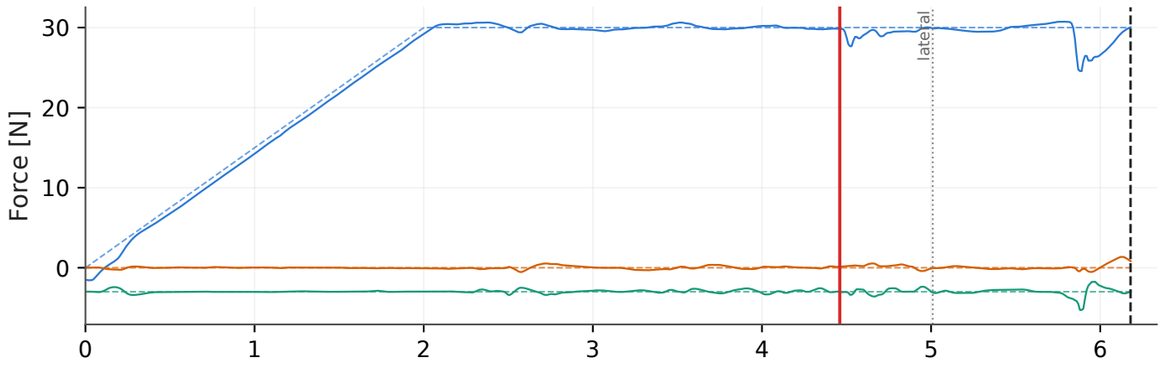
force and motion; re-engaged: True; review: False



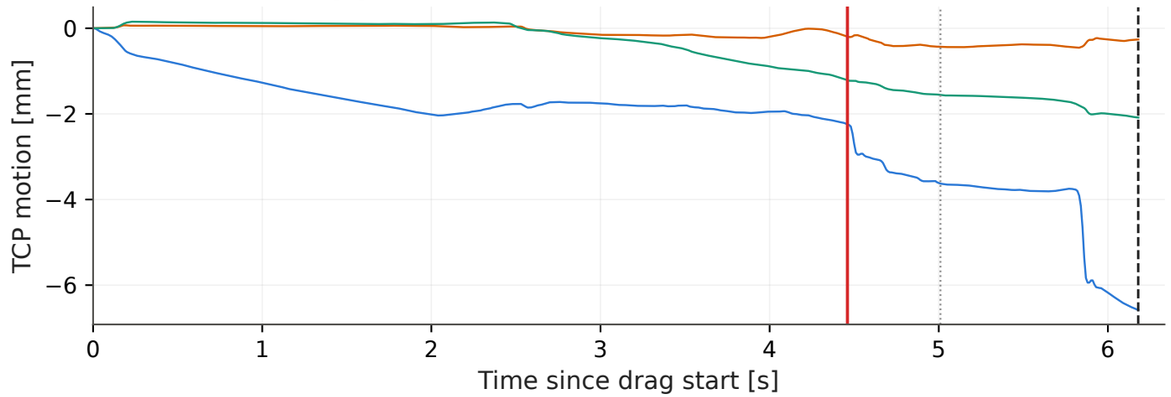
Trial audit | 52–54 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

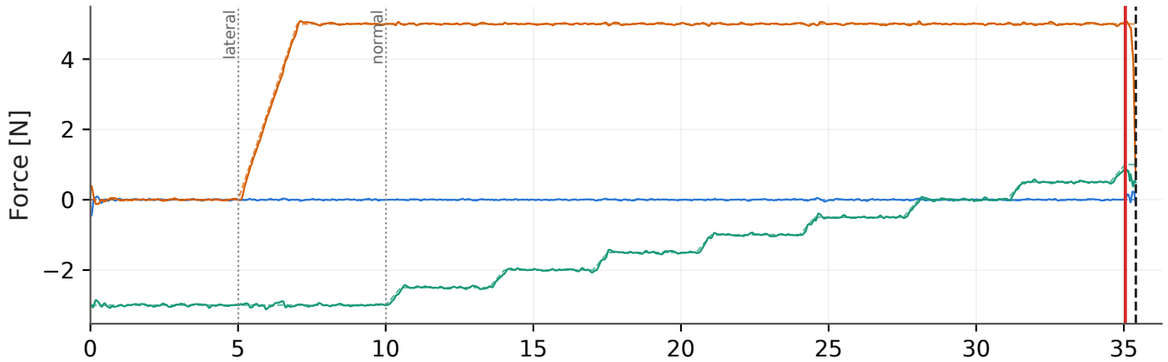
T52 R1 S=30, L=0 N | Forward shear / drag



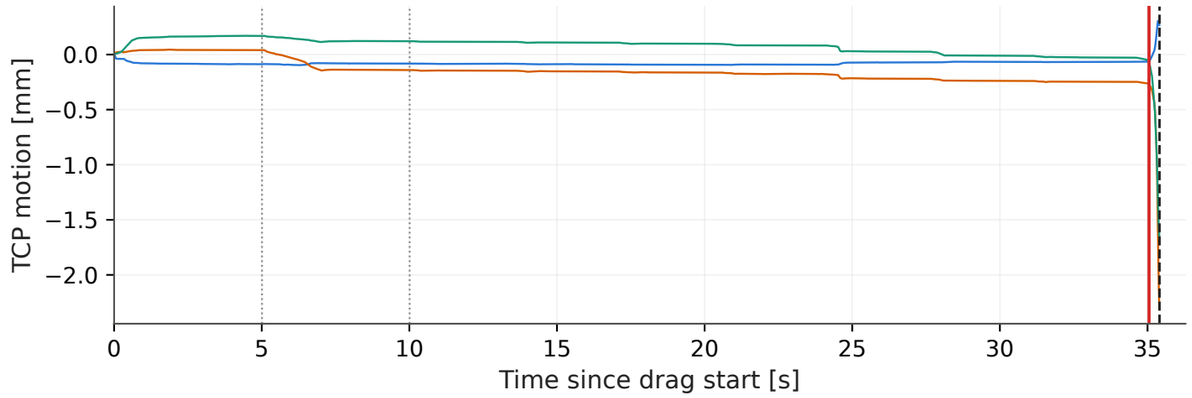
force and motion; re-engaged: False; review: False



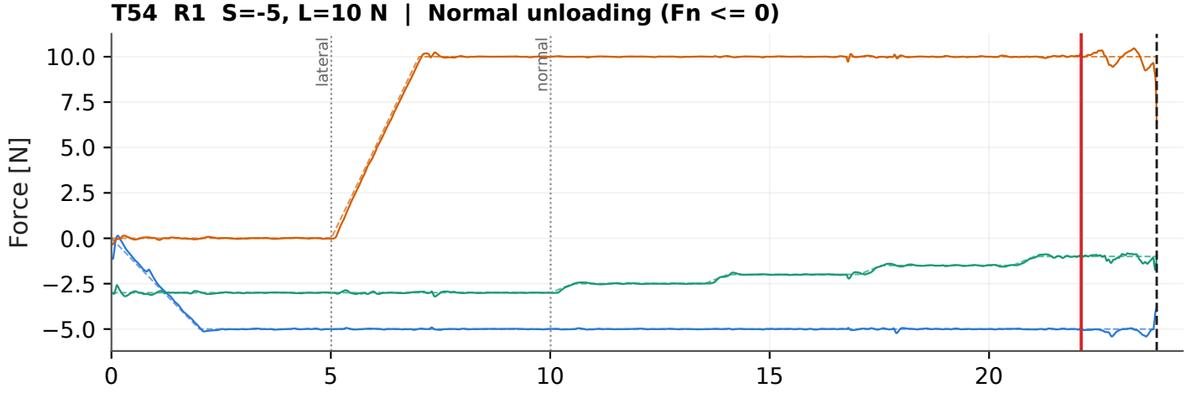
T53 R1 S=-0, L=5 N | Normal tension (Fn > 0)



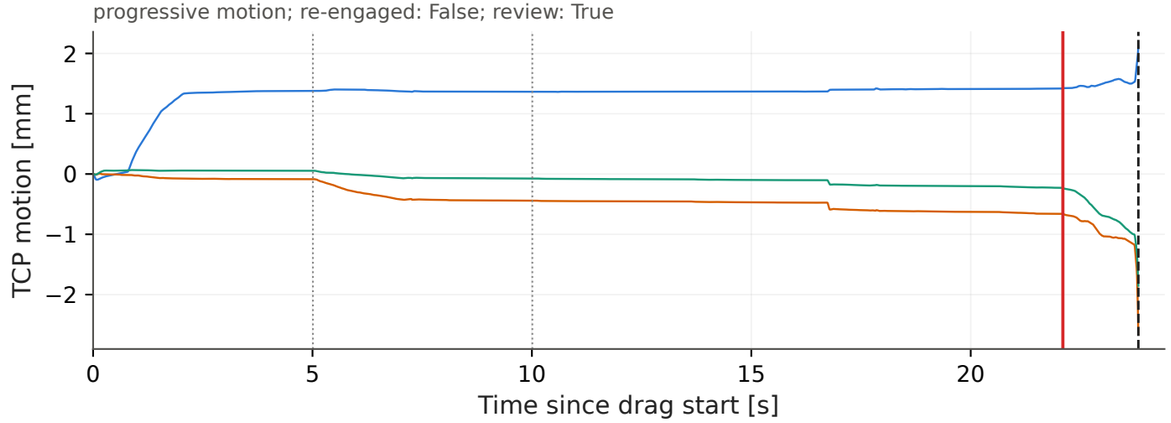
abrupt motion; re-engaged: False; review: False



T54 R1 S=-5, L=10 N | Normal unloading (Fn <= 0)



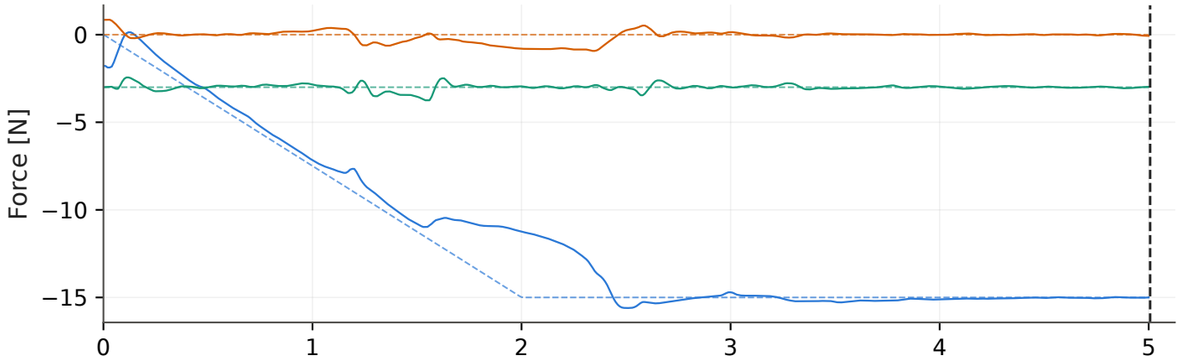
progressive motion; re-engaged: False; review: True



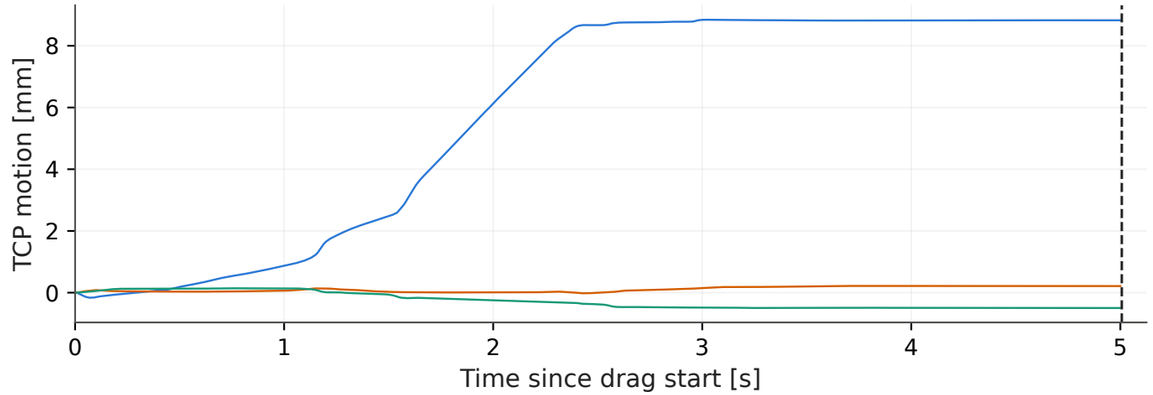
Trial audit | 55–57 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

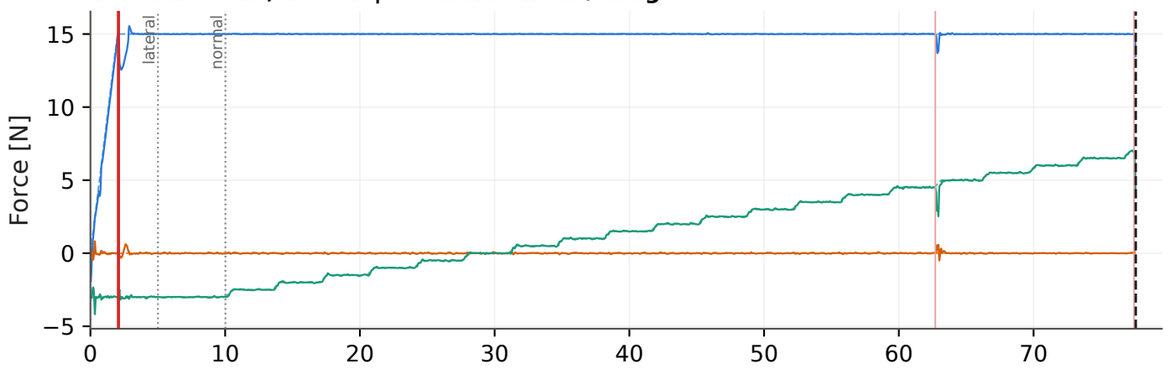
T55 R1 S=-15, L=15 N | Onset unresolved



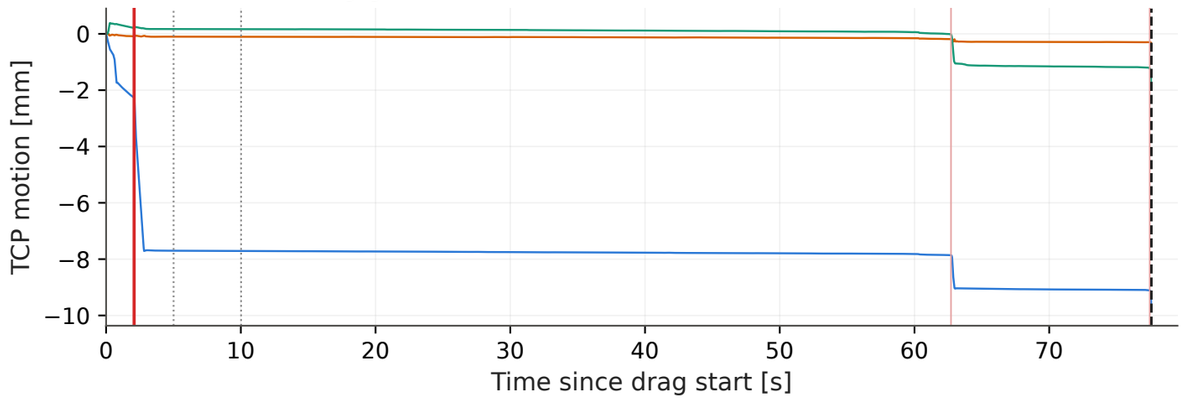
unresolved; re-engaged: False; review: True



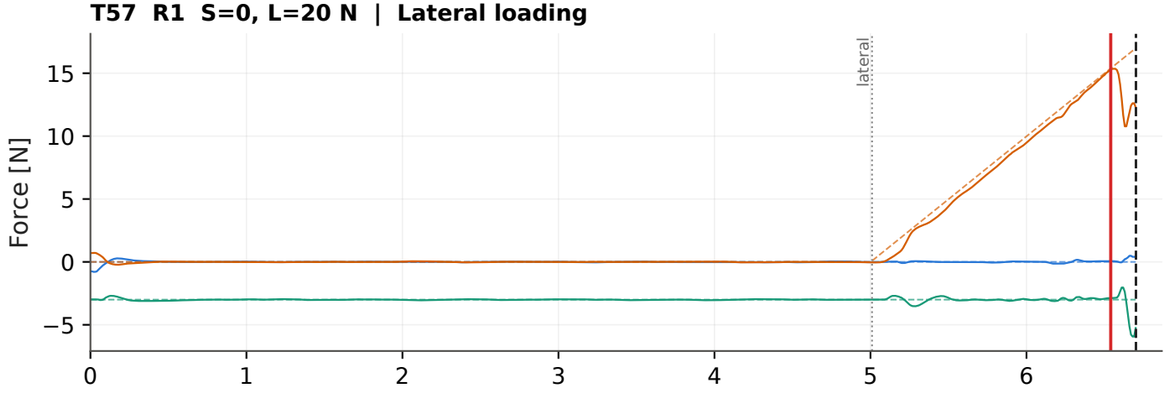
T56 R1 S=15, L=0 N | Forward shear / drag



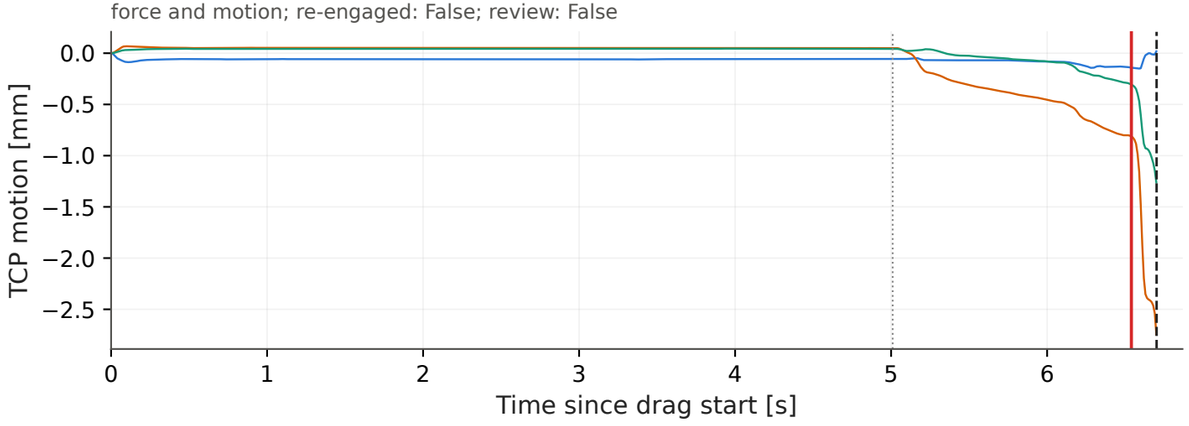
force and motion; re-engaged: True; review: False



T57 R1 S=0, L=20 N | Lateral loading



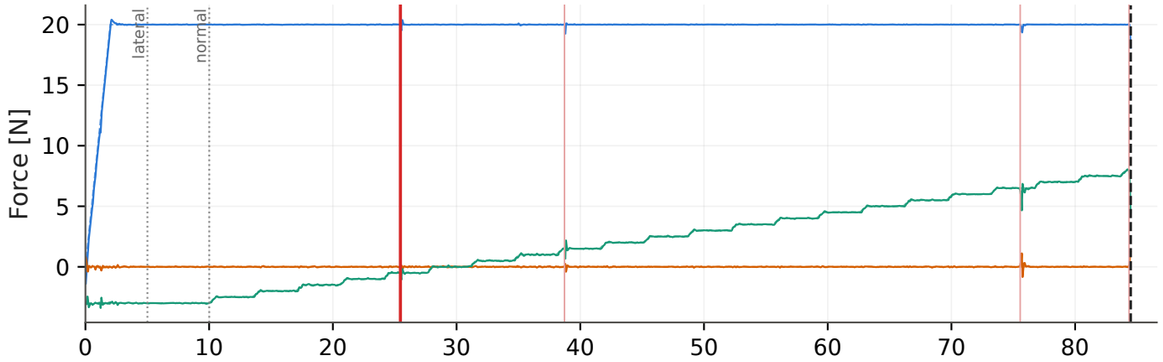
force and motion; re-engaged: False; review: False



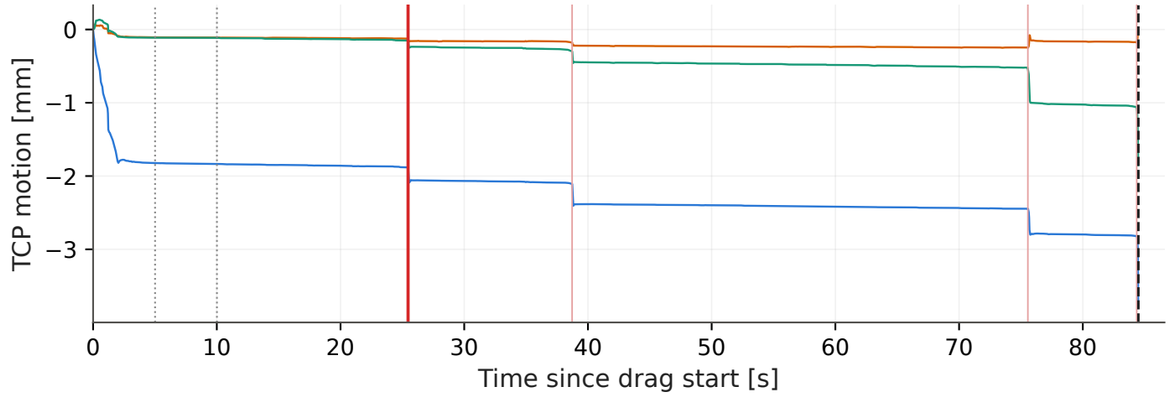
Trial audit | 58–60 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

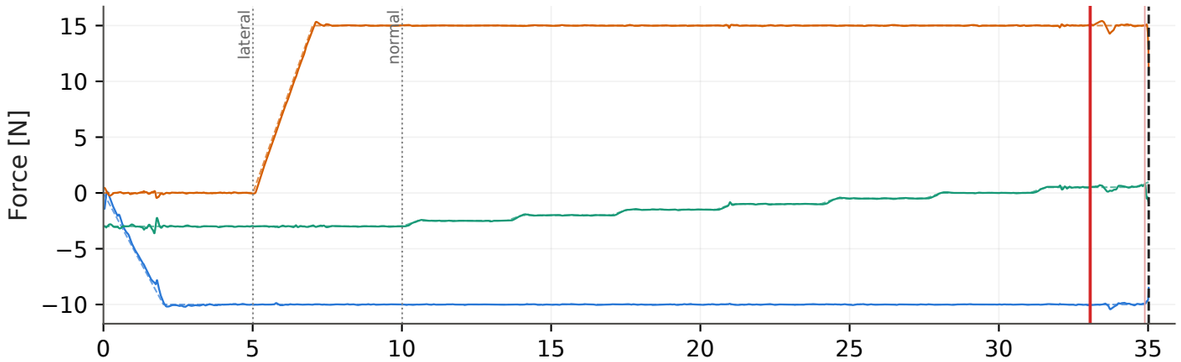
T58 R1 S=20, L=0 N | Normal unloading (Fn <= 0)



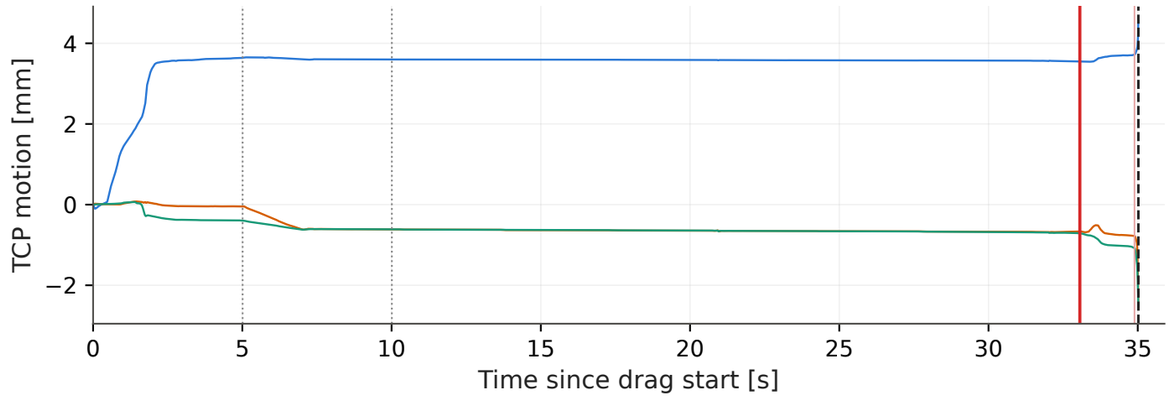
abrupt motion; re-engaged: True; review: False



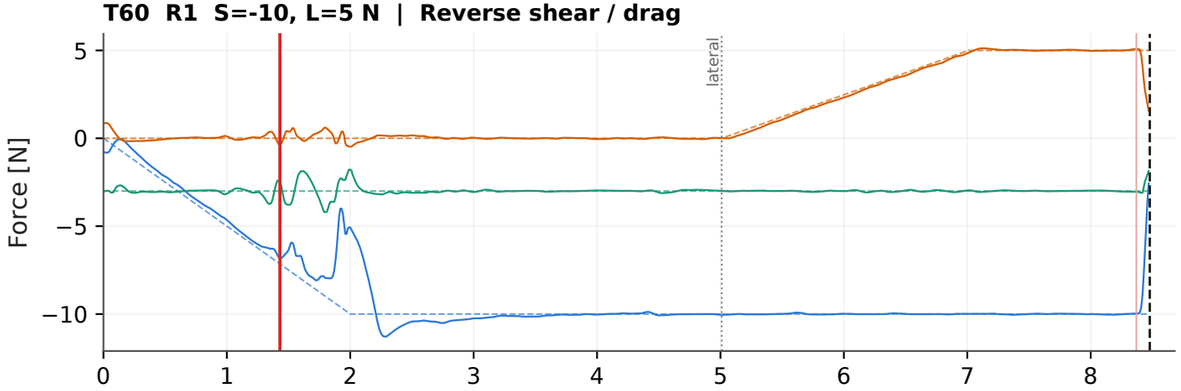
T59 R1 S=-10, L=15 N | Normal tension (Fn > 0)



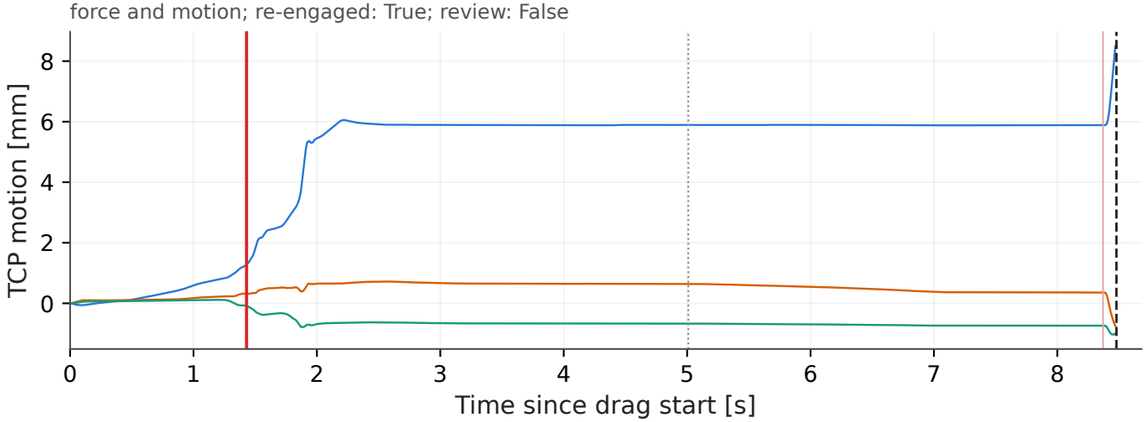
progressive motion; re-engaged: True; review: True



T60 R1 S=-10, L=5 N | Reverse shear / drag



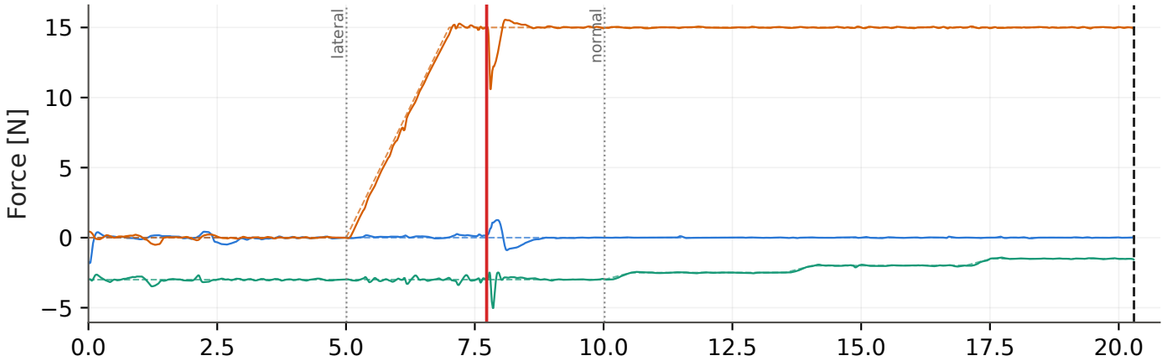
force and motion; re-engaged: True; review: False



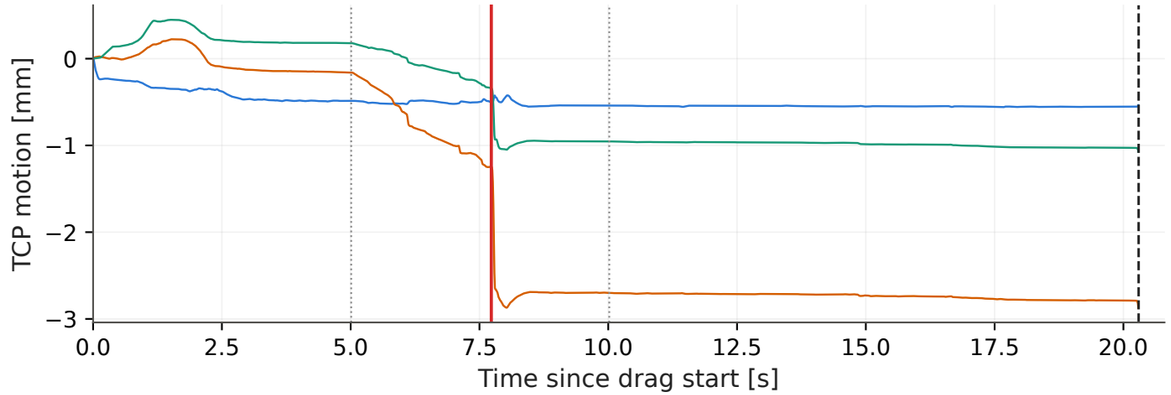
Trial audit | 61–63 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

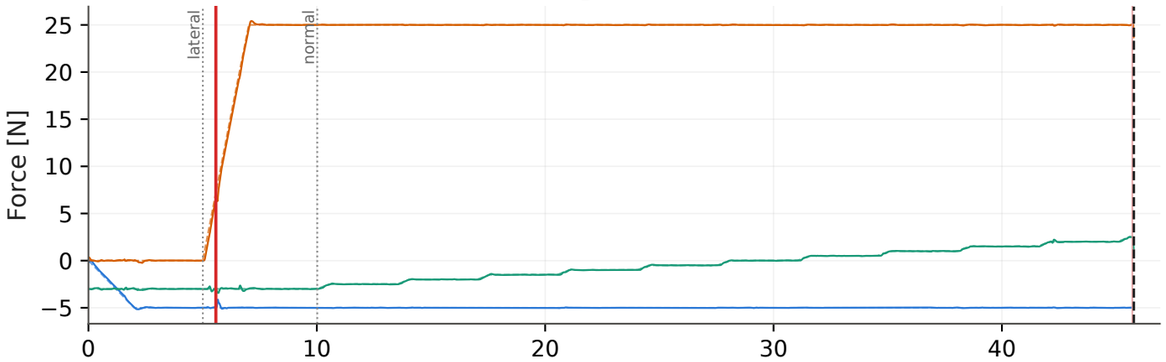
T61 R1 S=0, L=15 N | Lateral loading



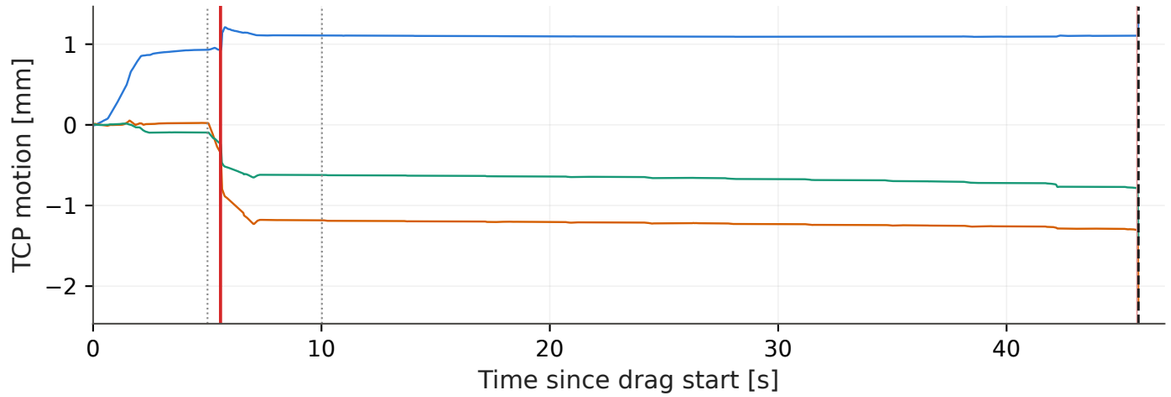
force and motion; re-engaged: True; review: False



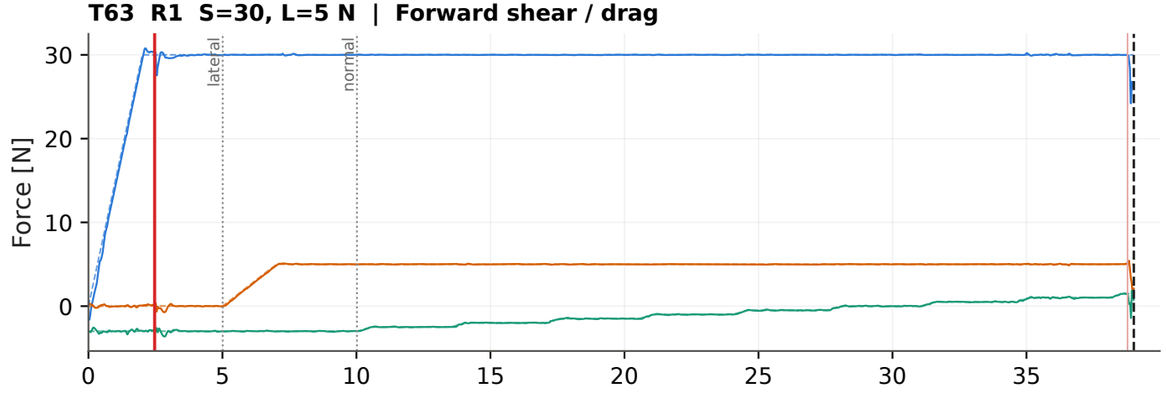
T62 R1 S=-5, L=25 N | Lateral loading



force and motion; re-engaged: True; review: True



T63 R1 S=30, L=5 N | Forward shear / drag



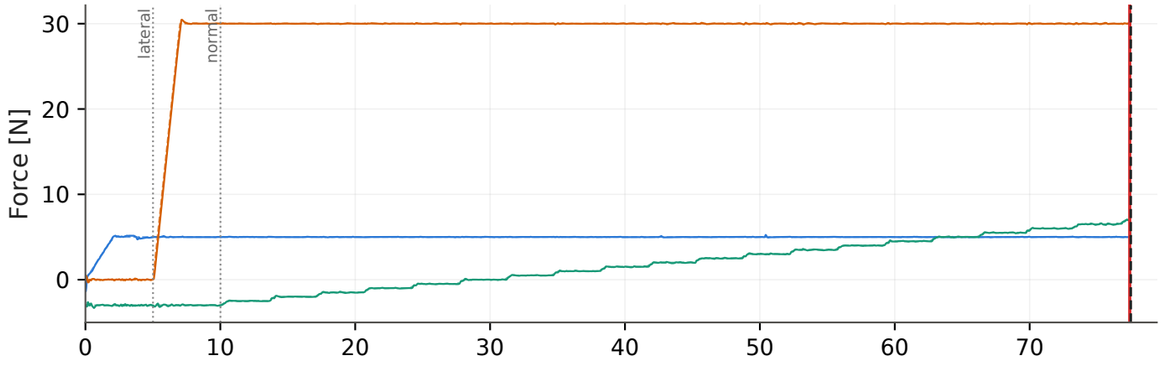
force and motion; re-engaged: True; review: False



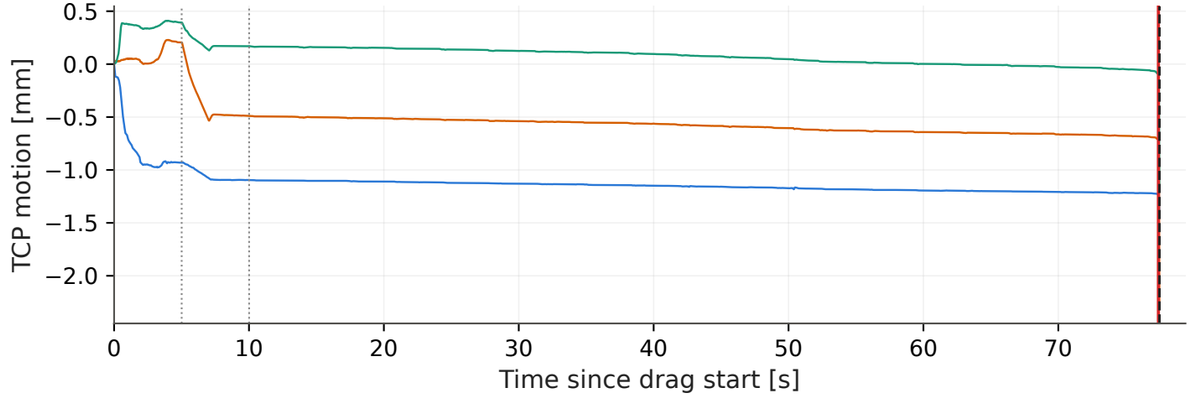
Trial audit | 64–66 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

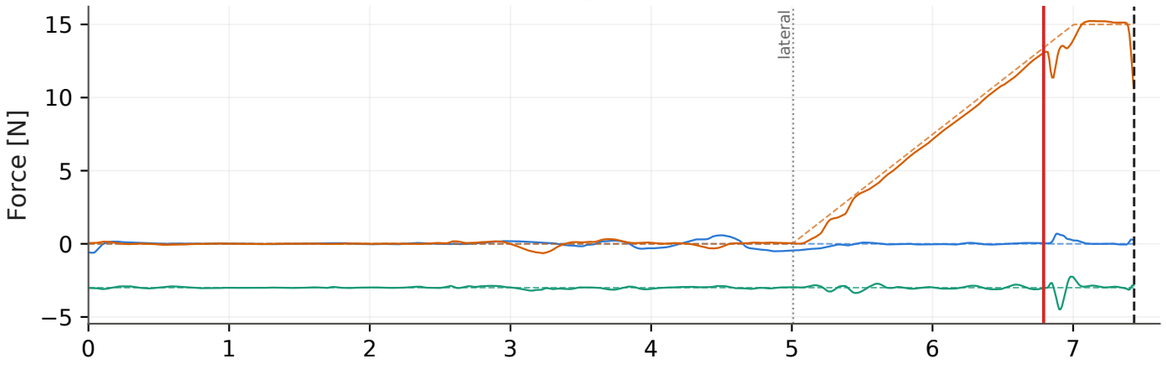
T64 R1 S=5, L=30 N | Normal tension (Fn > 0)



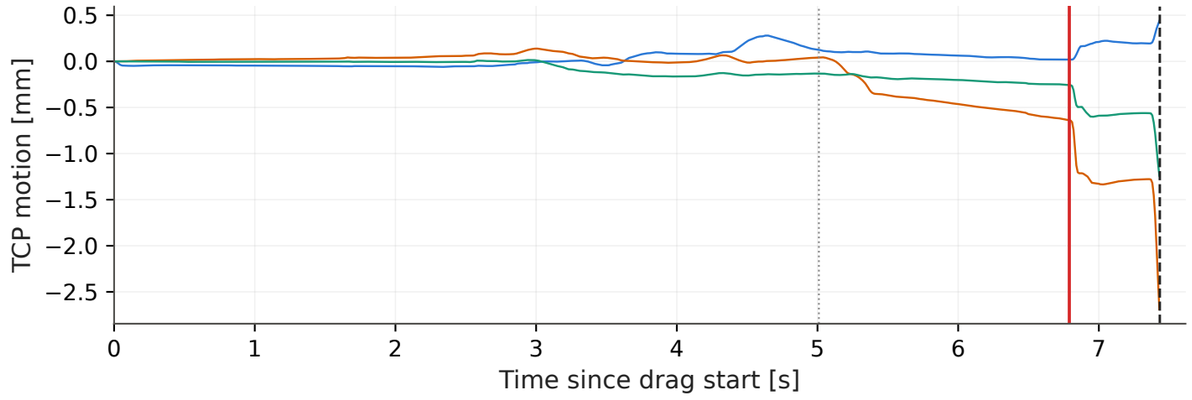
force and motion; re-engaged: False; review: False



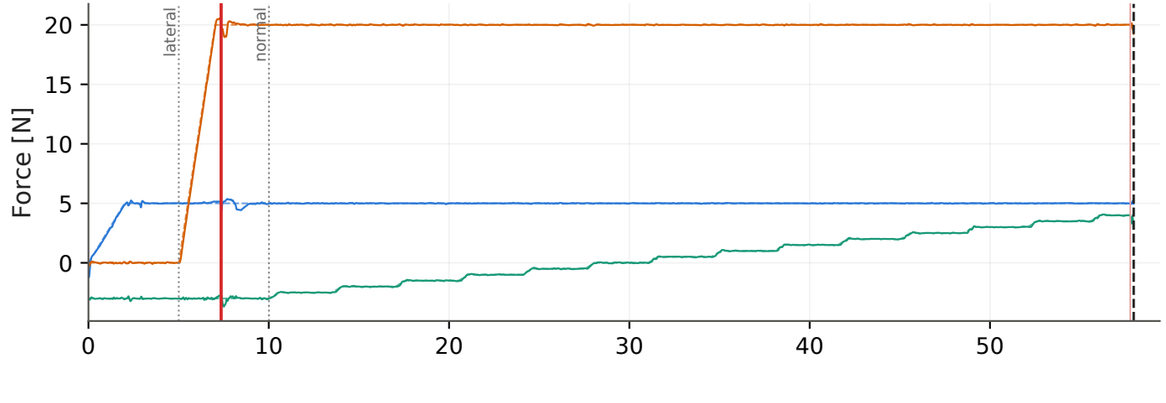
T65 R1 S=-0, L=15 N | Lateral loading



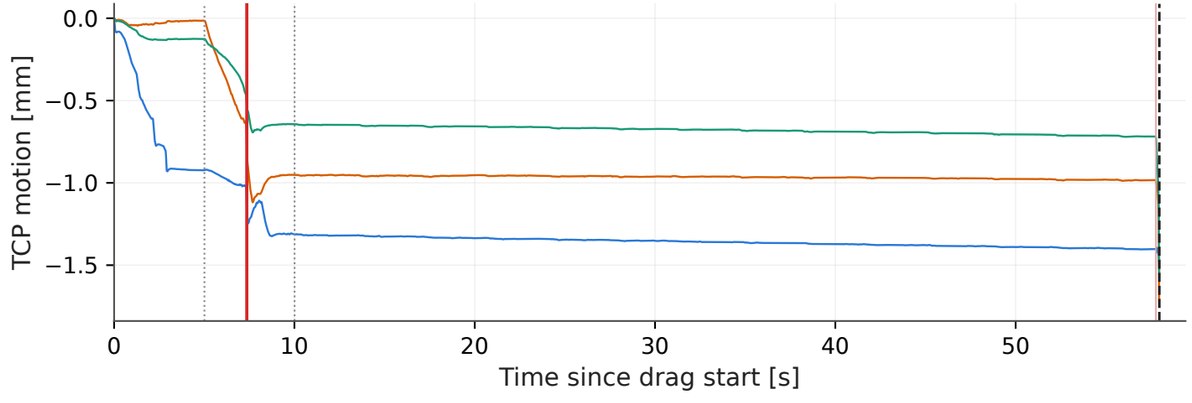
force and motion; re-engaged: False; review: False



T66 R1 S=5, L=20 N | Lateral loading



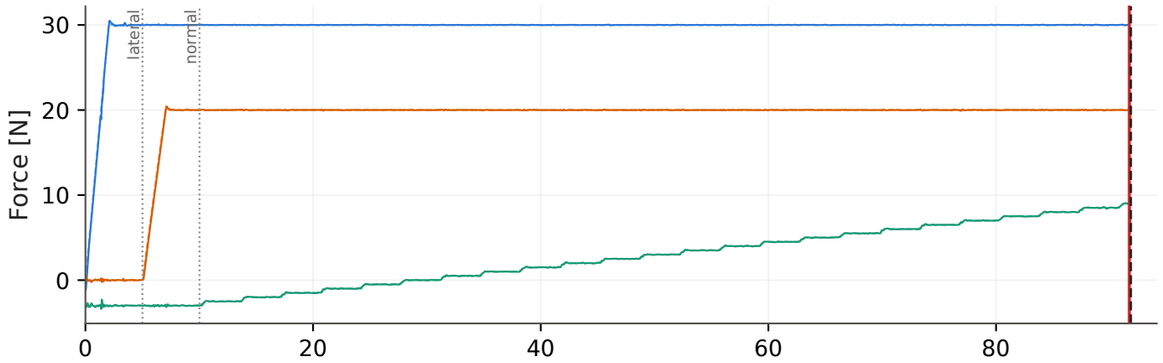
force and motion; re-engaged: True; review: True



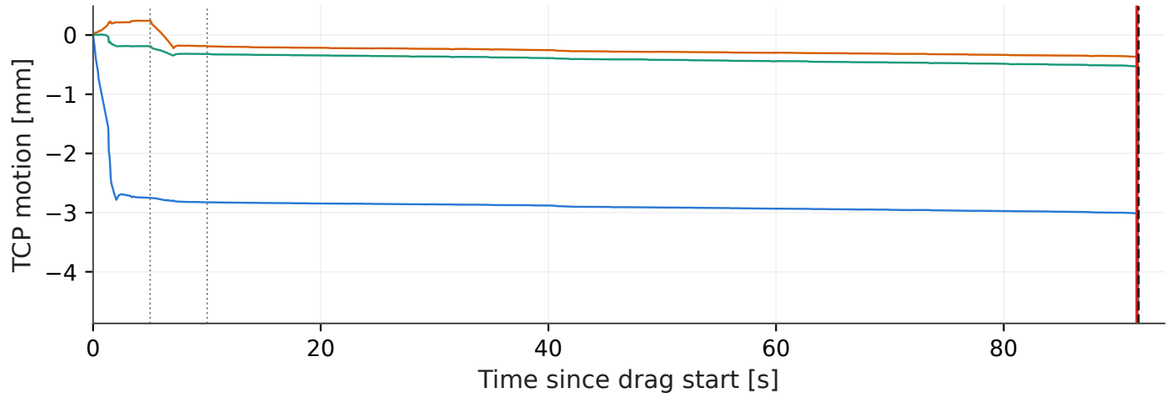
Trial audit | 67–69 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

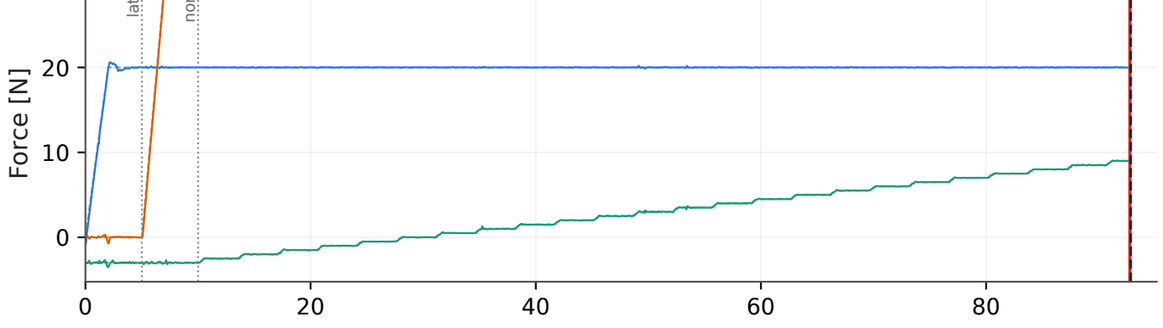
T67 R1 S=30, L=20 N | Normal tension (Fn > 0)



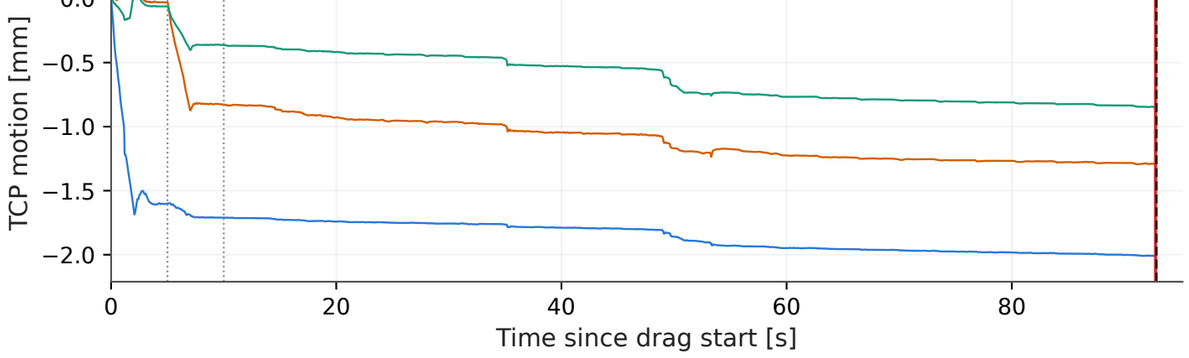
force and motion; re-engaged: False; review: False



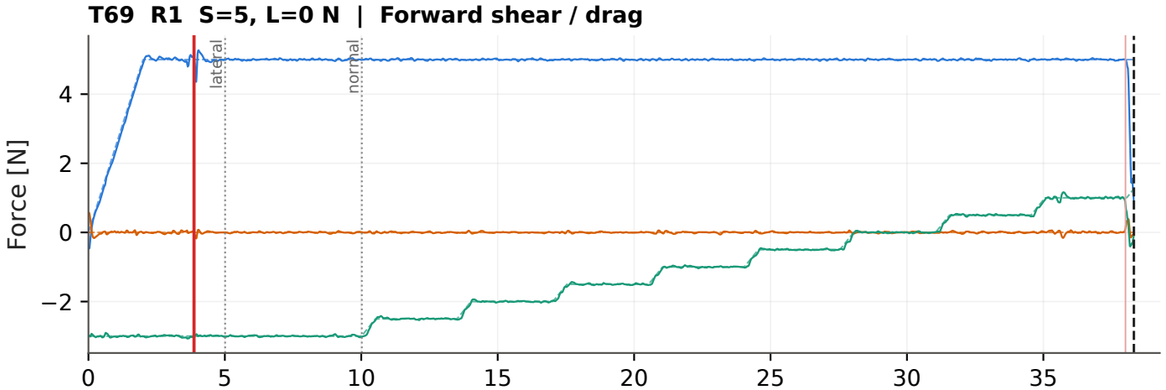
T68 R1 S=20, L=30 N | Normal tension (Fn > 0)



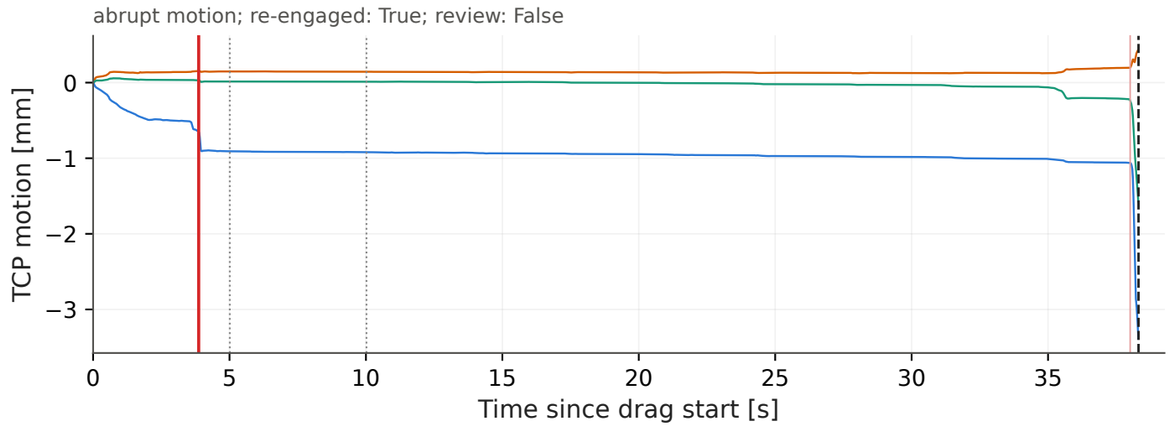
force and motion; re-engaged: False; review: False



T69 R1 S=5, L=0 N | Forward shear / drag



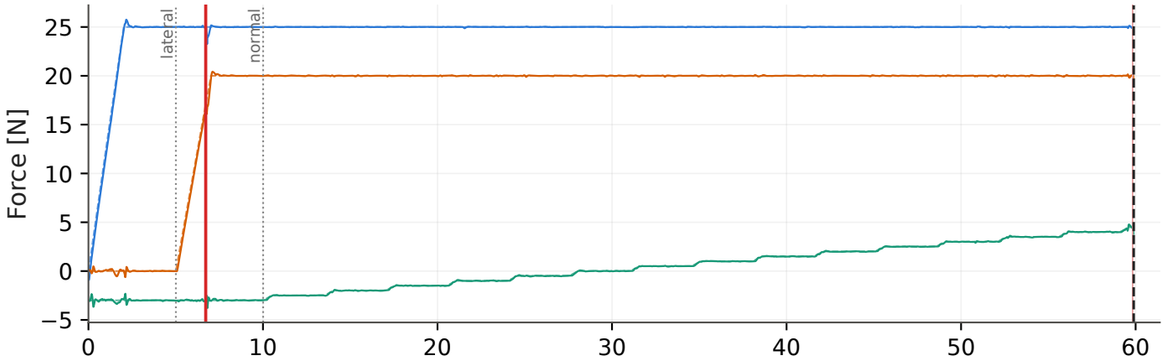
abrupt motion; re-engaged: True; review: False



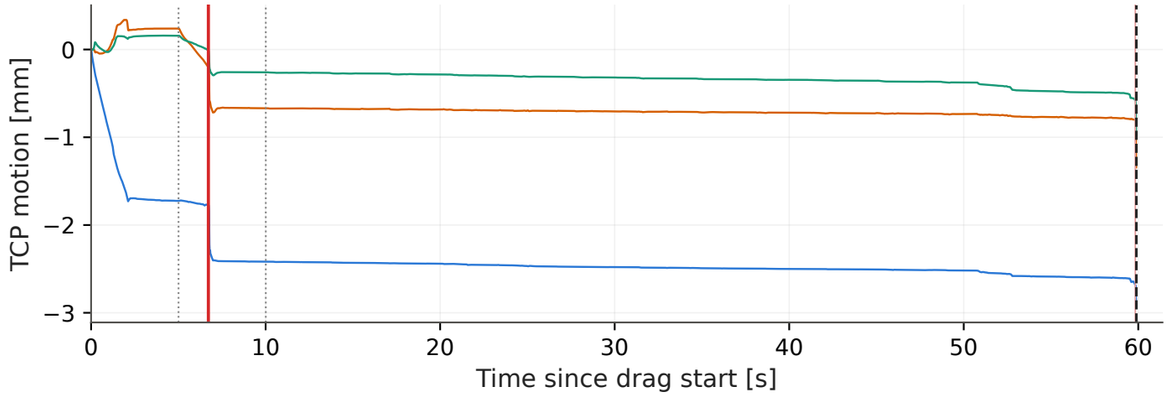
Trial audit | 70–72 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

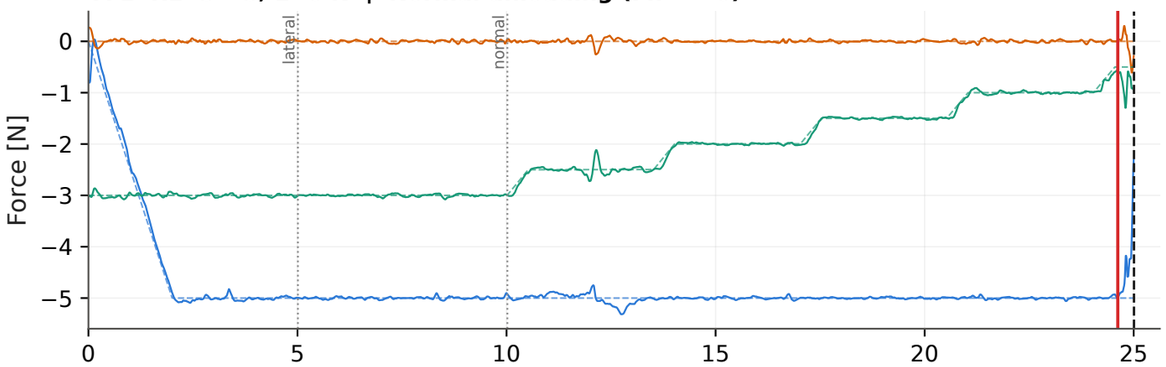
T70 R1 S=25, L=20 N | Lateral loading



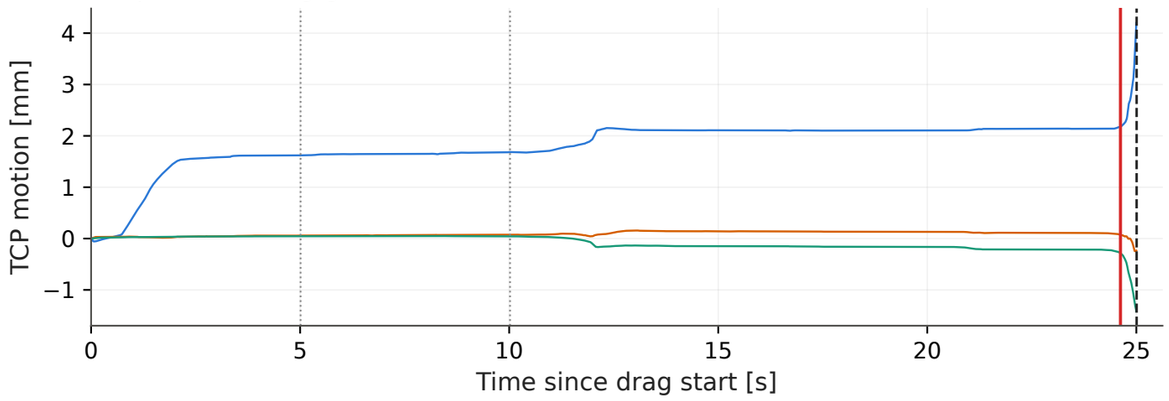
force and motion; re-engaged: True; review: False



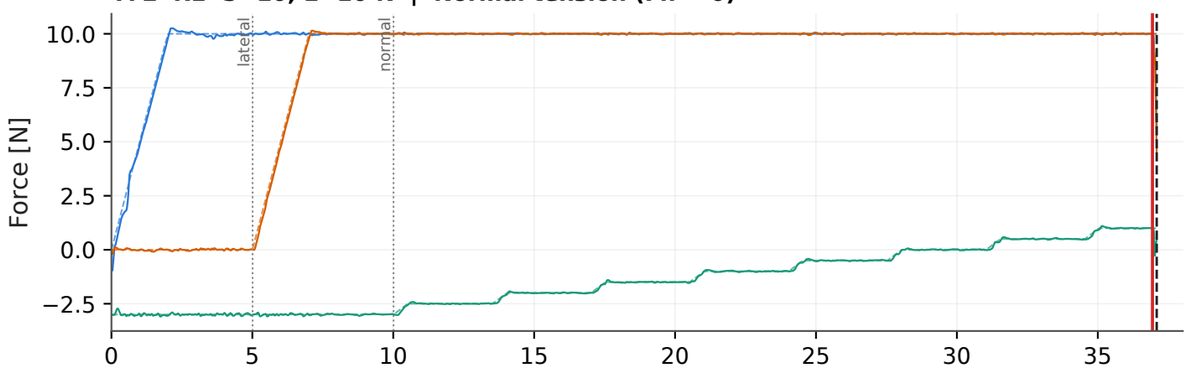
T71 R1 S=-5, L=0 N | Normal unloading (Fn <= 0)



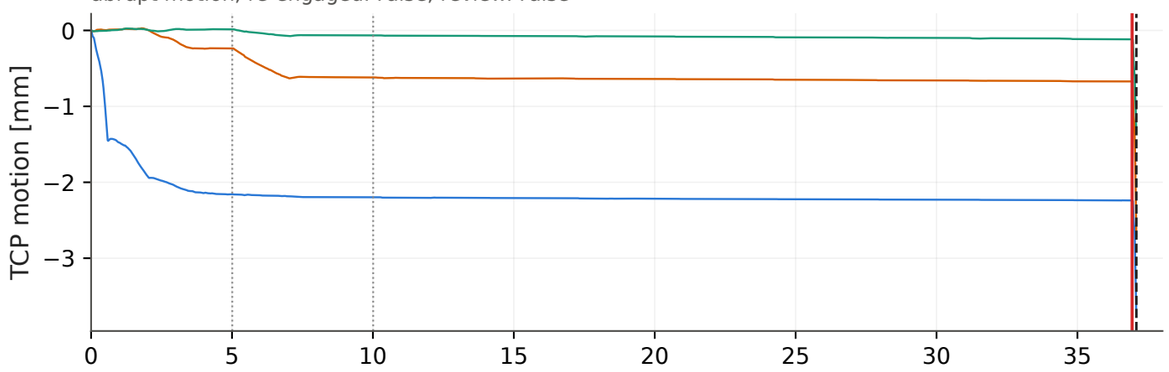
abrupt motion; re-engaged: False; review: False



T72 R1 S=10, L=10 N | Normal tension (Fn > 0)



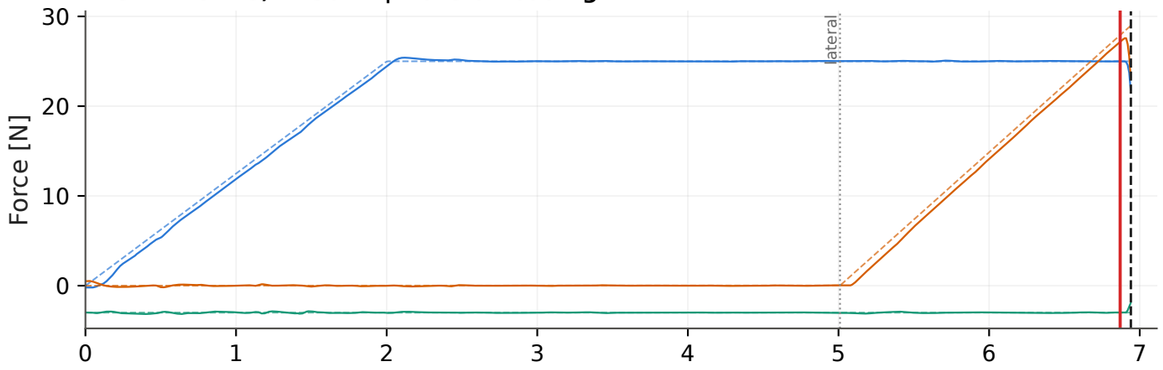
abrupt motion; re-engaged: False; review: False



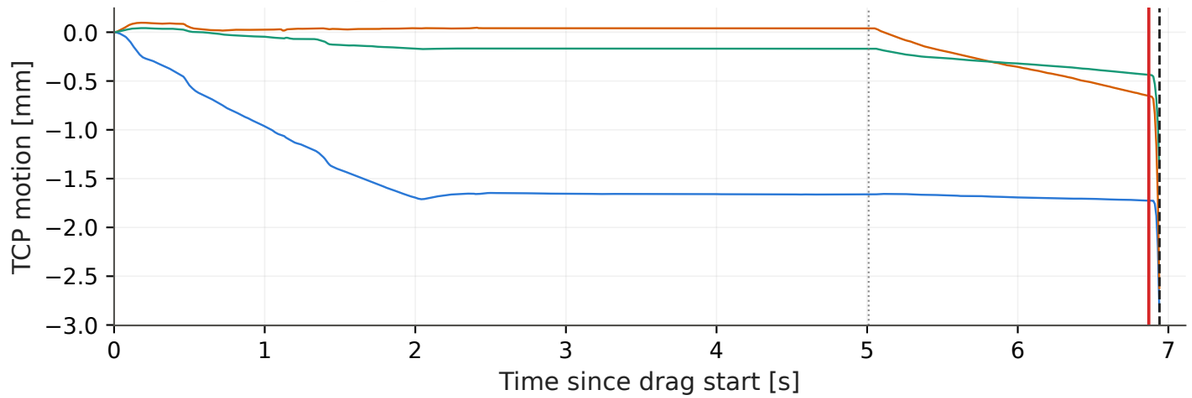
Trial audit | 73–75 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

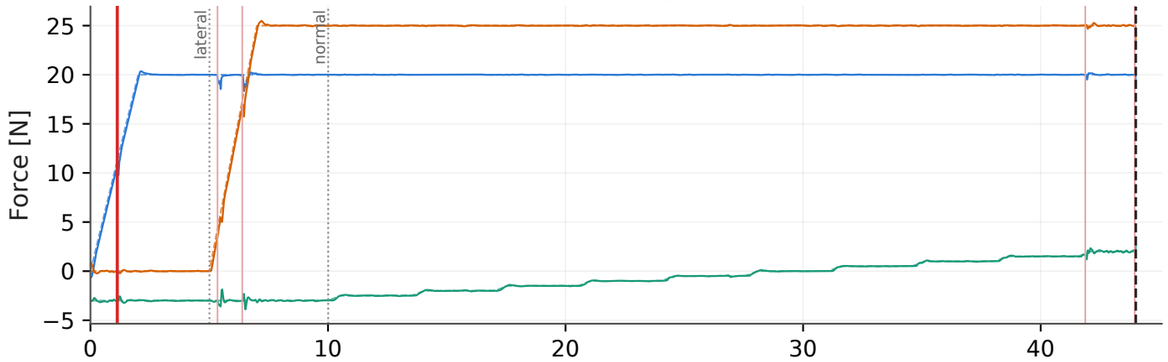
T73 R1 S=25, L=30 N | Lateral loading



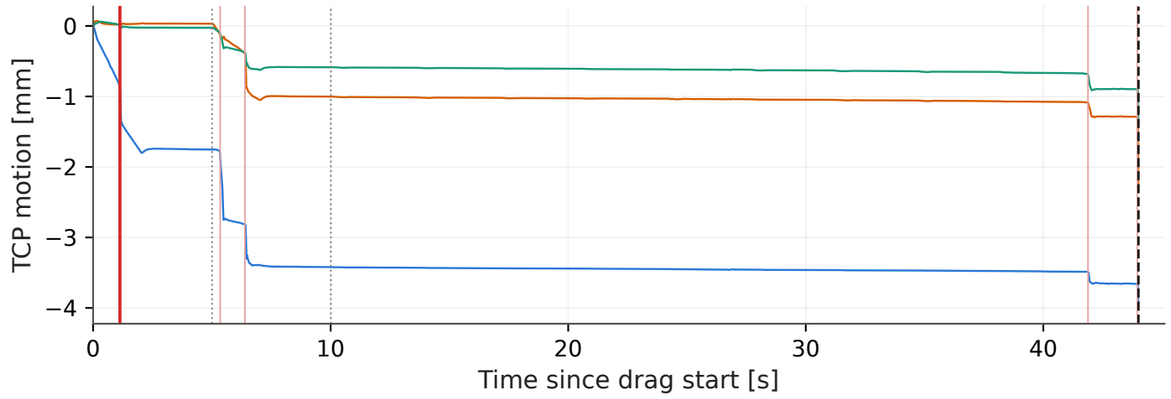
force and motion; re-engaged: False; review: False



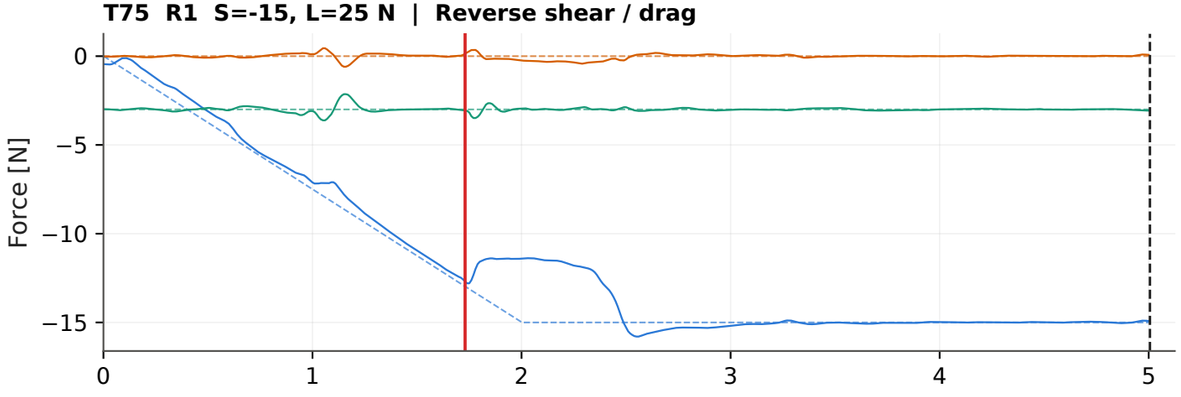
T74 R1 S=20, L=25 N | Forward shear / drag



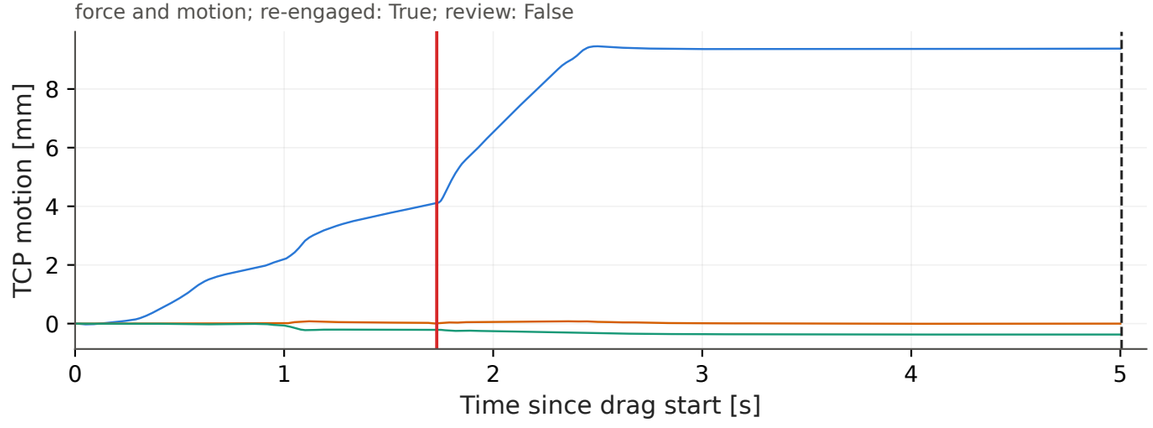
force and motion; re-engaged: True; review: True



T75 R1 S=-15, L=25 N | Reverse shear / drag



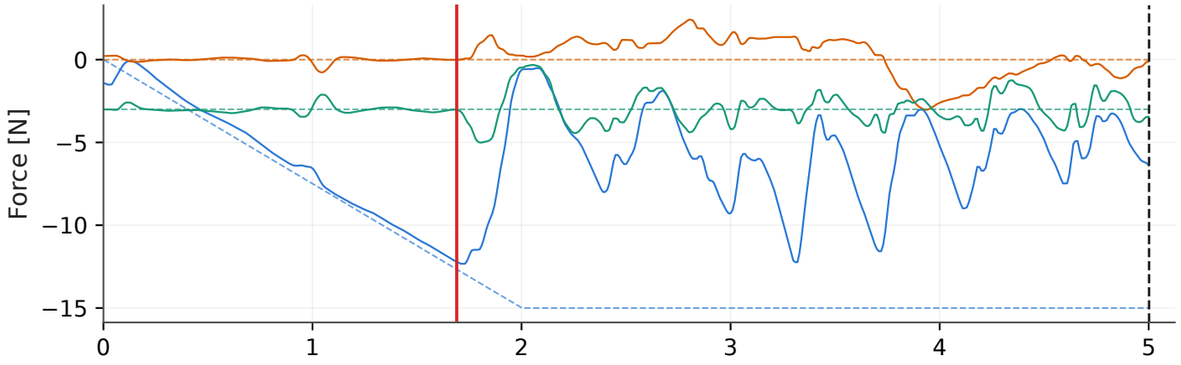
force and motion; re-engaged: True; review: False



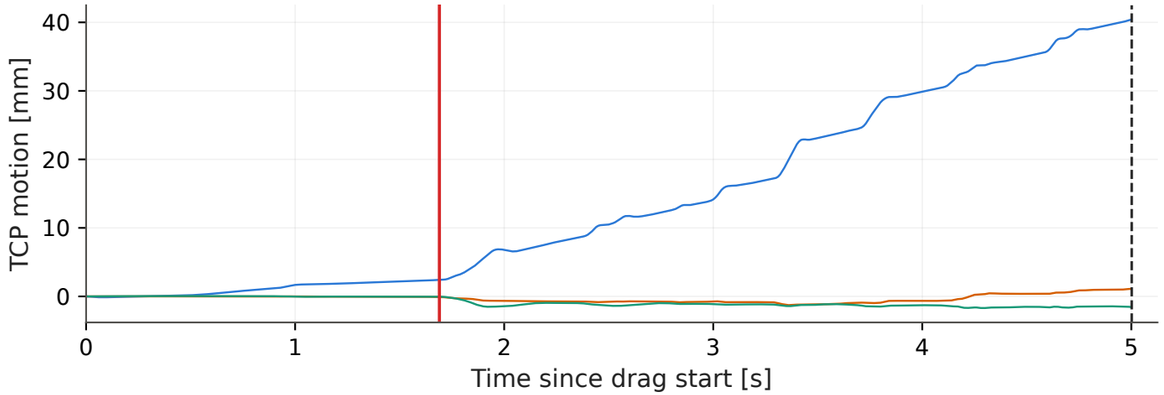
Trial audit | 76–78 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

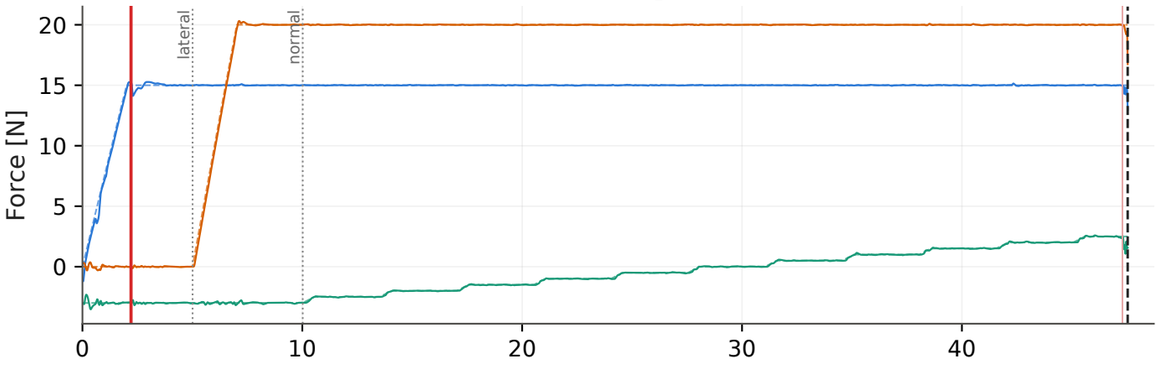
T76 R1 S=-15, L=0 N | Reverse shear / drag



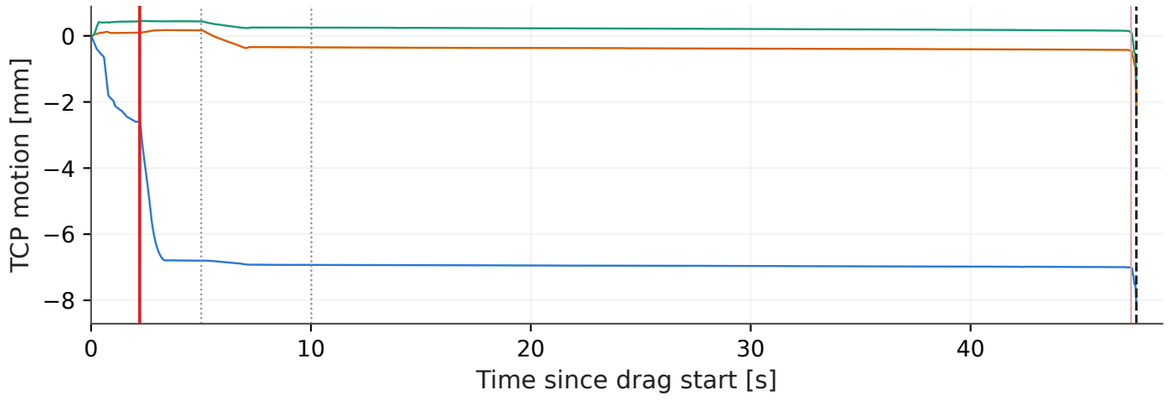
force and motion; re-engaged: False; review: False



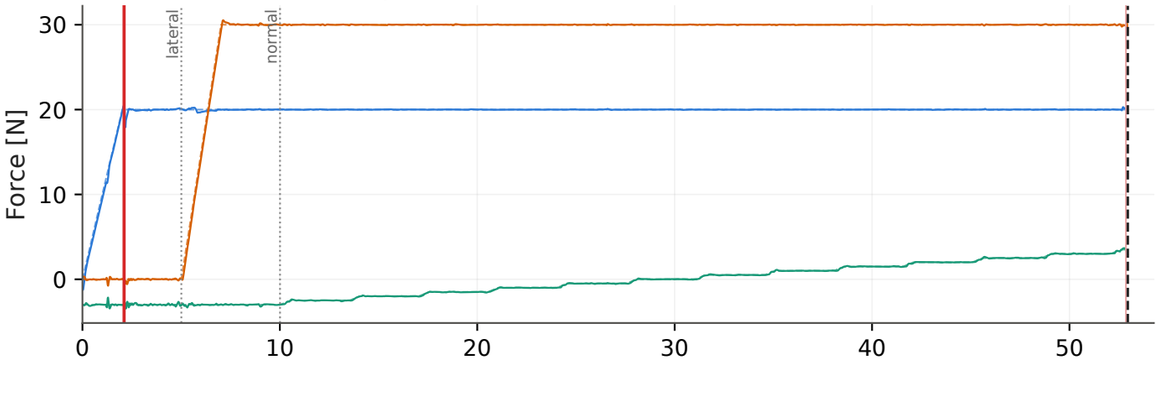
T77 R1 S=15, L=20 N | Forward shear / drag



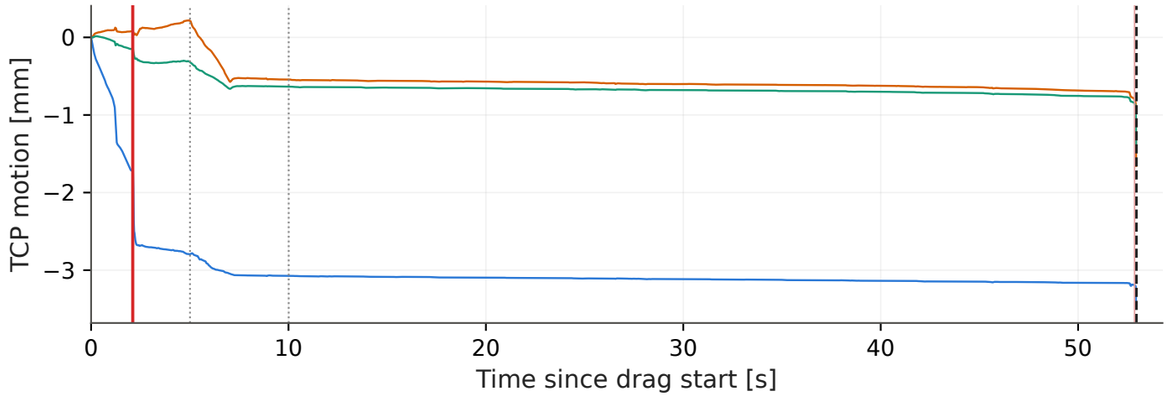
force and motion; re-engaged: True; review: False



T78 R2 S=20, L=30 N | Forward shear / drag



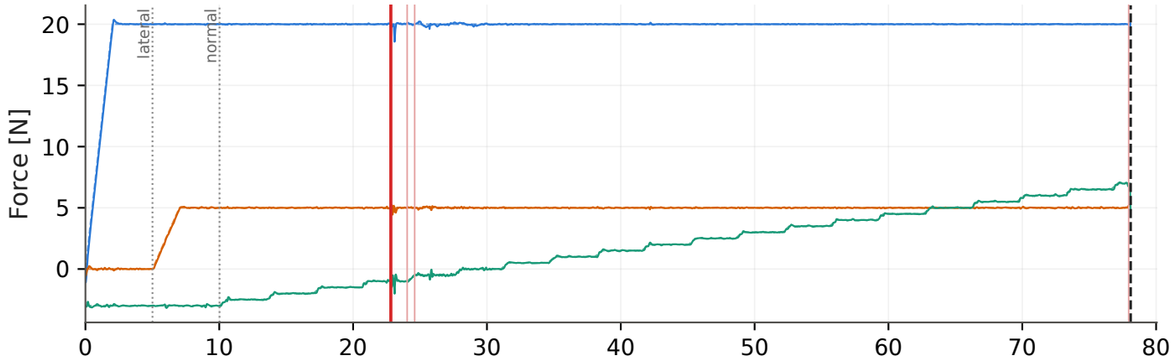
force and motion; re-engaged: True; review: False



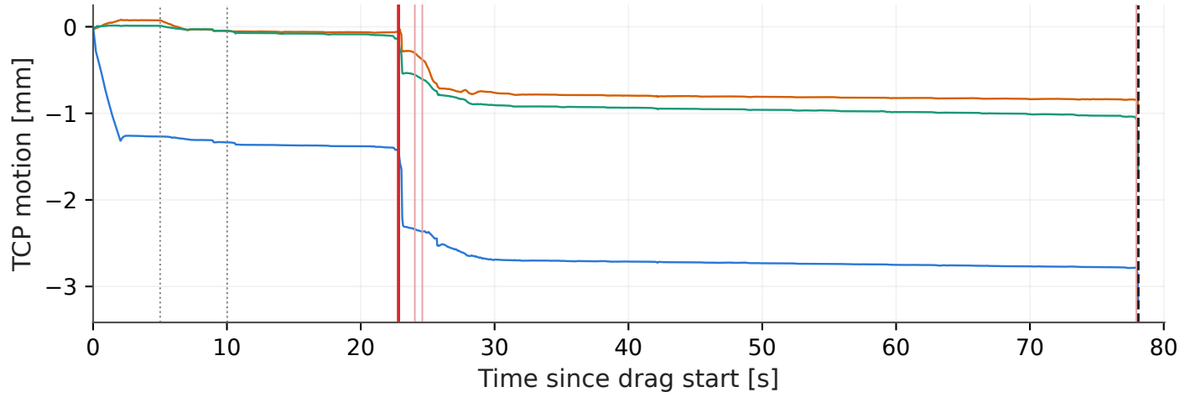
Trial audit | 79–81 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

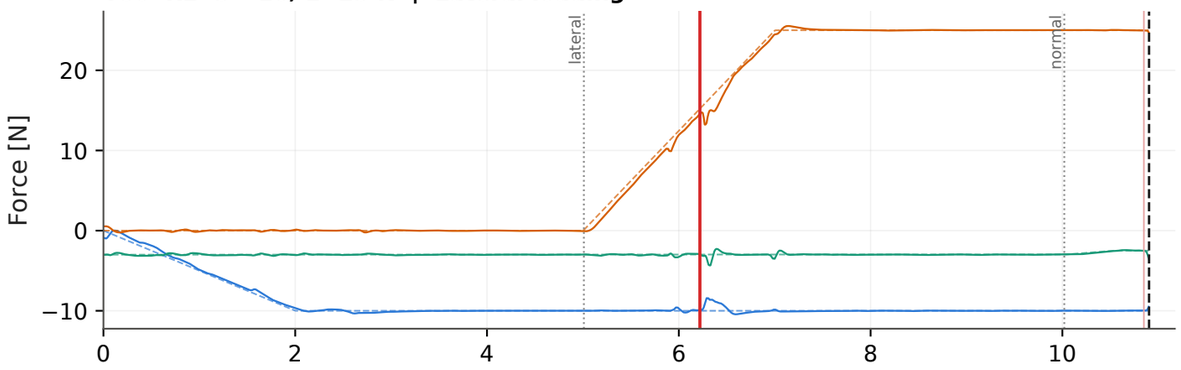
T79 R2 S=20, L=5 N | Normal unloading (Fn <= 0)



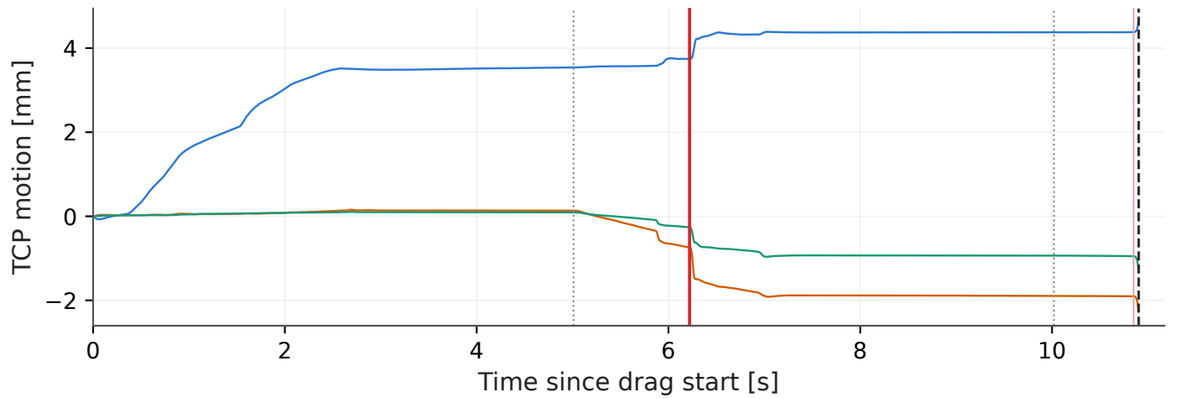
abrupt motion; re-engaged: True; review: False



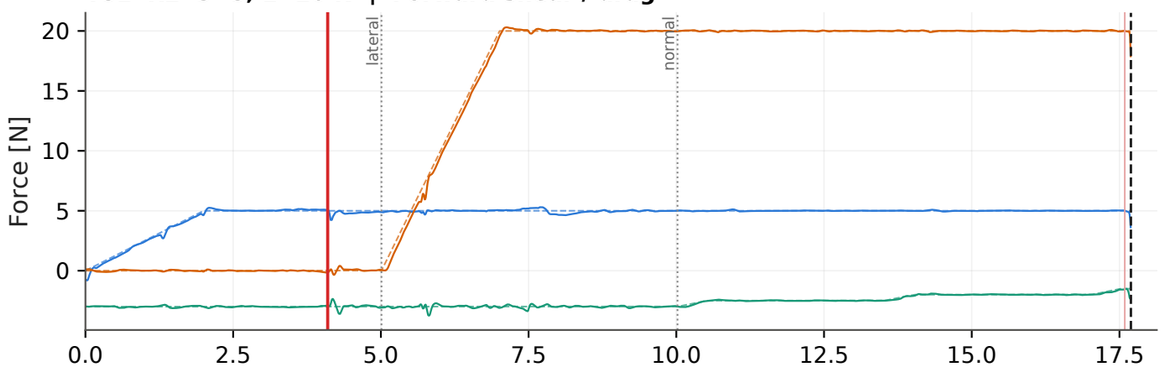
T80 R2 S=-10, L=25 N | Lateral loading



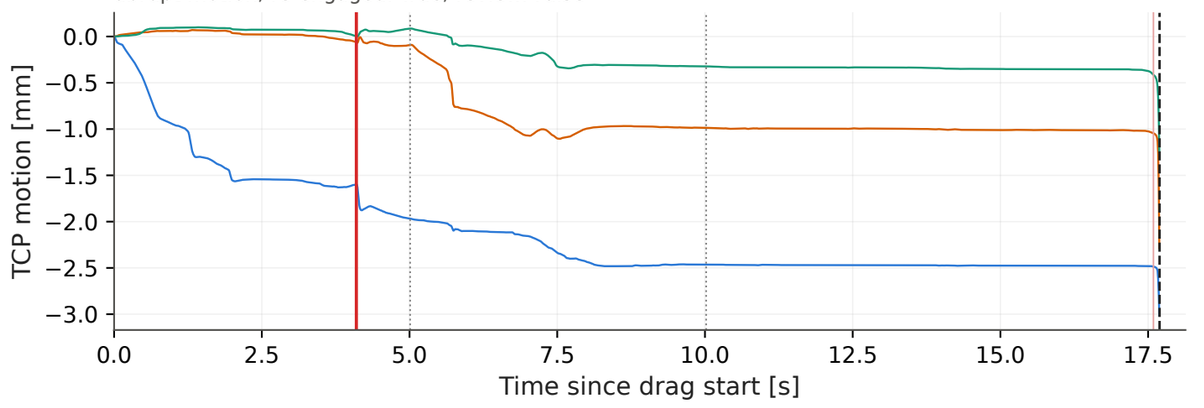
force and motion; re-engaged: True; review: False



T81 R2 S=5, L=20 N | Forward shear / drag



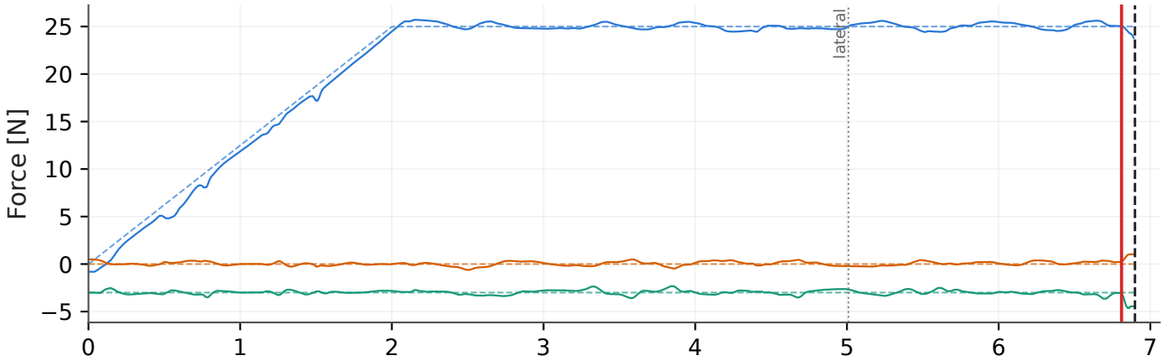
abrupt motion; re-engaged: True; review: False



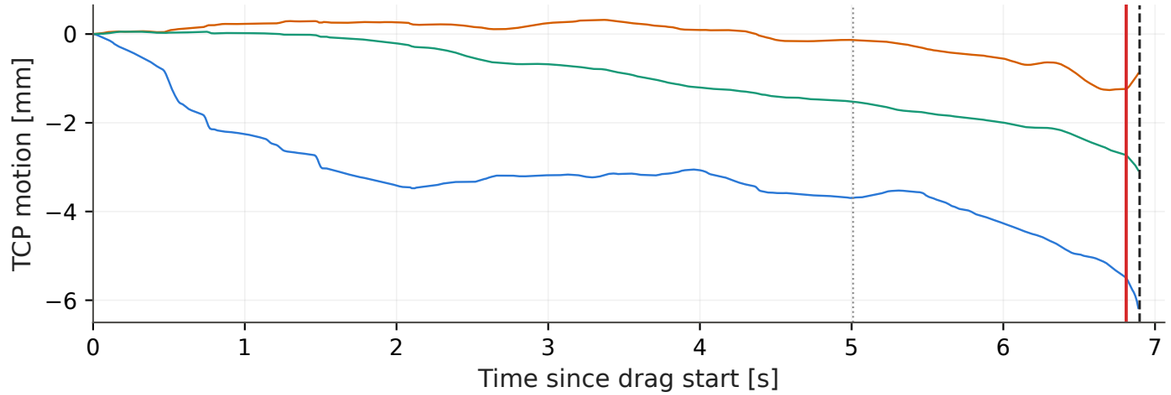
Trial audit | 82–84 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

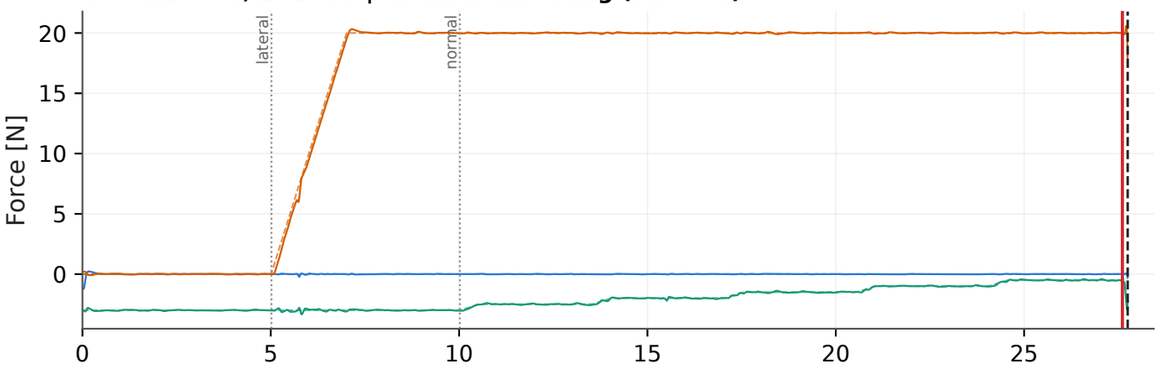
T82 R2 S=25, L=0 N | Lateral loading



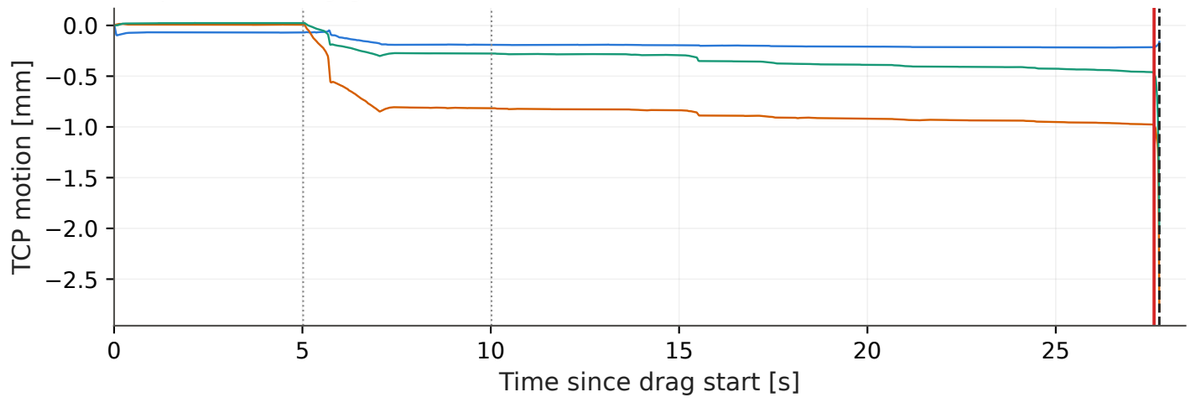
force and motion; re-engaged: False; review: False



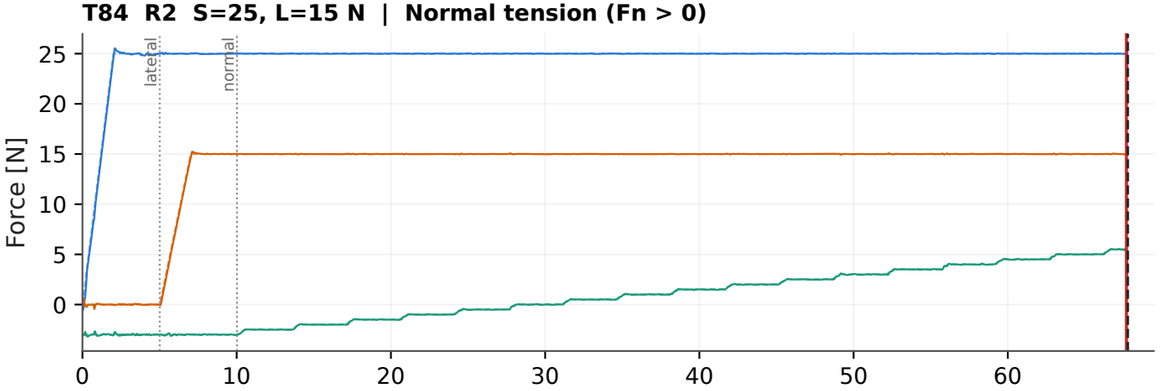
T83 R2 S=0, L=20 N | Normal unloading (Fn <= 0)



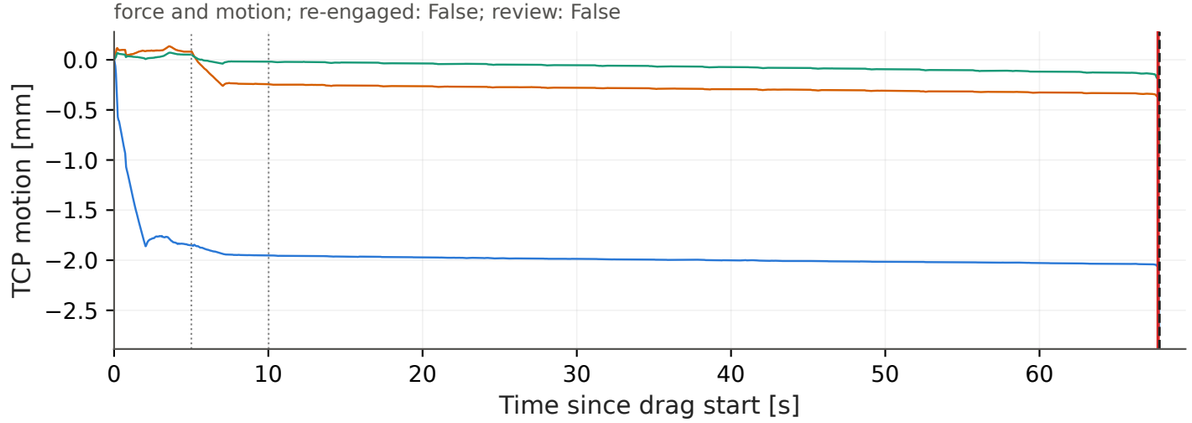
abrupt motion; re-engaged: False; review: False



T84 R2 S=25, L=15 N | Normal tension (Fn > 0)



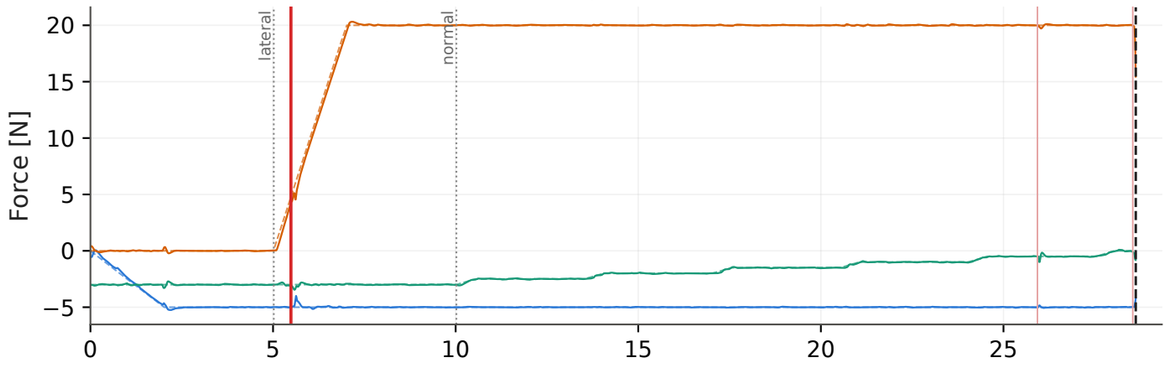
force and motion; re-engaged: False; review: False



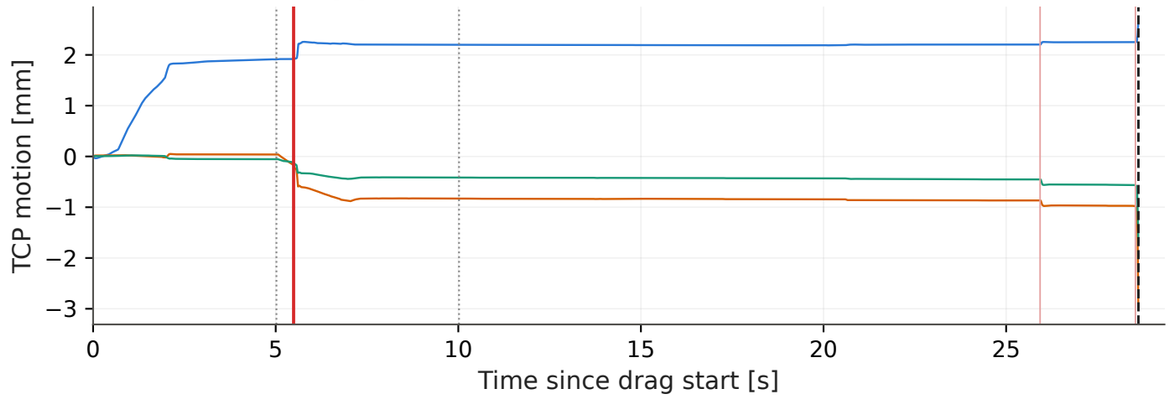
Trial audit | 85–87 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

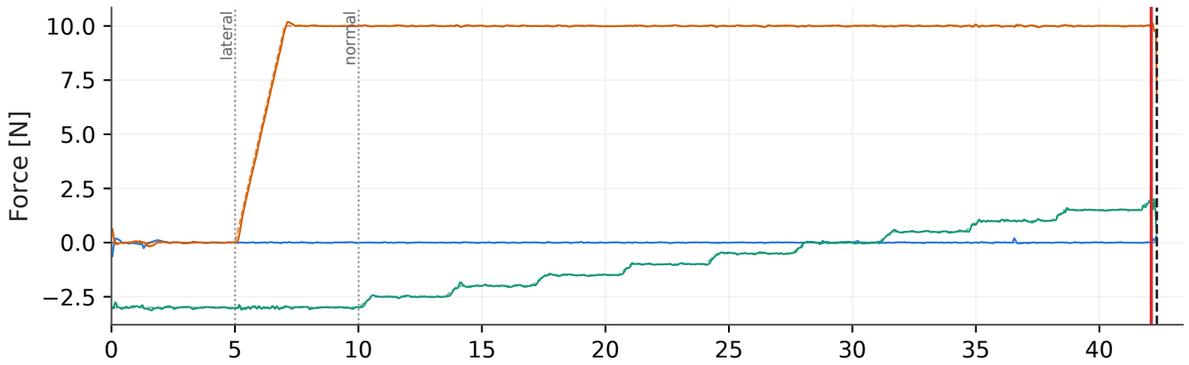
T85 R2 S=-5, L=20 N | Lateral loading



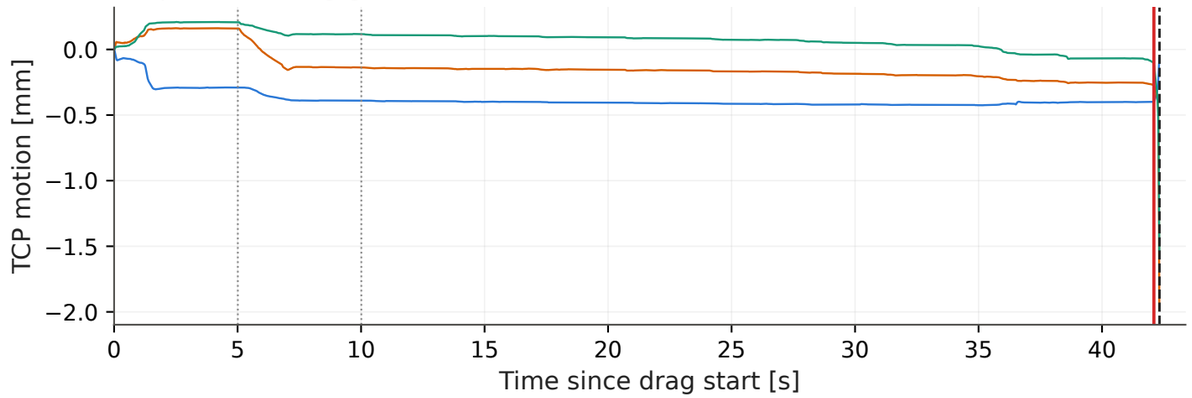
force and motion; re-engaged: True; review: False



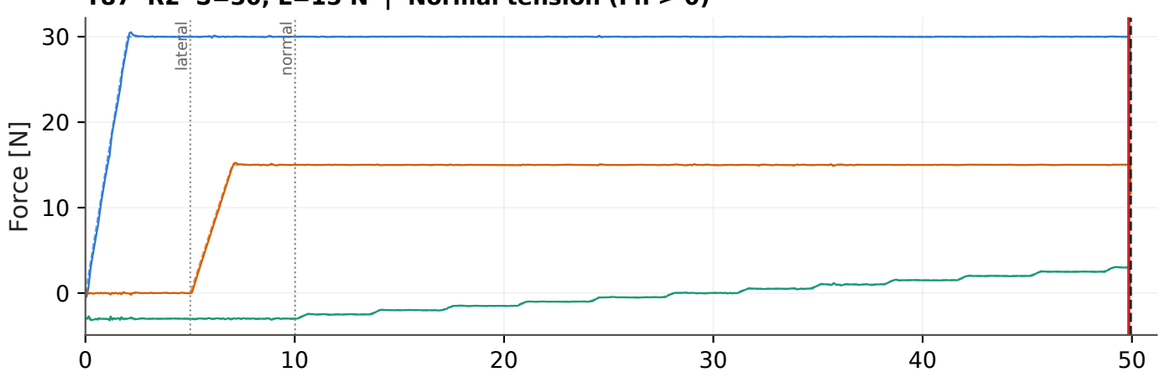
T86 R2 S=0, L=10 N | Normal tension (Fn > 0)



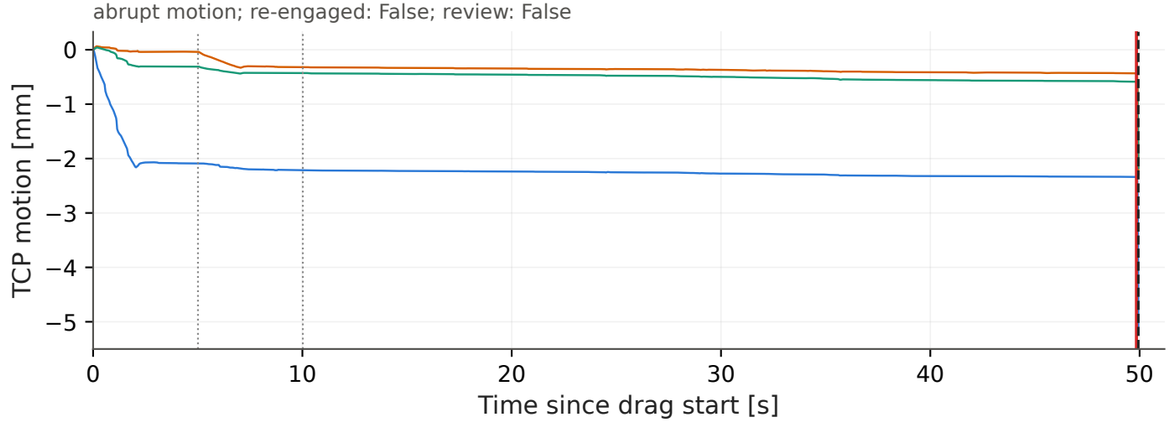
abrupt motion; re-engaged: False; review: False



T87 R2 S=30, L=15 N | Normal tension (Fn > 0)



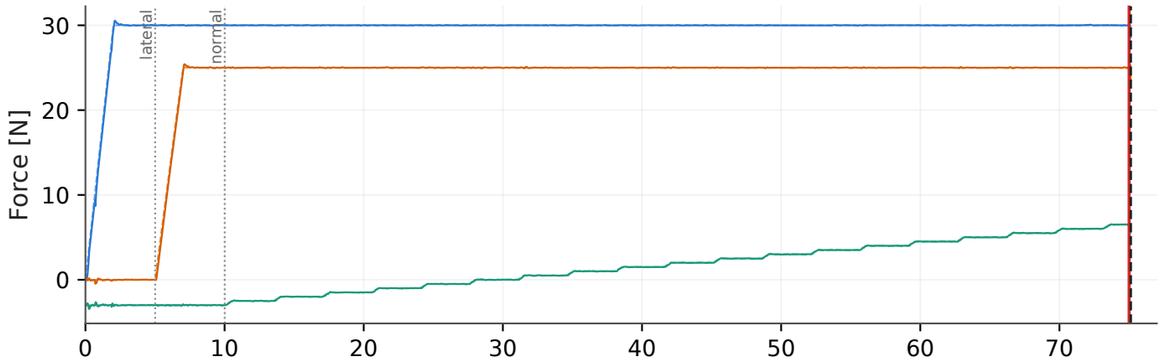
abrupt motion; re-engaged: False; review: False



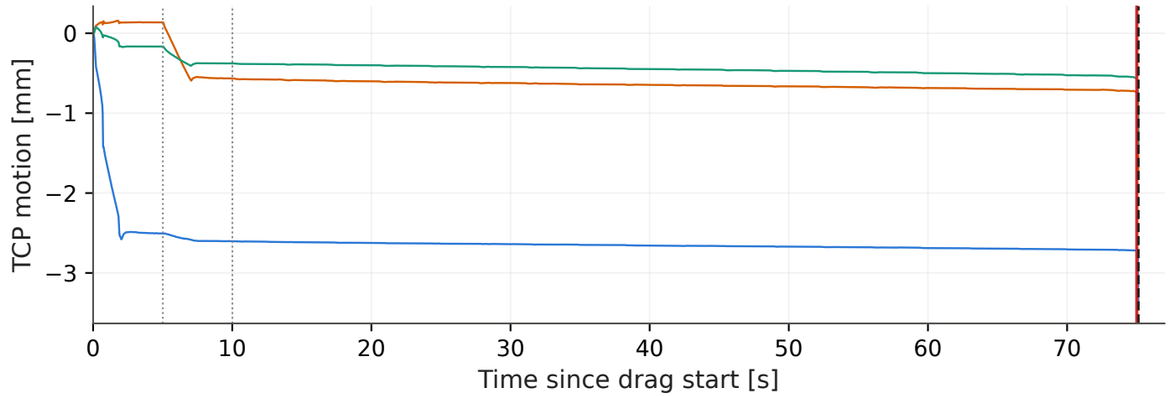
Trial audit | 88–90 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

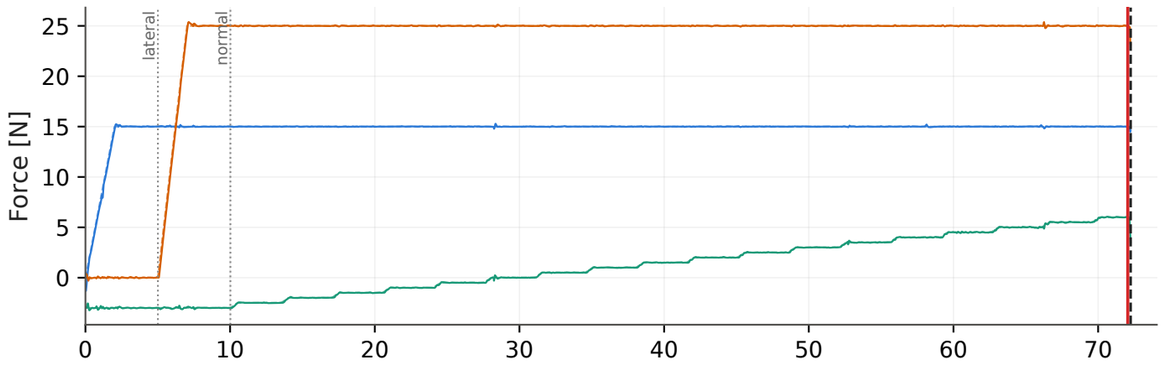
T88 R2 S=30, L=25 N | Normal tension (Fn > 0)



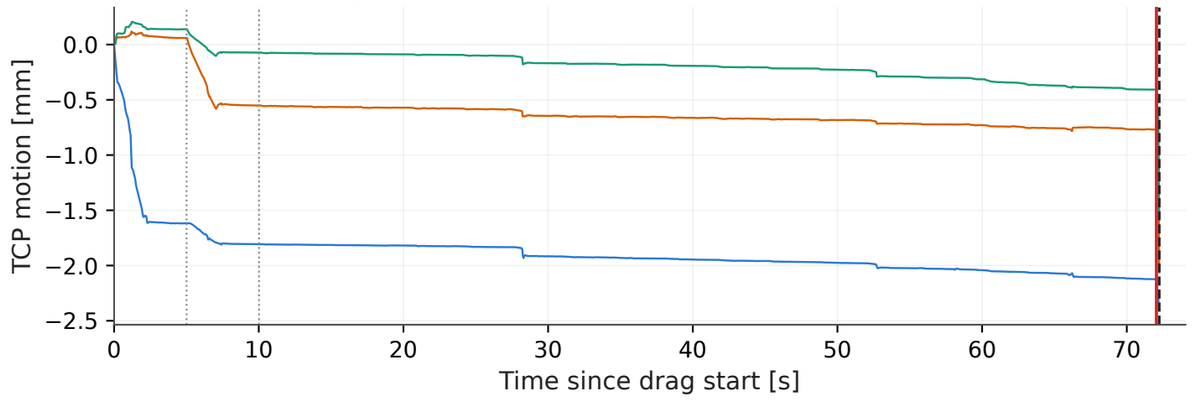
force and motion; re-engaged: False; review: True



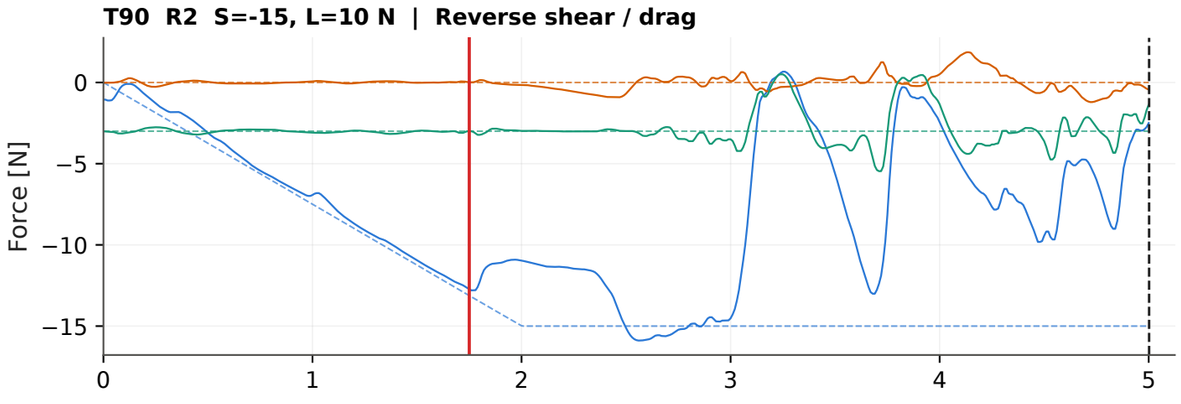
T89 R2 S=15, L=25 N | Normal tension (Fn > 0)



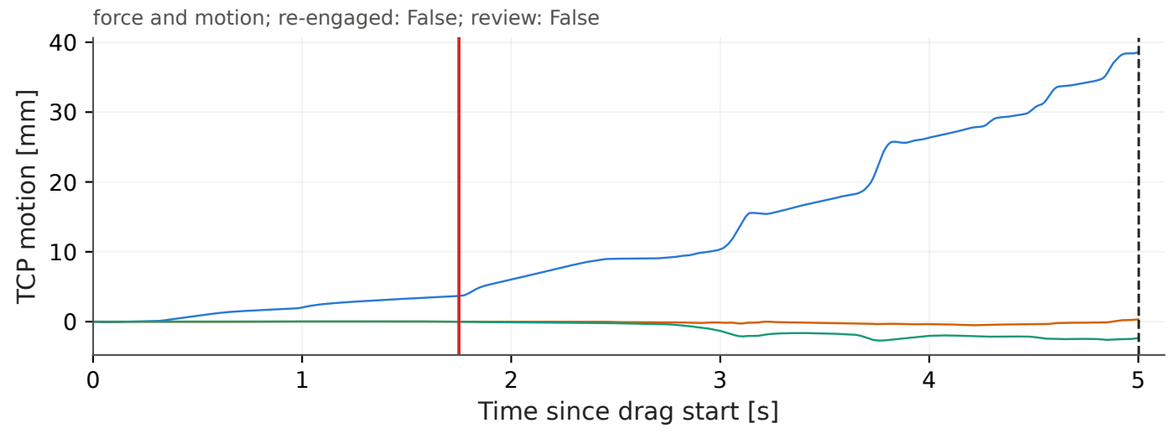
abrupt motion; re-engaged: False; review: True



T90 R2 S=-15, L=10 N | Reverse shear / drag



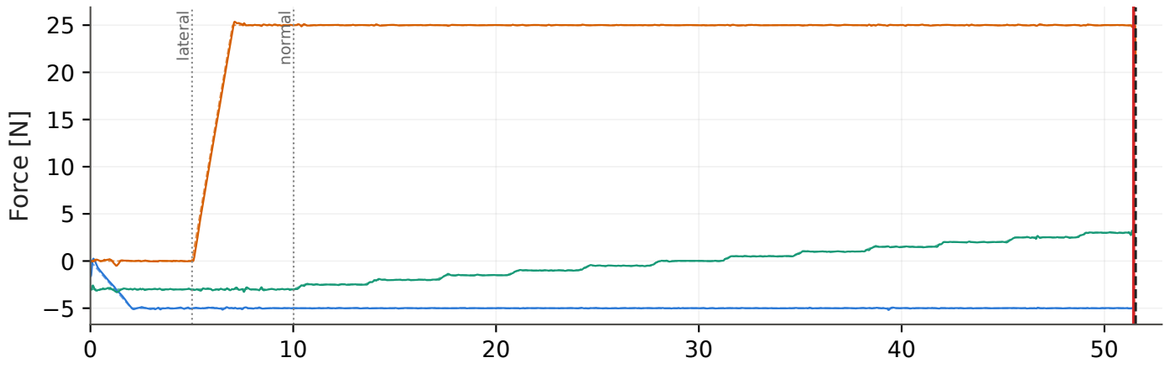
force and motion; re-engaged: False; review: False



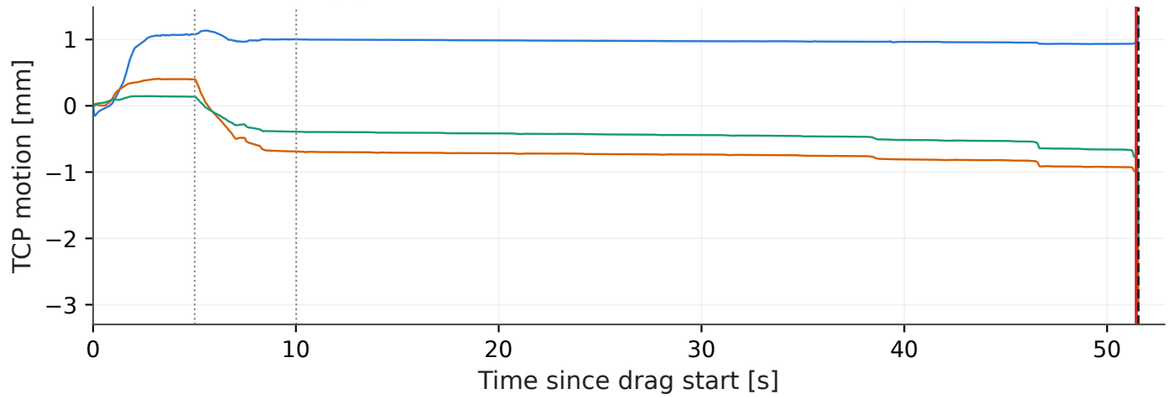
Trial audit | 91–93 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

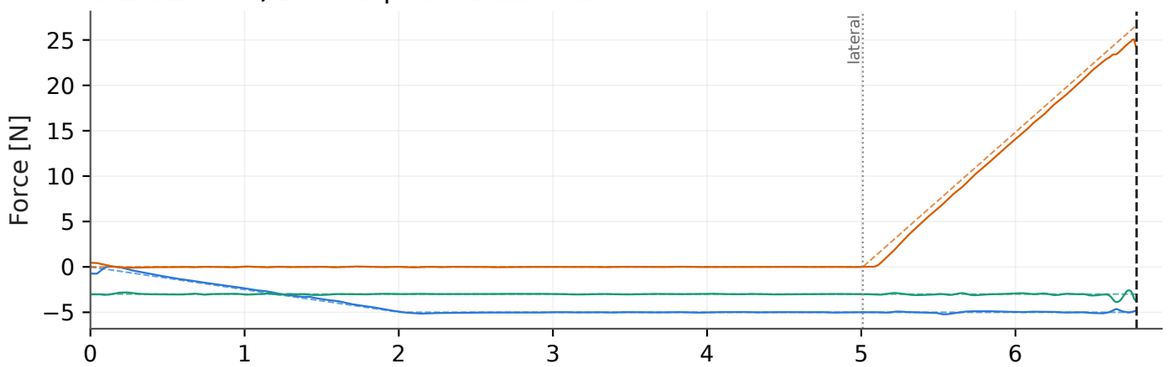
T91 R2 S=-5, L=25 N | Normal tension (Fn > 0)



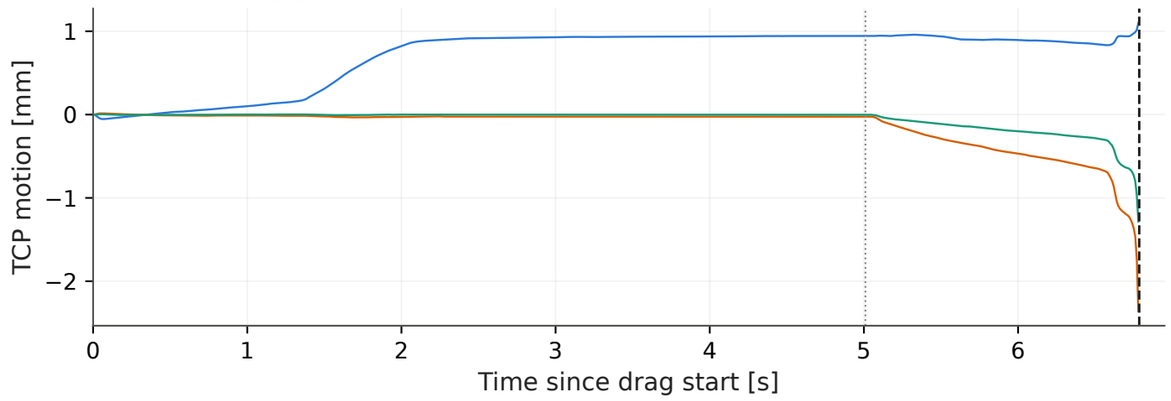
force and motion; re-engaged: False; review: False



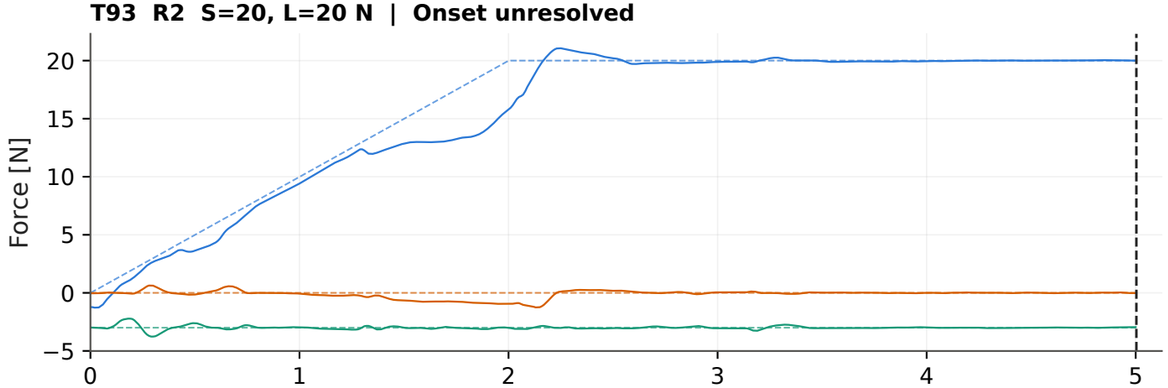
T92 R2 S=-5, L=30 N | Onset unresolved



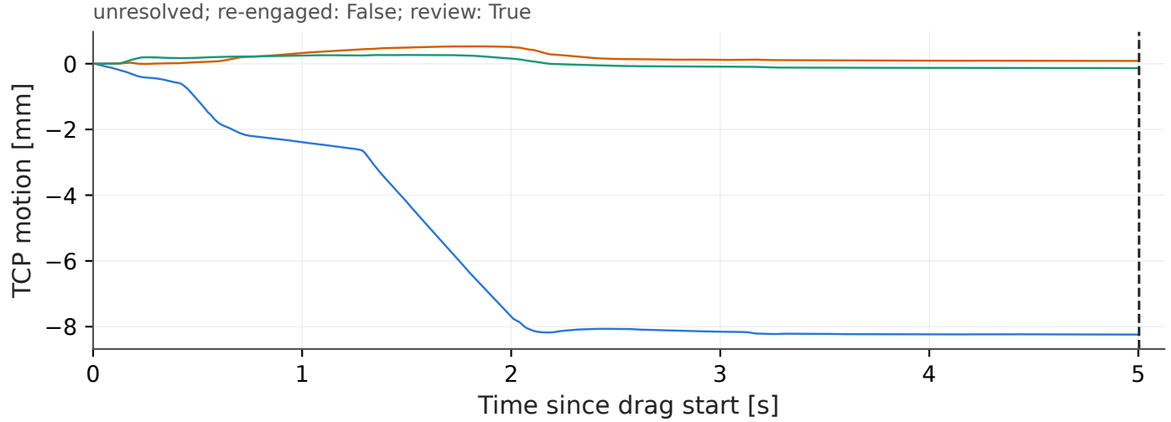
unresolved; re-engaged: False; review: True



T93 R2 S=20, L=20 N | Onset unresolved



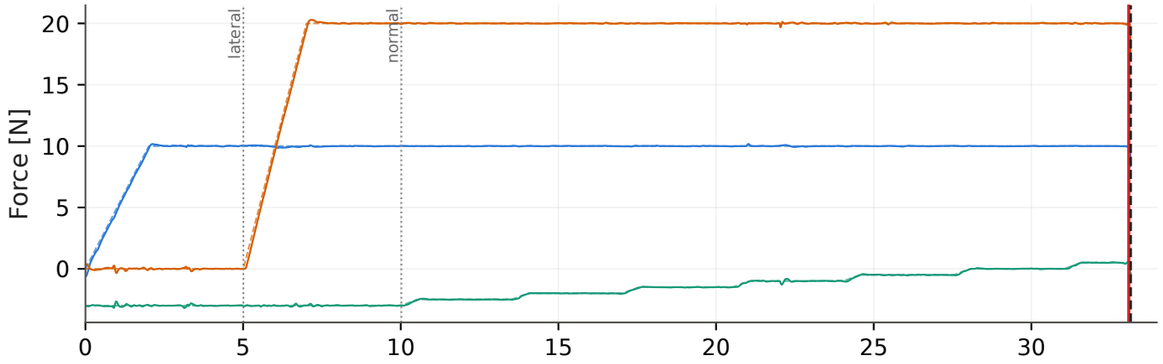
unresolved; re-engaged: False; review: True



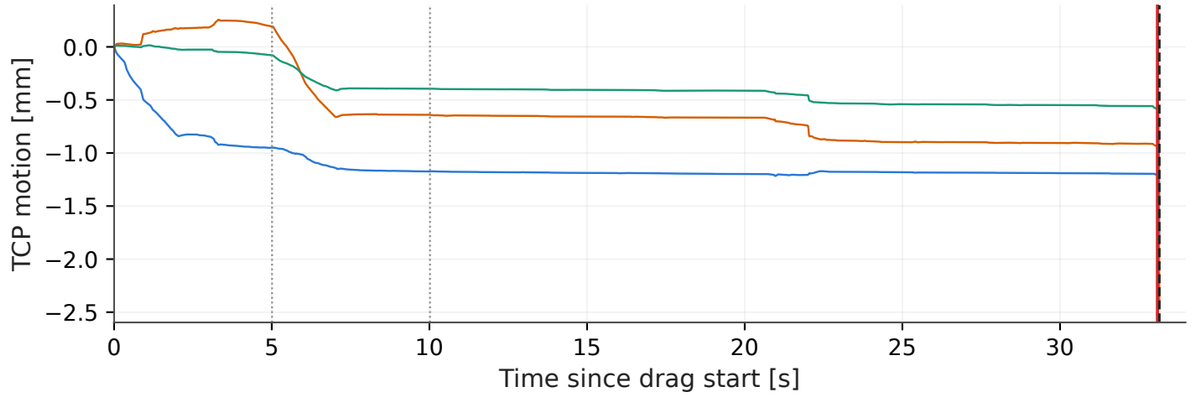
Trial audit | 94–96 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

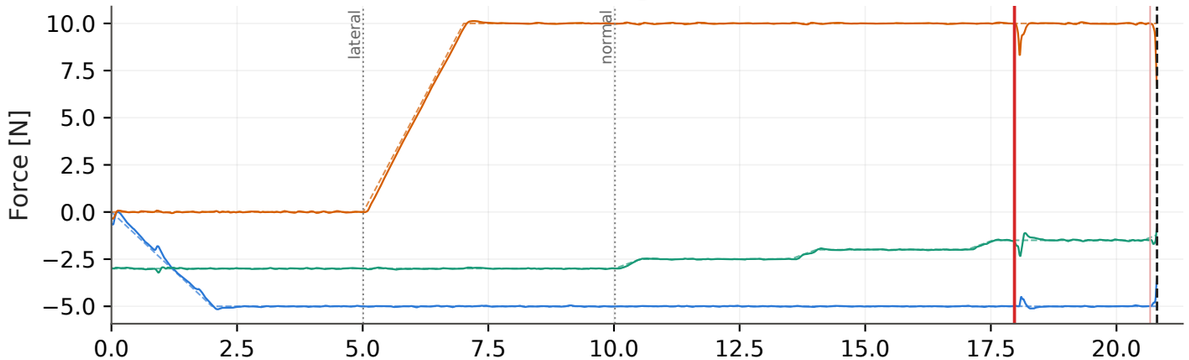
T94 R2 S=10, L=20 N | Normal tension (Fn > 0)



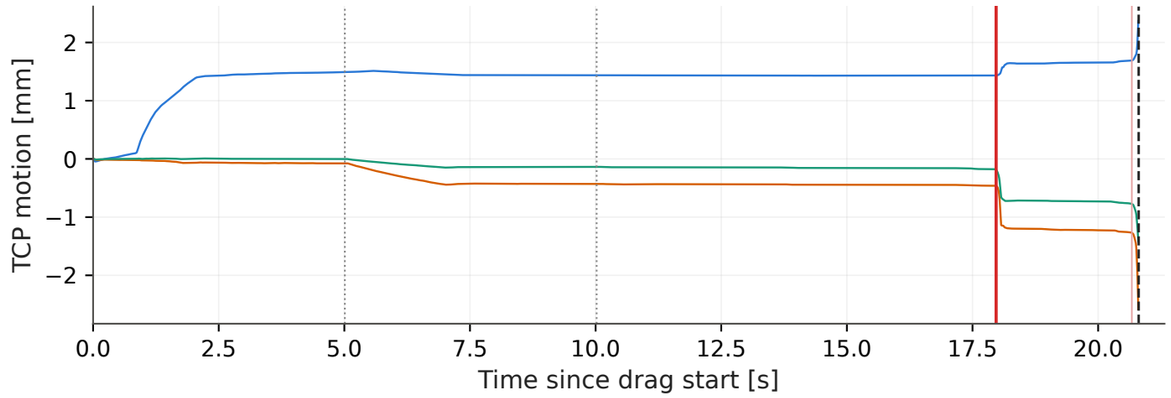
abrupt motion; re-engaged: False; review: True



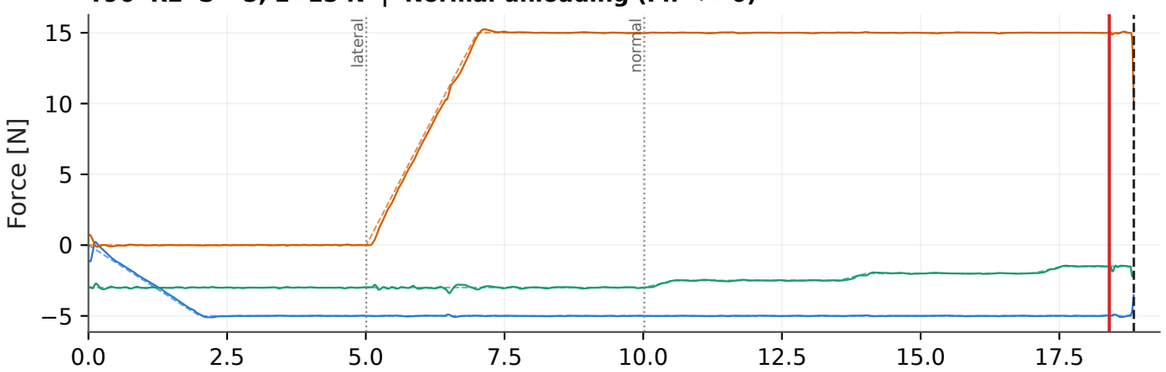
T95 R2 S=-5, L=10 N | Normal unloading (Fn <= 0)



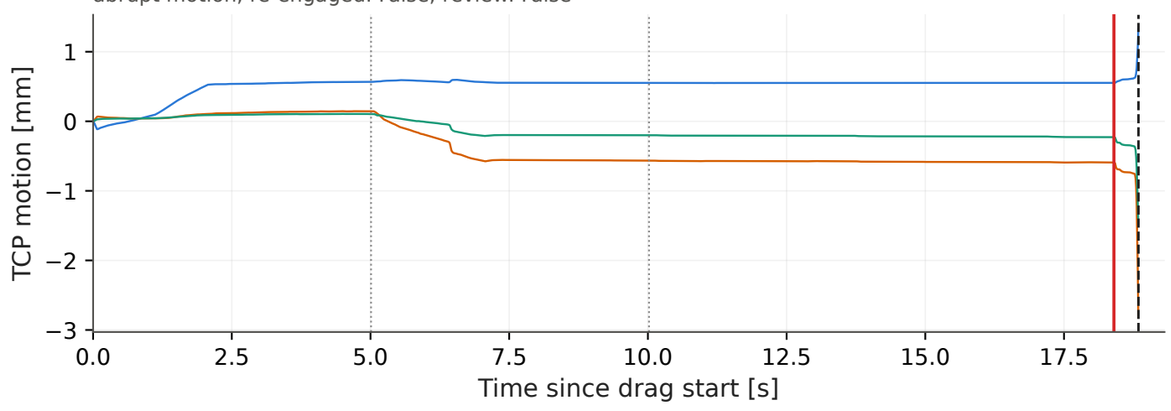
abrupt motion; re-engaged: True; review: False



T96 R2 S=-5, L=15 N | Normal unloading (Fn <= 0)



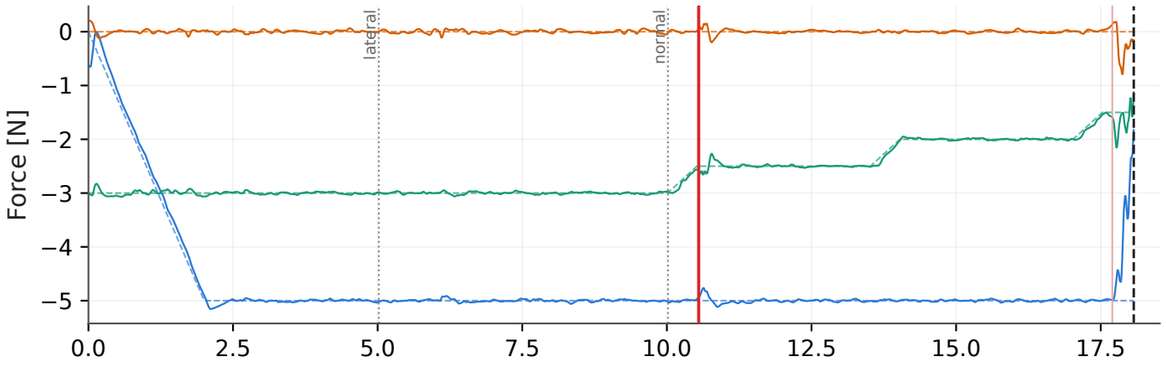
abrupt motion; re-engaged: False; review: False



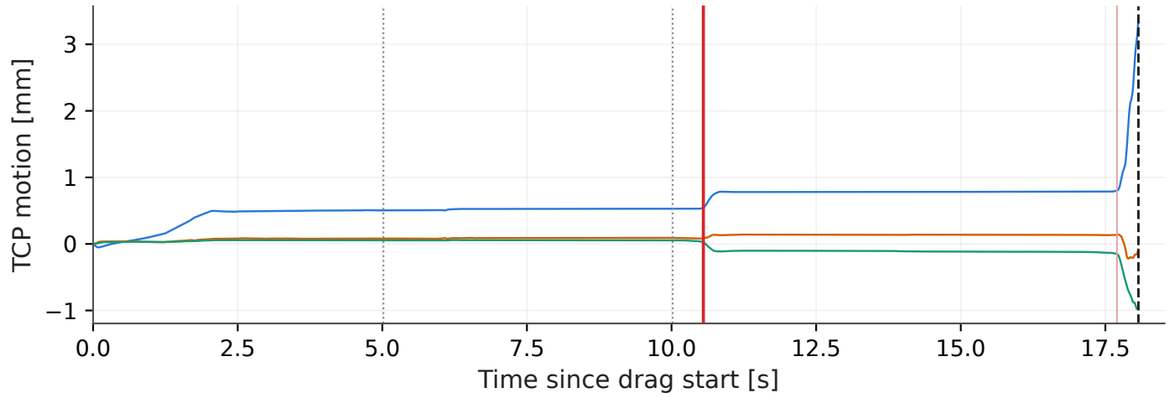
Trial audit | 97–99 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

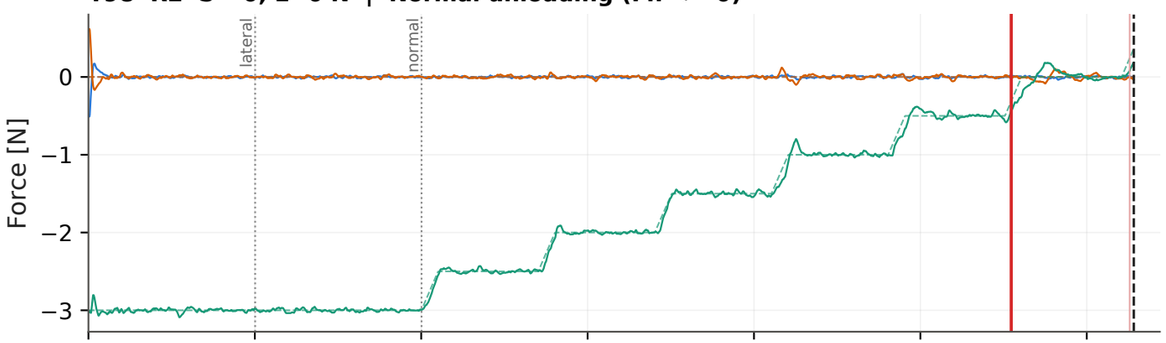
T97 R2 S=-5, L=0 N | Normal unloading (Fn <= 0)



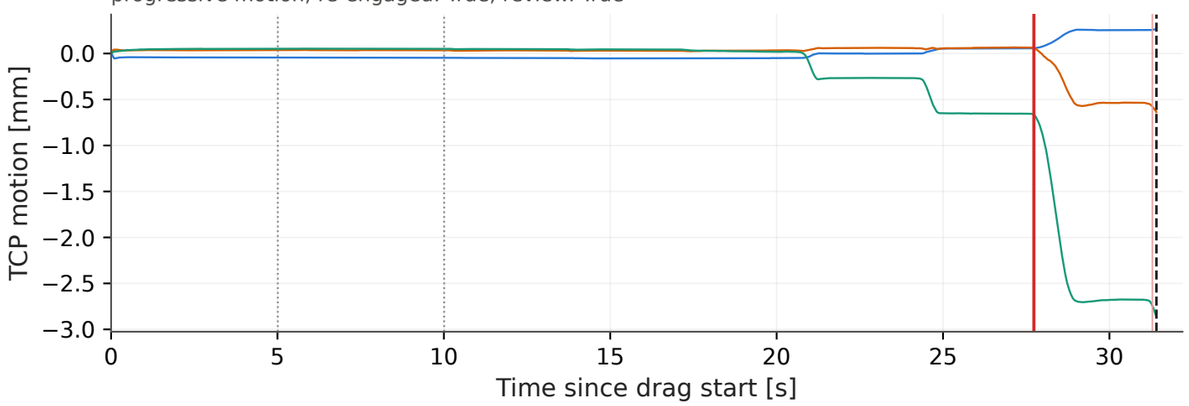
abrupt motion; re-engaged: True; review: False



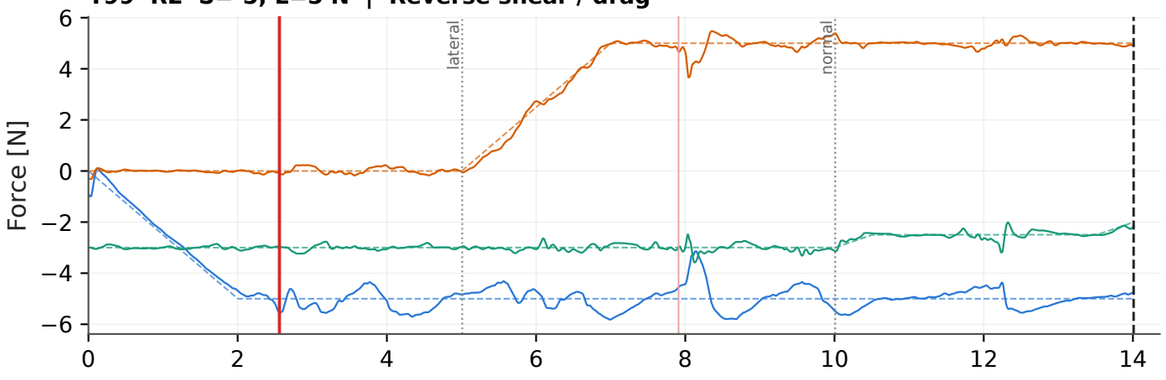
T98 R2 S=-0, L=0 N | Normal unloading (Fn <= 0)



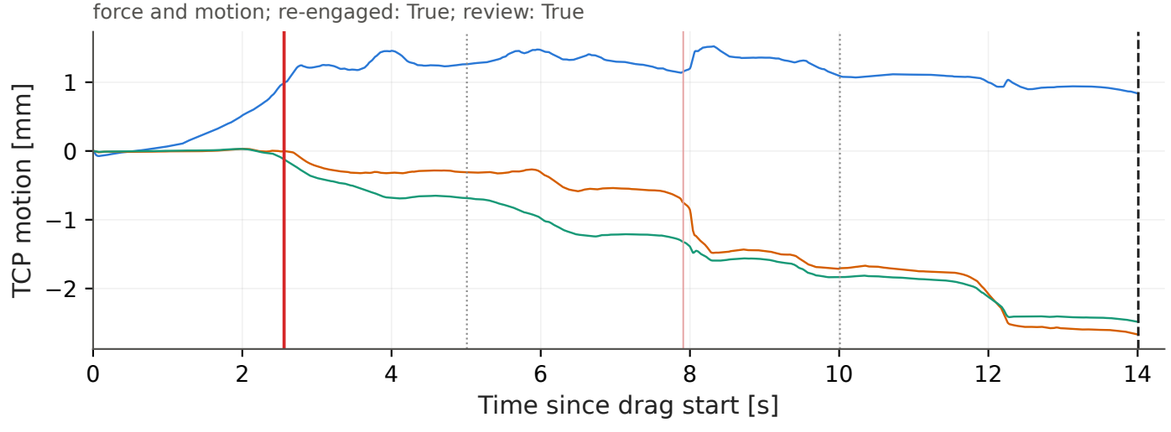
progressive motion; re-engaged: True; review: True



T99 R2 S=-5, L=5 N | Reverse shear / drag

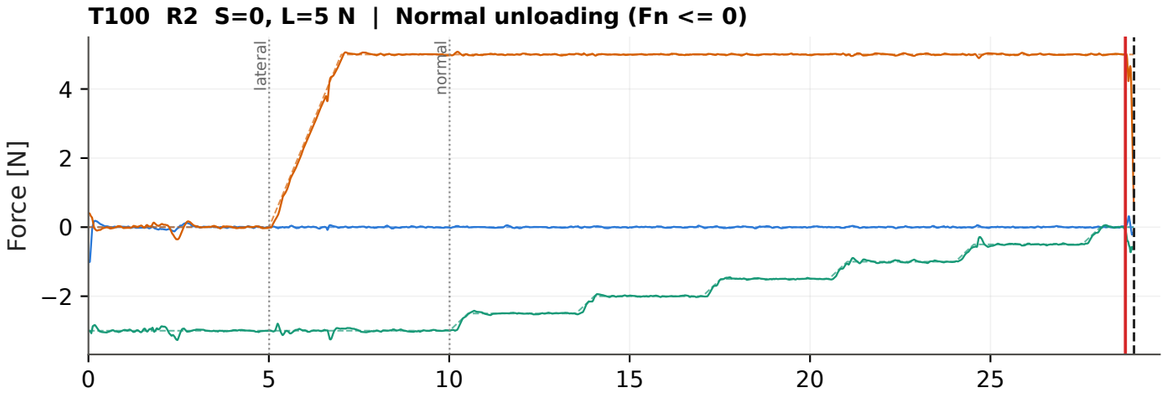


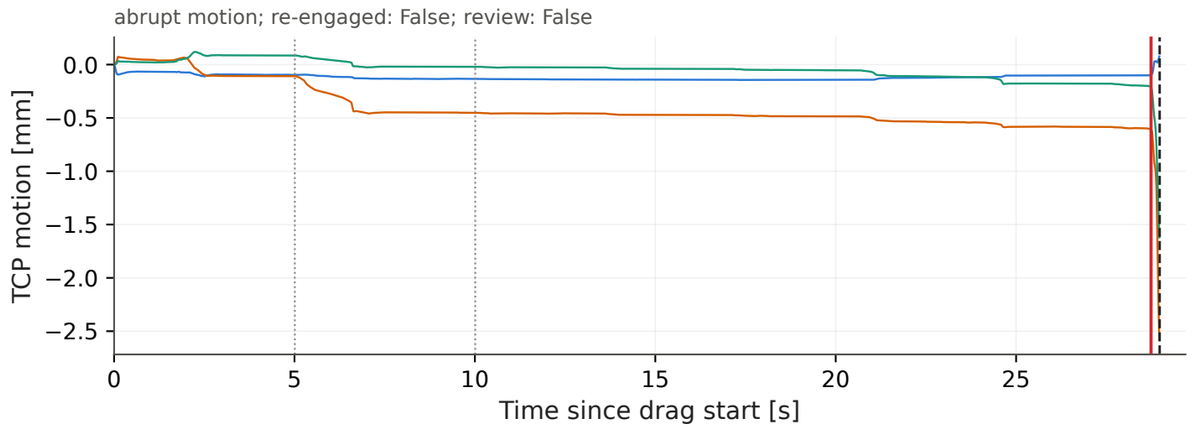
force and motion; re-engaged: True; review: True

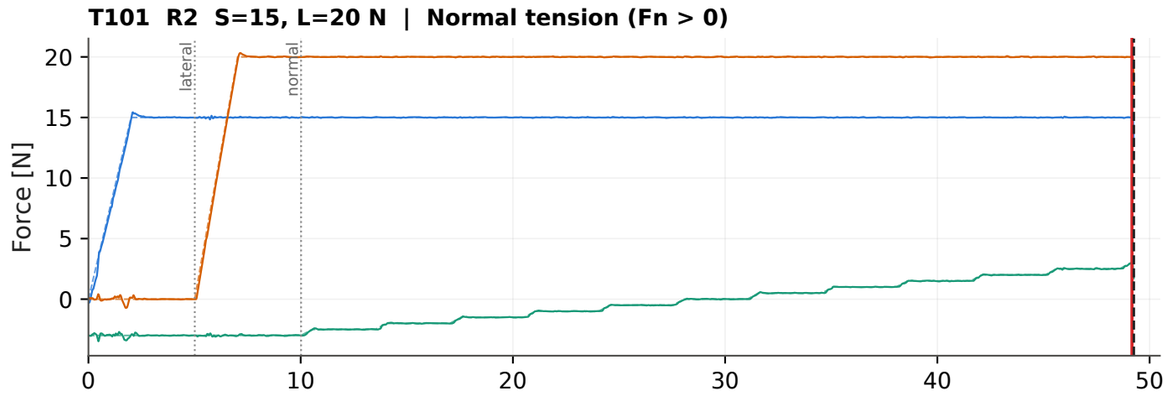


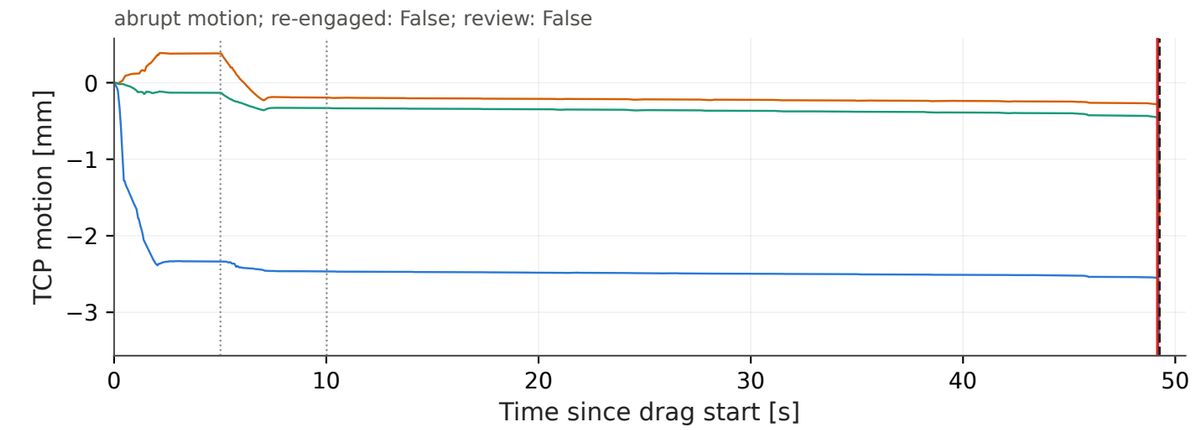
Trial audit | 100–102 / 231

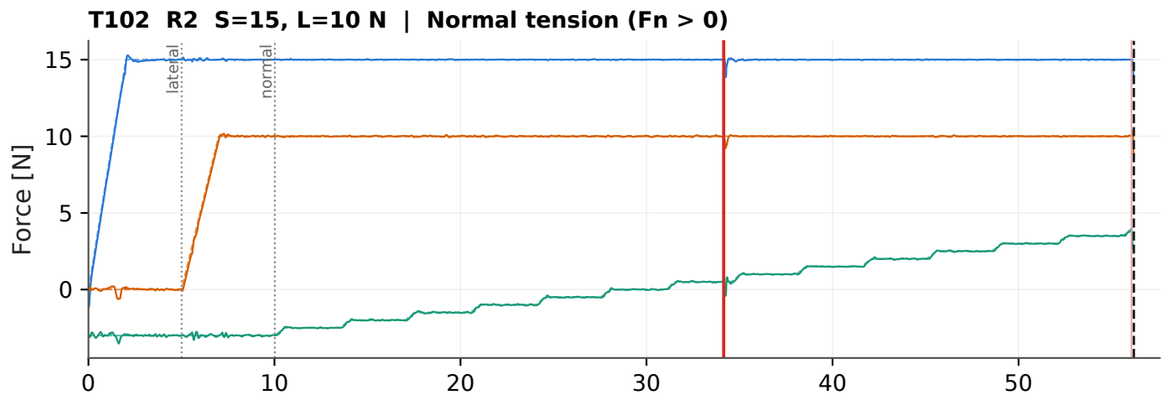
Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

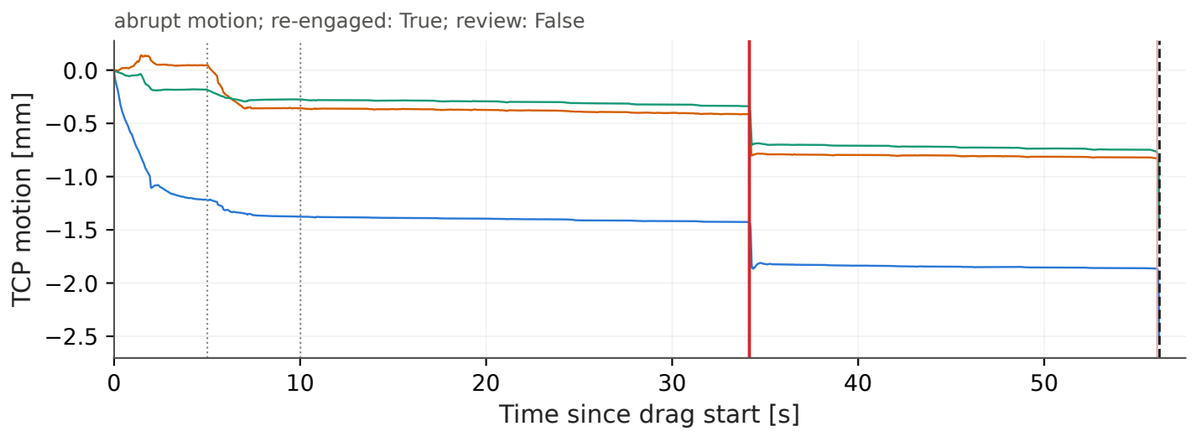






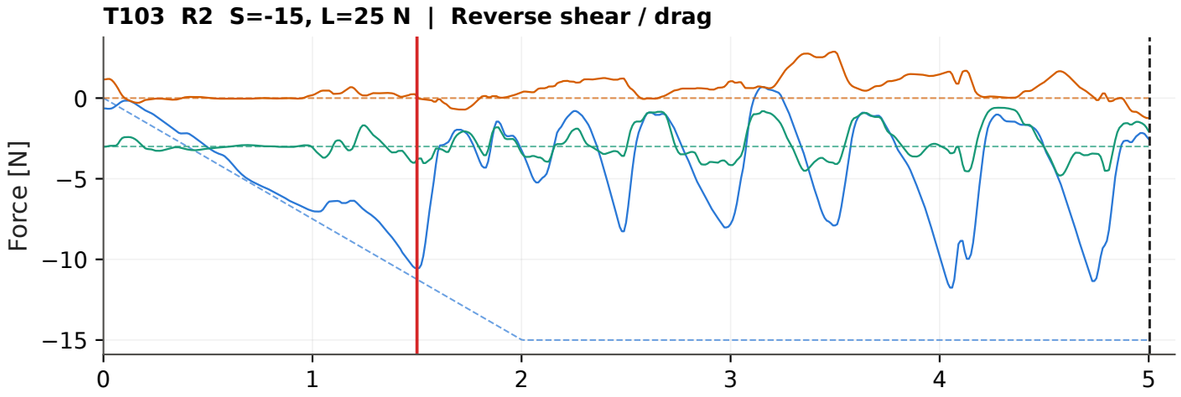


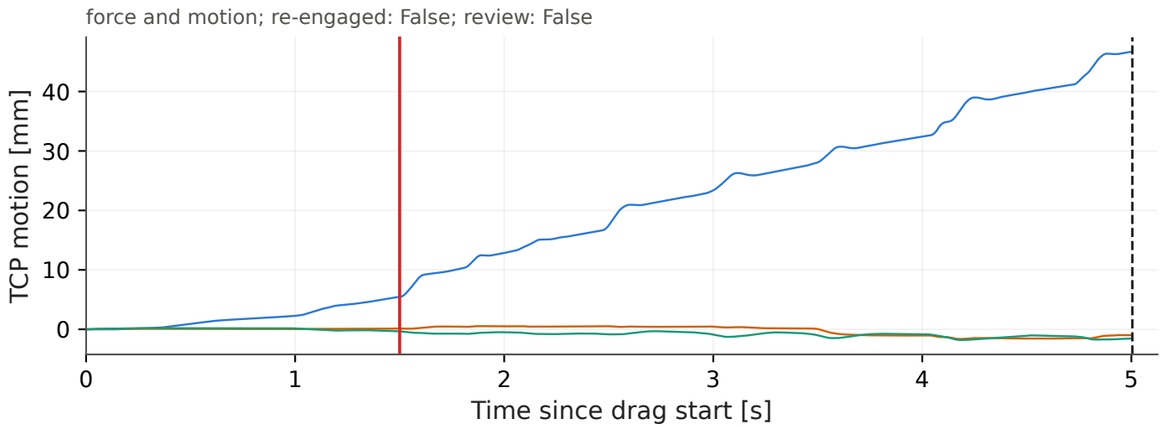


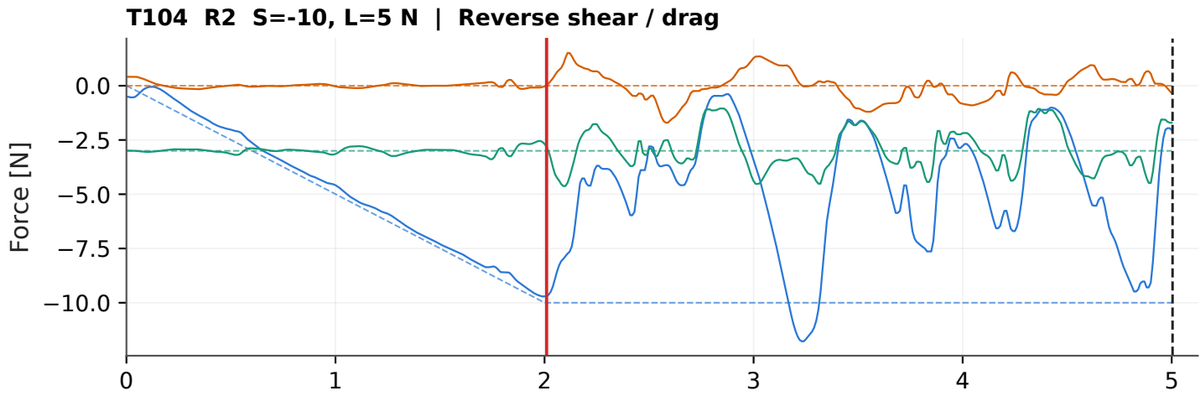


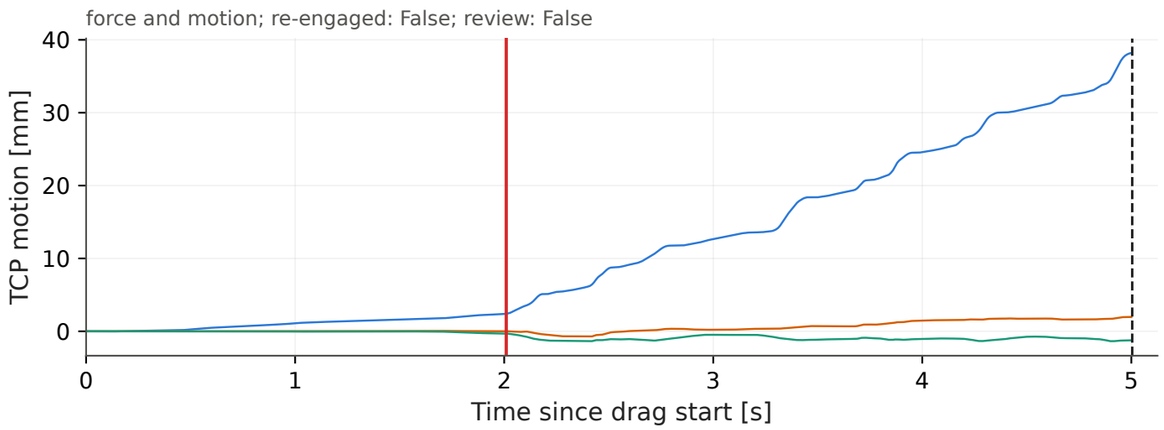
Trial audit | 103–105 / 231

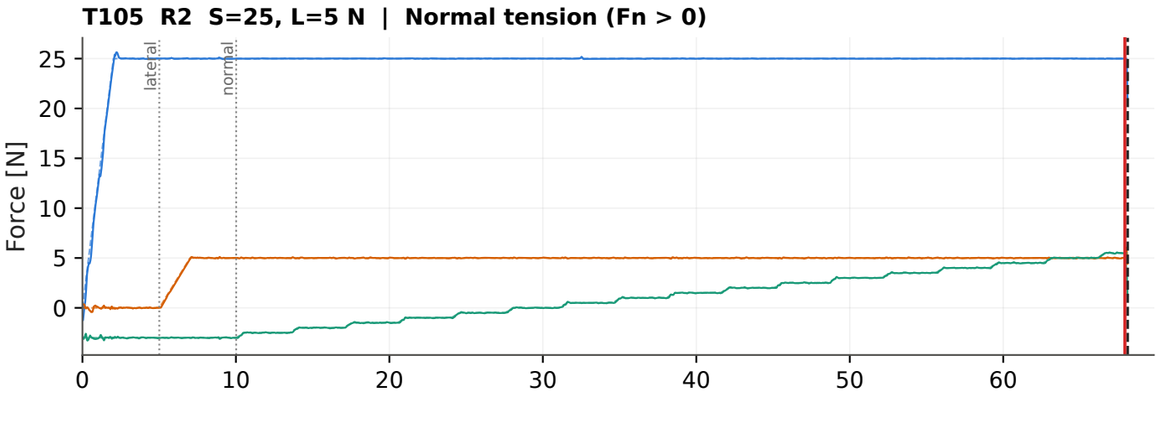
Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

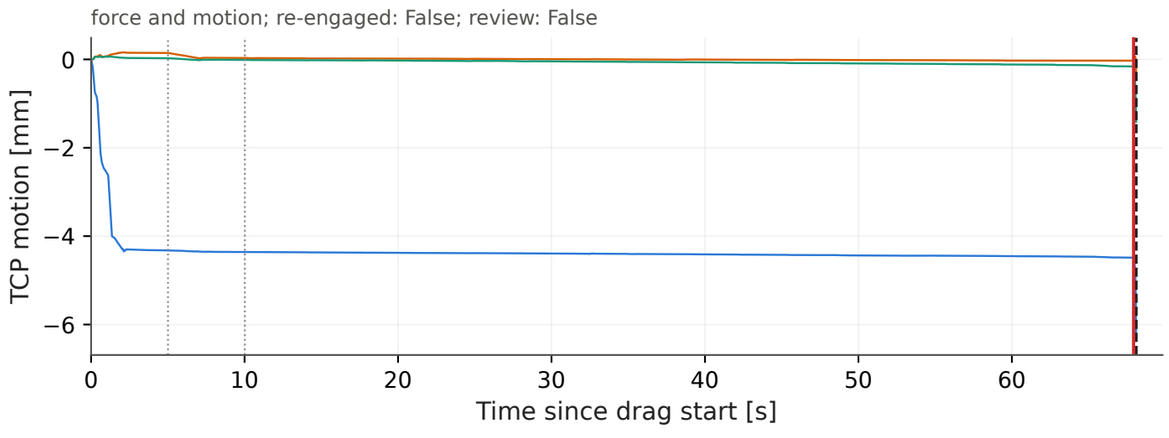








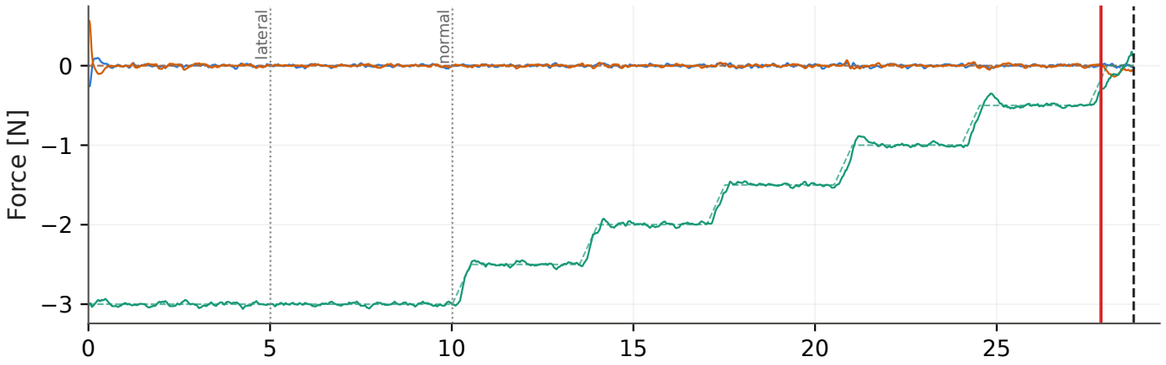




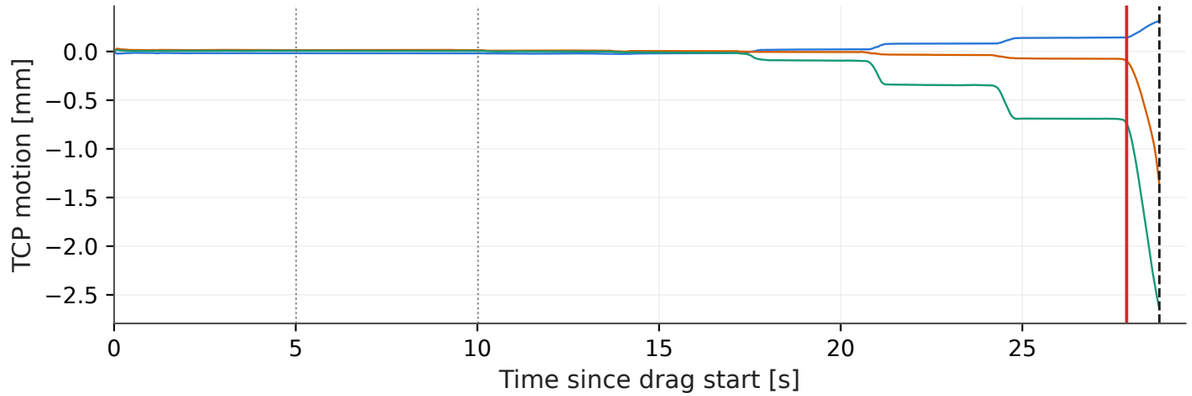
Trial audit | 106–108 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

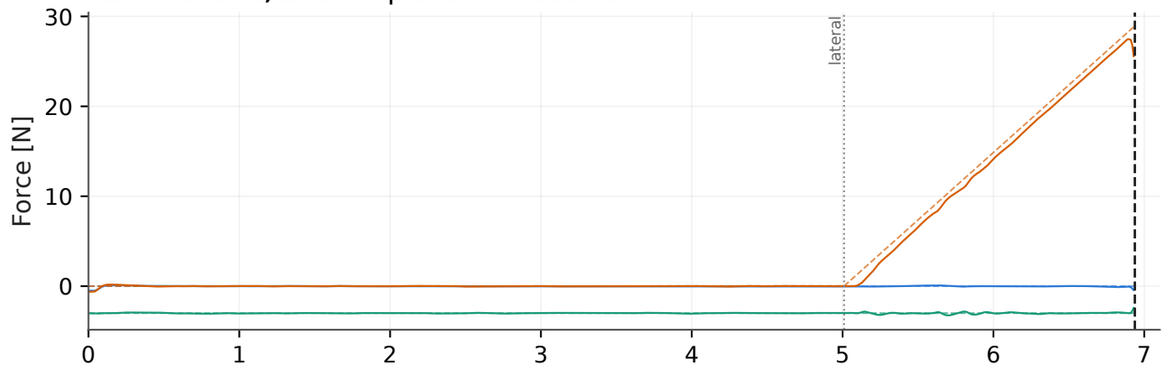
T106 R2 S=0, L=0 N | Normal unloading (Fn <= 0)



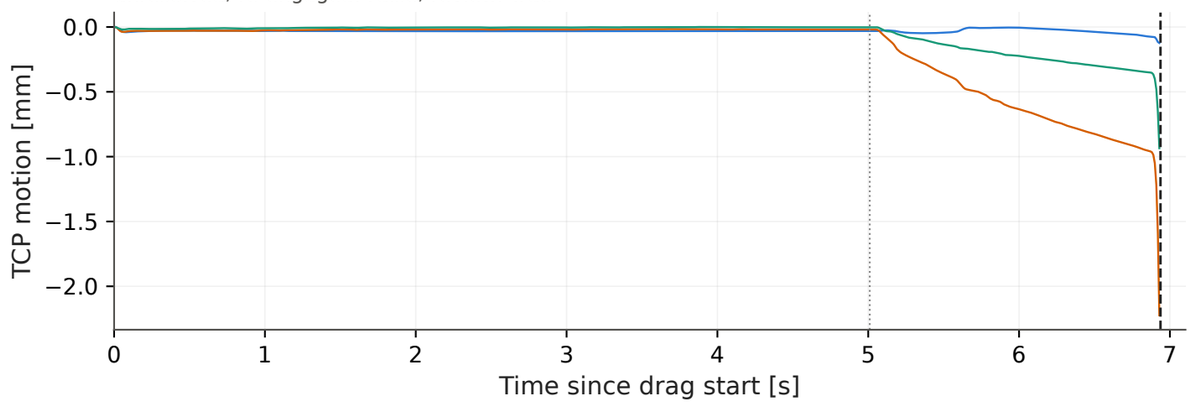
abrupt motion; re-engaged: False; review: False



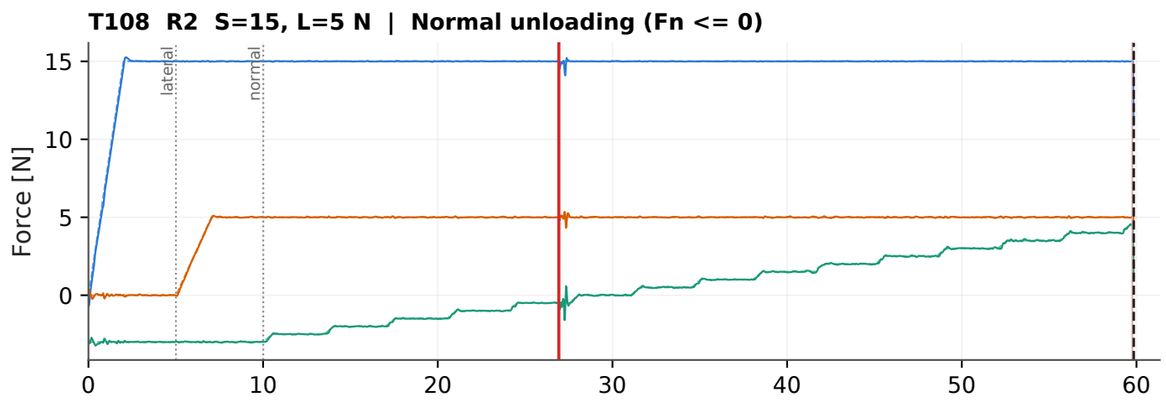
T107 R2 S=0, L=30 N | Onset unresolved



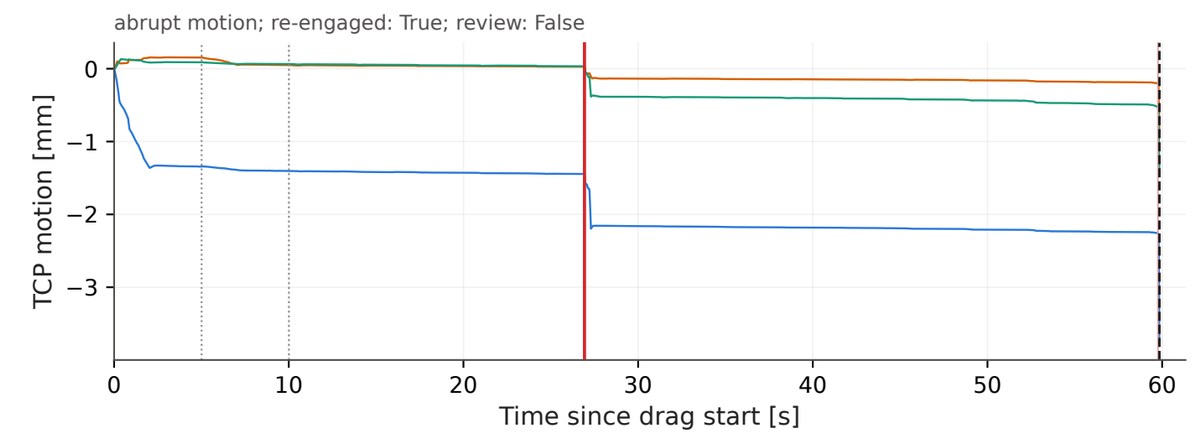
unresolved; re-engaged: False; review: True



T108 R2 S=15, L=5 N | Normal unloading (Fn <= 0)



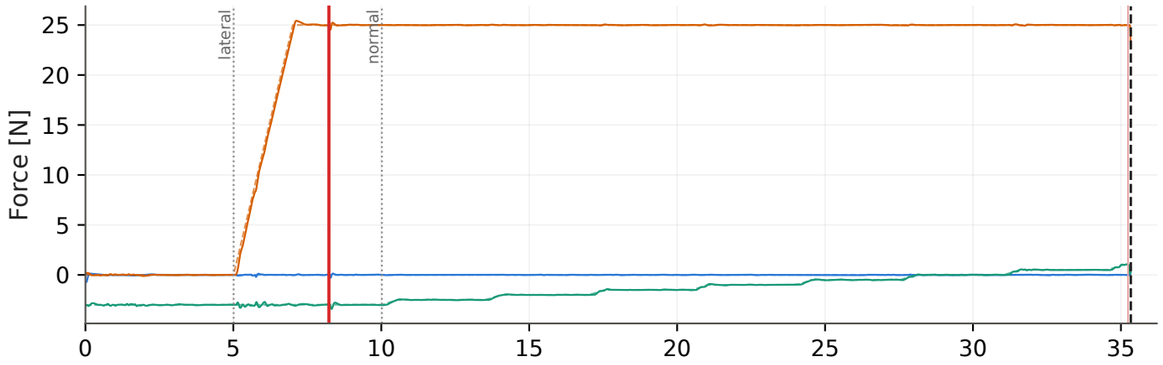
abrupt motion; re-engaged: True; review: False



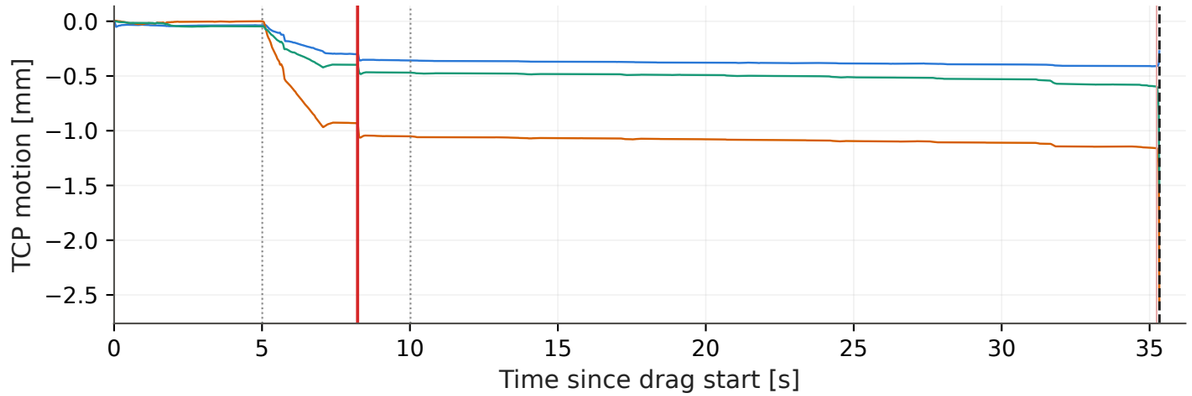
Trial audit | 109–111 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

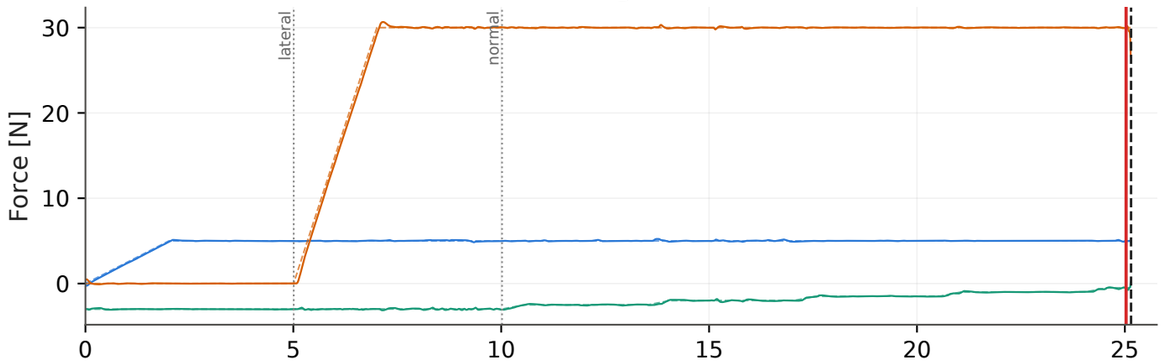
T109 R2 S=-0, L=25 N | Lateral loading



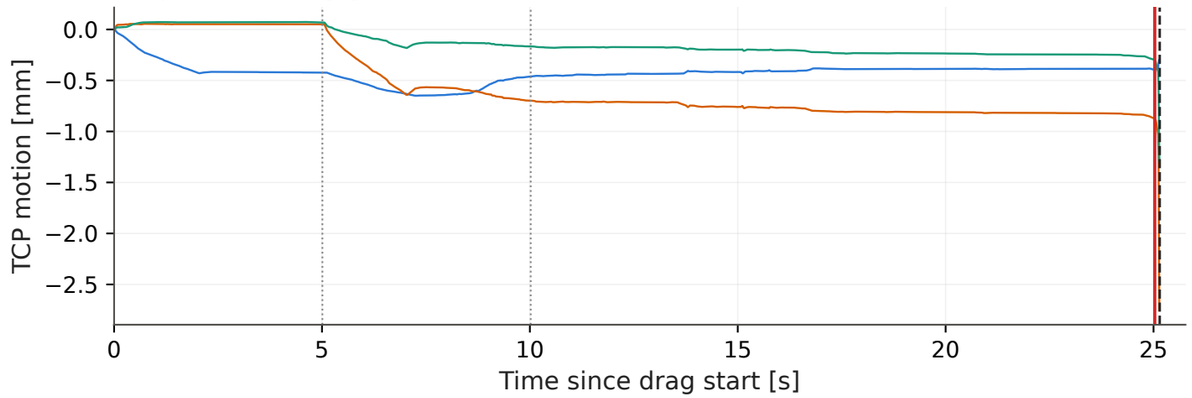
abrupt motion; re-engaged: True; review: True



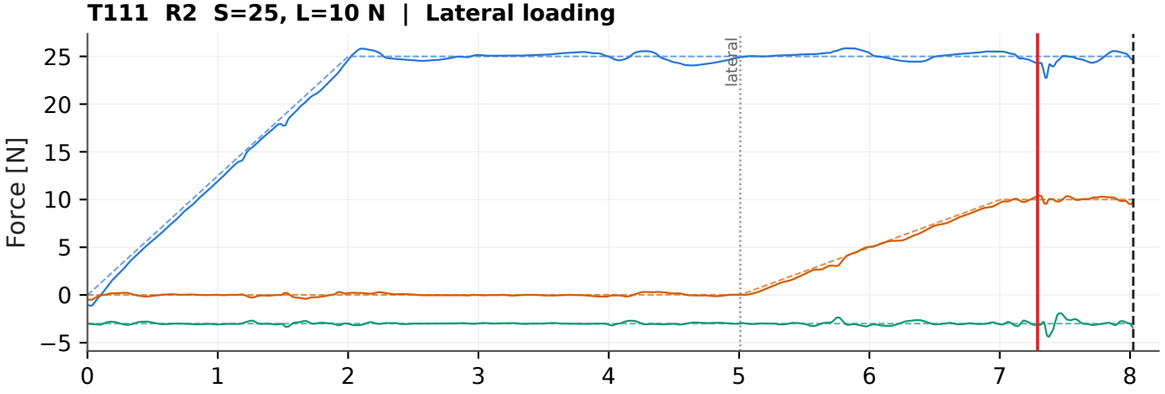
T110 R2 S=5, L=30 N | Normal unloading (Fn <= 0)



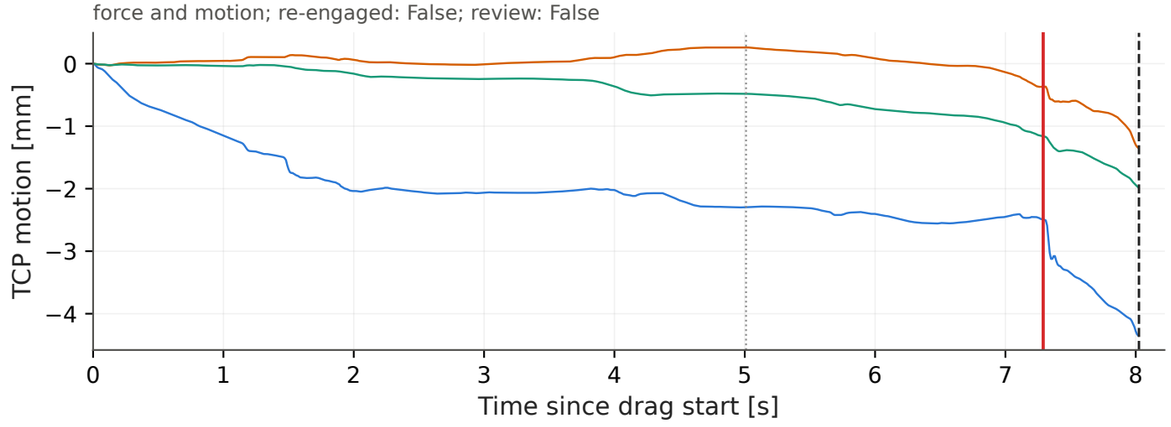
abrupt motion; re-engaged: False; review: False



T111 R2 S=25, L=10 N | Lateral loading



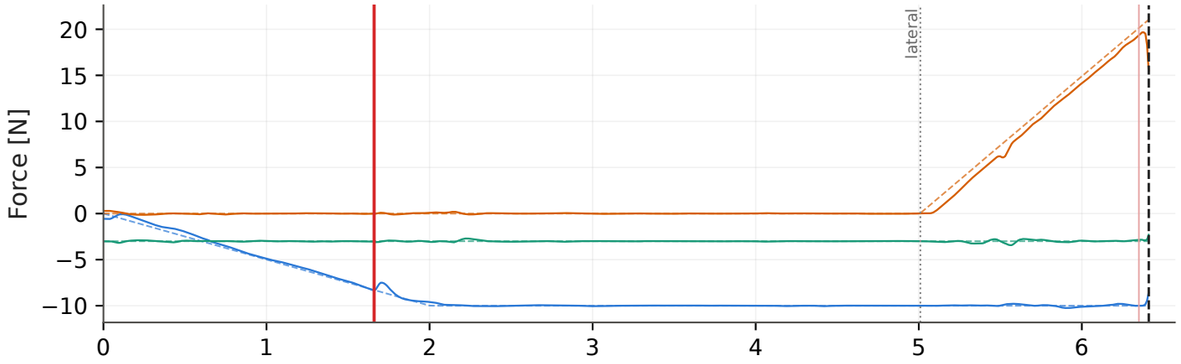
force and motion; re-engaged: False; review: False



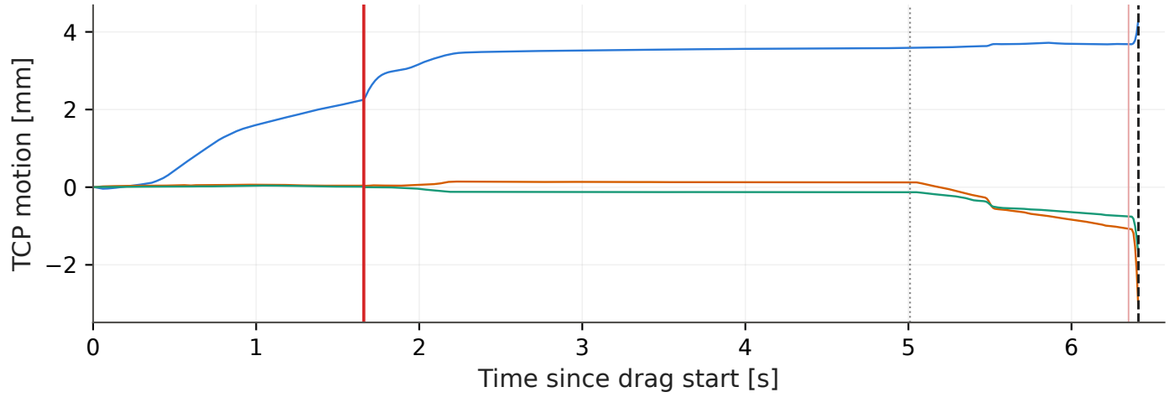
Trial audit | 112–114 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

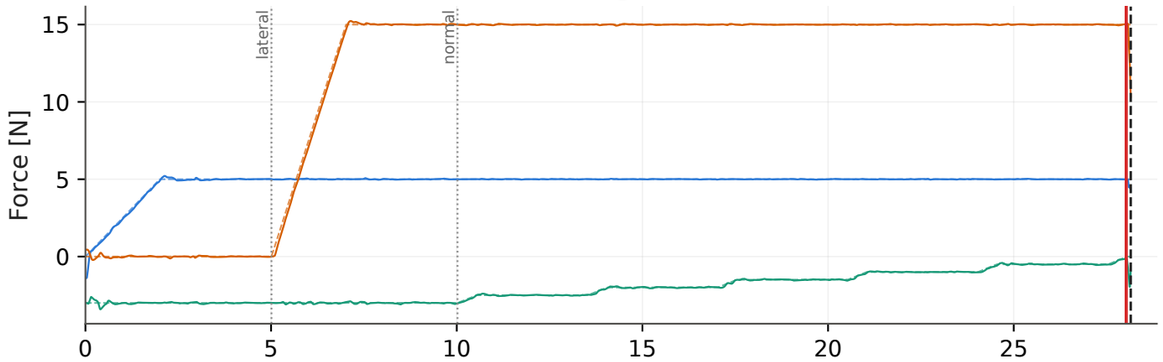
T112 R2 S=-10, L=30 N | Reverse shear / drag



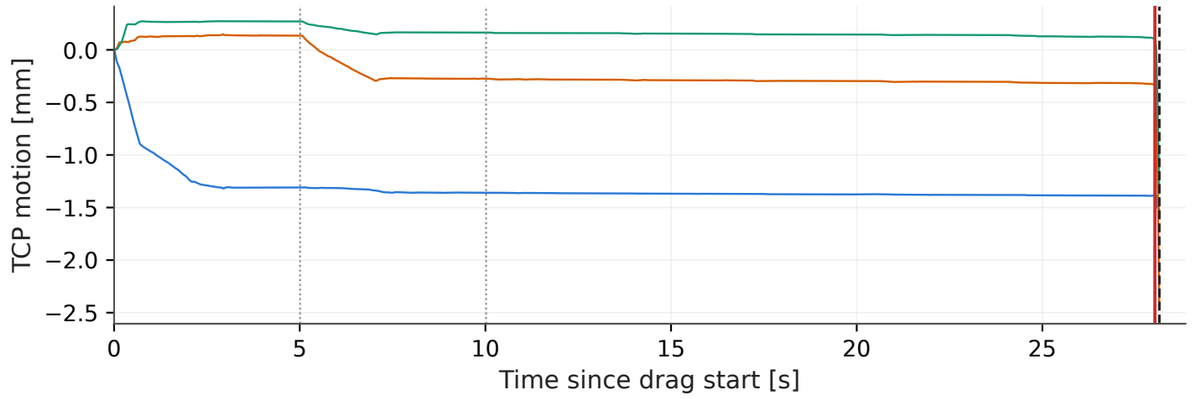
force and motion; re-engaged: True; review: True



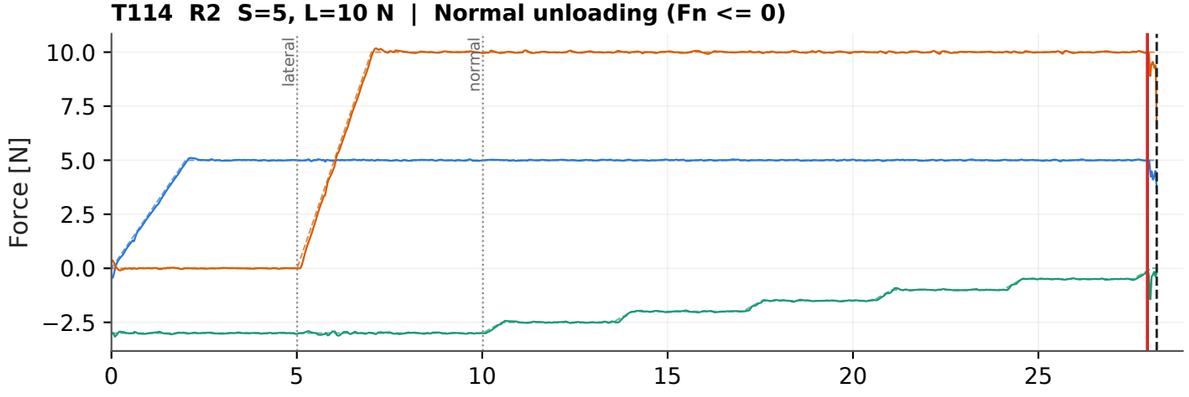
T113 R2 S=5, L=15 N | Normal unloading (Fn <= 0)



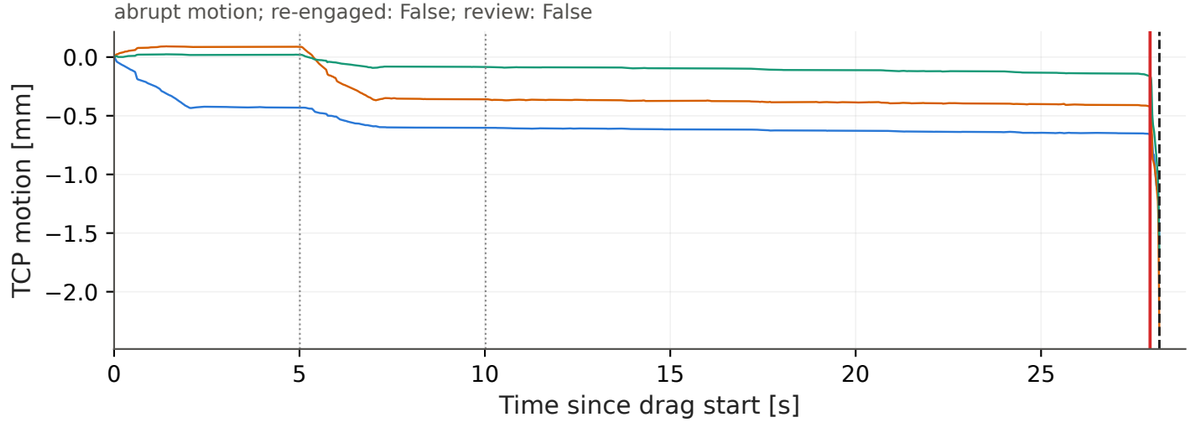
abrupt motion; re-engaged: False; review: False



T114 R2 S=5, L=10 N | Normal unloading (Fn <= 0)



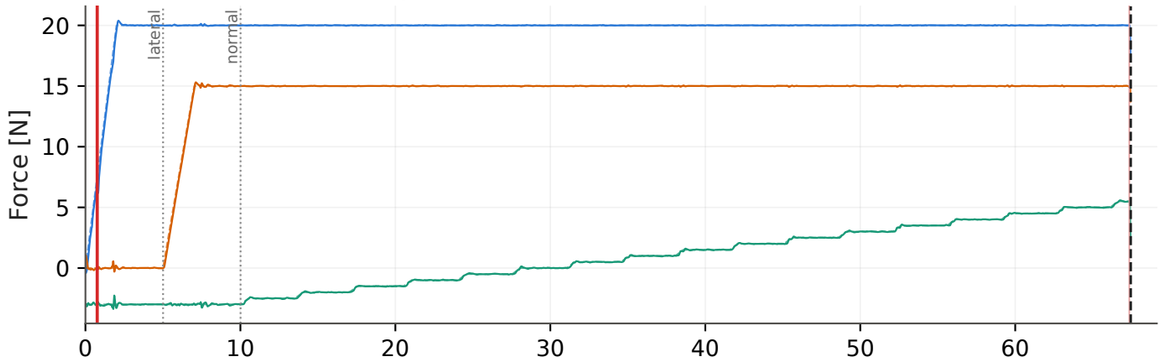
abrupt motion; re-engaged: False; review: False



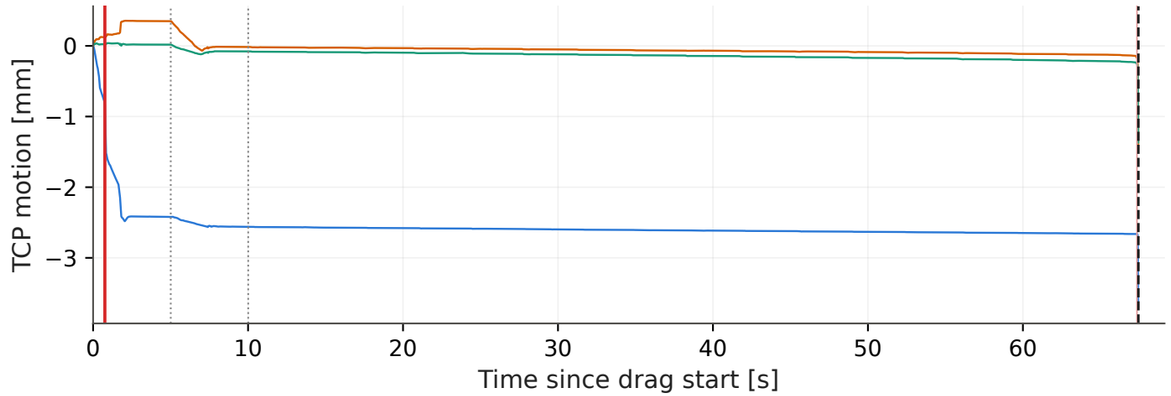
Trial audit | 115–117 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

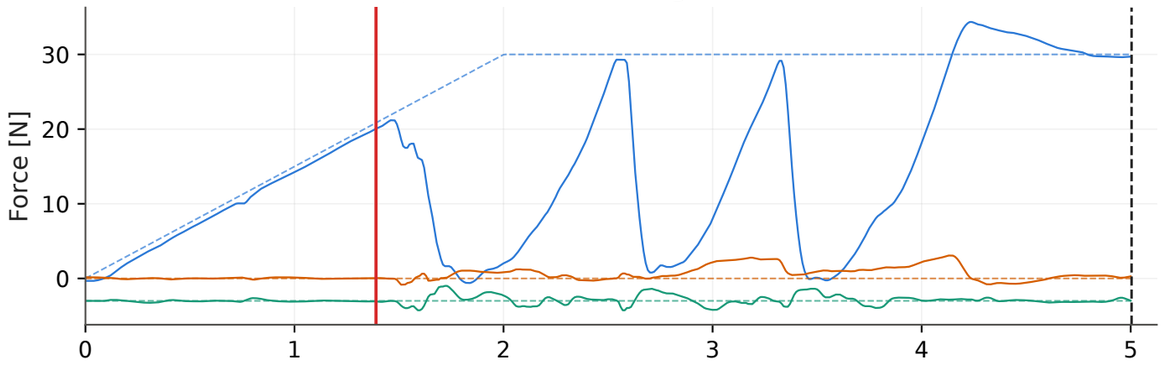
T115 R2 S=20, L=15 N | Forward shear / drag



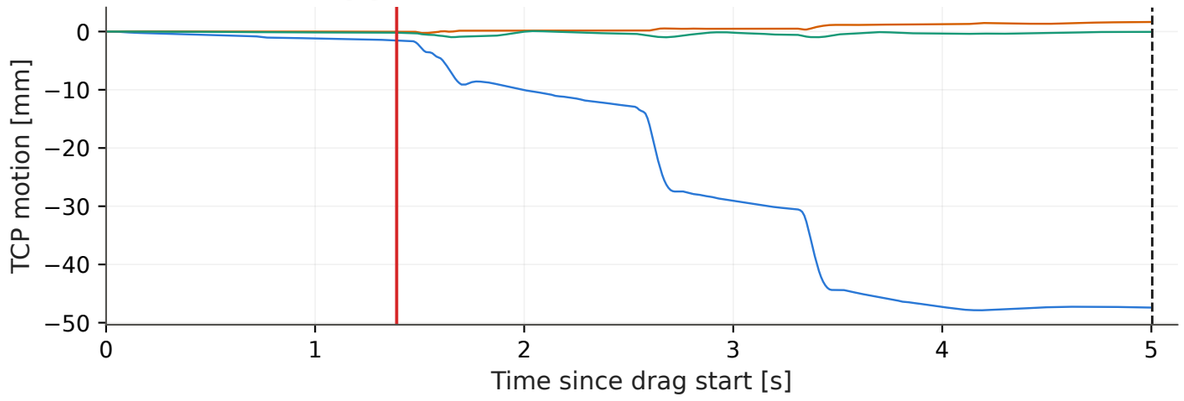
force and motion; re-engaged: True; review: True



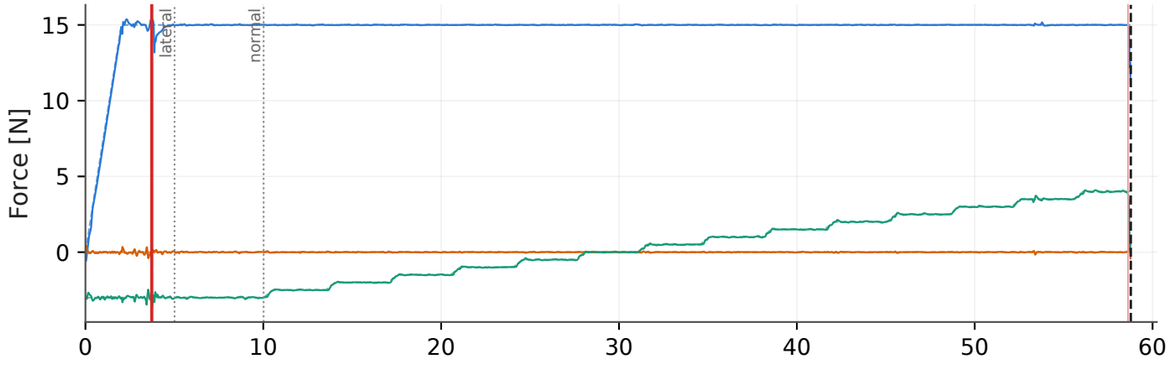
T116 R2 S=30, L=10 N | Forward shear / drag



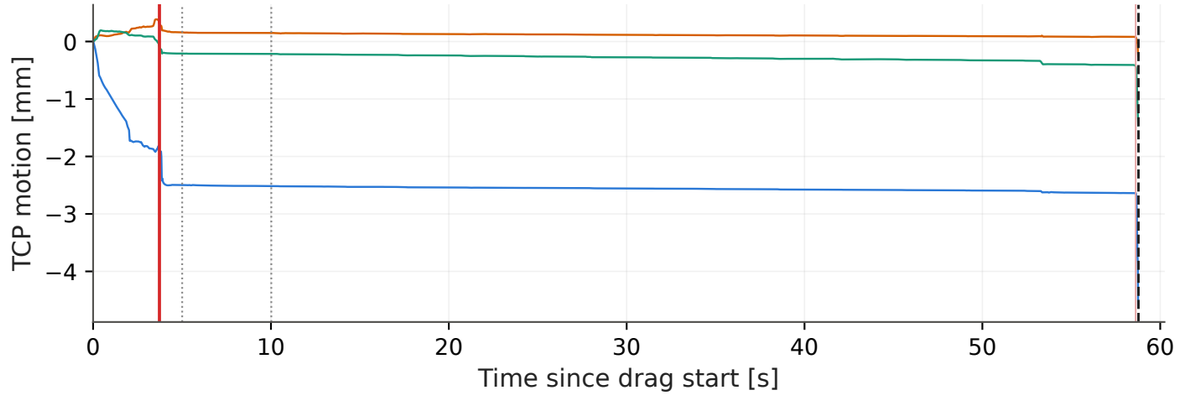
force and motion; re-engaged: False; review: False



T117 R2 S=15, L=0 N | Forward shear / drag



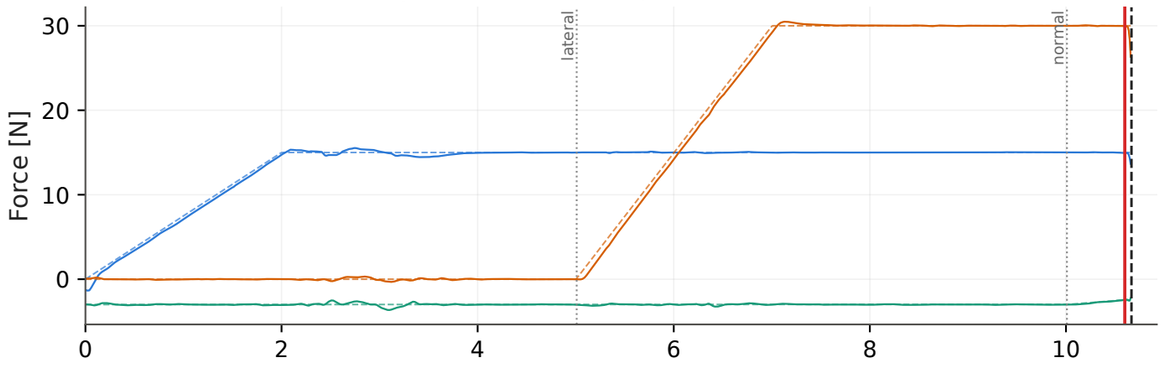
force and motion; re-engaged: True; review: False



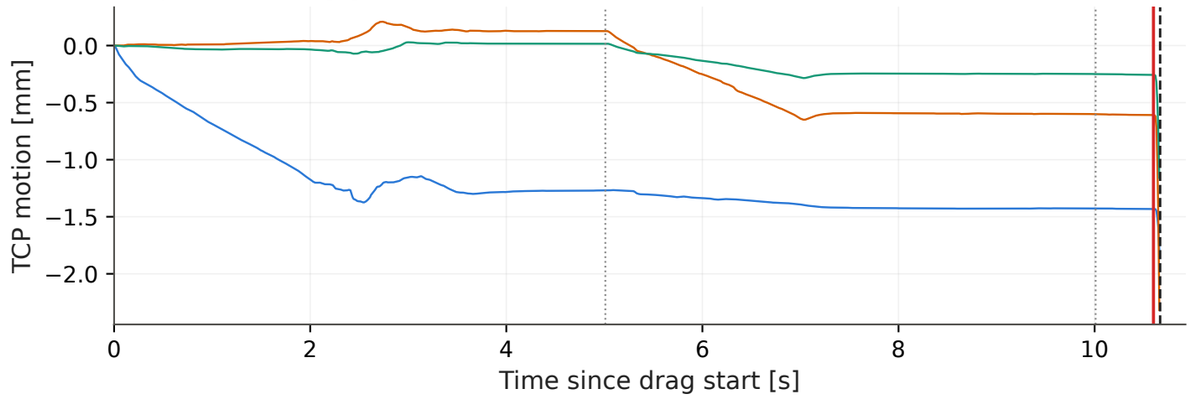
Trial audit | 118–120 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

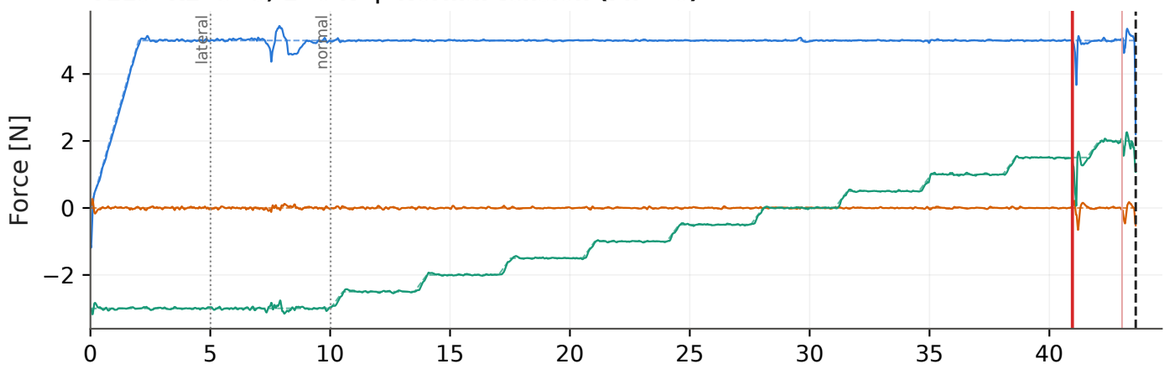
T118 R2 S=15, L=30 N | Normal unloading (Fn <= 0)



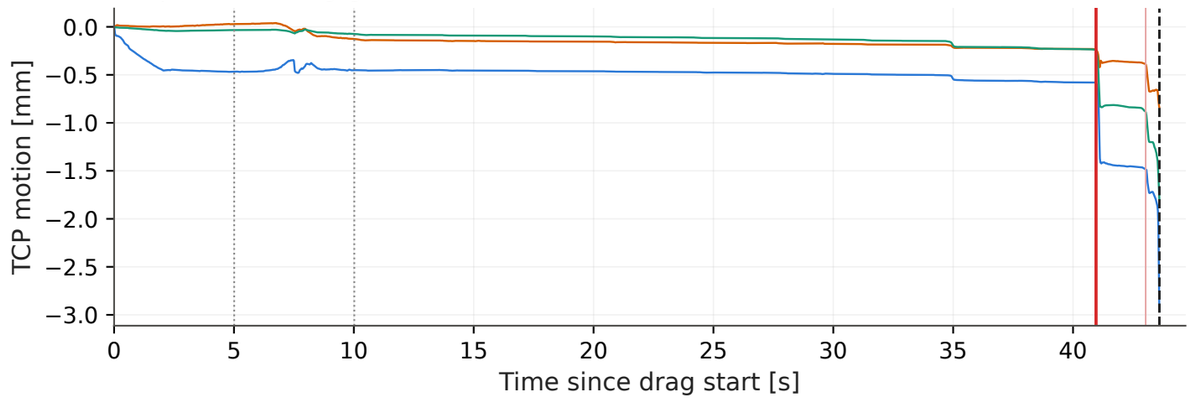
abrupt motion; re-engaged: False; review: False



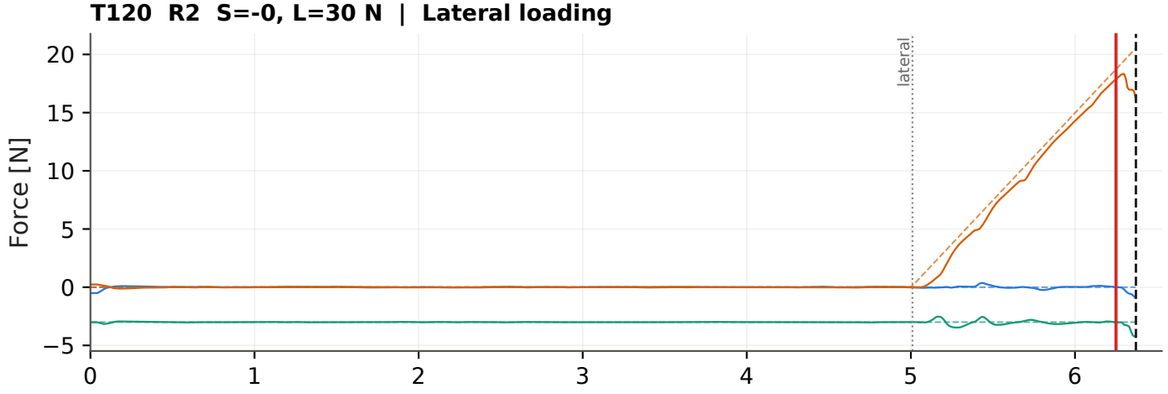
T119 R2 S=5, L=0 N | Normal tension (Fn > 0)



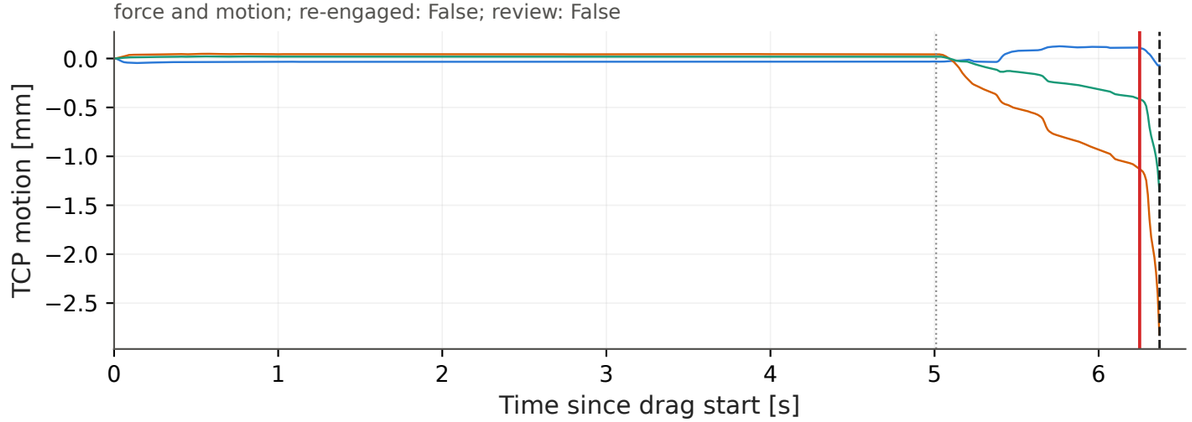
abrupt motion; re-engaged: True; review: True



T120 R2 S=-0, L=30 N | Lateral loading



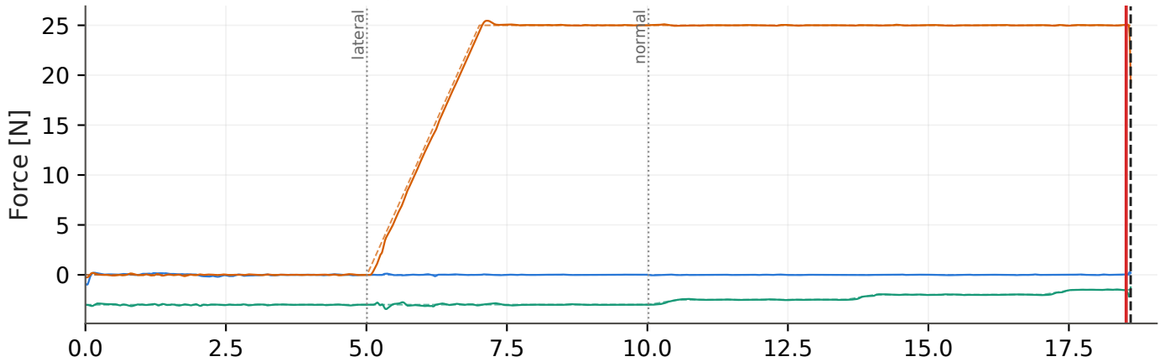
force and motion; re-engaged: False; review: False



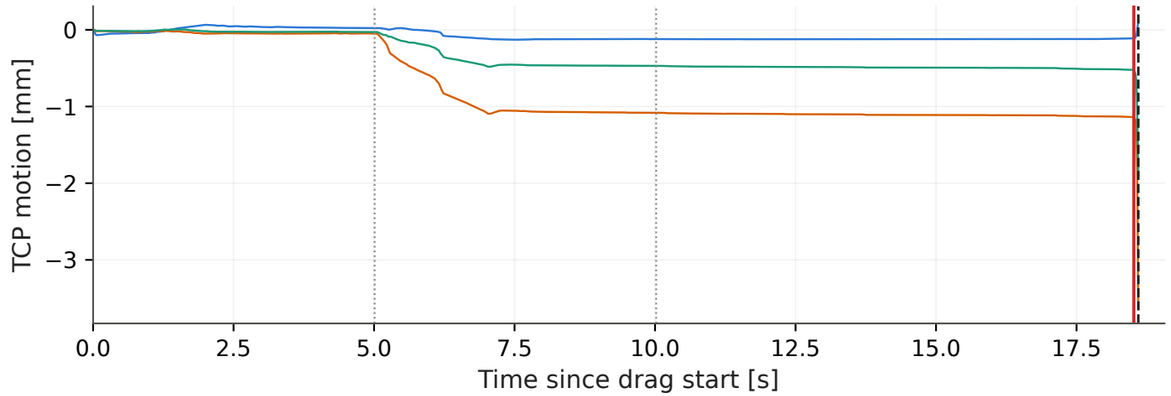
Trial audit | 121–123 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

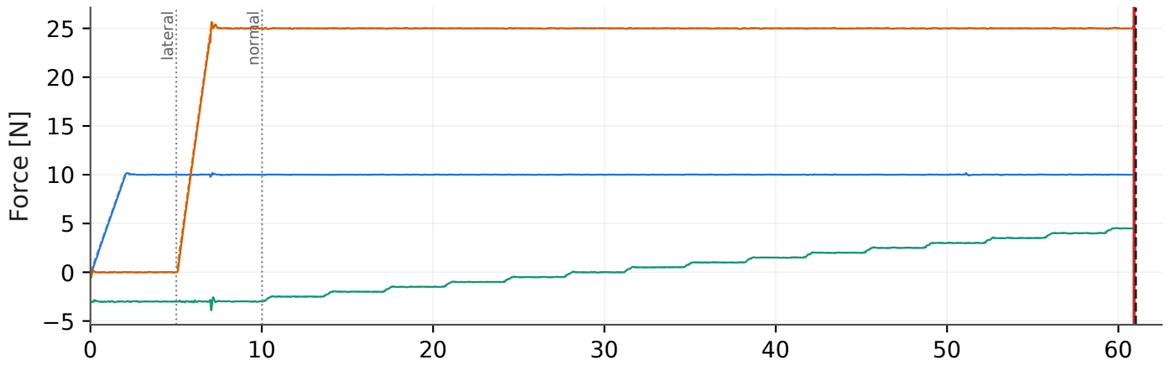
T121 R2 S=0, L=25 N | Normal unloading (Fn <= 0)



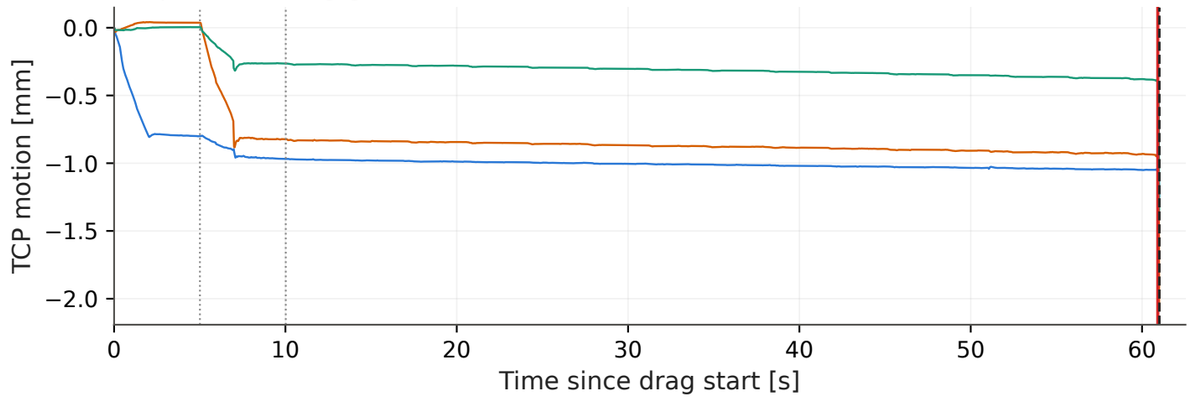
abrupt motion; re-engaged: False; review: False



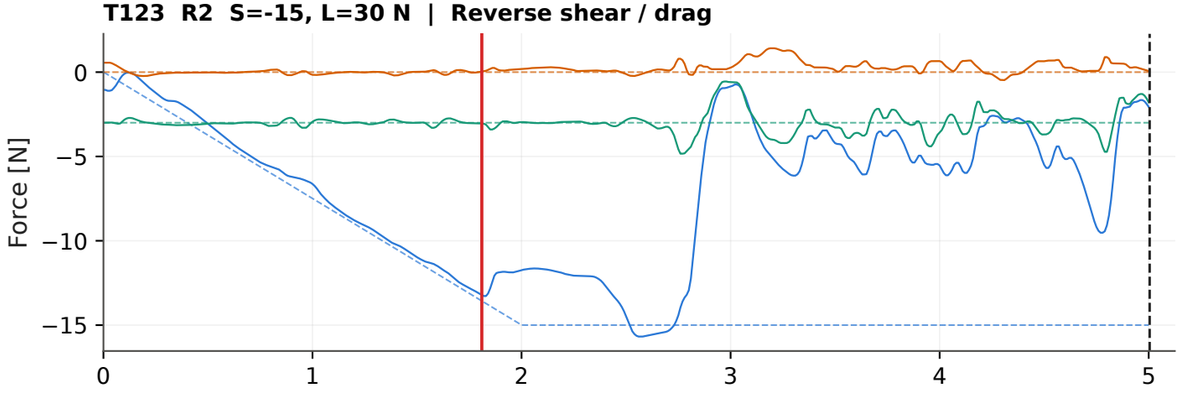
T122 R2 S=10, L=25 N | Normal tension (Fn > 0)



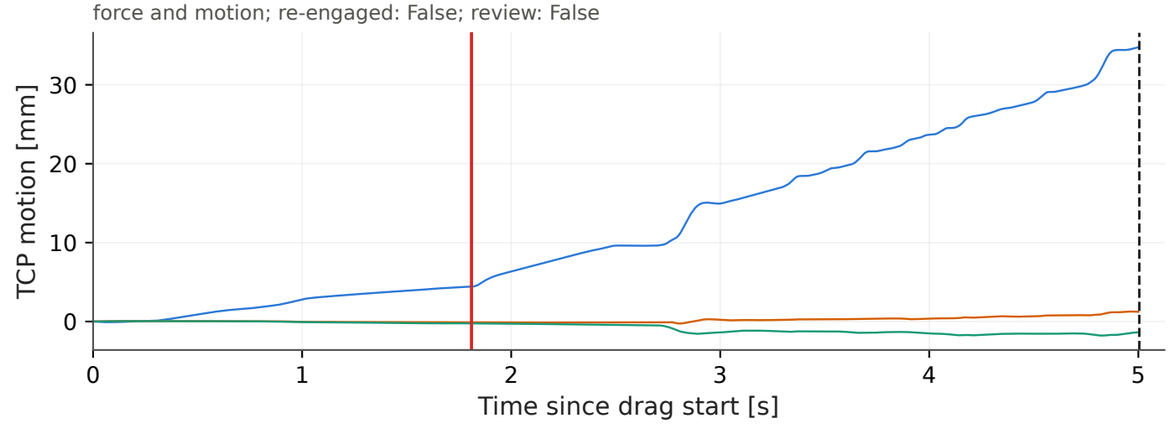
abrupt motion; re-engaged: False; review: False



T123 R2 S=-15, L=30 N | Reverse shear / drag



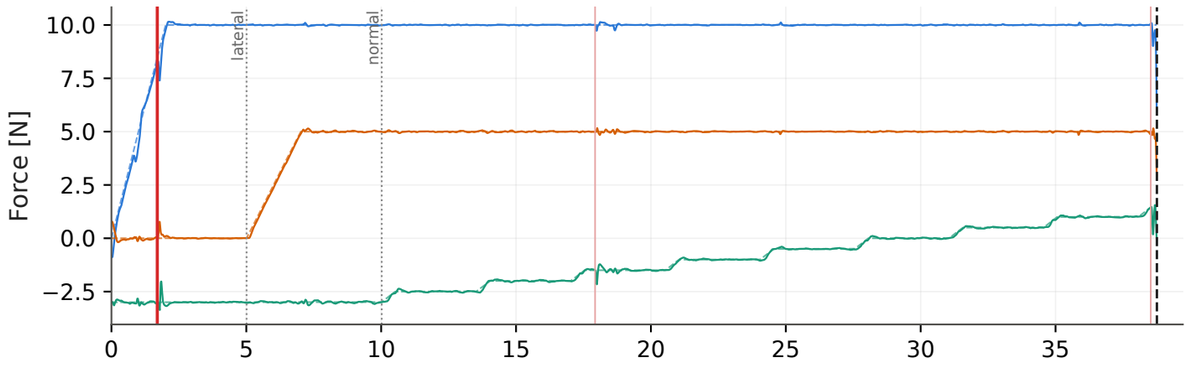
force and motion; re-engaged: False; review: False



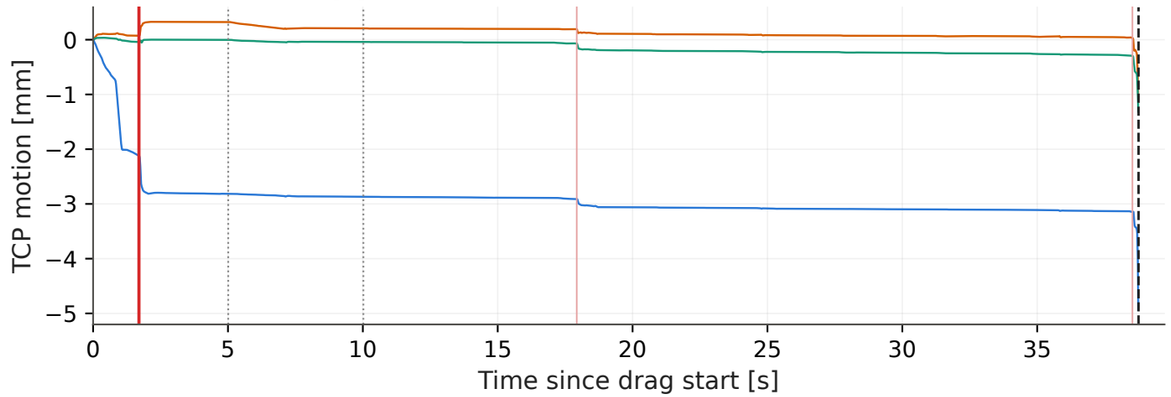
Trial audit | 124–126 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

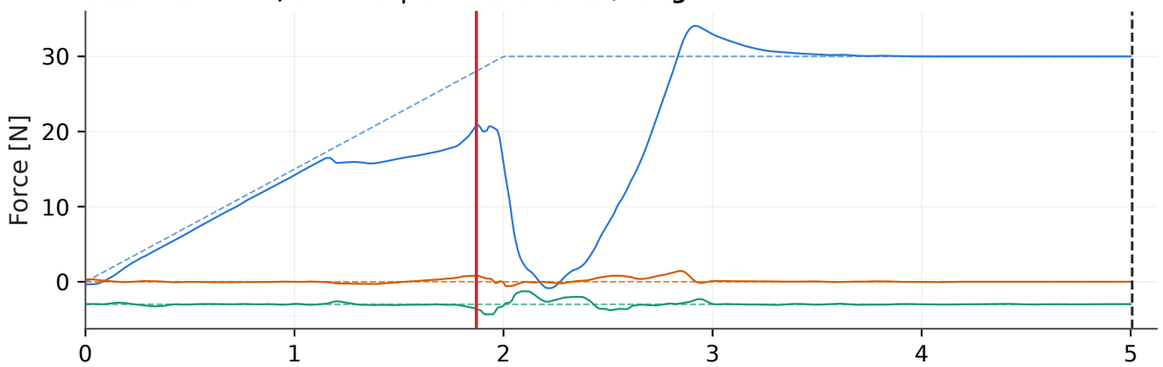
T124 R2 S=10, L=5 N | Forward shear / drag



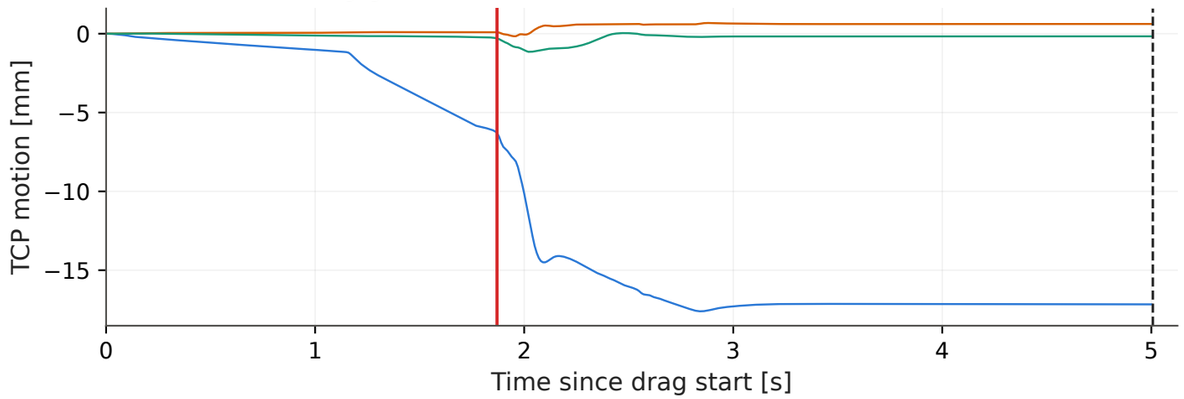
force and motion; re-engaged: True; review: True



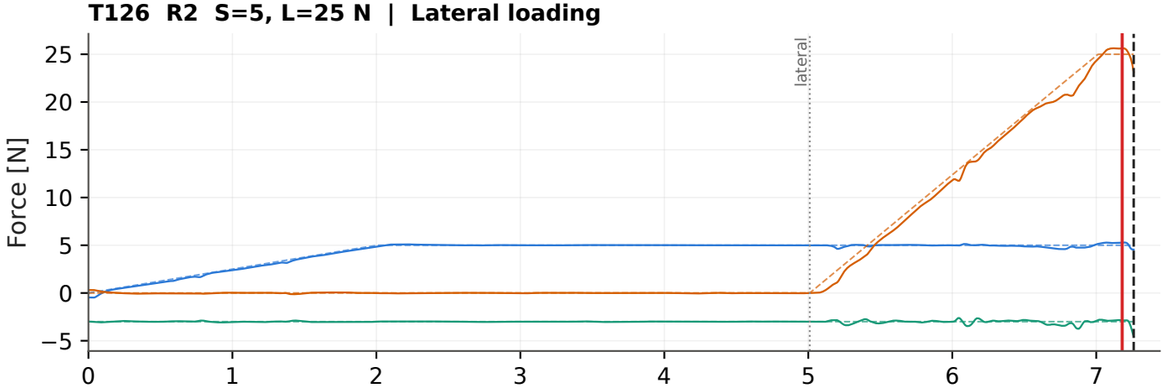
T125 R2 S=30, L=30 N | Forward shear / drag



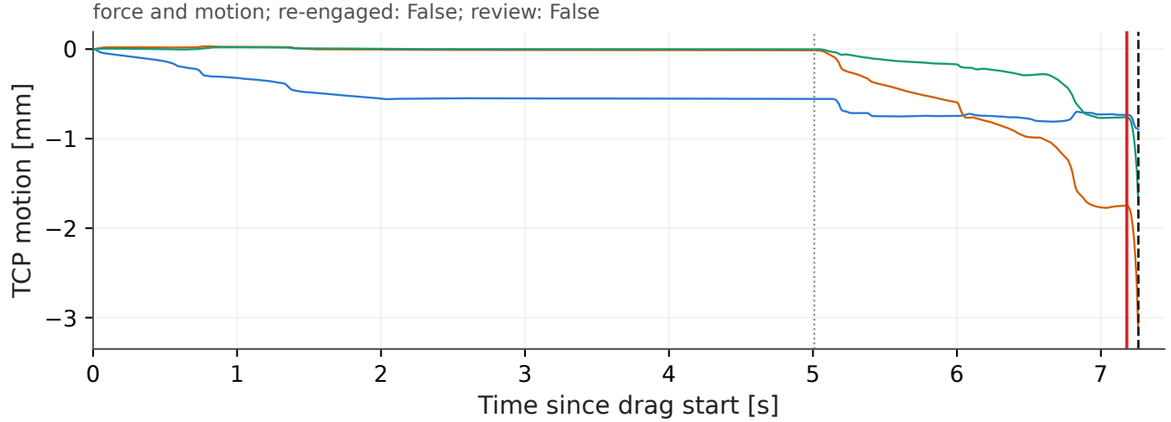
force and motion; re-engaged: True; review: False



T126 R2 S=5, L=25 N | Lateral loading



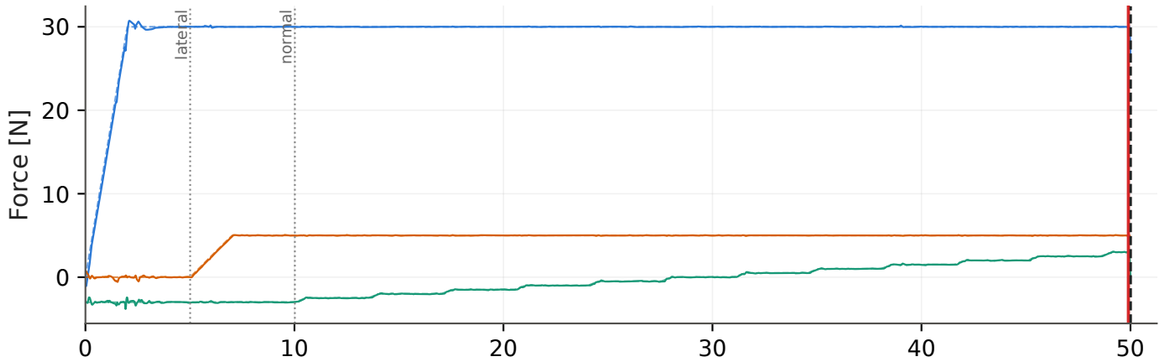
force and motion; re-engaged: False; review: False



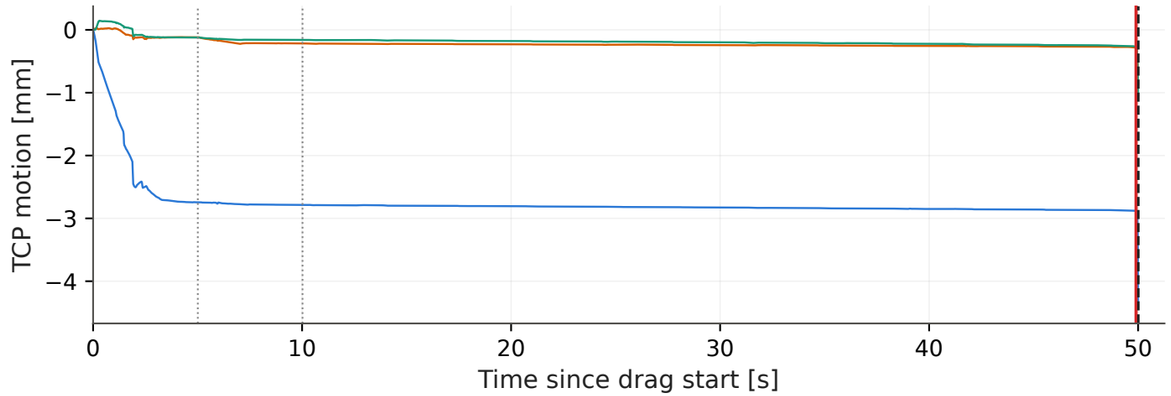
Trial audit | 127–129 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

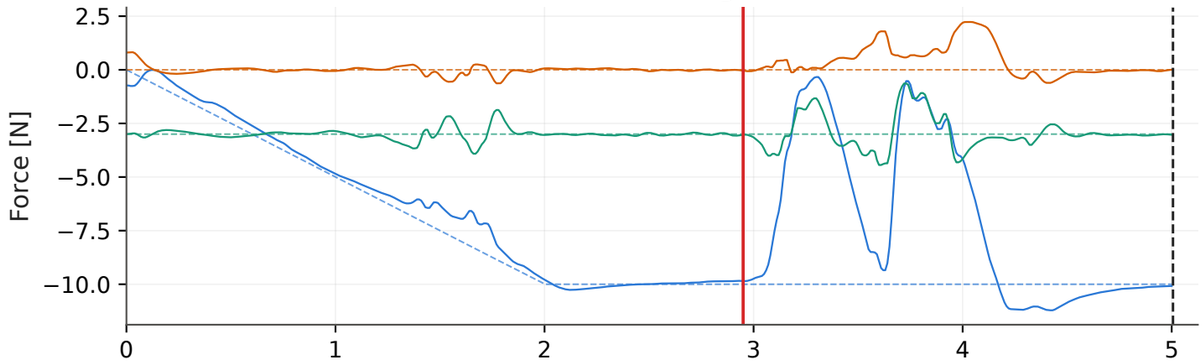
T127 R2 S=30, L=5 N | Normal tension (Fn > 0)



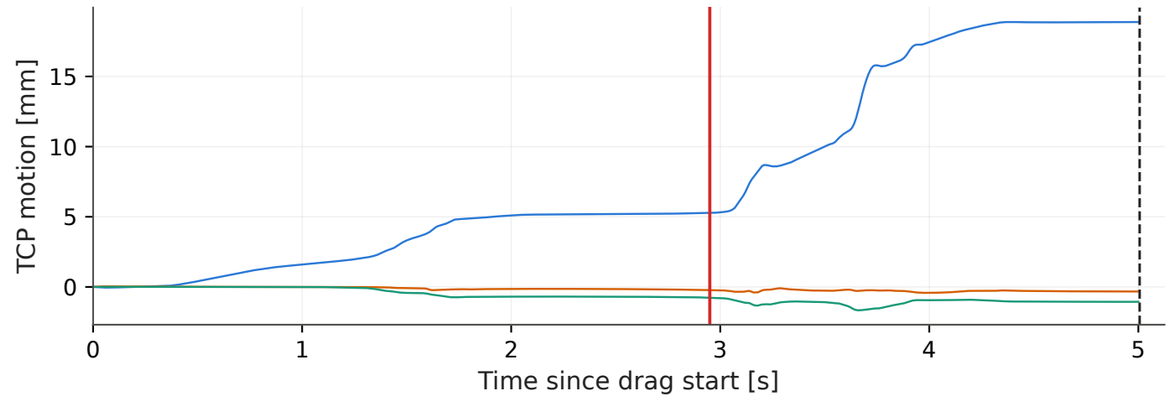
abrupt motion; re-engaged: False; review: False



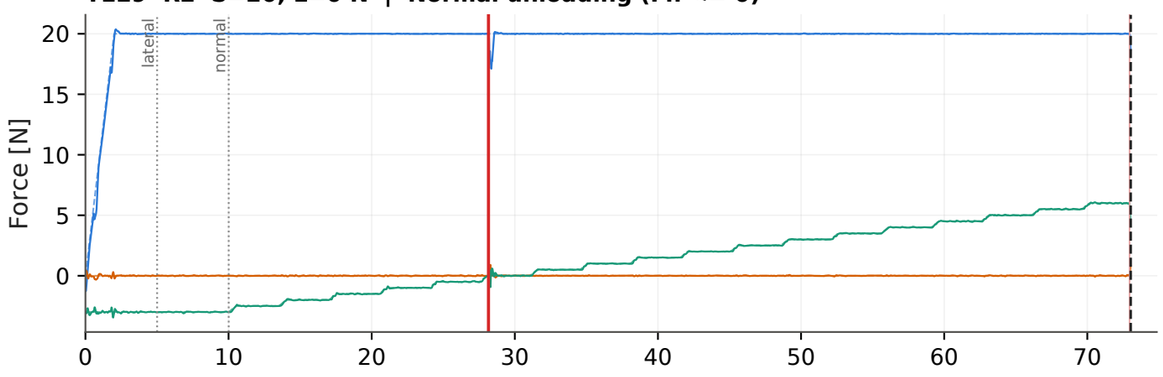
T128 R2 S=-10, L=15 N | Reverse shear / drag



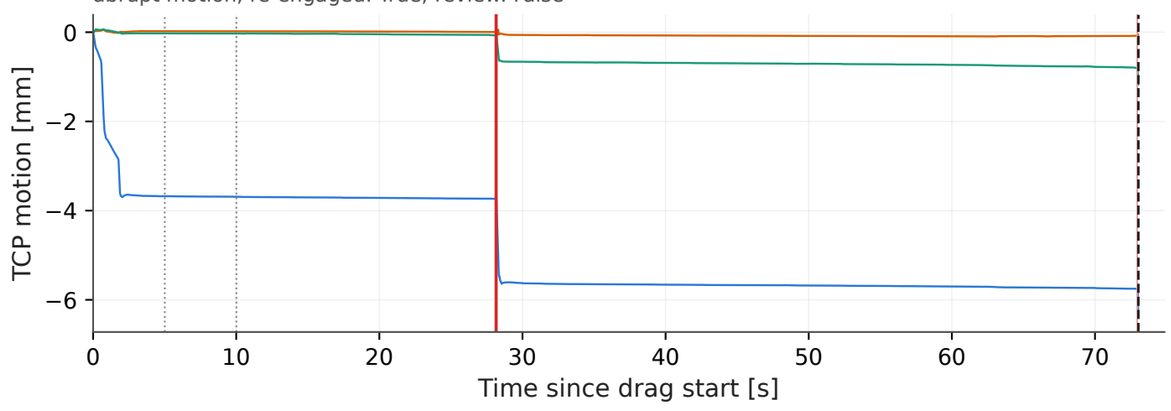
force and motion; re-engaged: True; review: False



T129 R2 S=20, L=0 N | Normal unloading (Fn <= 0)



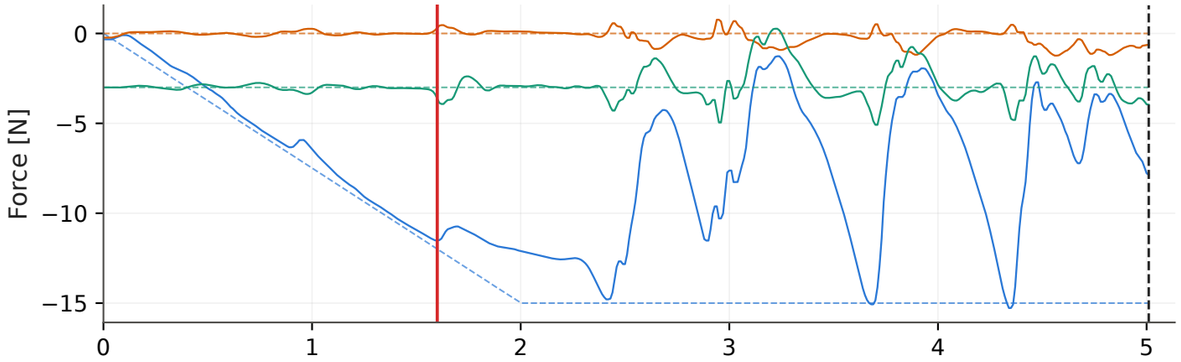
abrupt motion; re-engaged: True; review: False



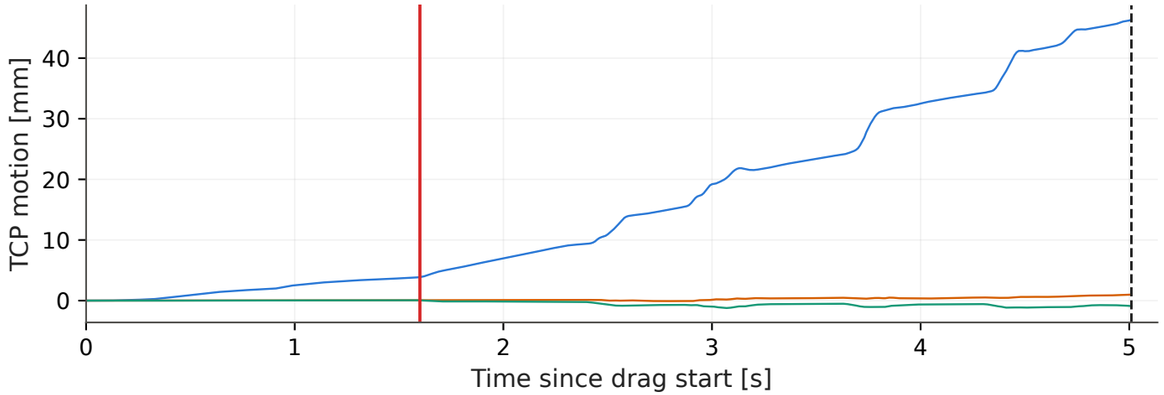
Trial audit | 130–132 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

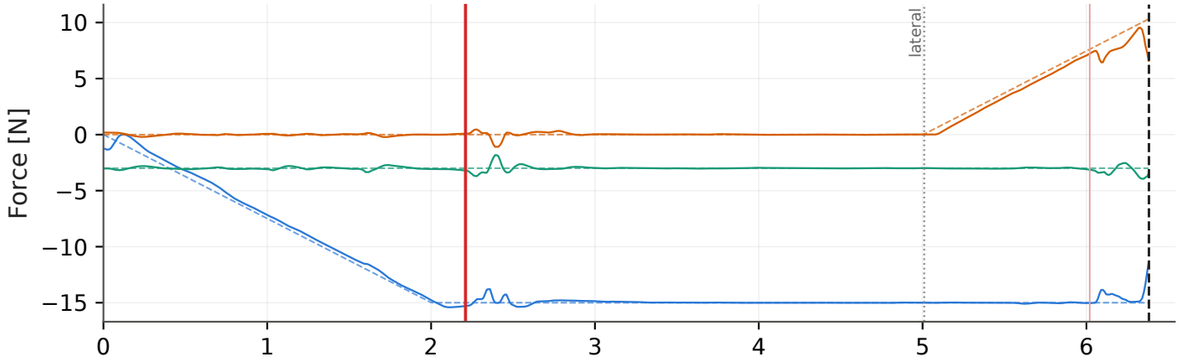
T130 R2 S=-15, L=5 N | Reverse shear / drag



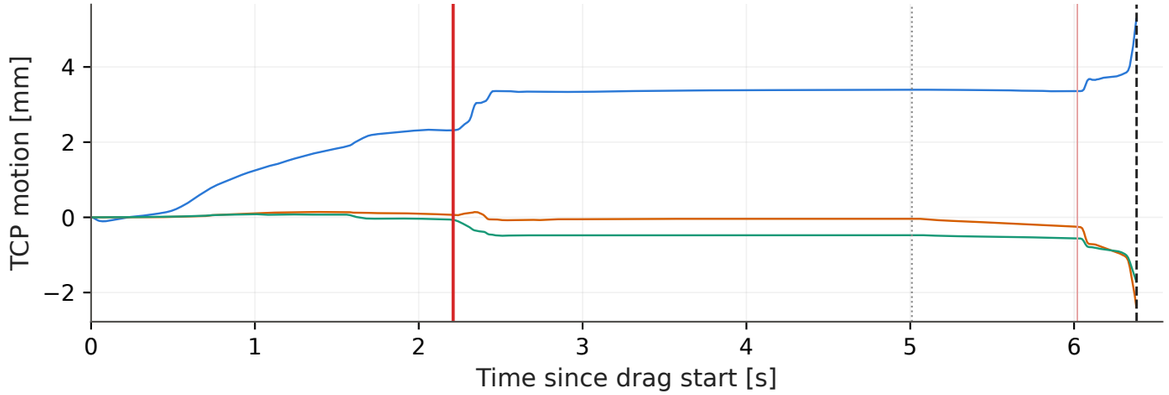
force and motion; re-engaged: False; review: False



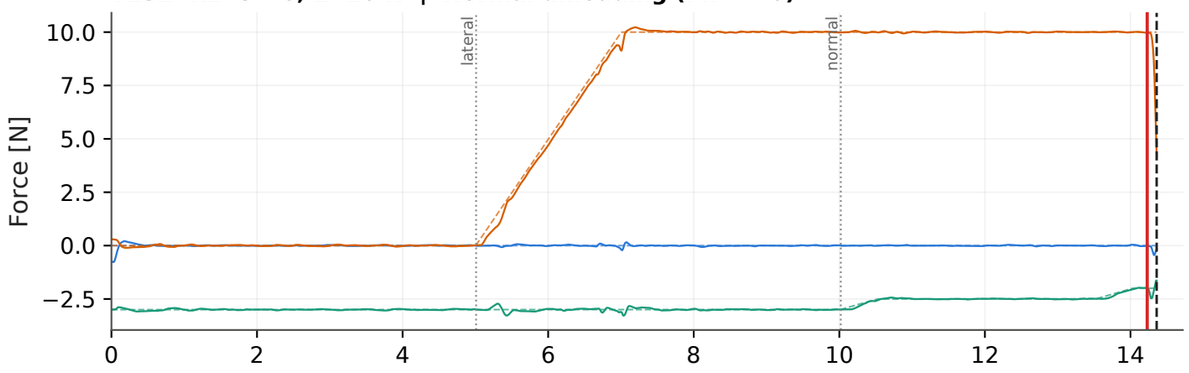
T131 R2 S=-15, L=15 N | Reverse shear / drag



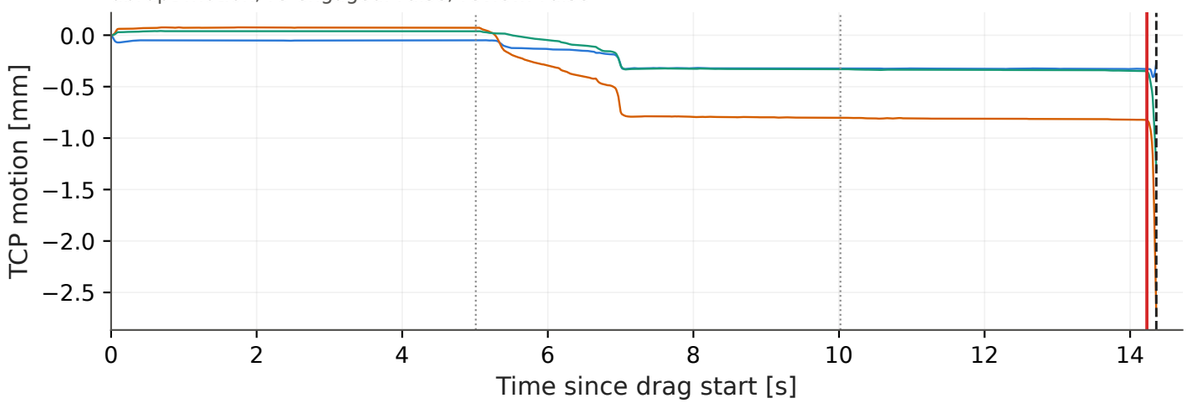
force and motion; re-engaged: True; review: False



T132 R2 S=-0, L=10 N | Normal unloading (Fn <= 0)



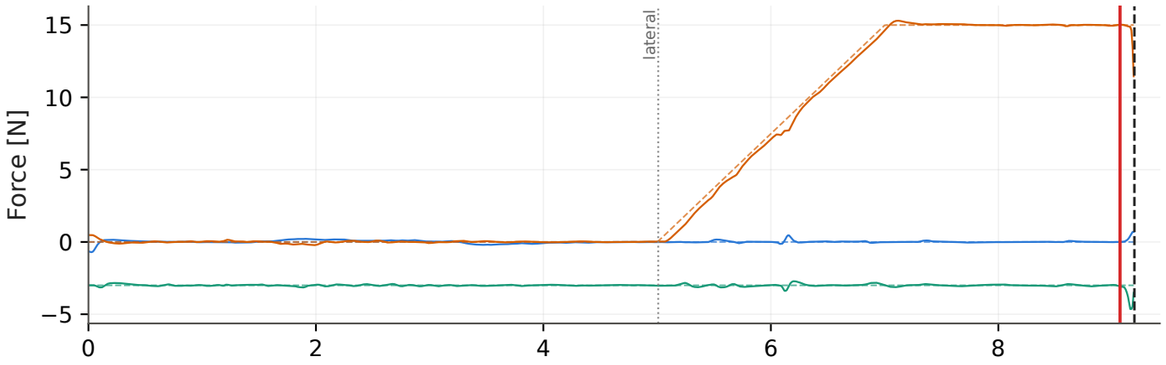
abrupt motion; re-engaged: False; review: False

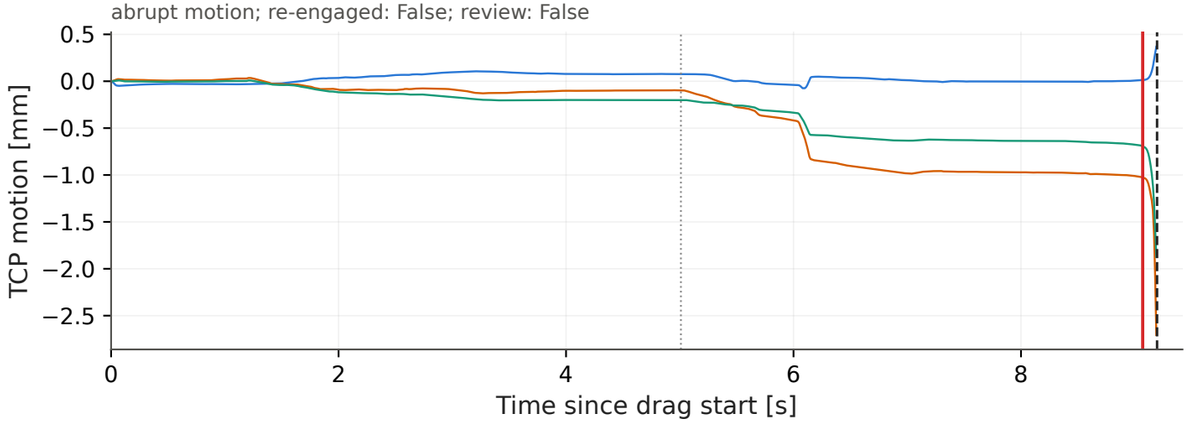


Trial audit | 133–135 / 231

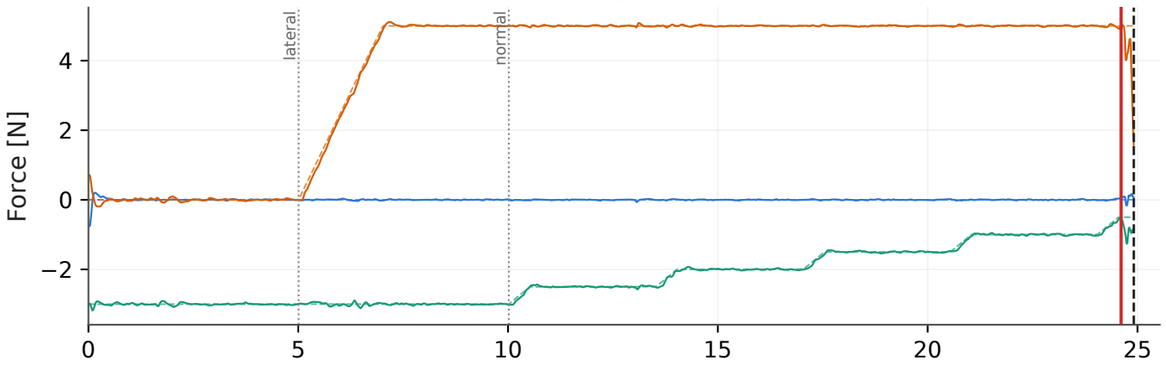
Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

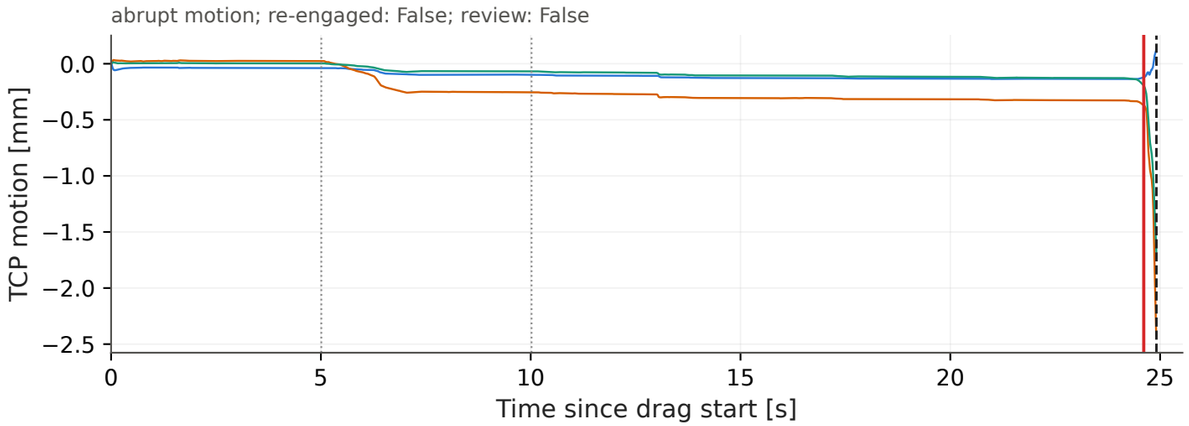
T133 R2 S=-0, L=15 N | Lateral loading



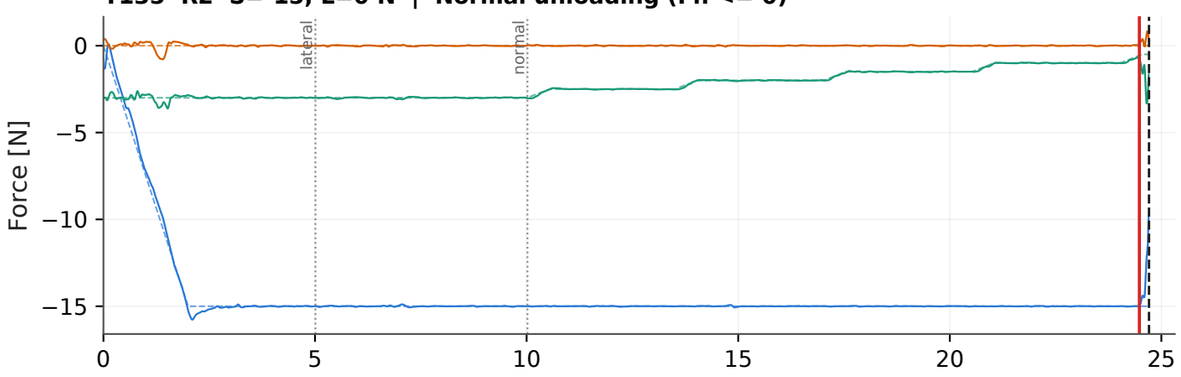


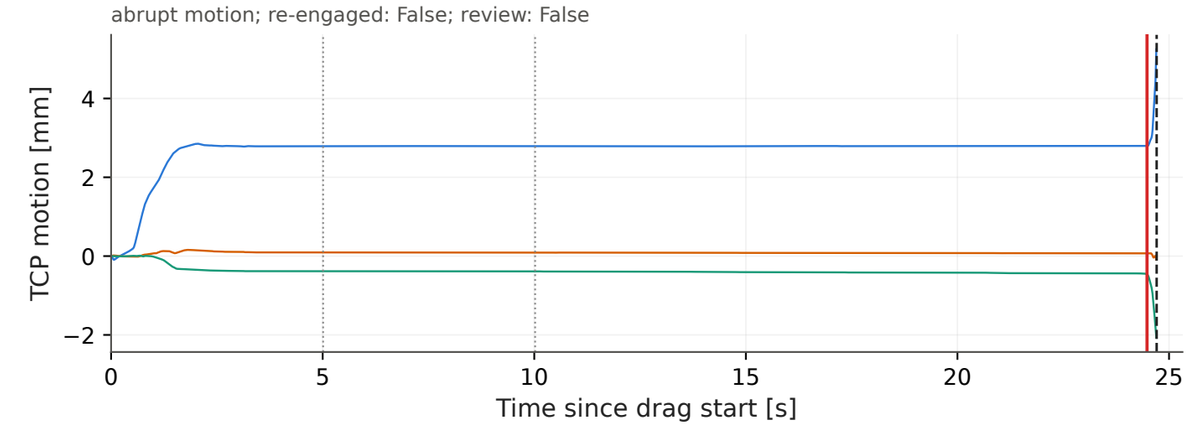
T134 R2 S=-0, L=5 N | Normal unloading (Fn <= 0)





T135 R2 S=-15, L=0 N | Normal unloading (Fn <= 0)

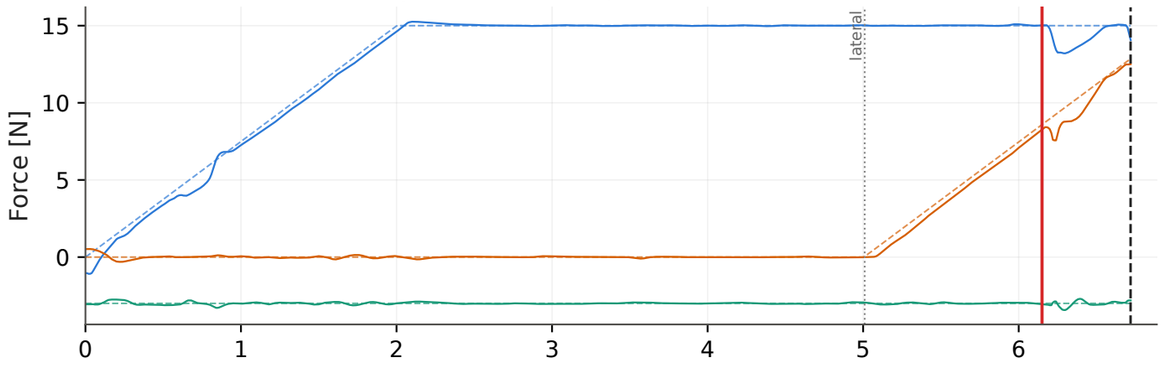




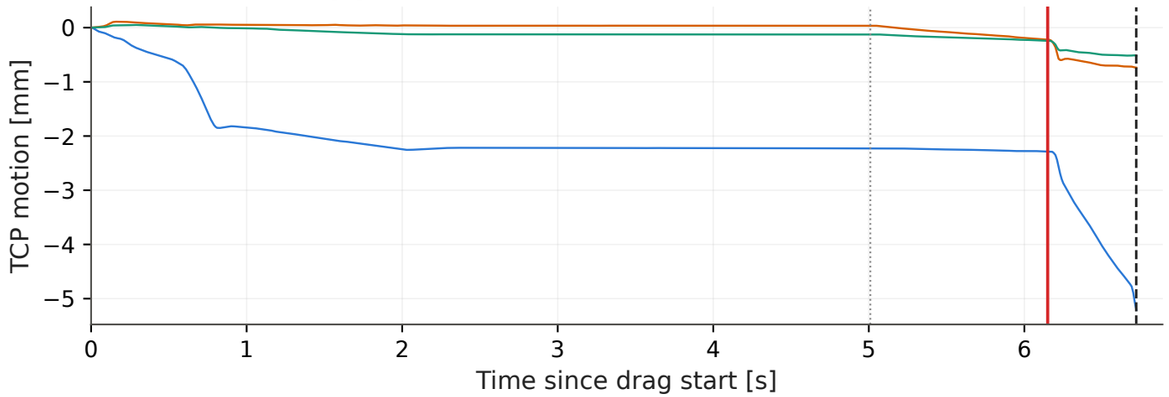
Trial audit | 136–138 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

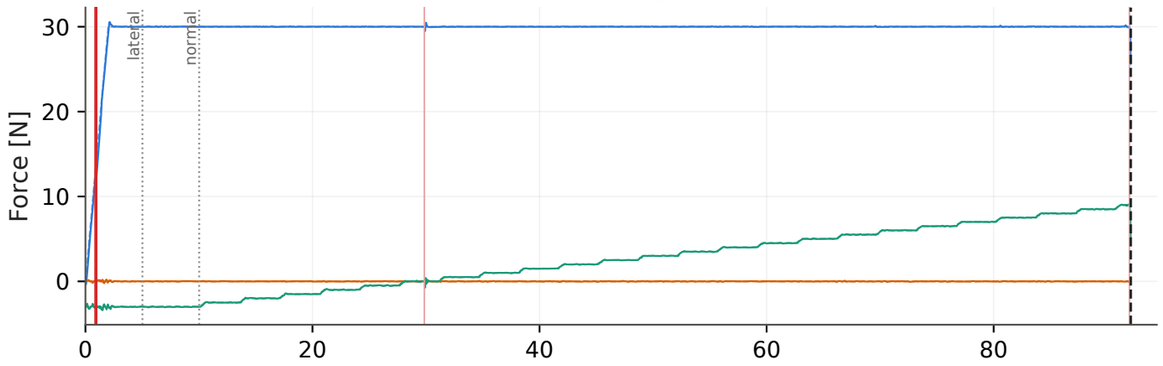
T136 R2 S=15, L=15 N | Lateral loading



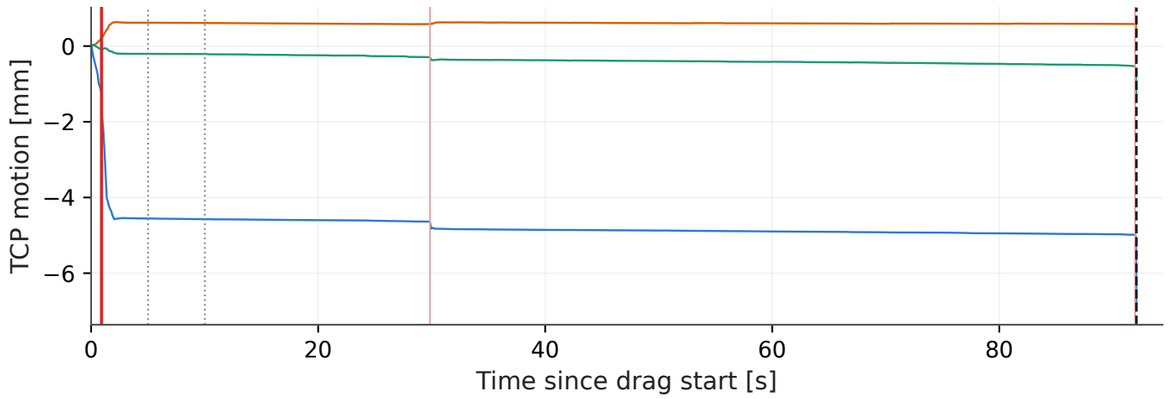
force and motion; re-engaged: False; review: False



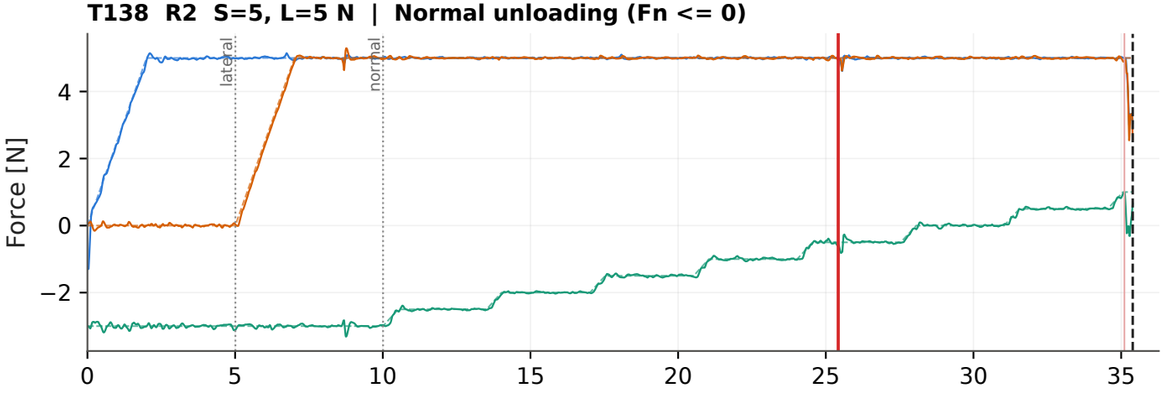
T137 R2 S=30, L=0 N | Forward shear / drag



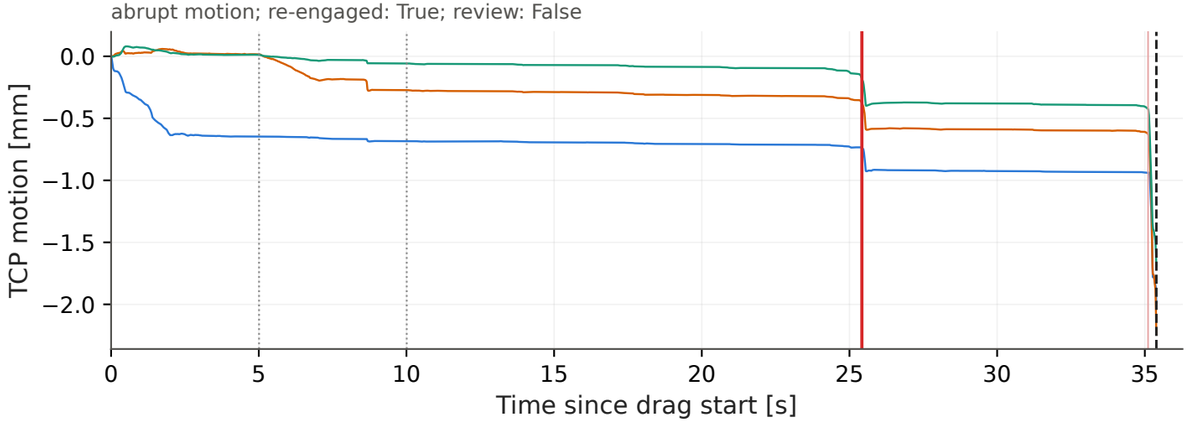
force and motion; re-engaged: True; review: True



T138 R2 S=5, L=5 N | Normal unloading (Fn <= 0)



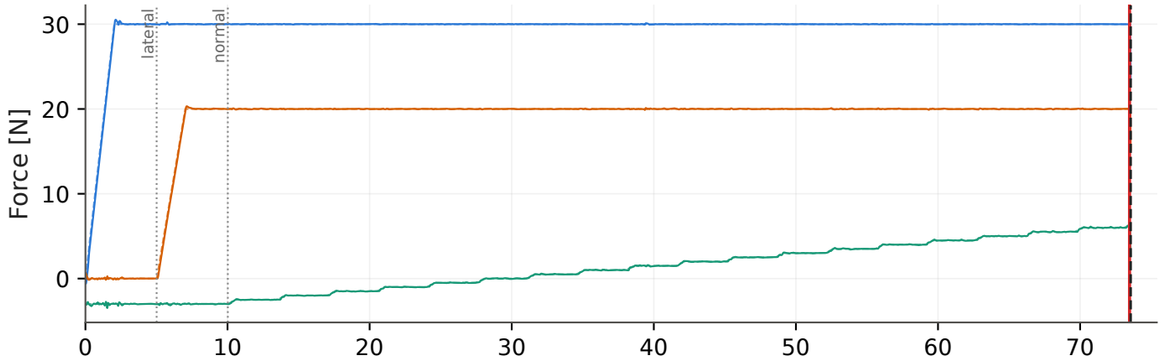
abrupt motion; re-engaged: True; review: False



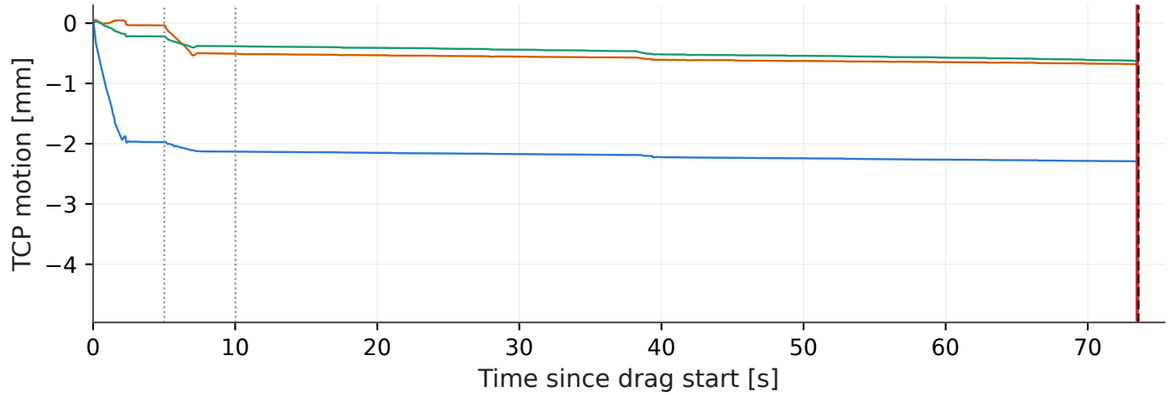
Trial audit | 139–141 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

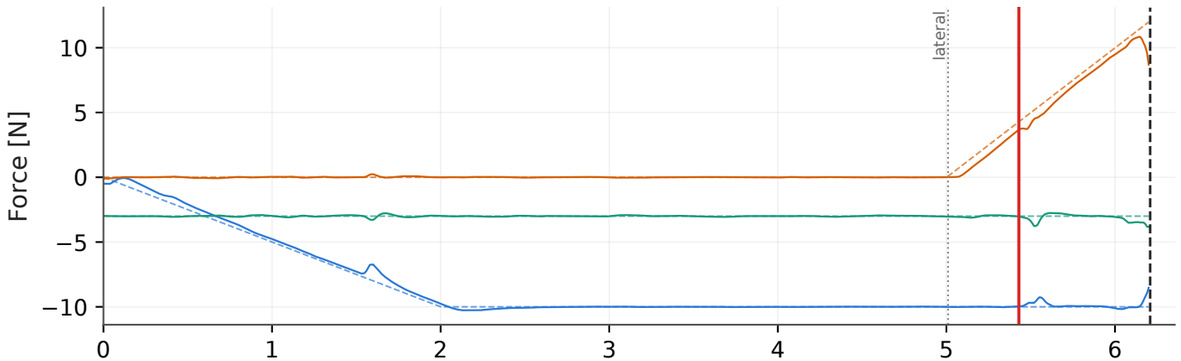
T139 R2 S=30, L=20 N | Normal tension (Fn > 0)



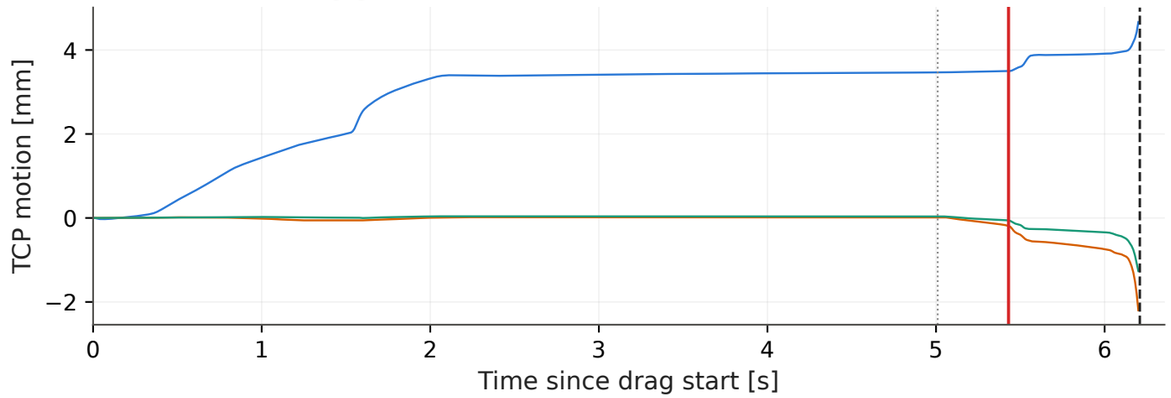
abrupt motion; re-engaged: False; review: False



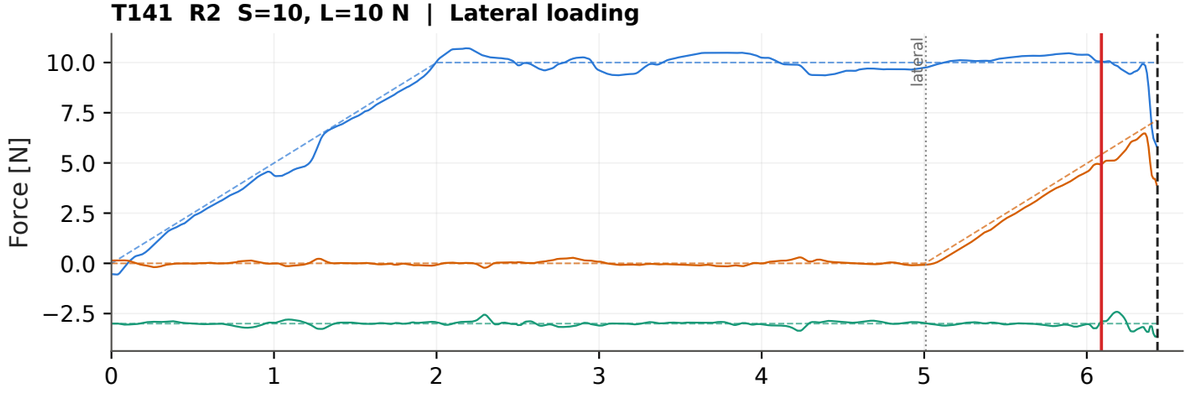
T140 R2 S=-10, L=20 N | Lateral loading



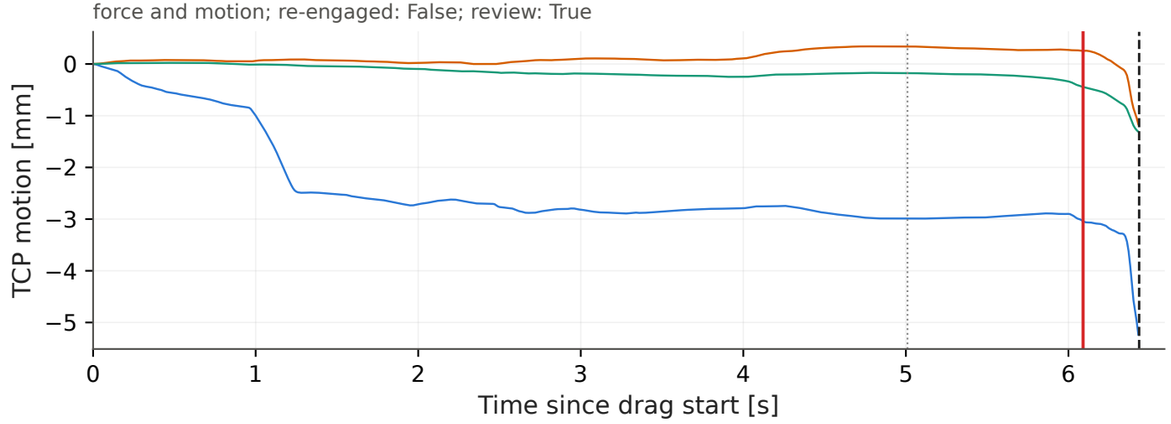
force and motion; re-engaged: False; review: True



T141 R2 S=10, L=10 N | Lateral loading



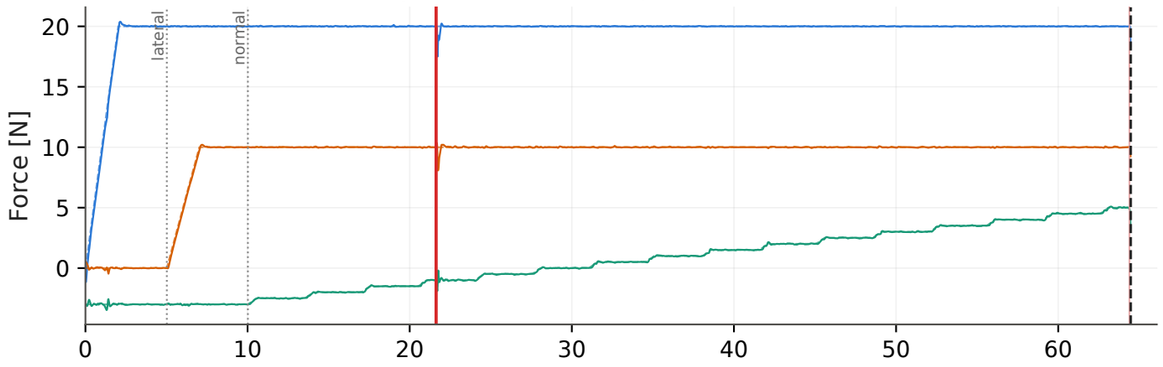
force and motion; re-engaged: False; review: True



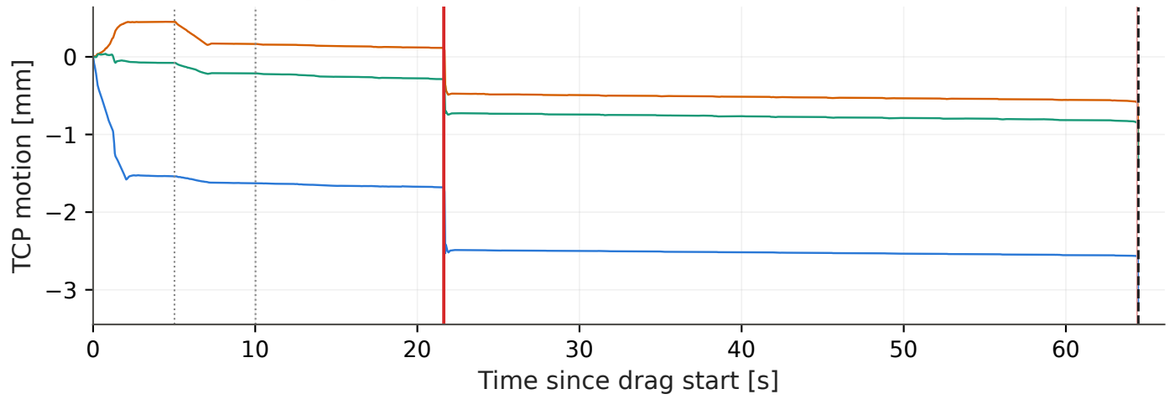
Trial audit | 142–144 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

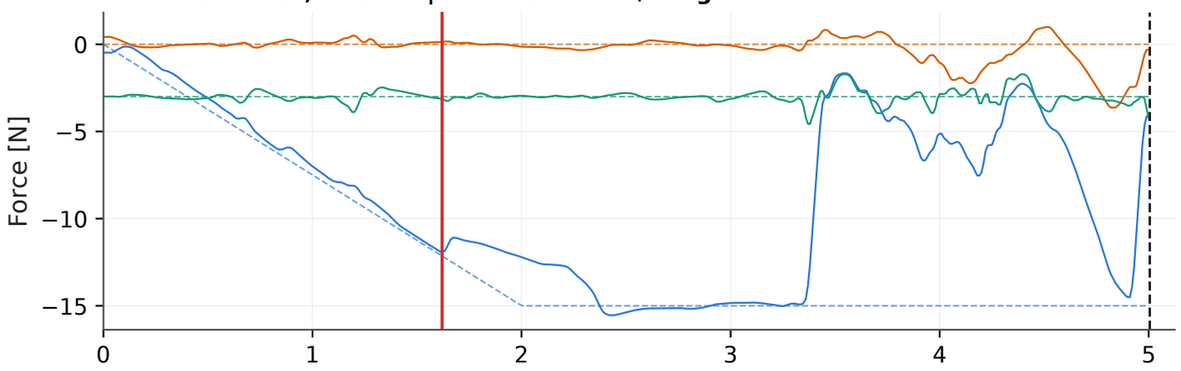
T142 R2 S=20, L=10 N | Normal unloading (Fn <= 0)



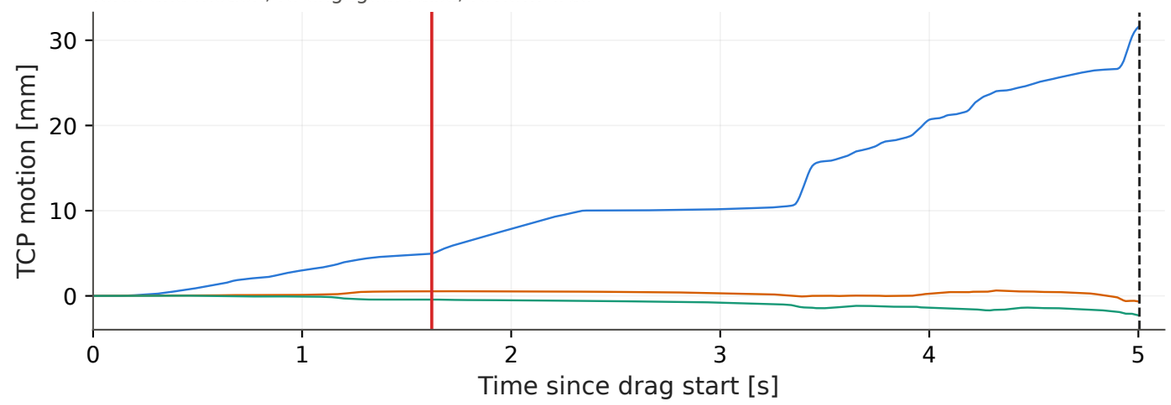
abrupt motion; re-engaged: True; review: False



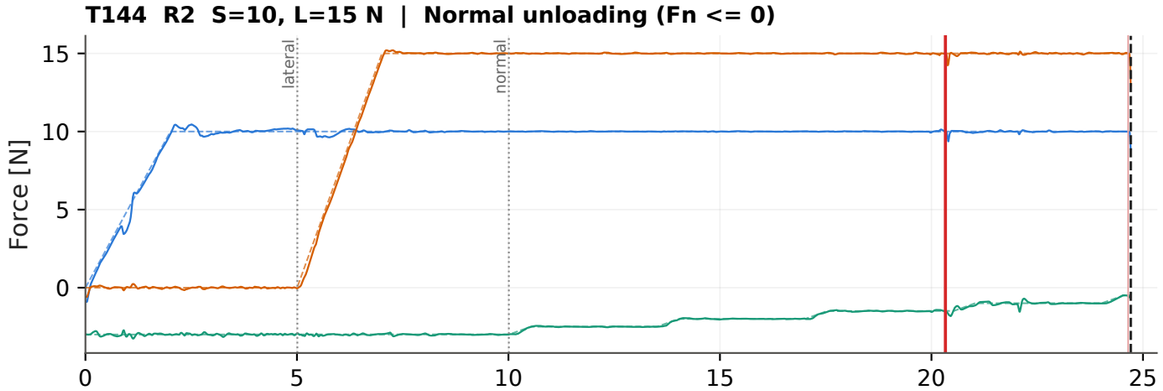
T143 R2 S=-15, L=20 N | Reverse shear / drag



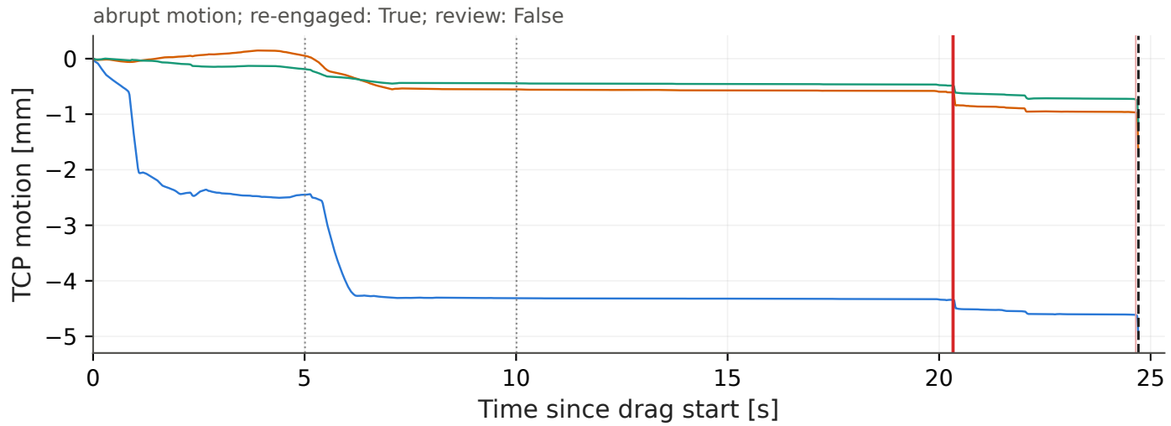
force and motion; re-engaged: False; review: True



T144 R2 S=10, L=15 N | Normal unloading (Fn <= 0)



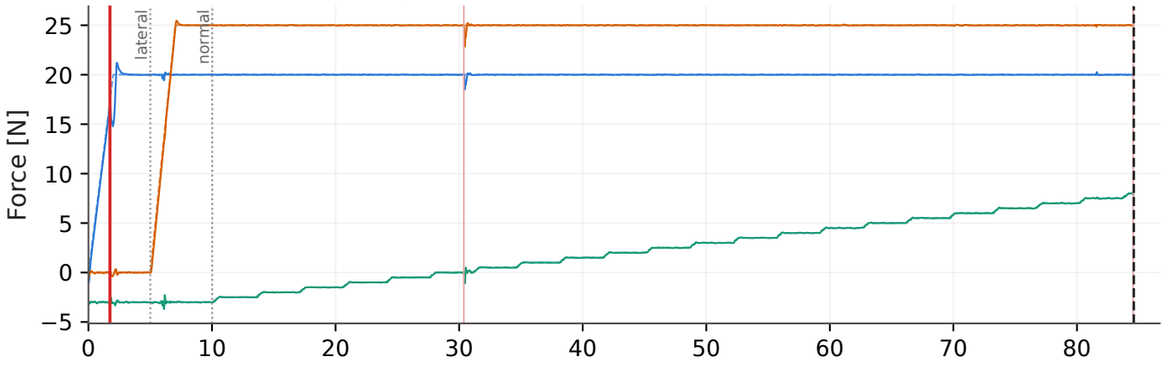
abrupt motion; re-engaged: True; review: False



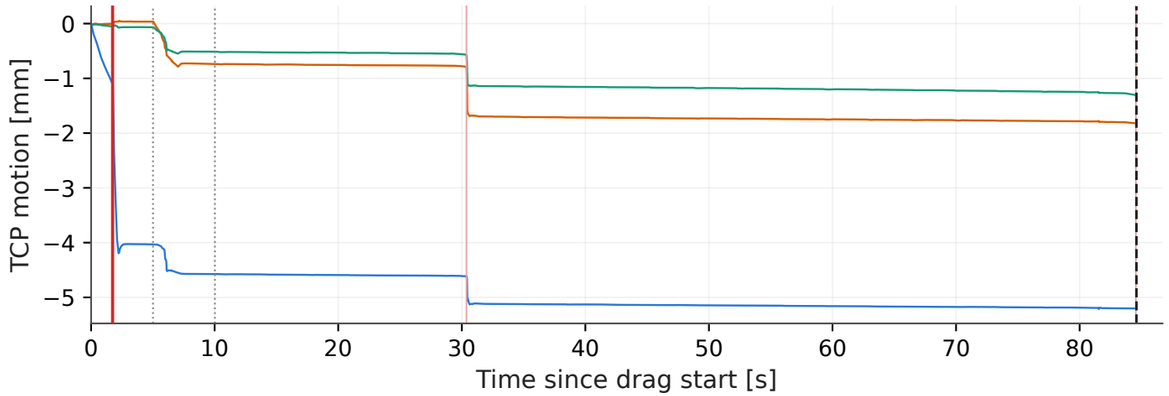
Trial audit | 145–147 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

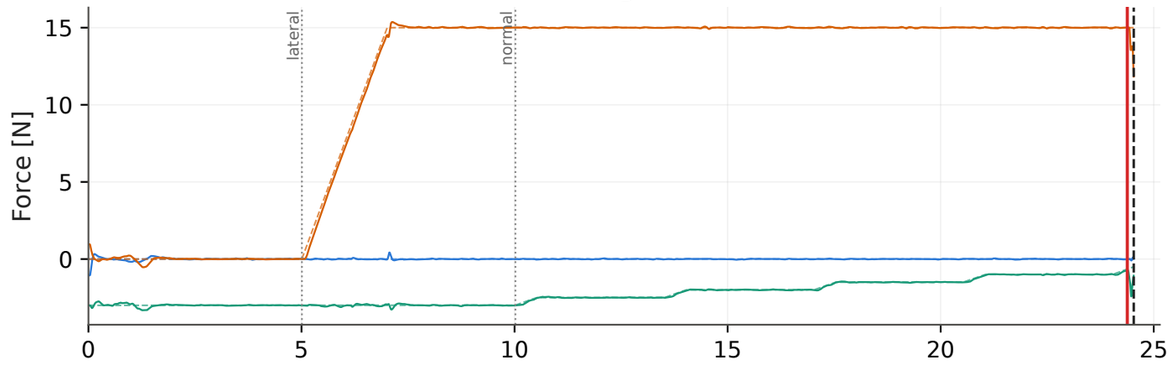
T145 R2 S=20, L=25 N | Forward shear / drag



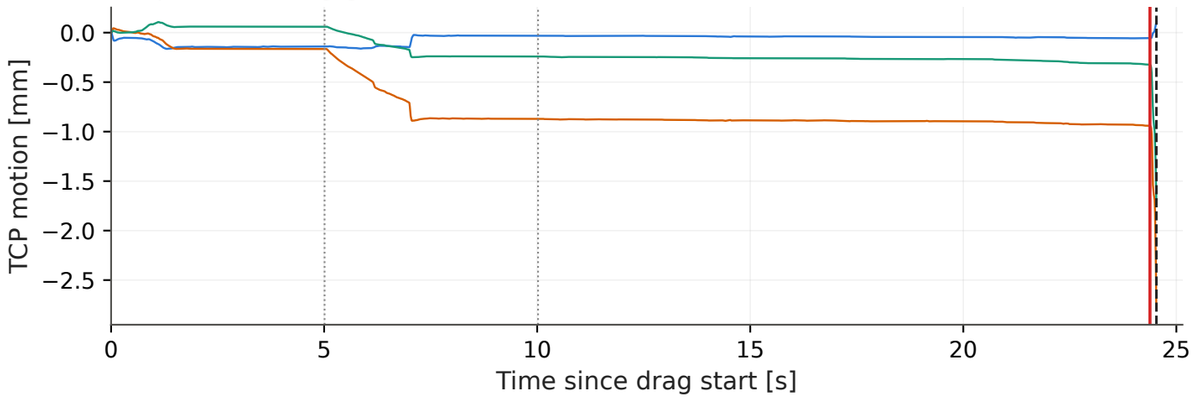
force and motion; re-engaged: True; review: False



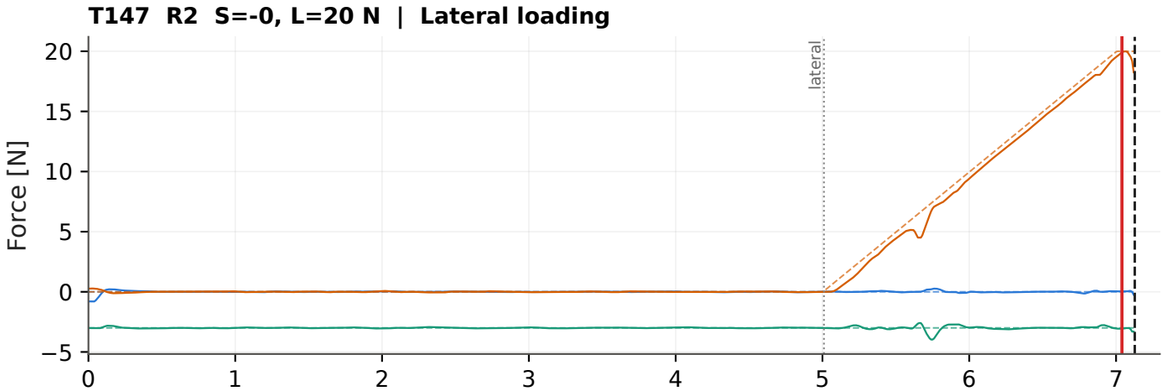
T146 R2 S=0, L=15 N | Normal unloading (Fn <= 0)



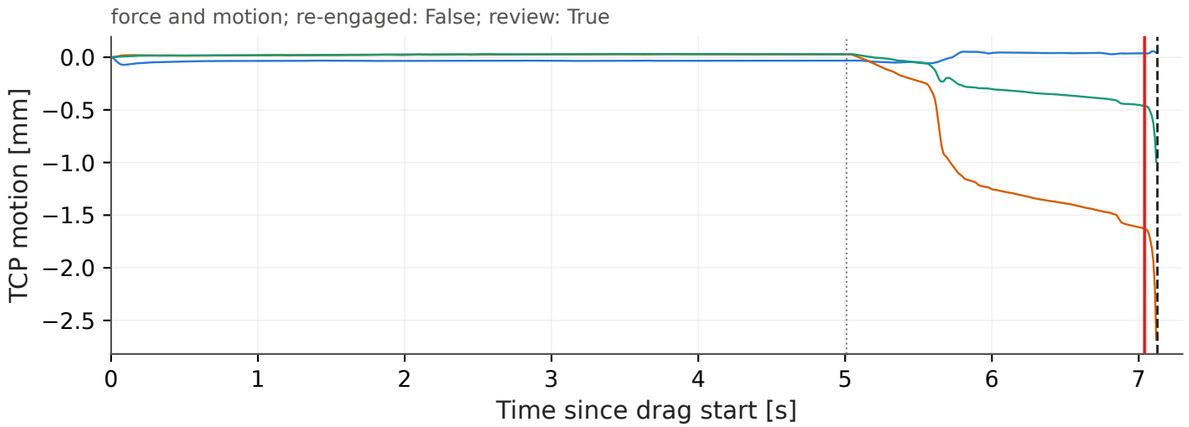
abrupt motion; re-engaged: False; review: False



T147 R2 S=-0, L=20 N | Lateral loading



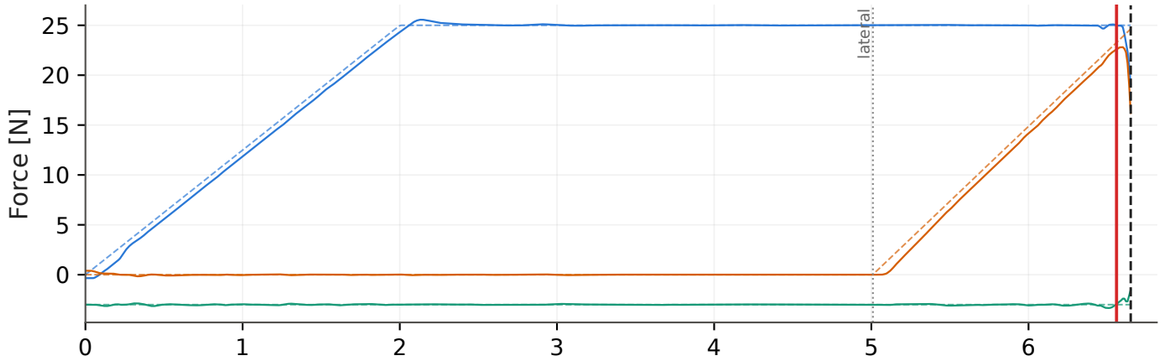
force and motion; re-engaged: False; review: True



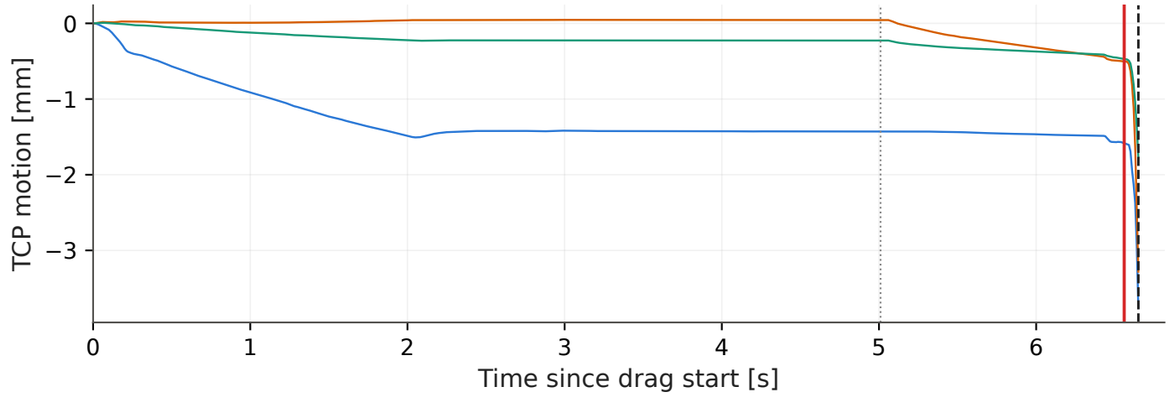
Trial audit | 148–150 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

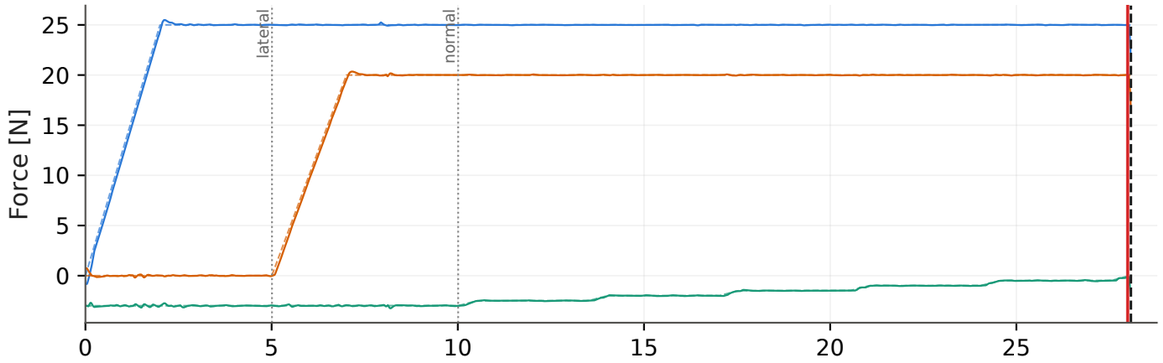
T148 R2 S=25, L=30 N | Lateral loading



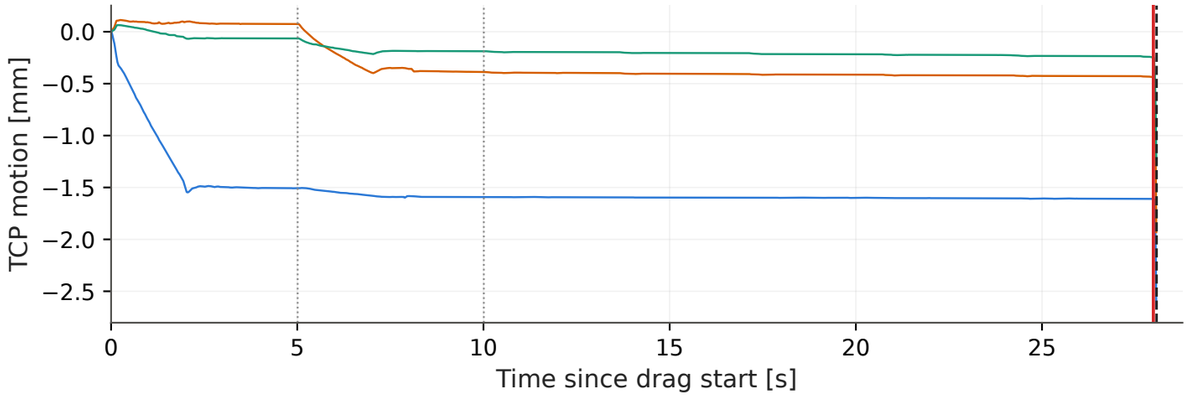
force and motion; re-engaged: False; review: False



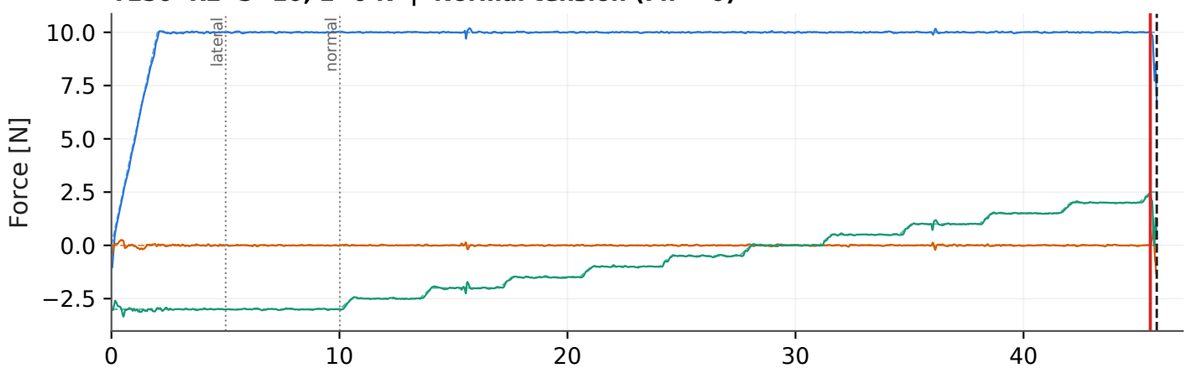
T149 R2 S=25, L=20 N | Normal unloading (Fn <= 0)



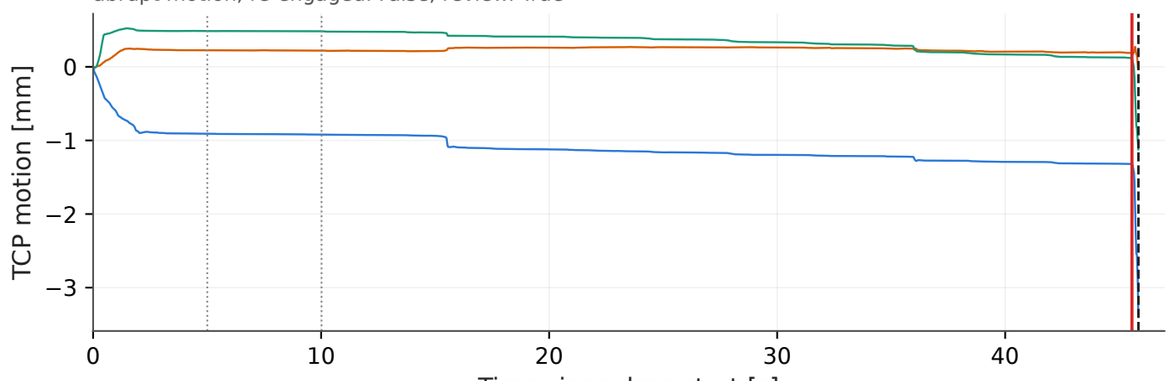
abrupt motion; re-engaged: False; review: False



T150 R2 S=10, L=0 N | Normal tension (Fn > 0)



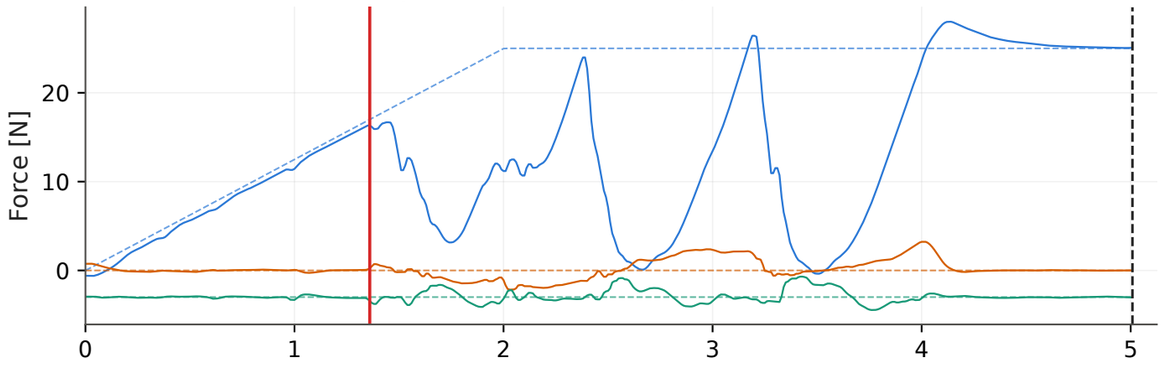
abrupt motion; re-engaged: False; review: True



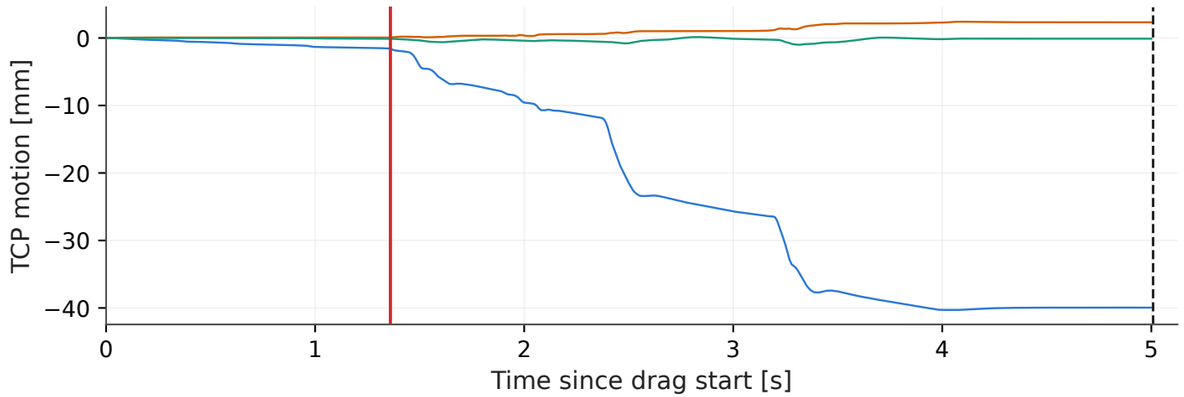
Trial audit | 151–153 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

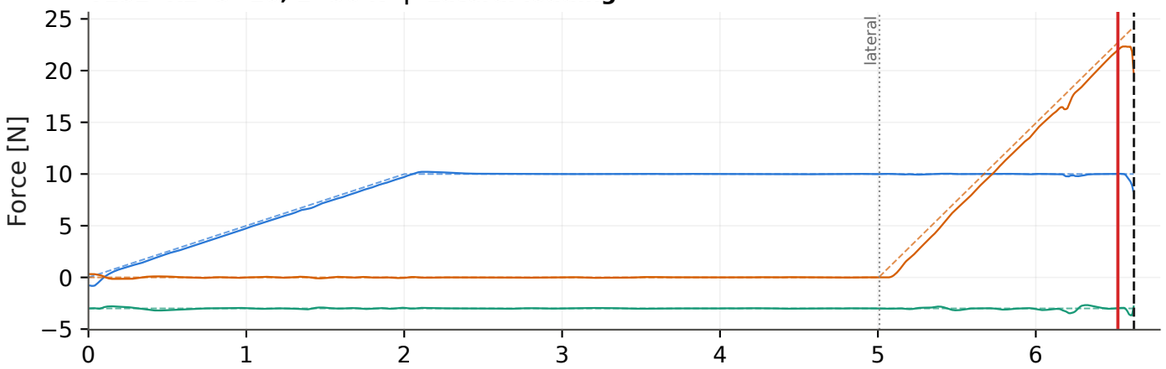
T151 R2 S=25, L=25 N | Forward shear / drag



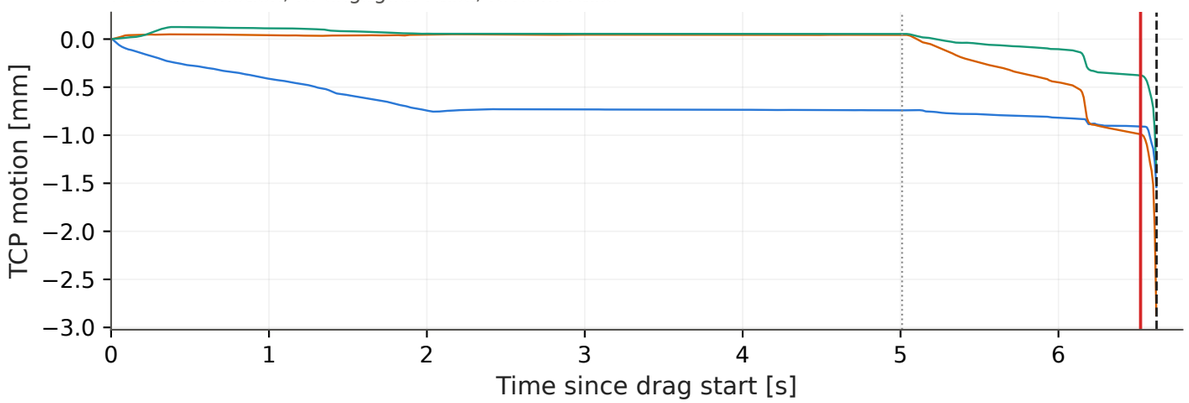
force and motion; re-engaged: True; review: False



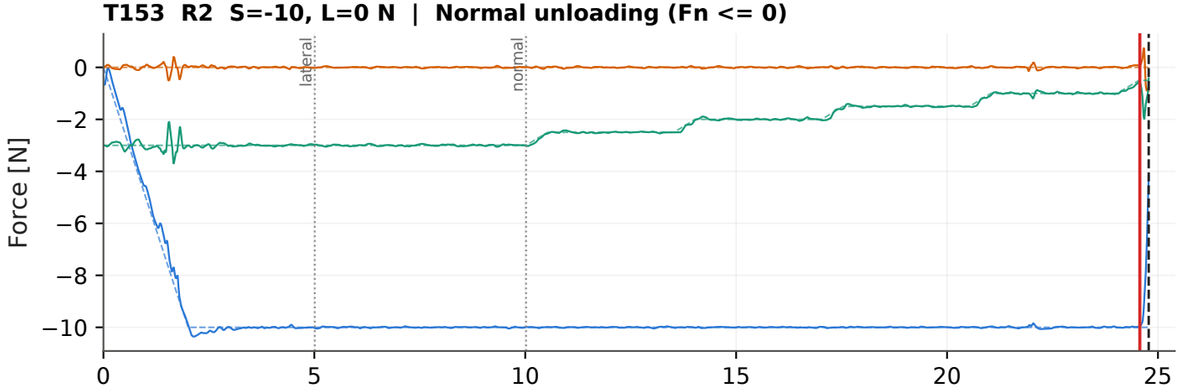
T152 R2 S=10, L=30 N | Lateral loading



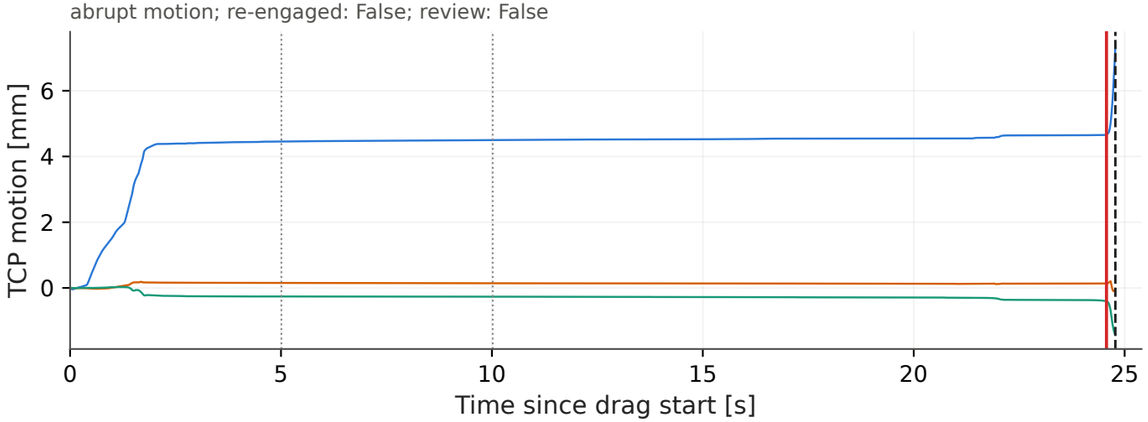
force and motion; re-engaged: False; review: False



T153 R2 S=-10, L=0 N | Normal unloading (Fn <= 0)



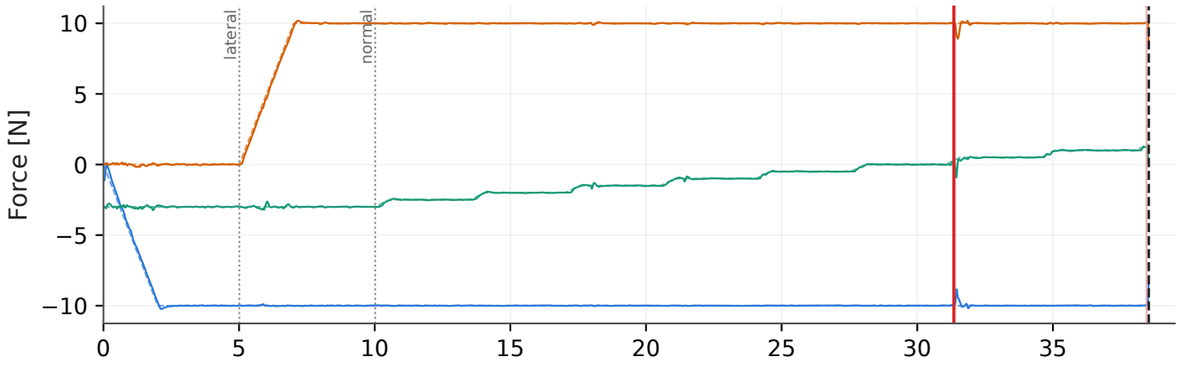
abrupt motion; re-engaged: False; review: False



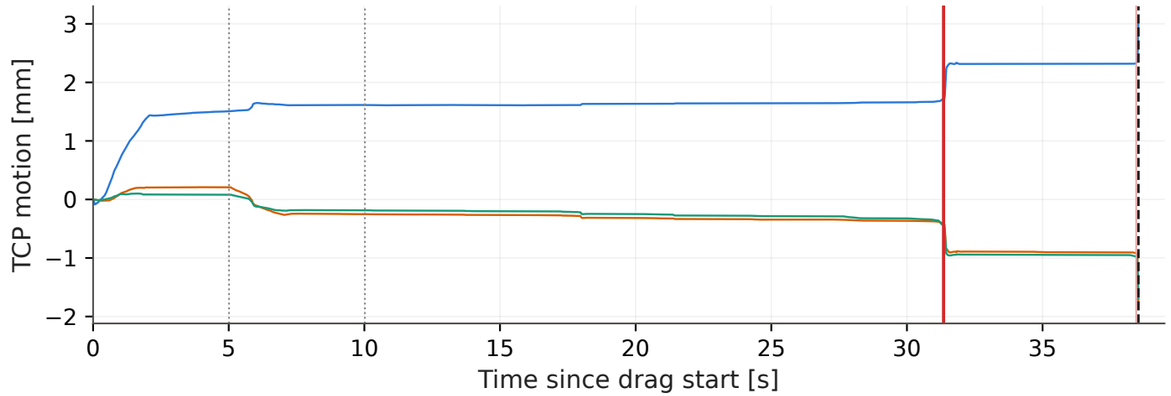
Trial audit | 154–156 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

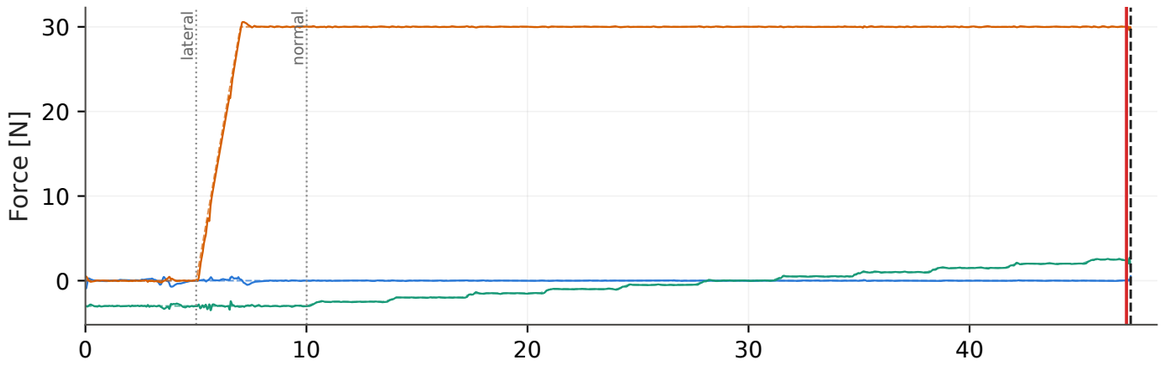
T154 R2 S=-10, L=10 N | Normal tension (Fn > 0)



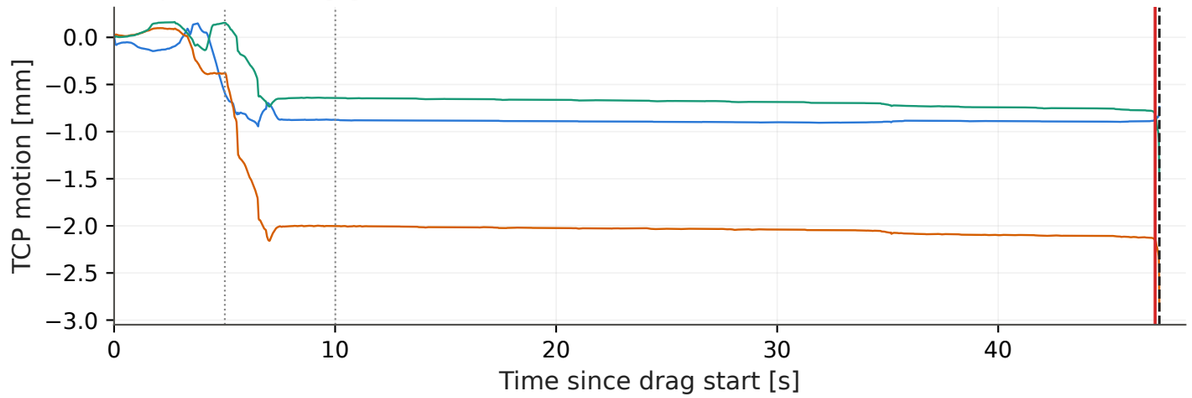
abrupt motion; re-engaged: True; review: False



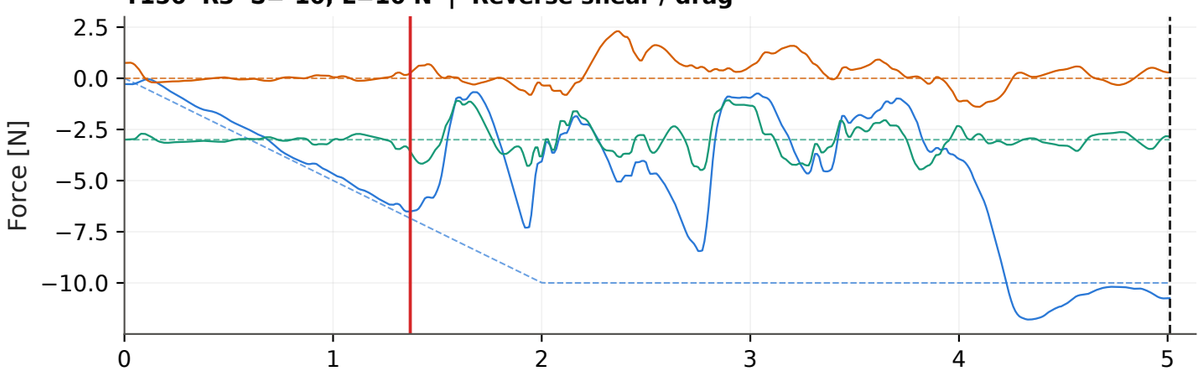
T155 R3 S=0, L=30 N | Normal tension (Fn > 0)



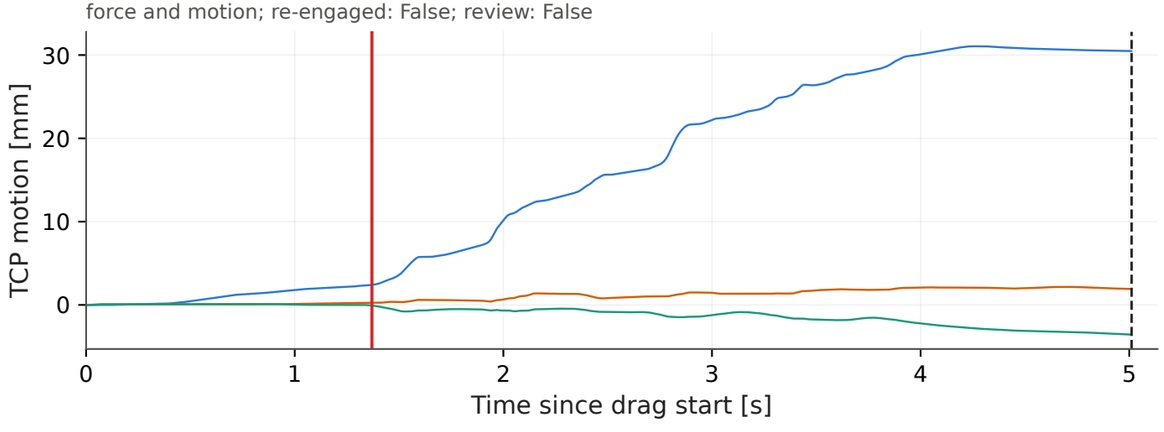
abrupt motion; re-engaged: False; review: False



T156 R3 S=-10, L=10 N | Reverse shear / drag



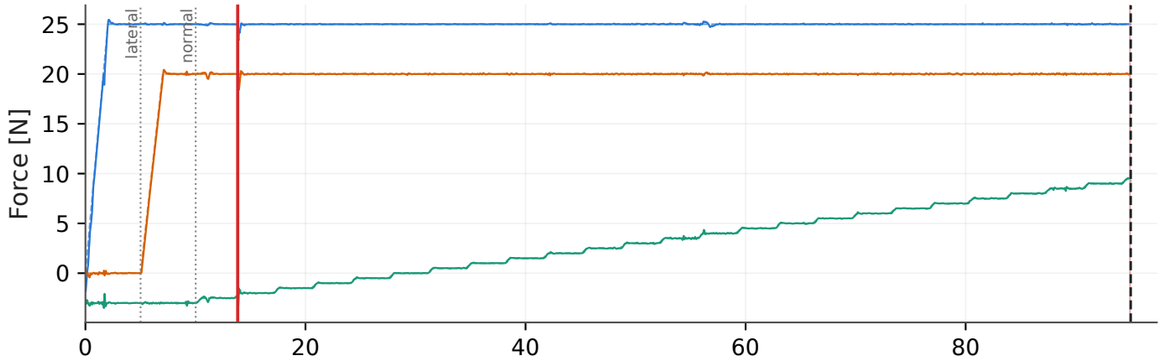
force and motion; re-engaged: False; review: False



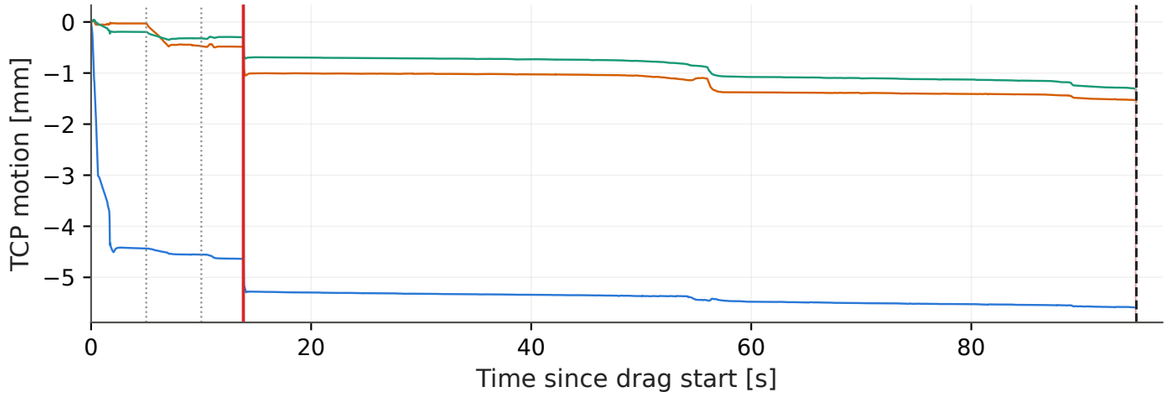
Trial audit | 157–159 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

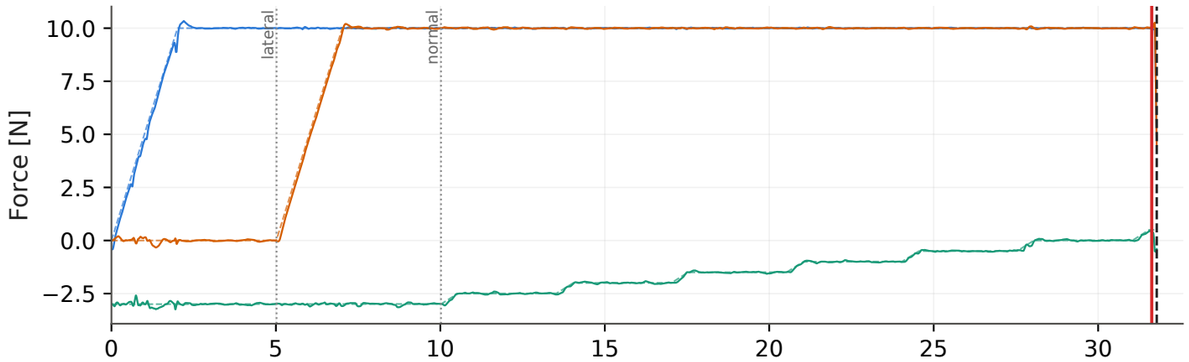
T157 R3 S=25, L=20 N | Normal unloading (Fn <= 0)



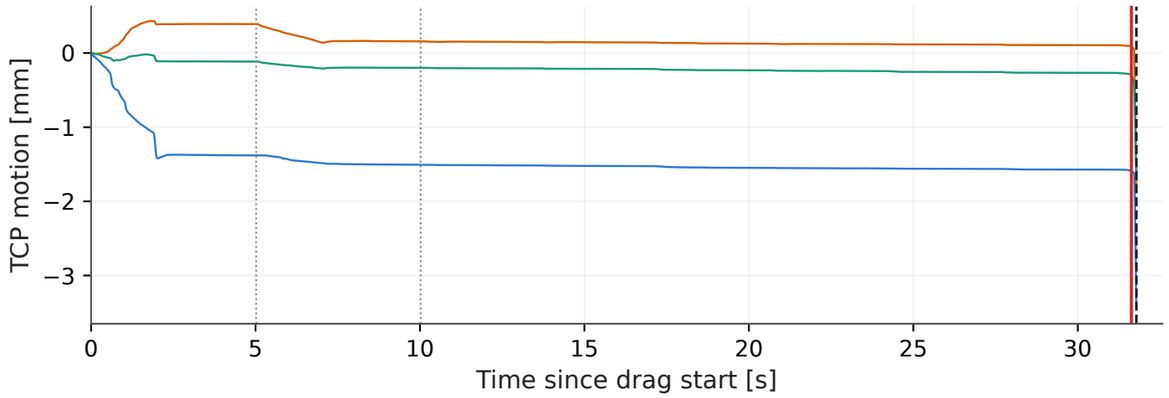
abrupt motion; re-engaged: True; review: False



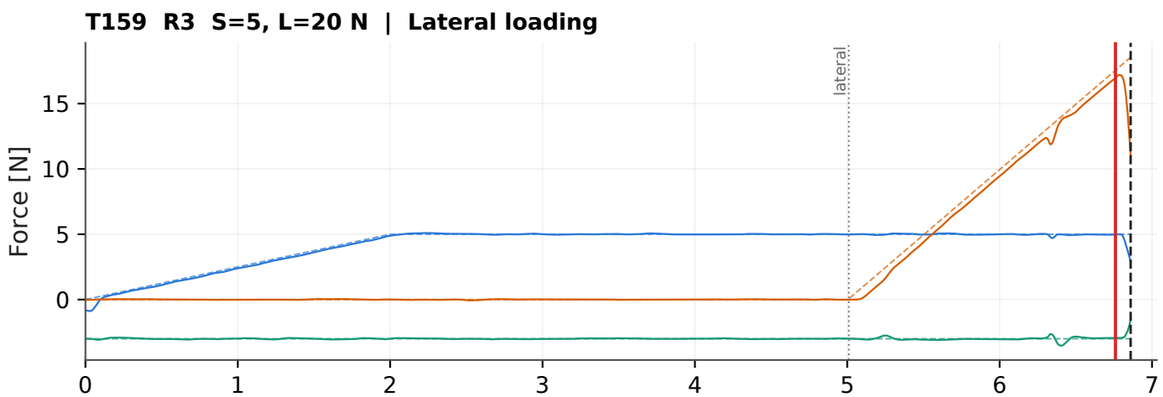
T158 R3 S=10, L=10 N | Normal tension (Fn > 0)



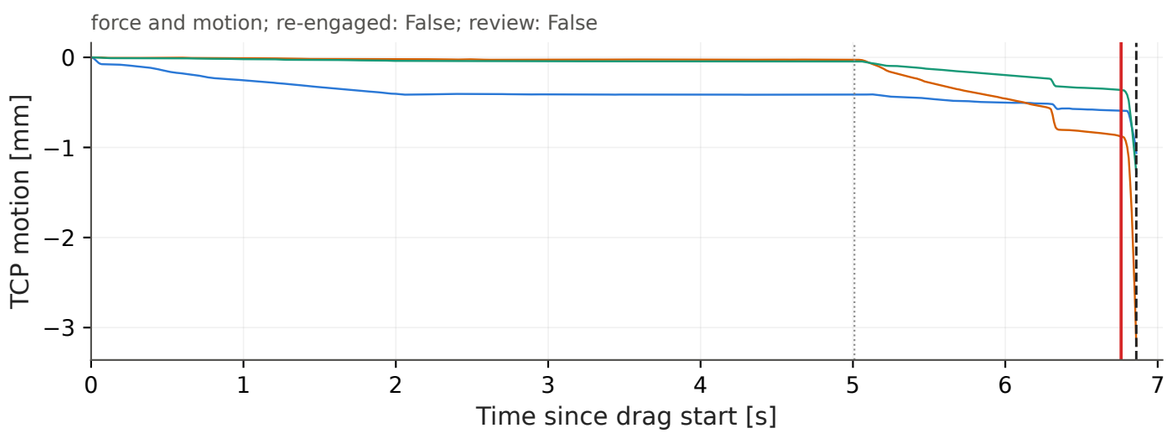
abrupt motion; re-engaged: False; review: False



T159 R3 S=5, L=20 N | Lateral loading



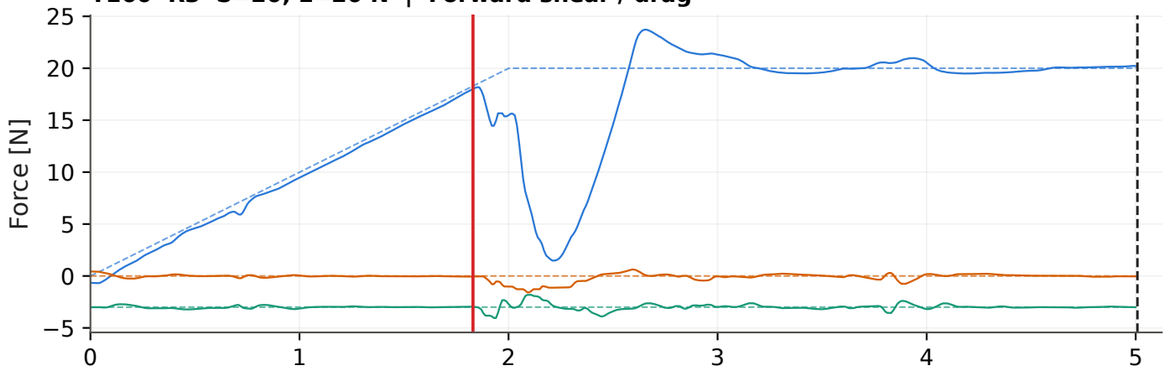
force and motion; re-engaged: False; review: False



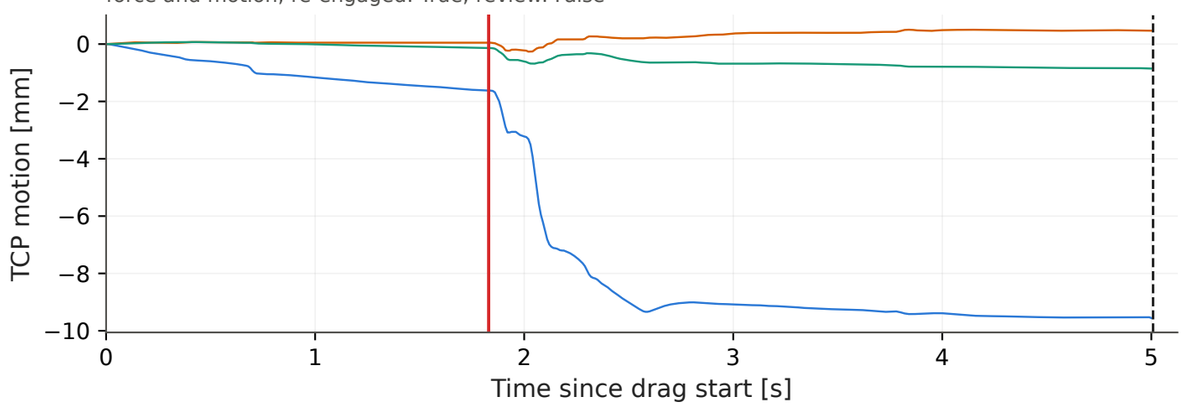
Trial audit | 160–162 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

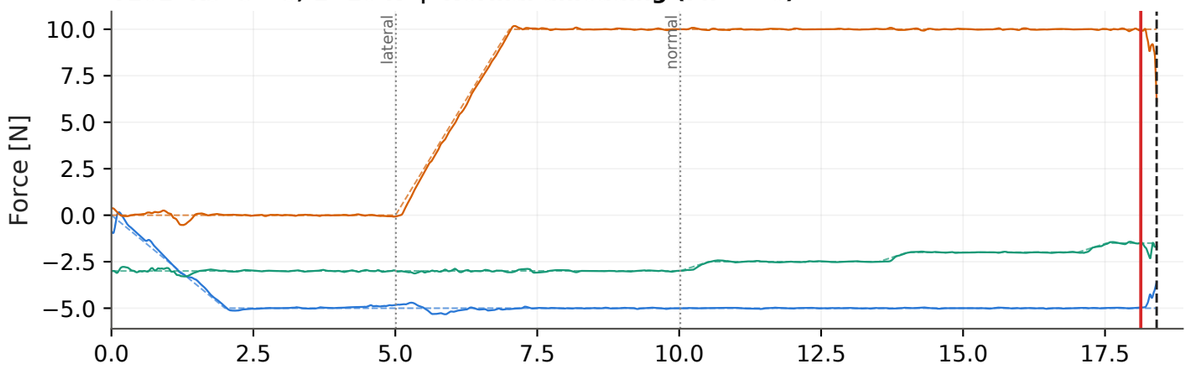
T160 R3 S=20, L=20 N | Forward shear / drag



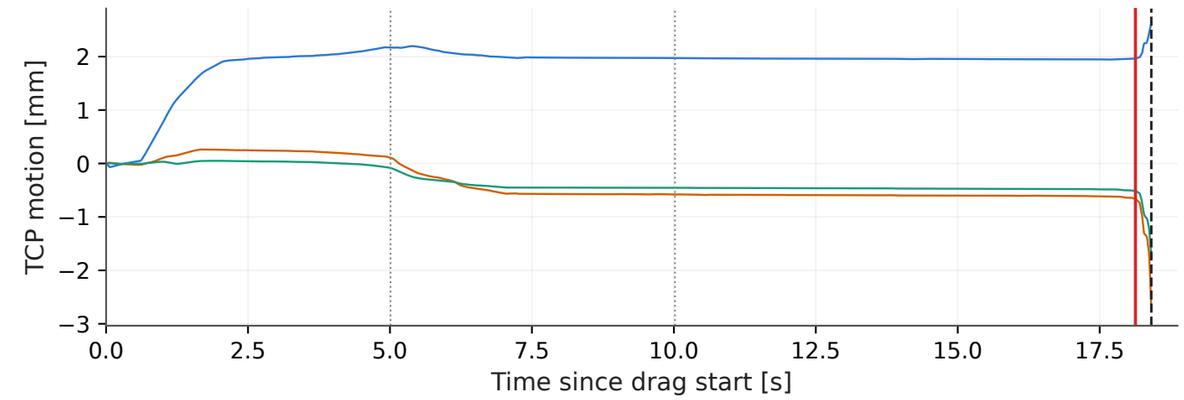
force and motion; re-engaged: True; review: False



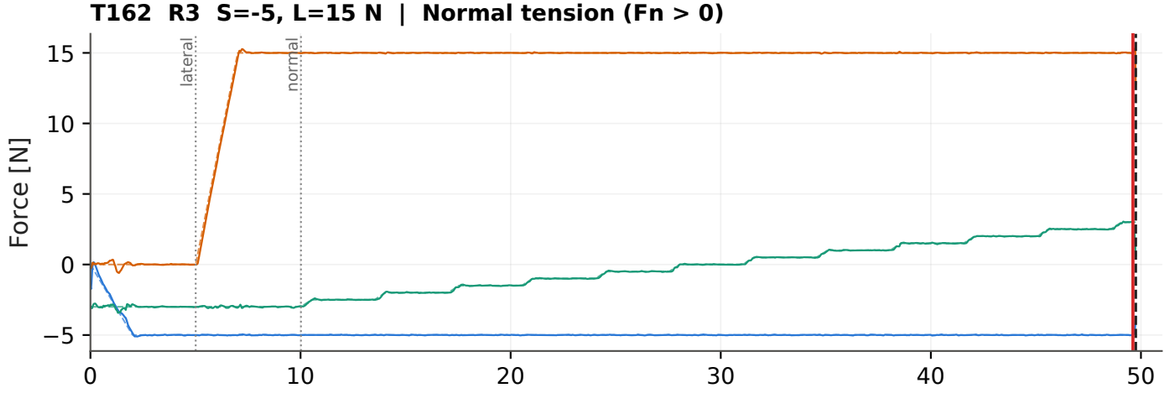
T161 R3 S=-5, L=10 N | Normal unloading (Fn <= 0)



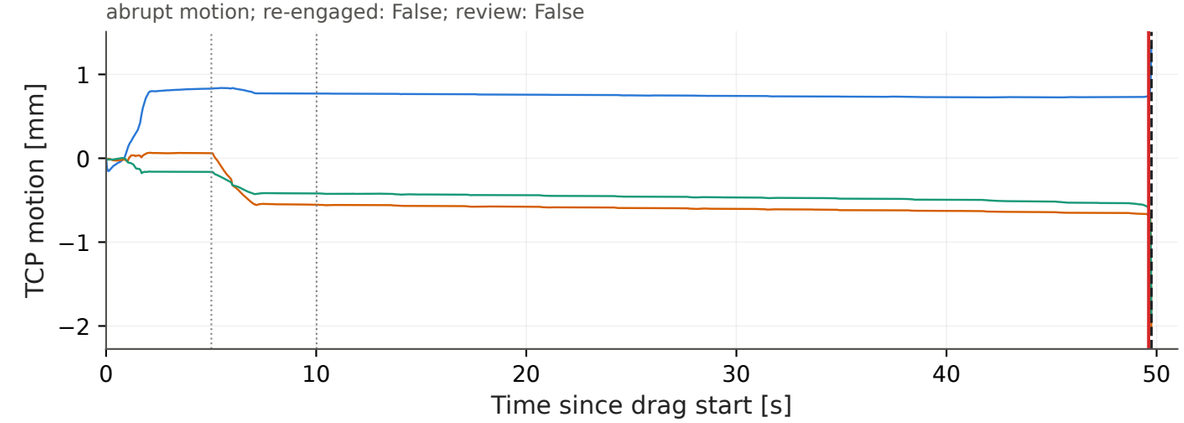
abrupt motion; re-engaged: False; review: False



T162 R3 S=-5, L=15 N | Normal tension (Fn > 0)



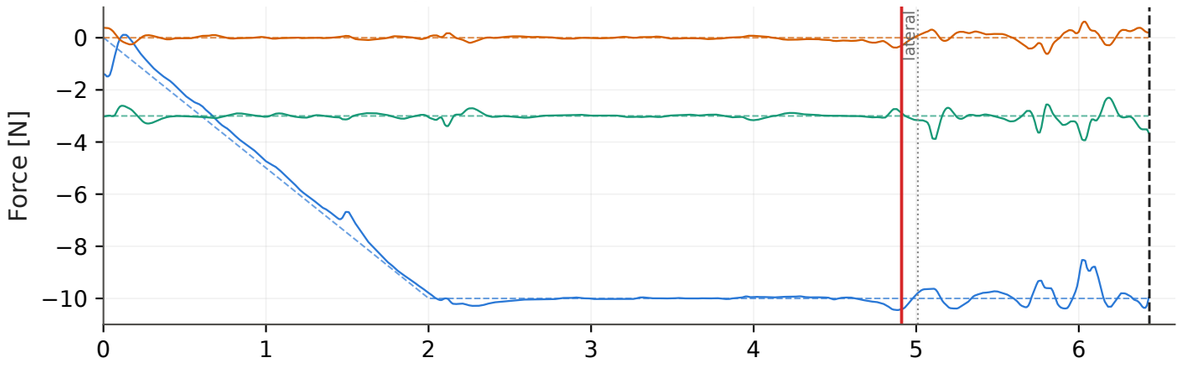
abrupt motion; re-engaged: False; review: False



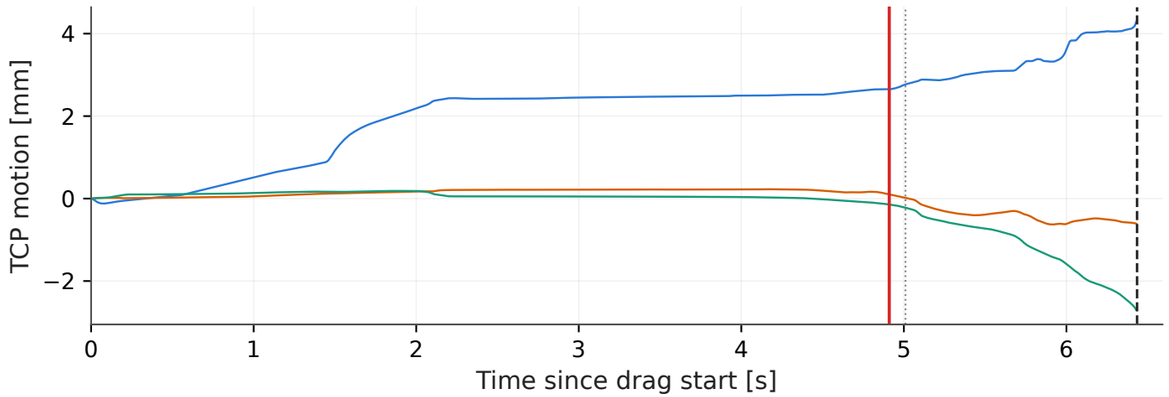
Trial audit | 163–165 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

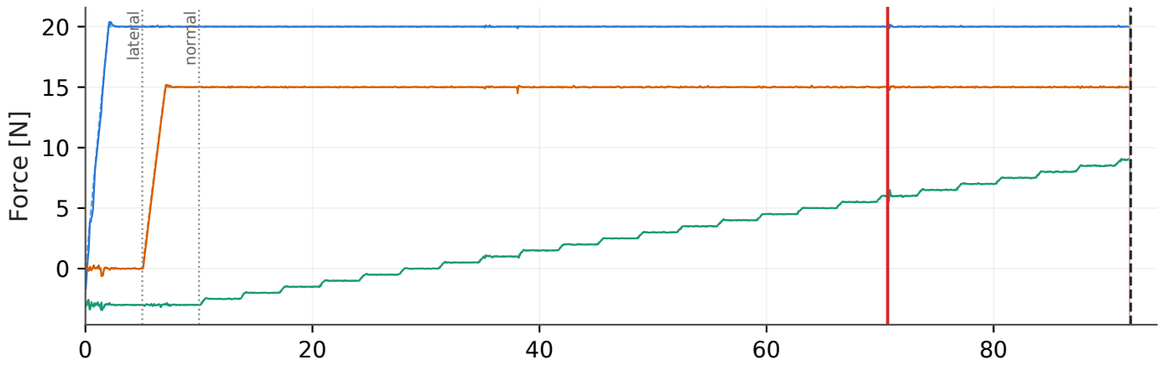
T163 R3 S=-10, L=0 N | Reverse shear / drag



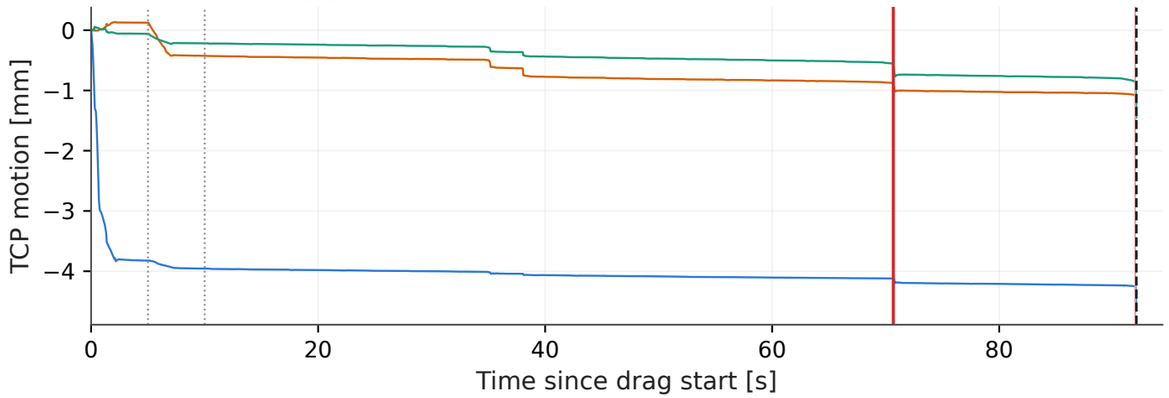
force and motion; re-engaged: False; review: True



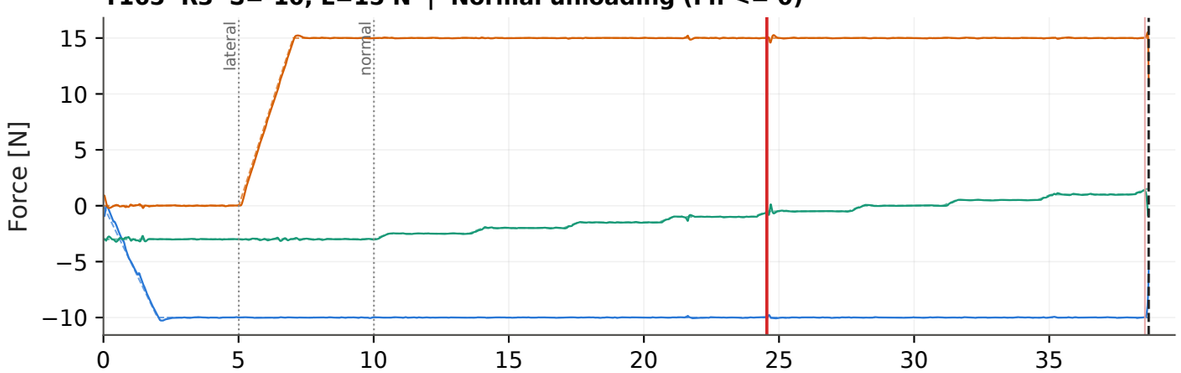
T164 R3 S=20, L=15 N | Normal tension (Fn > 0)



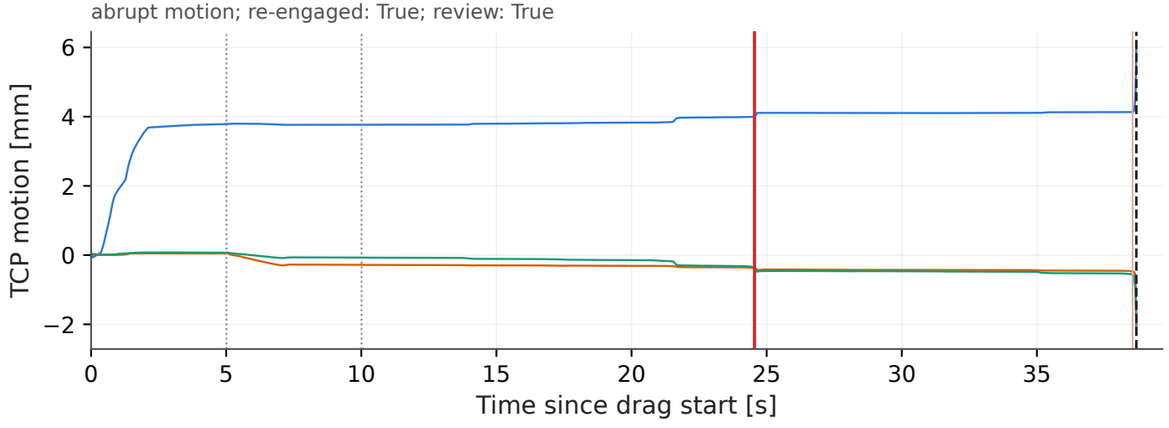
abrupt motion; re-engaged: True; review: True



T165 R3 S=-10, L=15 N | Normal unloading (Fn <= 0)



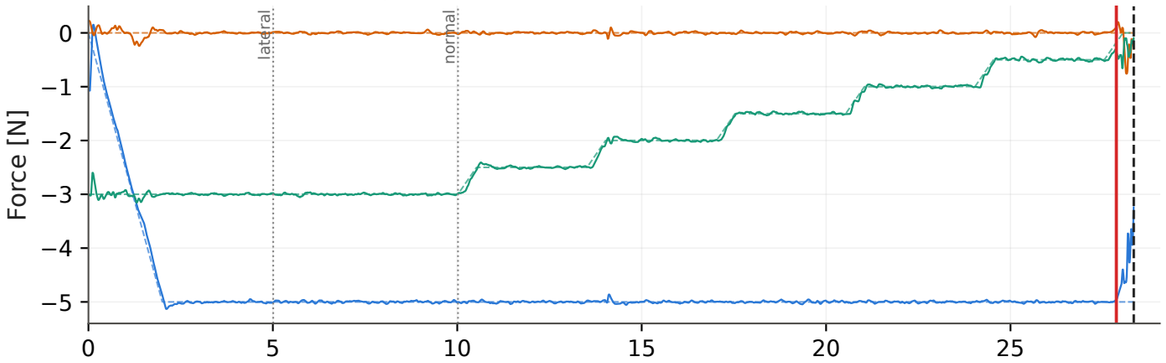
abrupt motion; re-engaged: True; review: True



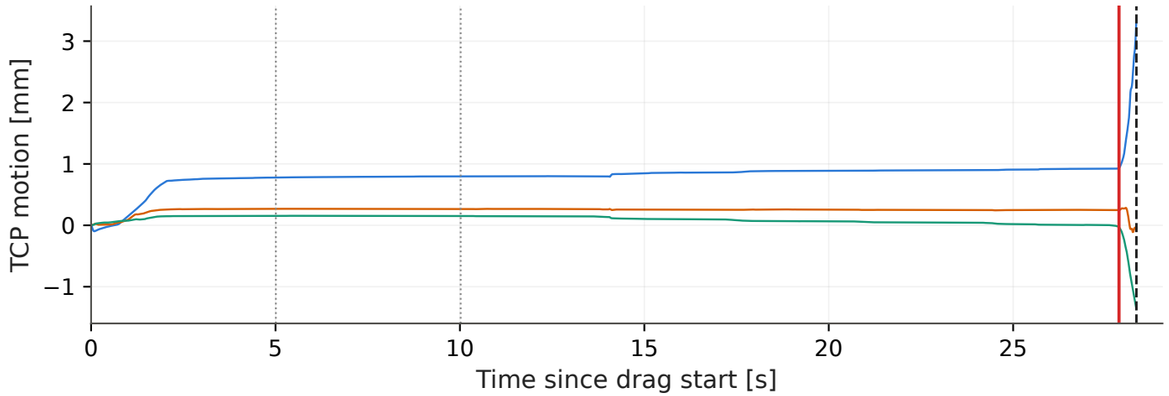
Trial audit | 166–168 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

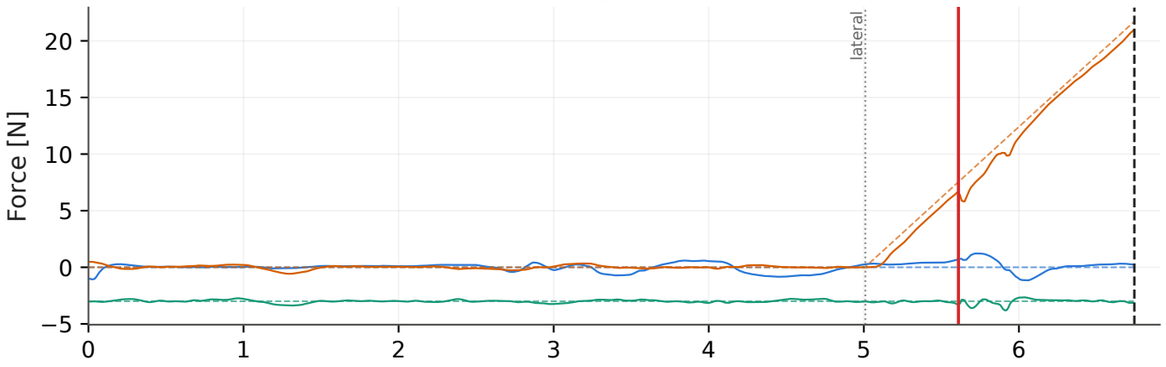
T166 R3 S=-5, L=0 N | Normal unloading (Fn <= 0)



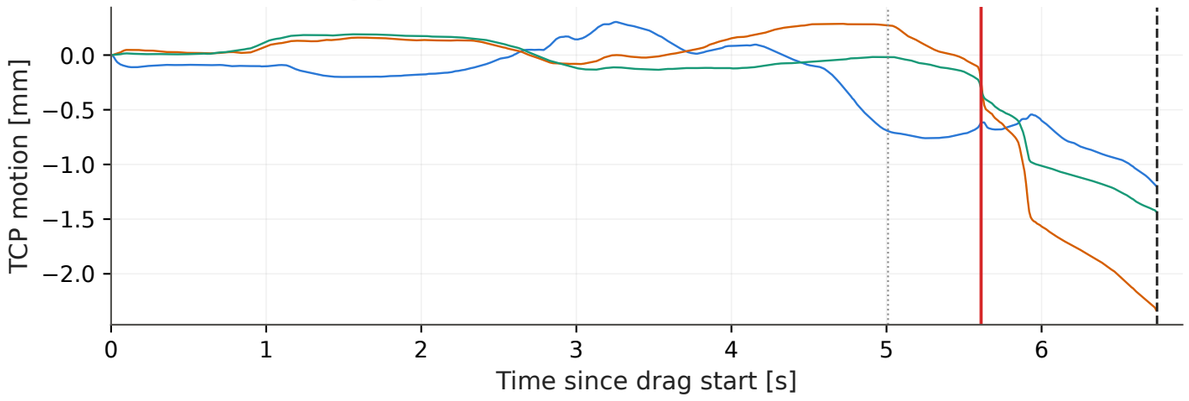
abrupt motion; re-engaged: False; review: False



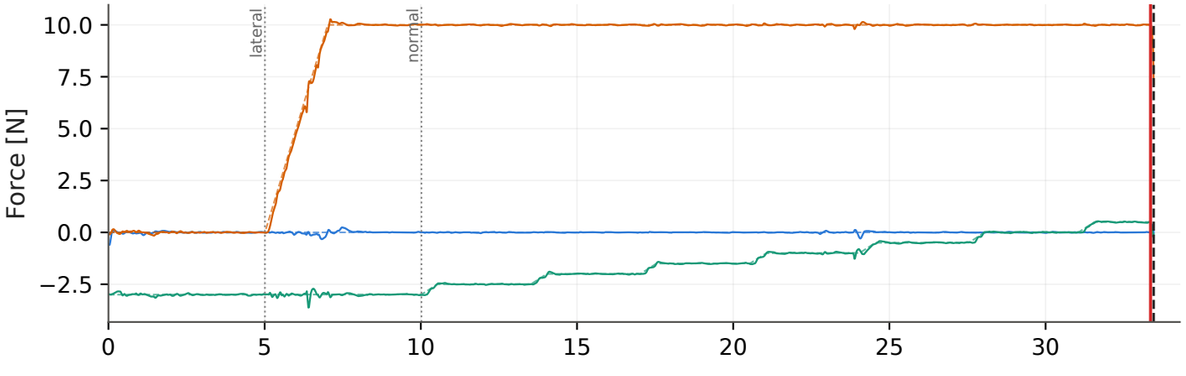
T167 R3 S=-0, L=25 N | Lateral loading



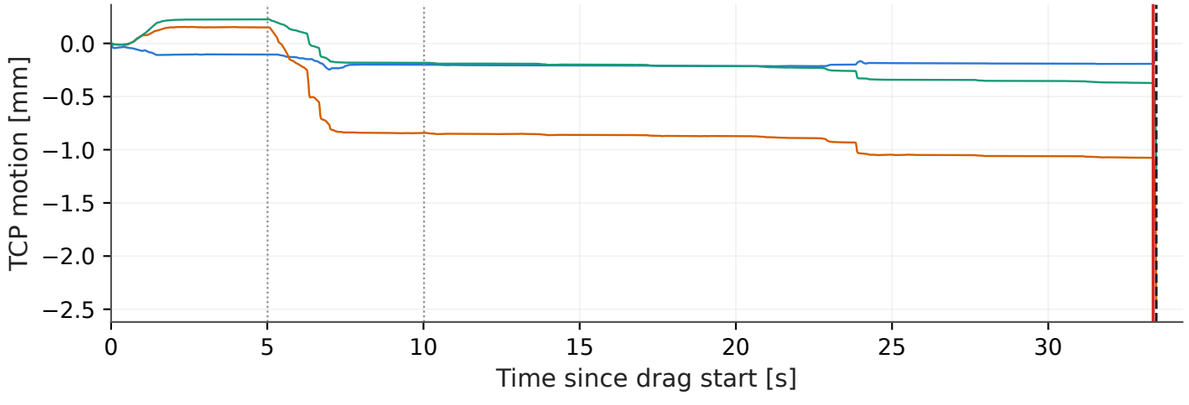
force and motion; re-engaged: False; review: True



T168 R3 S=0, L=10 N | Normal tension (Fn > 0)



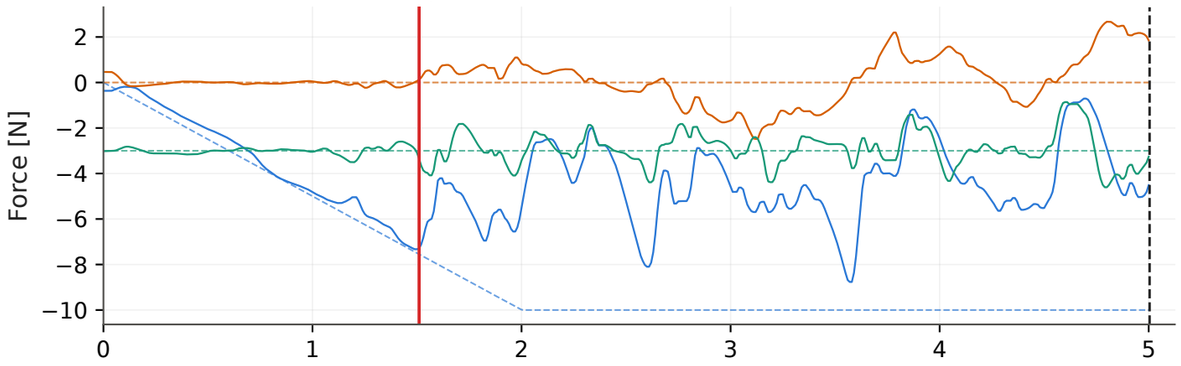
abrupt motion; re-engaged: False; review: True



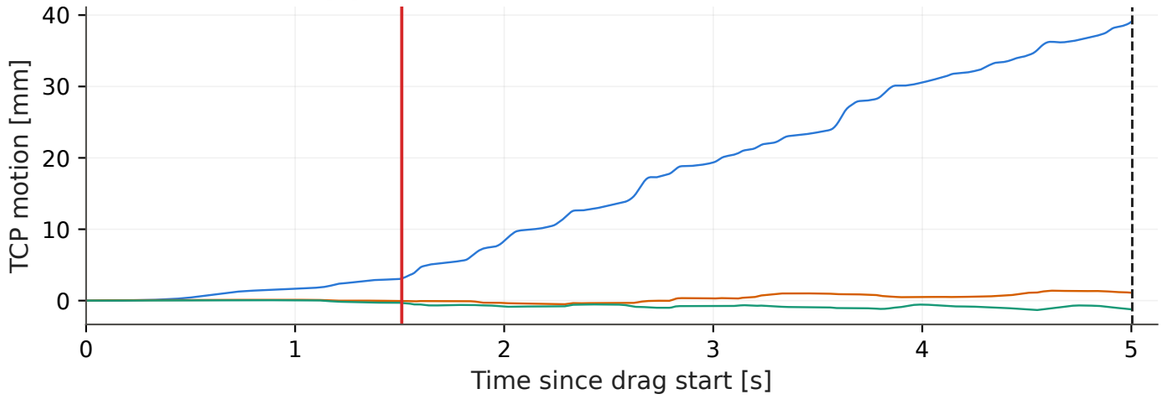
Trial audit | 169–171 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

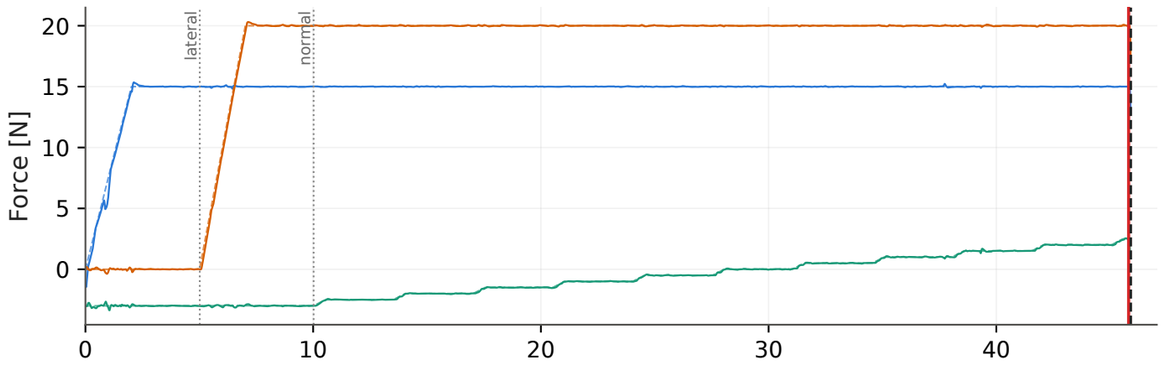
T169 R3 S=-10, L=30 N | Reverse shear / drag



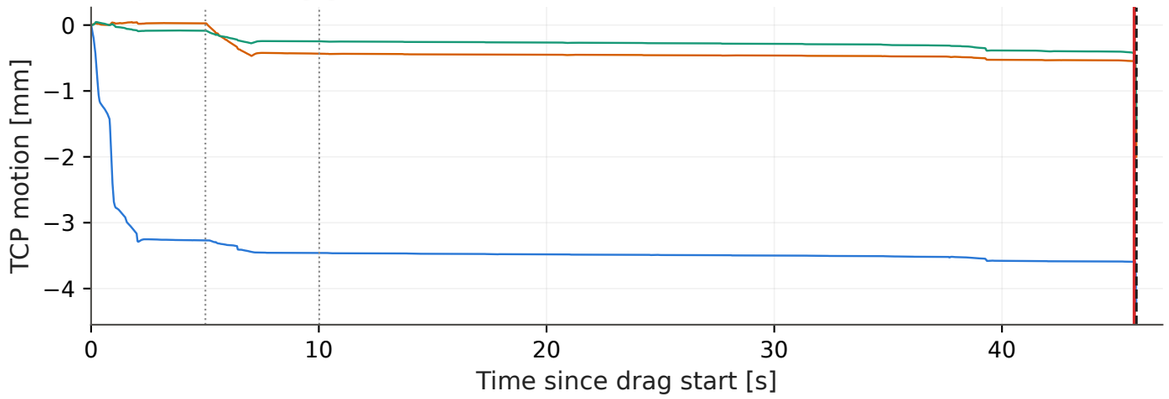
force and motion; re-engaged: False; review: False



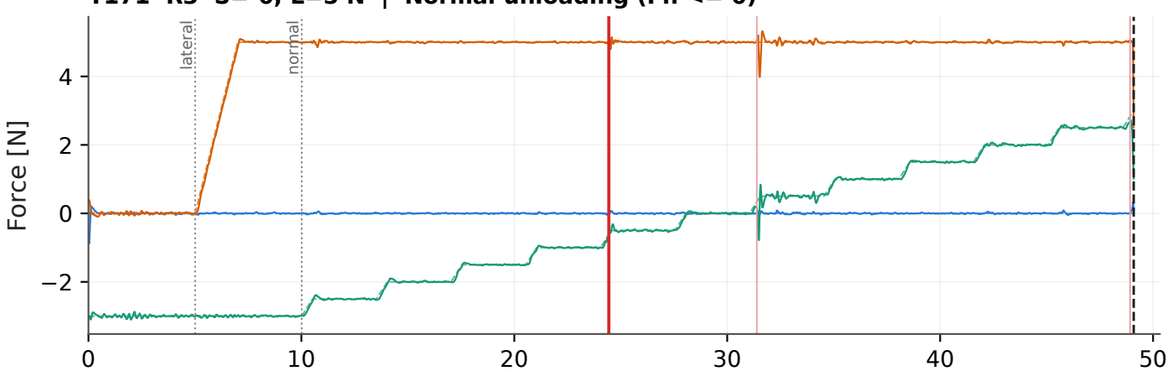
T170 R3 S=15, L=20 N | Normal tension (Fn > 0)



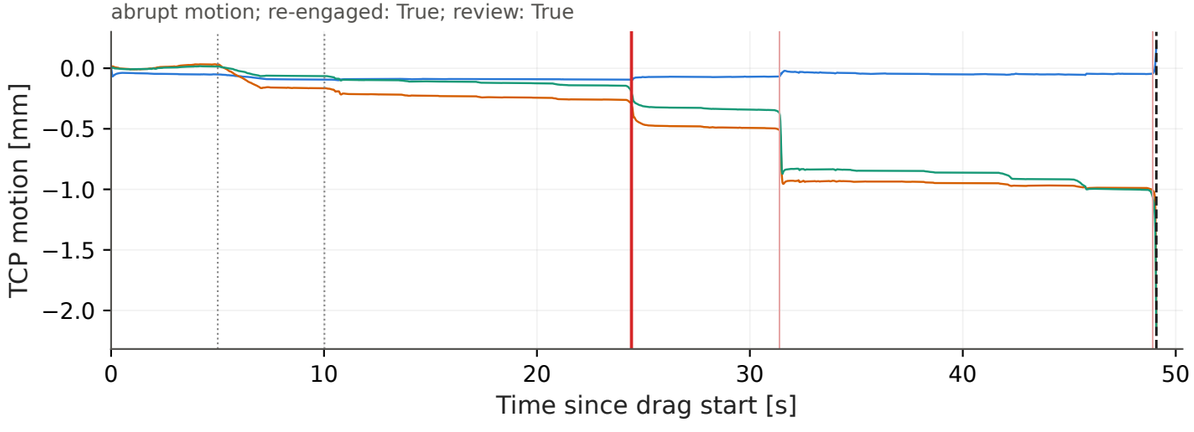
abrupt motion; re-engaged: False; review: True



T171 R3 S=-0, L=5 N | Normal unloading (Fn <= 0)



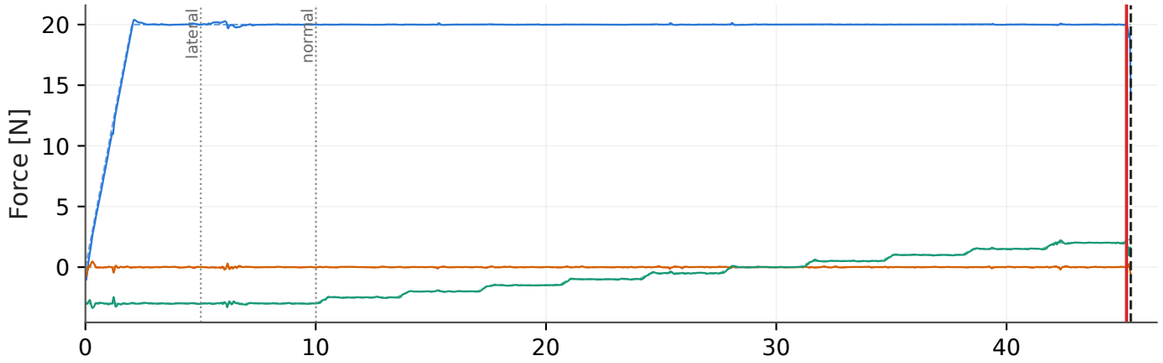
abrupt motion; re-engaged: True; review: True



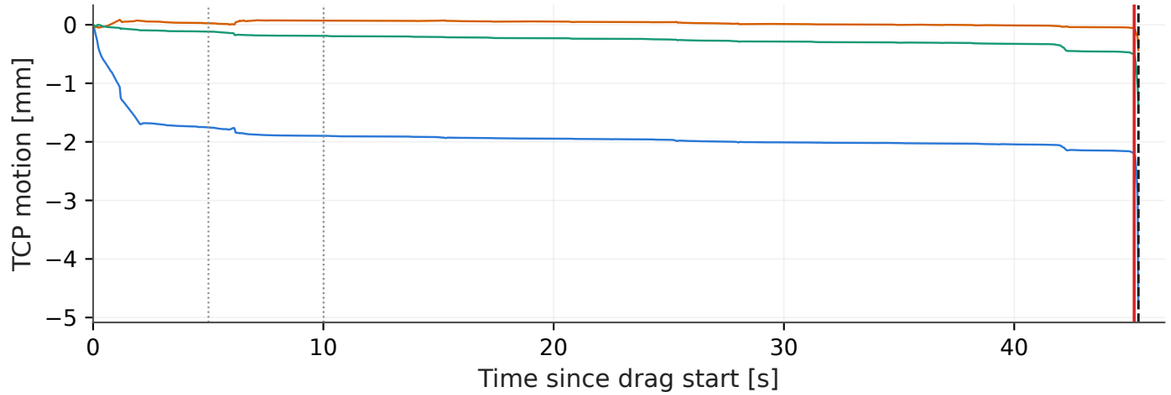
Trial audit | 172–174 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

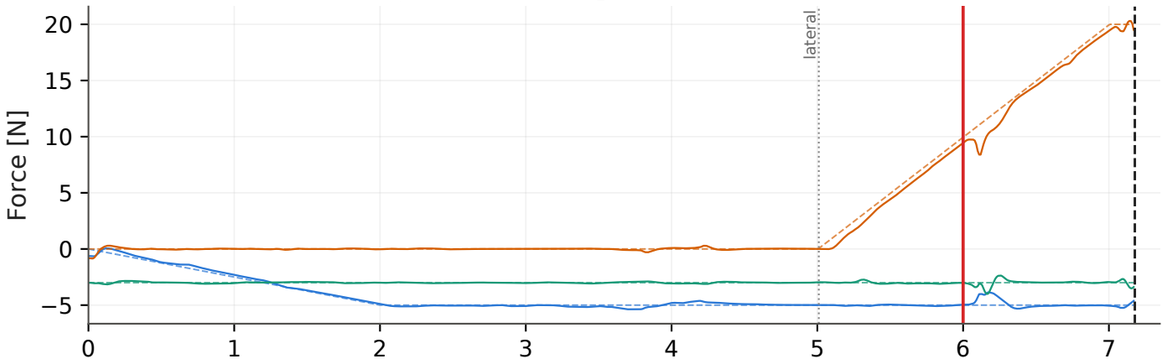
T172 R3 S=20, L=0 N | Normal tension (Fn > 0)



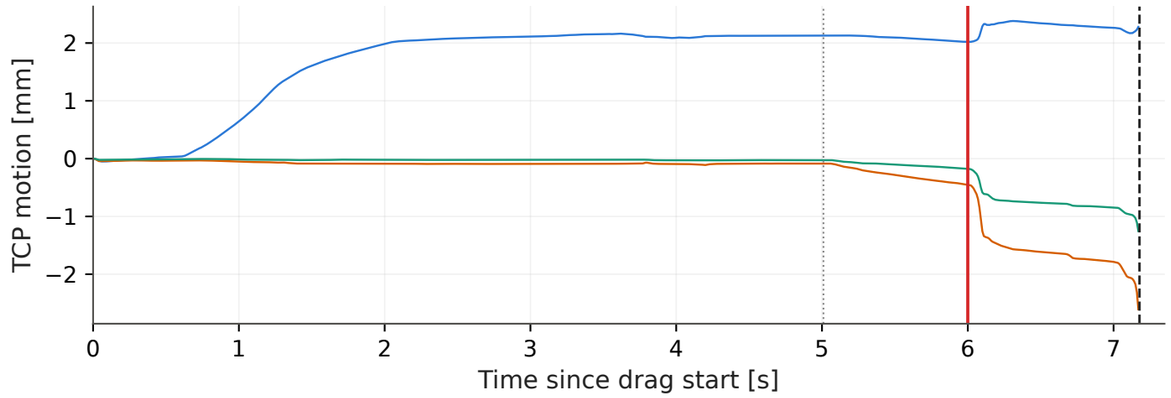
abrupt motion; re-engaged: False; review: False



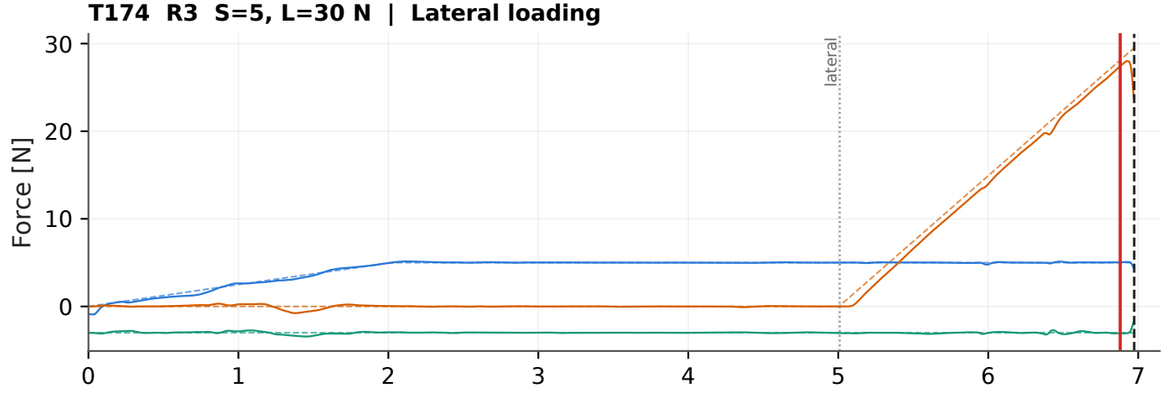
T173 R3 S=-5, L=20 N | Lateral loading



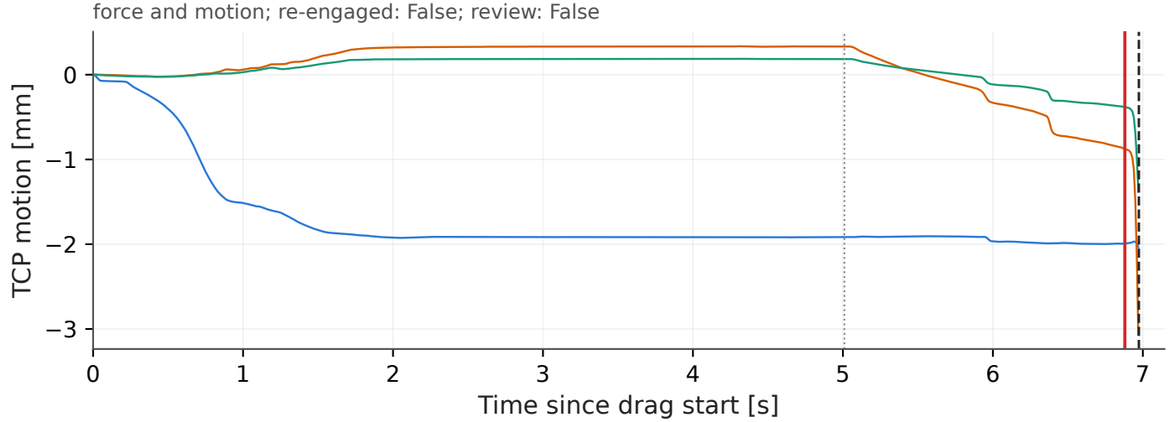
force and motion; re-engaged: False; review: False



T174 R3 S=5, L=30 N | Lateral loading



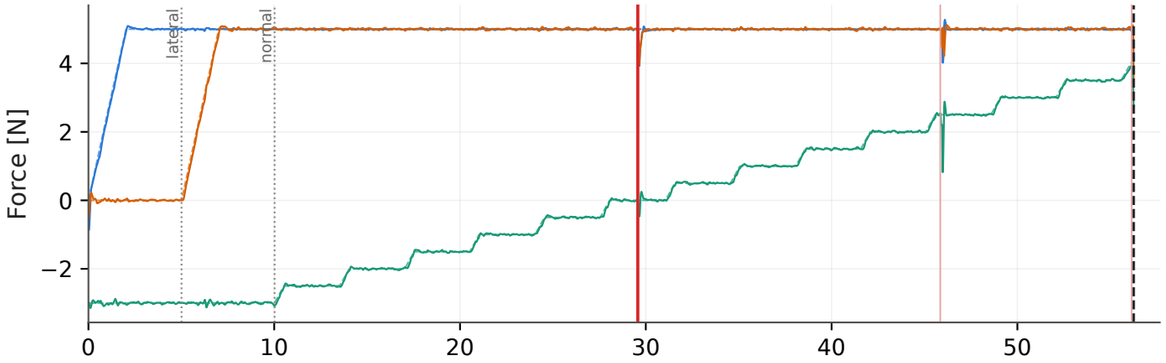
force and motion; re-engaged: False; review: False



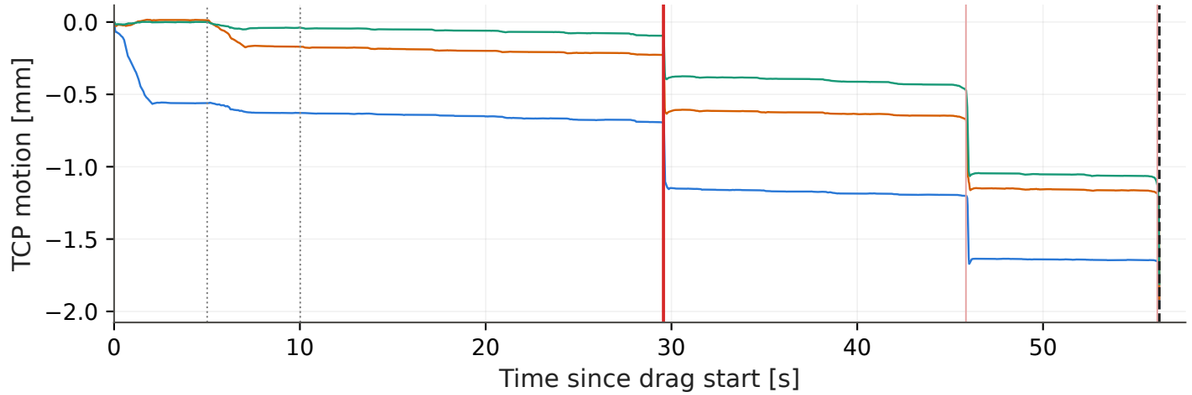
Trial audit | 175–177 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

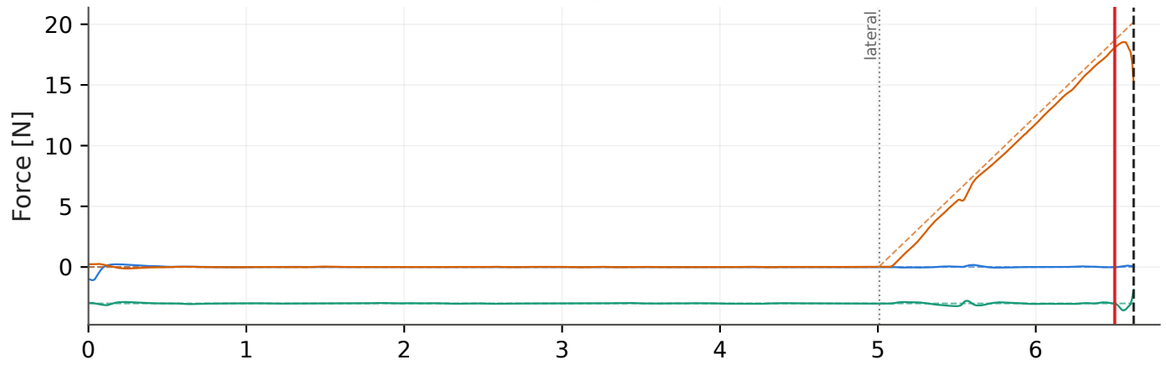
T175 R3 S=5, L=5 N | Normal unloading (Fn <= 0)



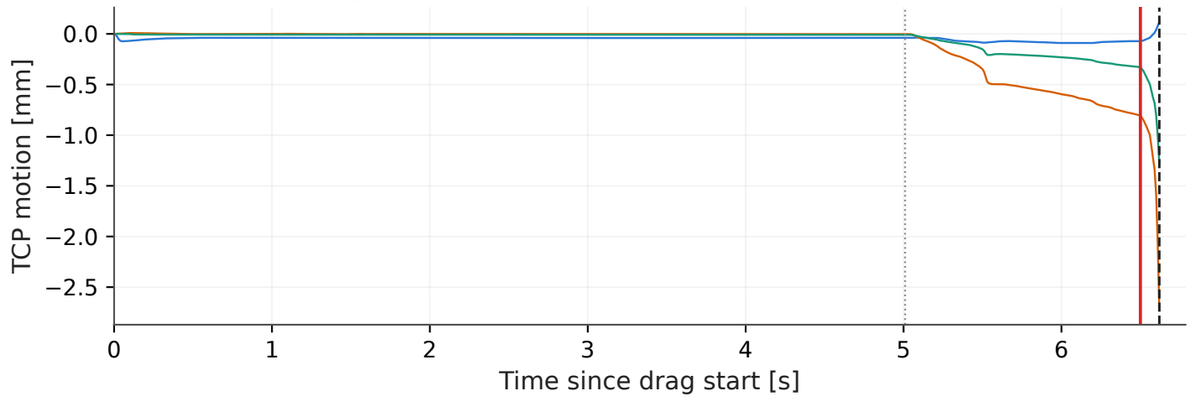
abrupt motion; re-engaged: True; review: False



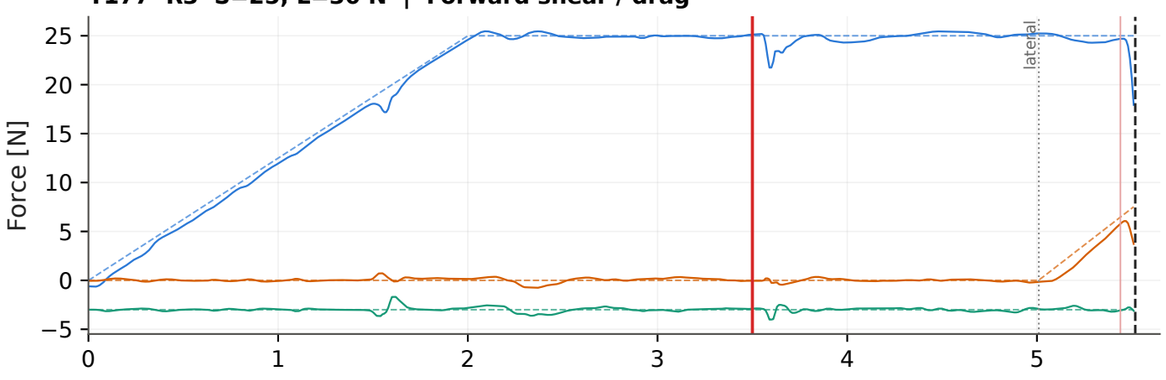
T176 R3 S=0, L=25 N | Lateral loading



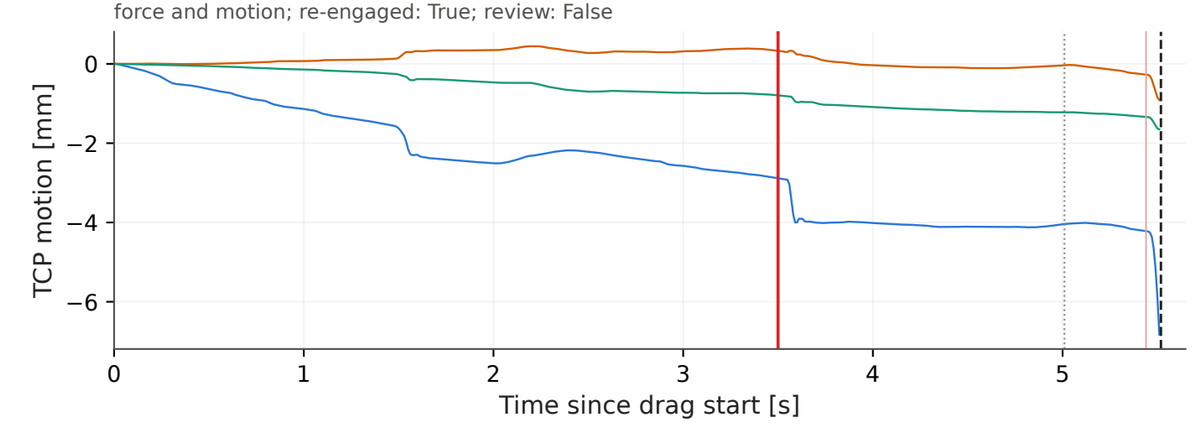
force and motion; re-engaged: False; review: False



T177 R3 S=25, L=30 N | Forward shear / drag



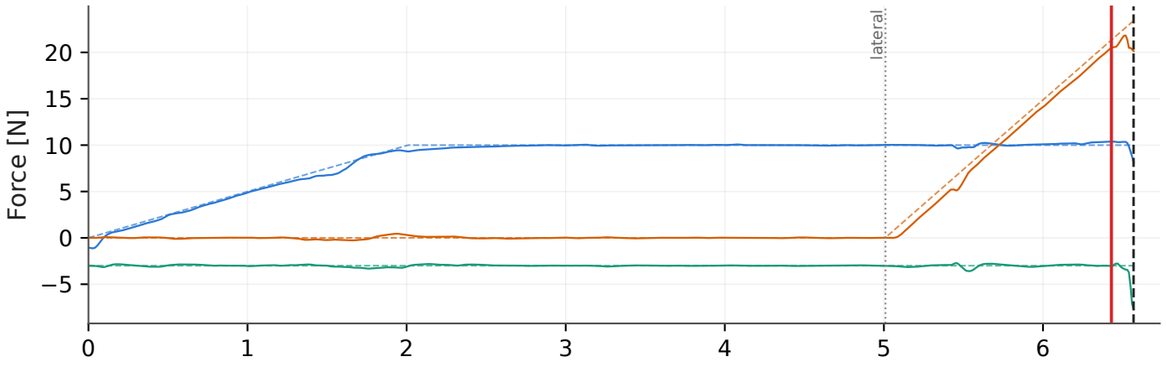
force and motion; re-engaged: True; review: False

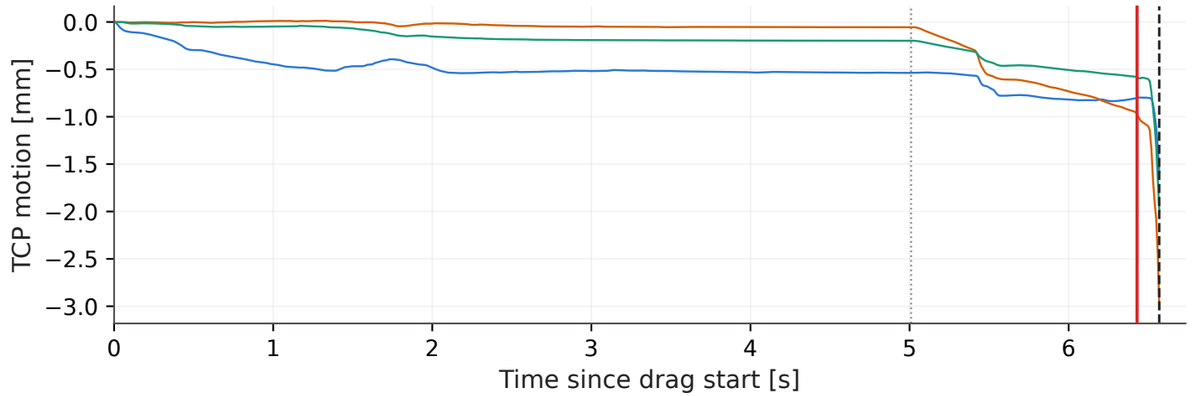


Trial audit | 178–180 / 231

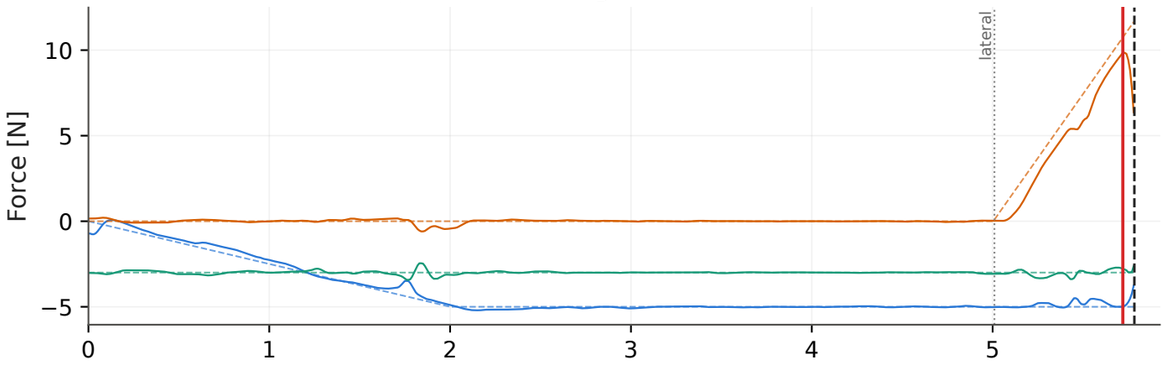
Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

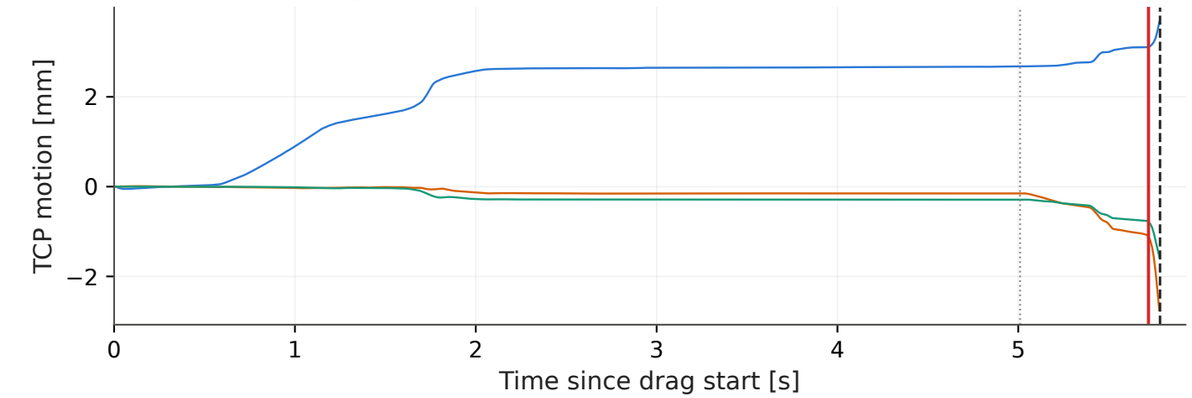
T178 R3 S=10, L=30 N | Lateral loading



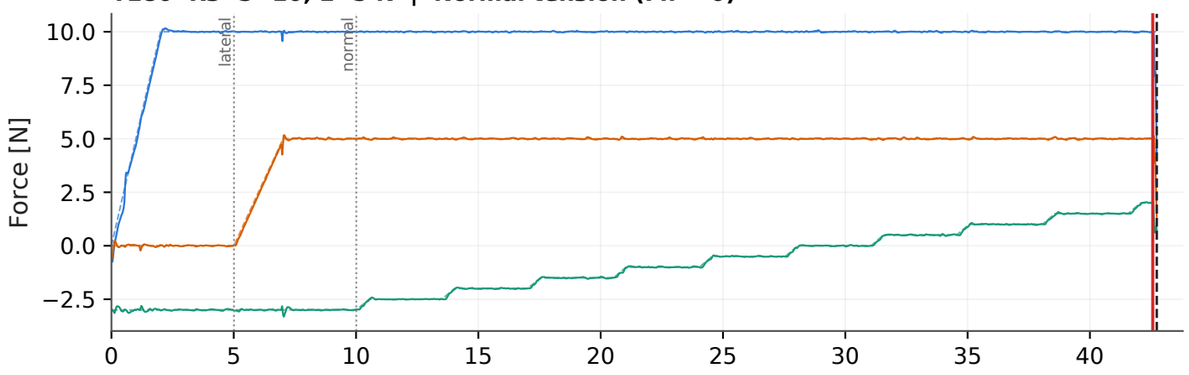


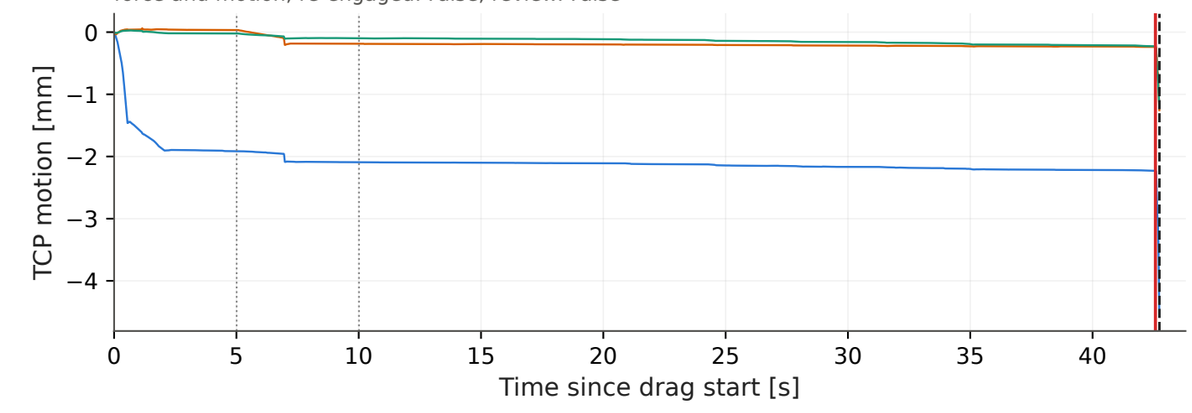
T179 R3 S=-5, L=30 N | Lateral loading





T180 R3 S=10, L=5 N | Normal tension (Fn > 0)

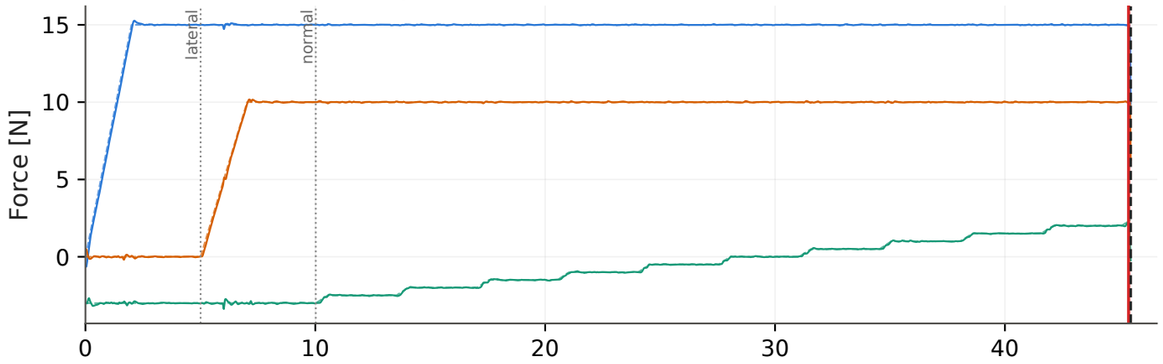




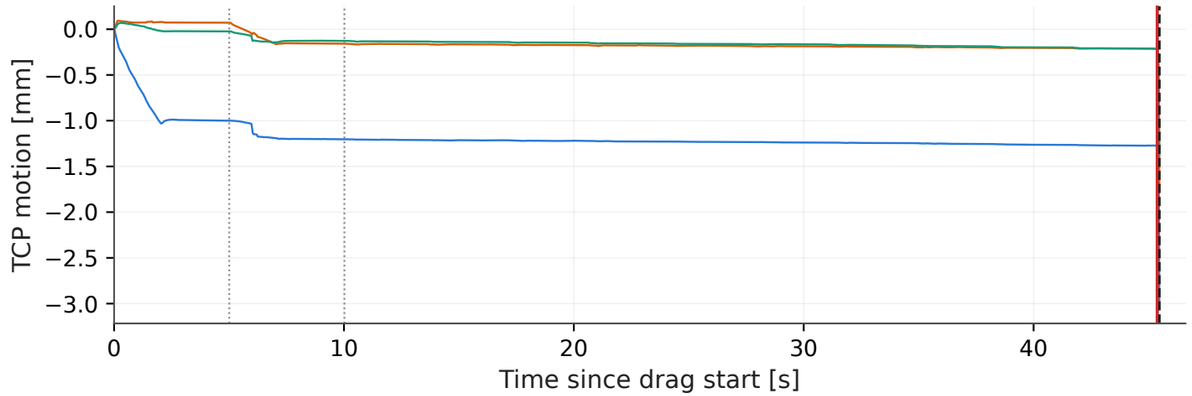
Trial audit | 181–183 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

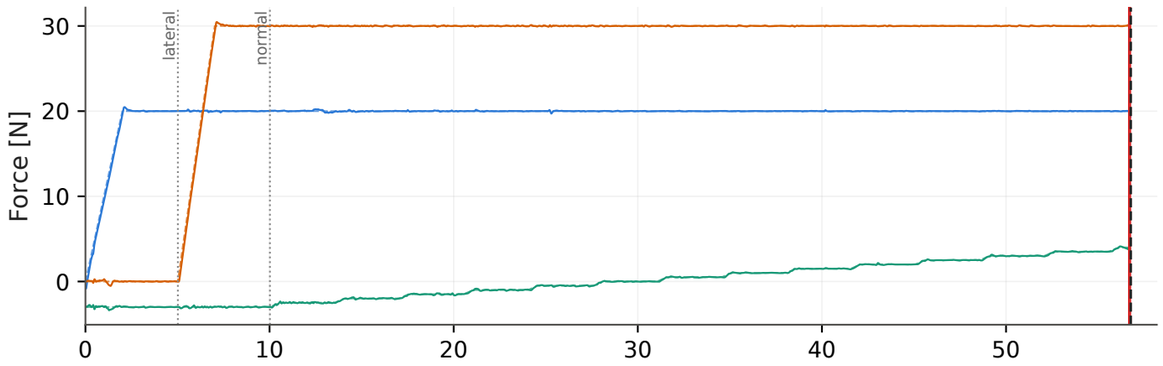
T181 R3 S=15, L=10 N | Normal tension (Fn > 0)



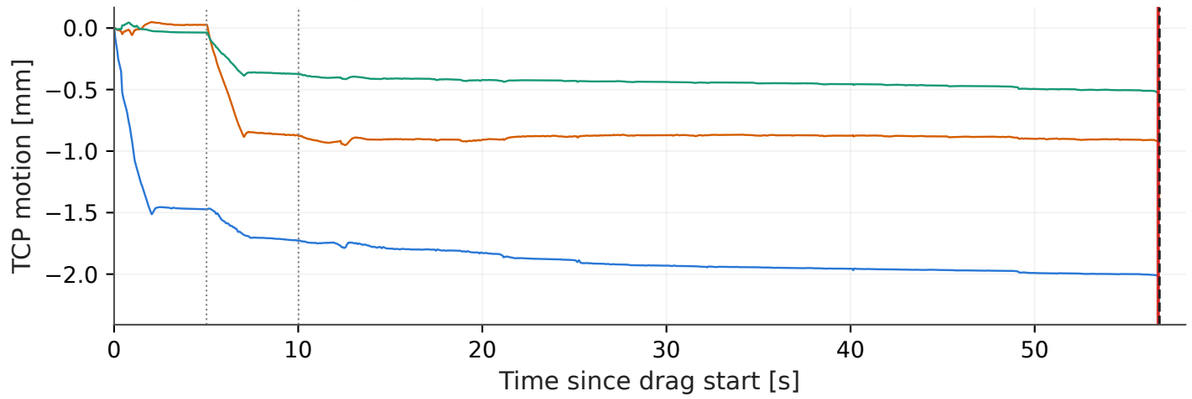
force and motion; re-engaged: False; review: False



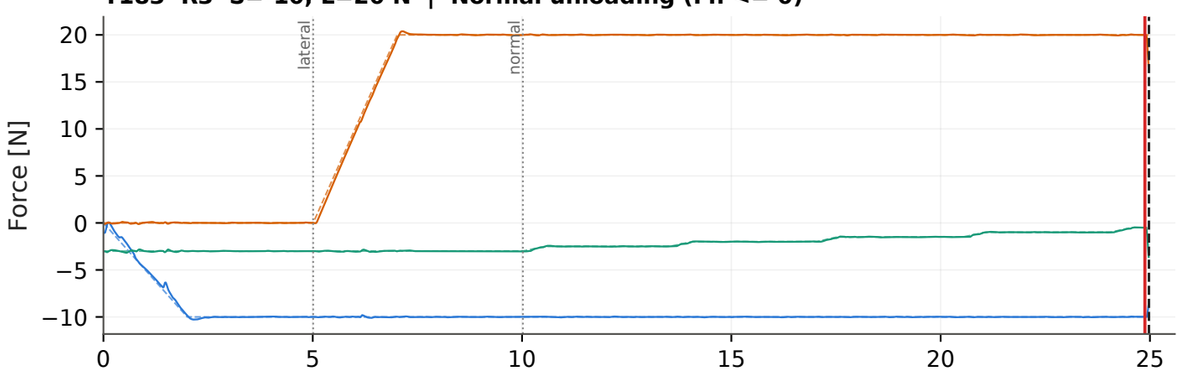
T182 R3 S=20, L=30 N | Normal tension (Fn > 0)



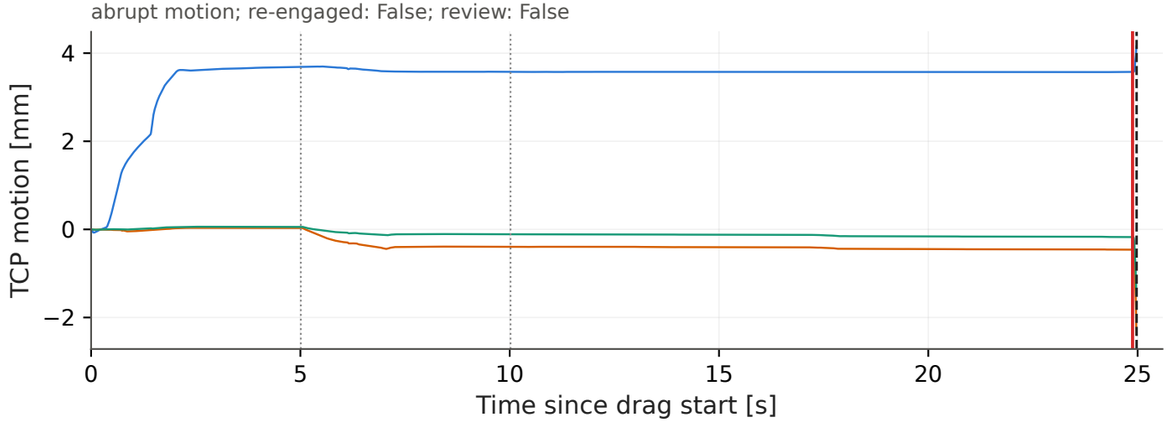
abrupt motion; re-engaged: False; review: False



T183 R3 S=-10, L=20 N | Normal unloading (Fn <= 0)



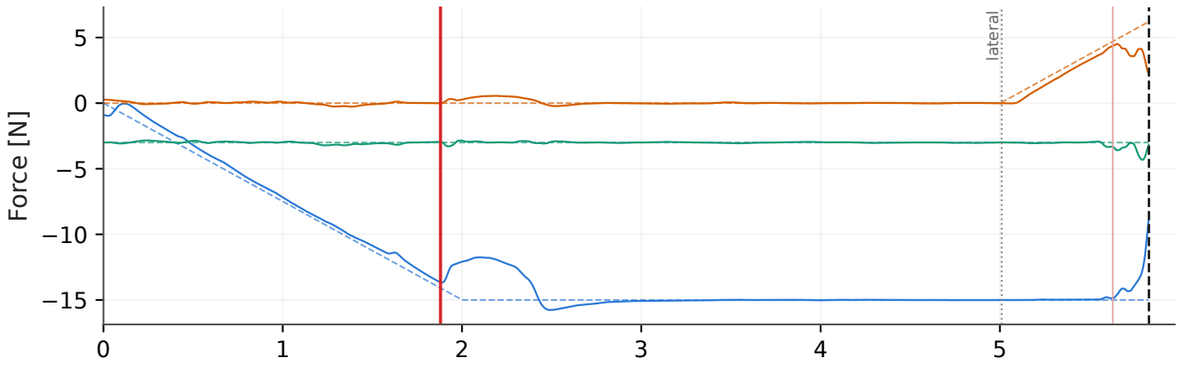
abrupt motion; re-engaged: False; review: False



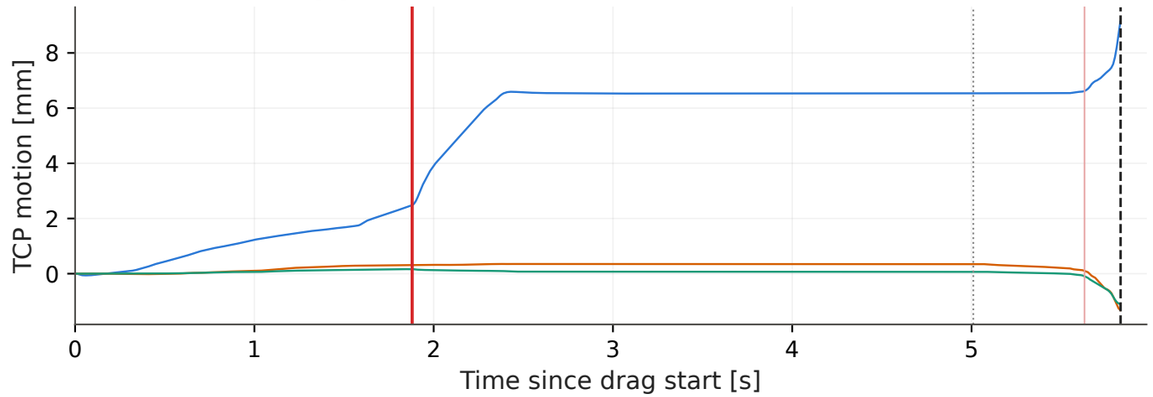
Trial audit | 184–186 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

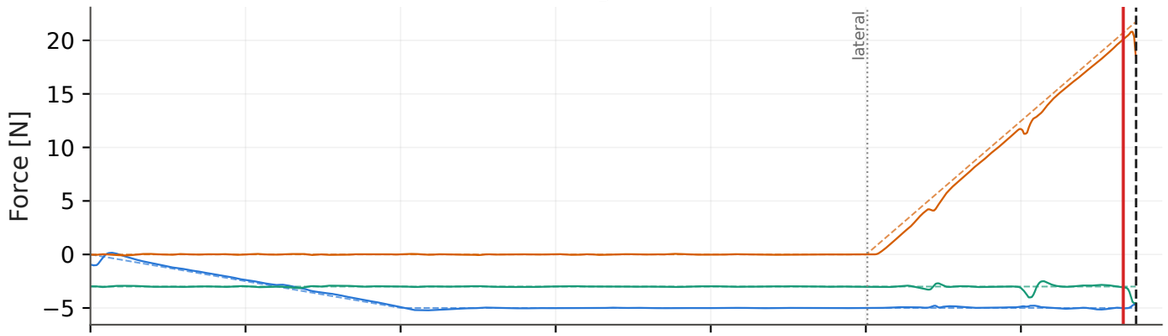
T184 R3 S=-15, L=15 N | Reverse shear / drag



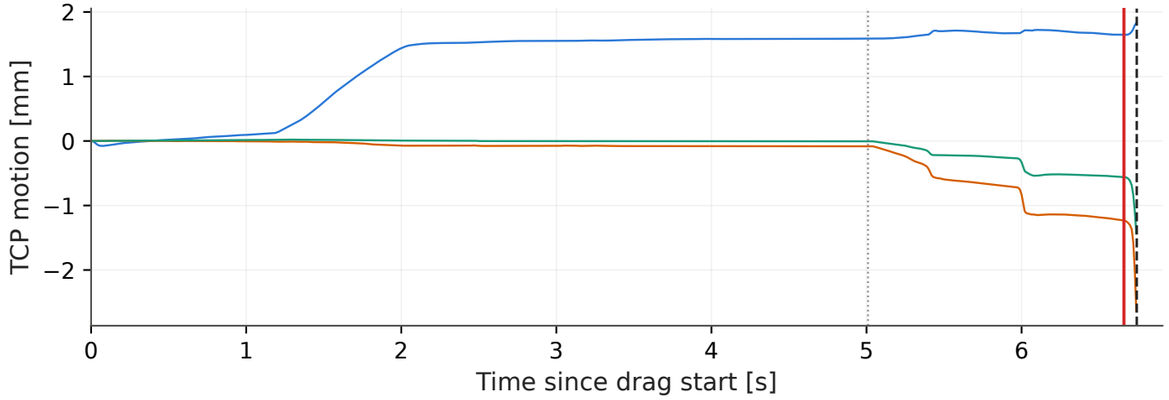
force and motion; re-engaged: True; review: False



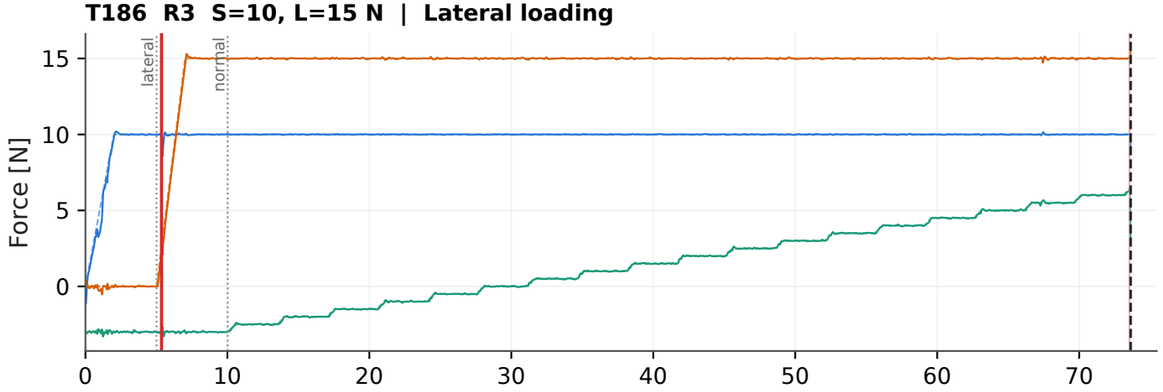
T185 R3 S=-5, L=25 N | Lateral loading



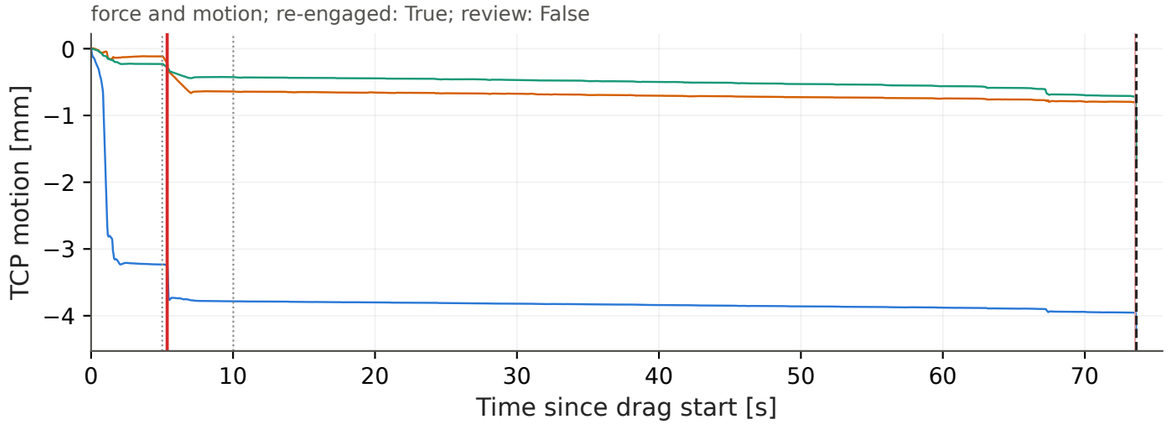
force and motion; re-engaged: False; review: False



T186 R3 S=10, L=15 N | Lateral loading



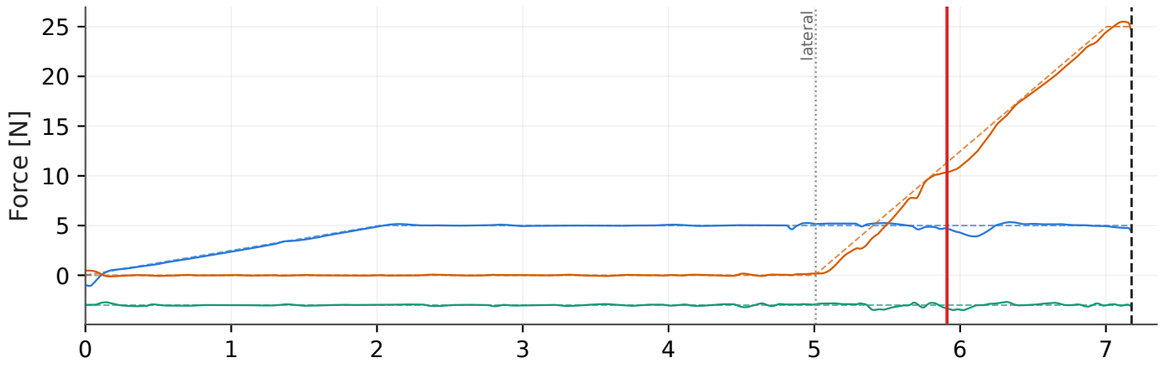
force and motion; re-engaged: True; review: False



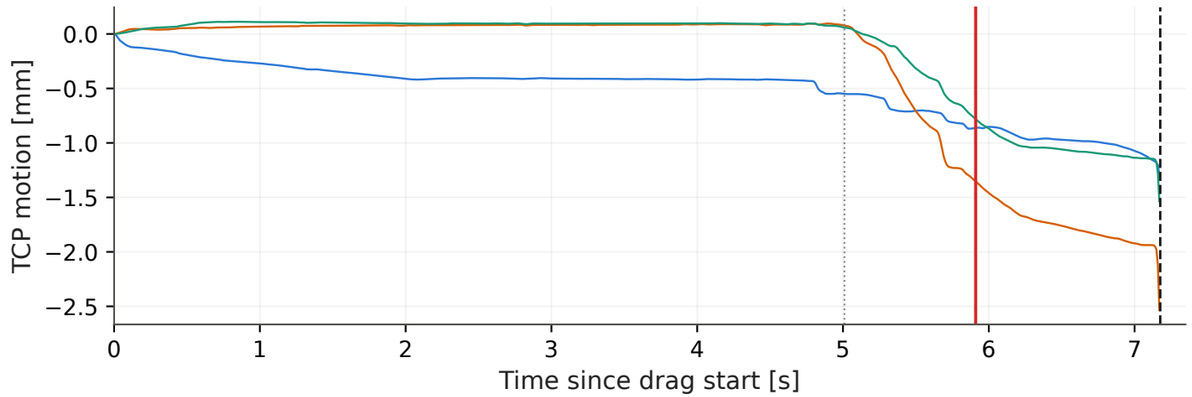
Trial audit | 187–189 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

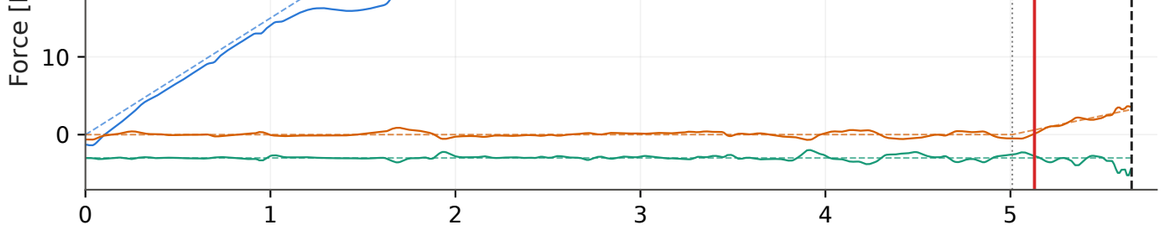
T187 R3 S=5, L=25 N | Lateral loading



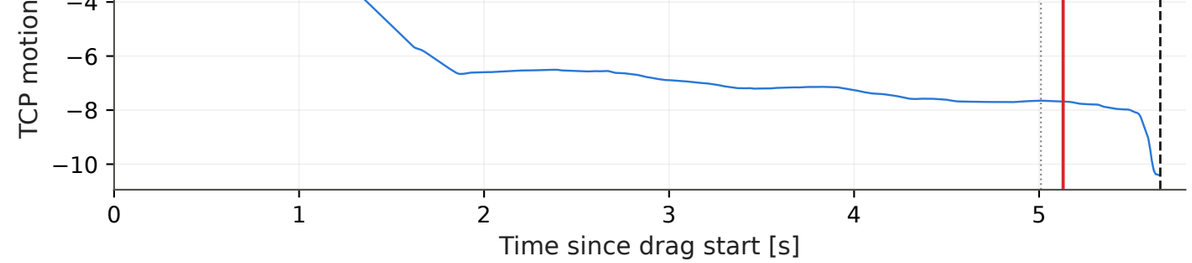
force and motion; re-engaged: False; review: True



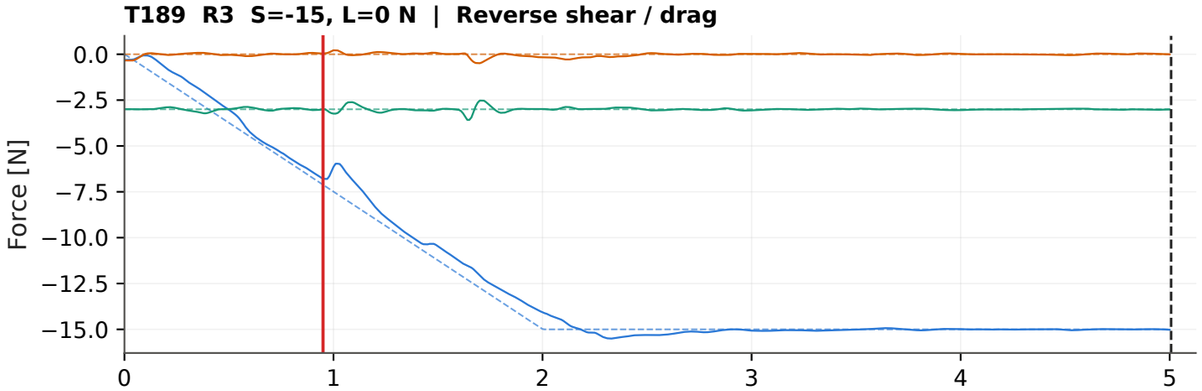
T188 R3 S=30, L=10 N | Lateral loading



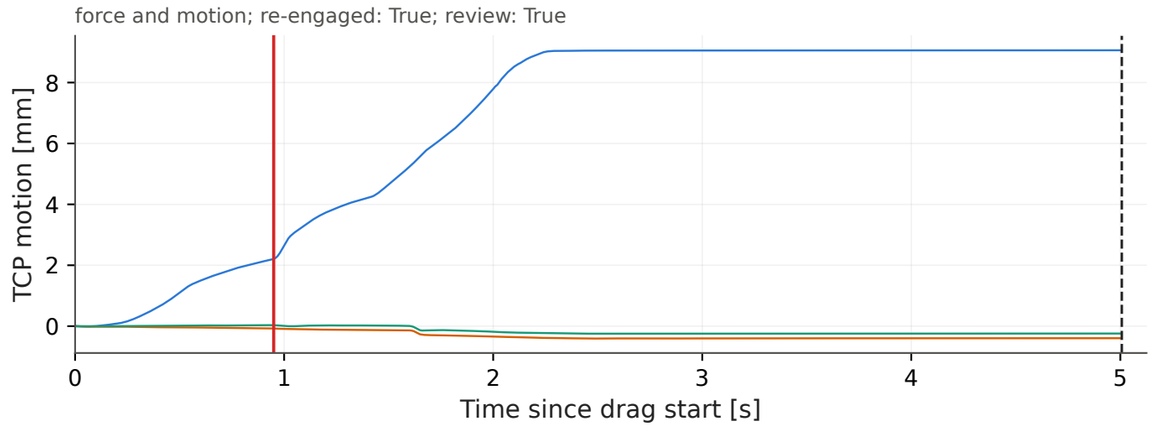
force and motion; re-engaged: False; review: False



T189 R3 S=-15, L=0 N | Reverse shear / drag



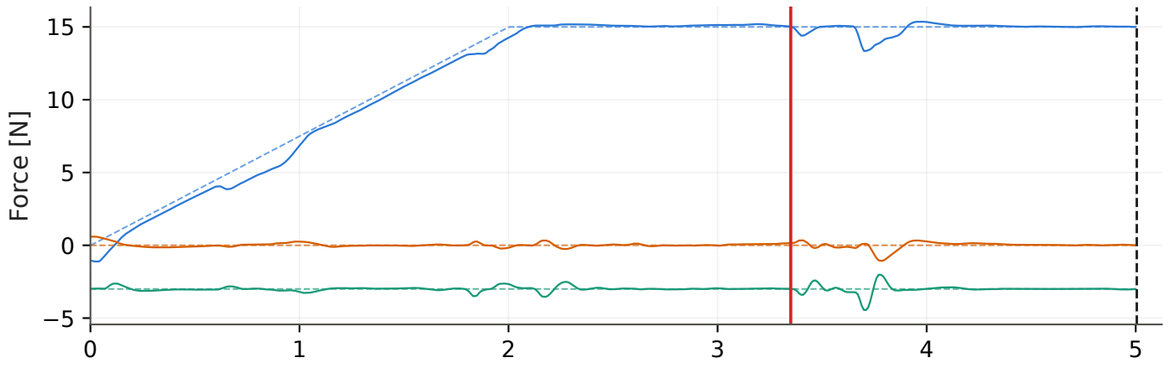
force and motion; re-engaged: True; review: True



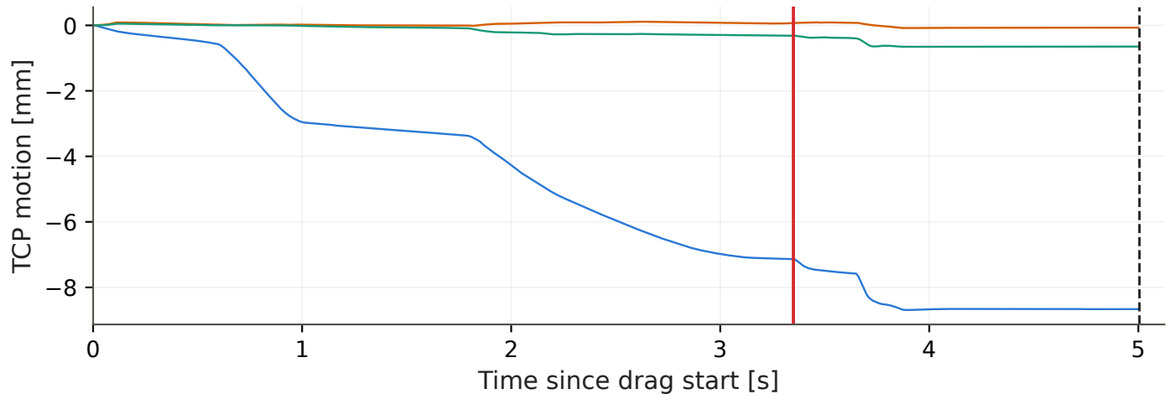
Trial audit | 190–192 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

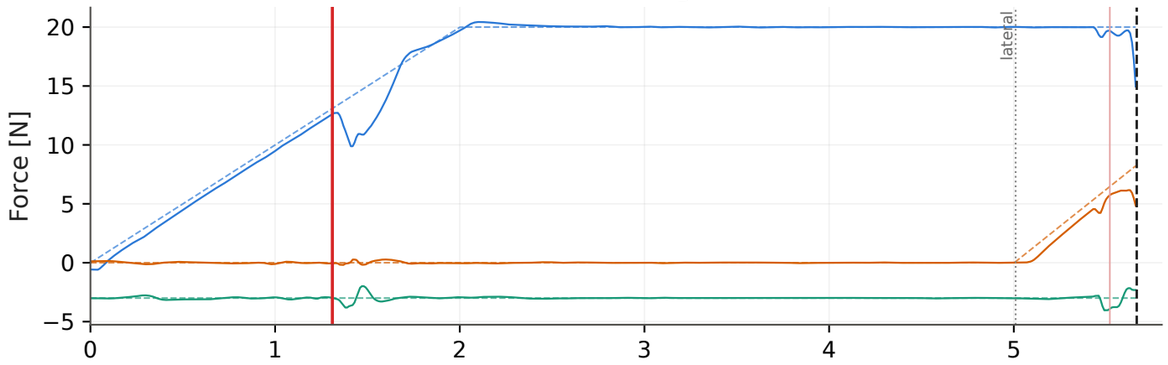
T190 R3 S=15, L=0 N | Forward shear / drag



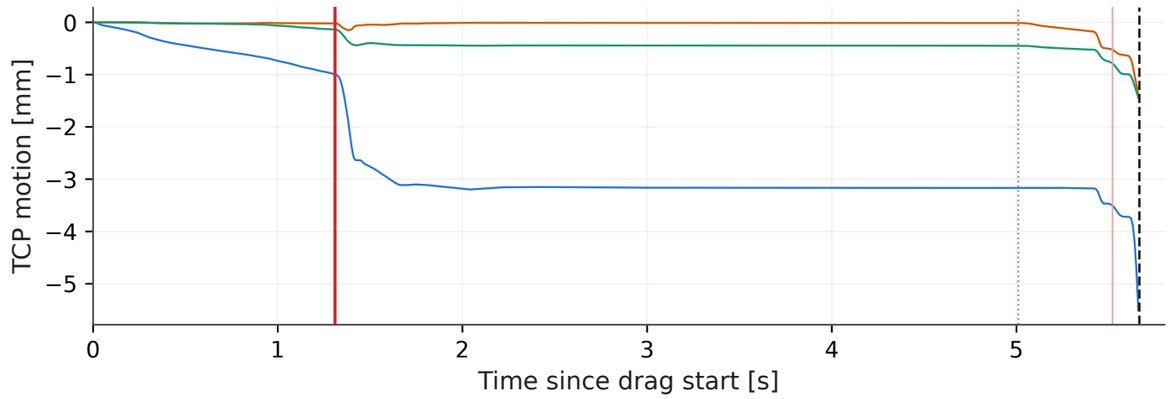
abrupt motion; re-engaged: True; review: False



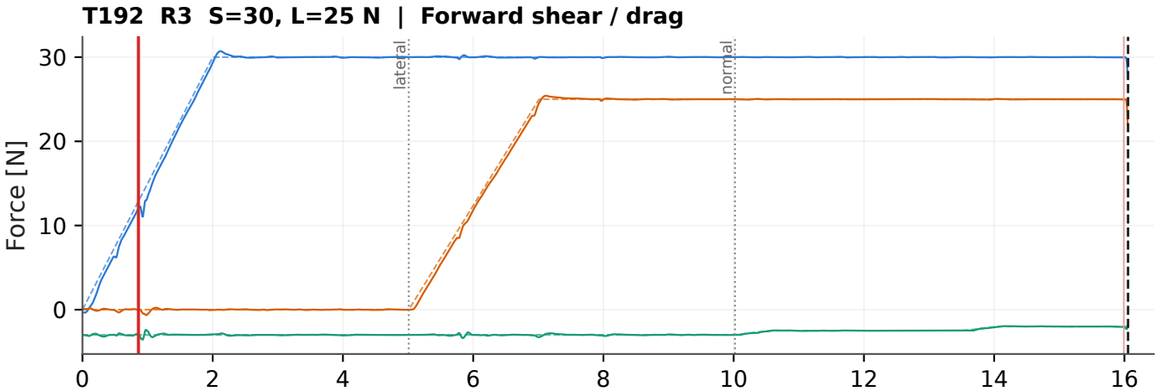
T191 R3 S=20, L=25 N | Forward shear / drag



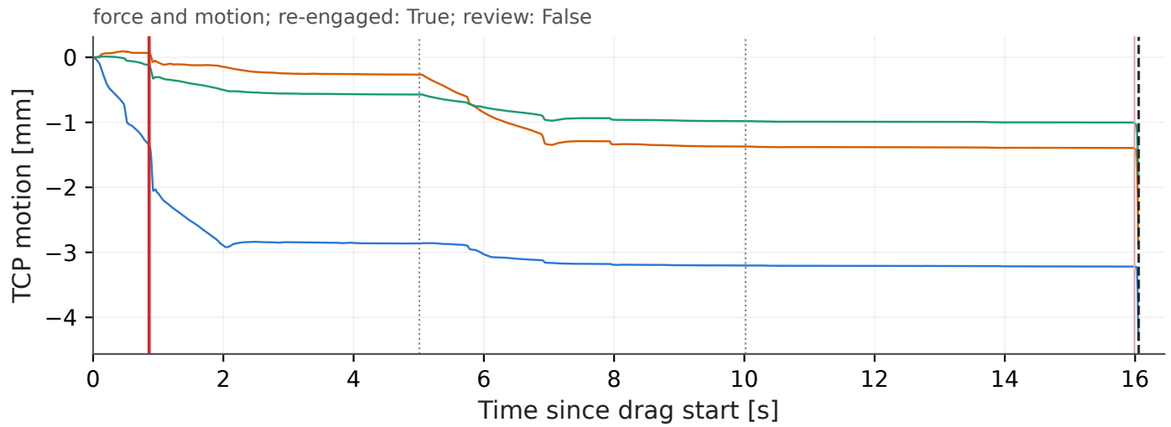
force and motion; re-engaged: True; review: False



T192 R3 S=30, L=25 N | Forward shear / drag



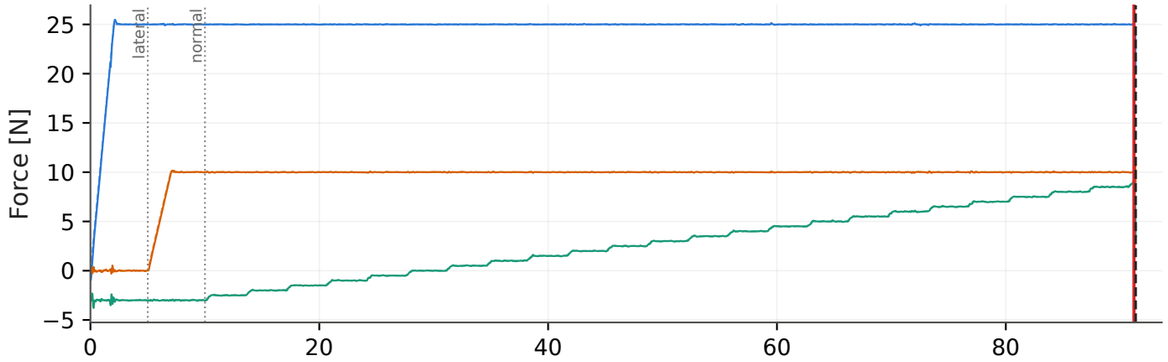
force and motion; re-engaged: True; review: False



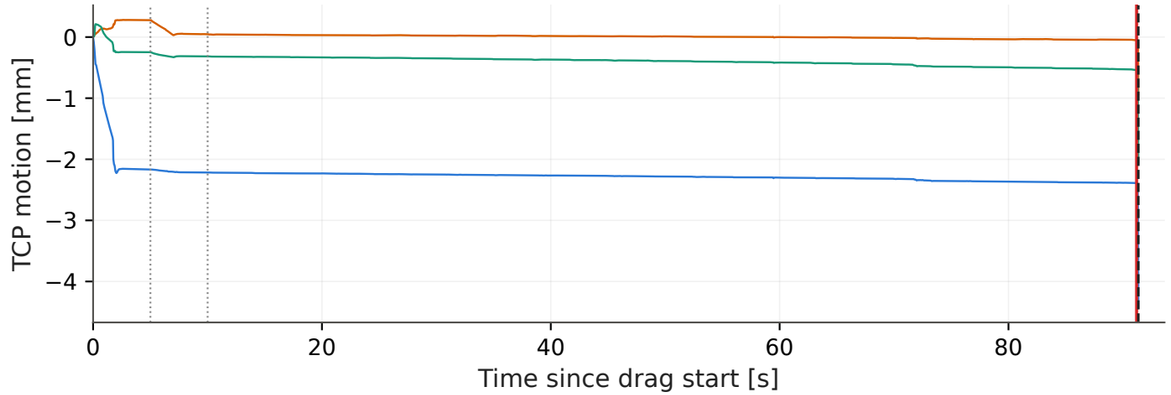
Trial audit | 193–195 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

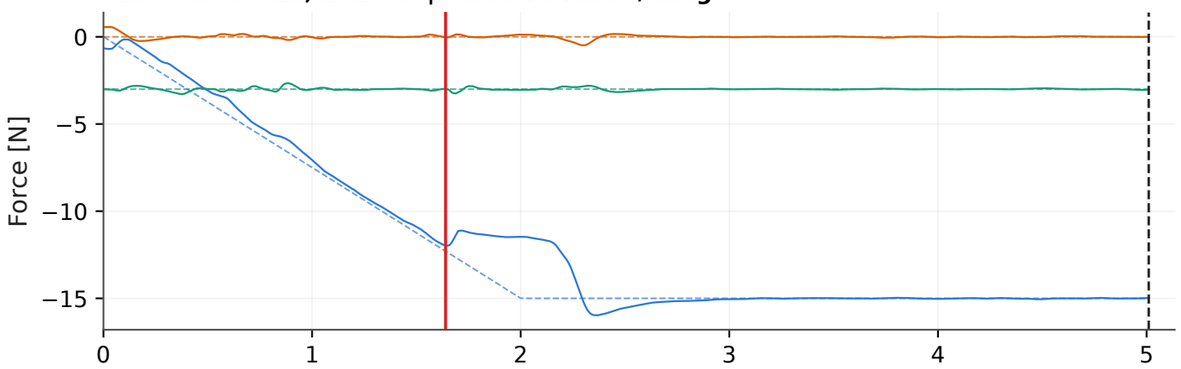
T193 R3 S=25, L=10 N | Normal tension (Fn > 0)



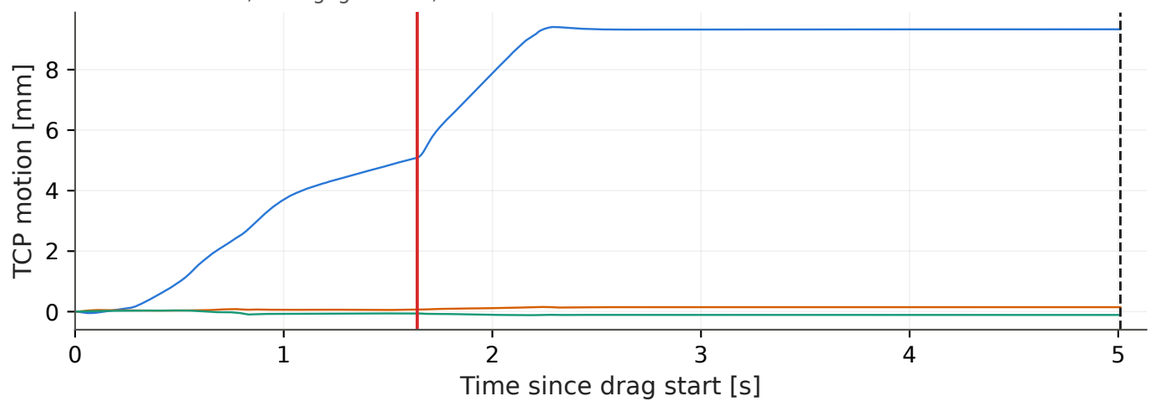
force and motion; re-engaged: False; review: False



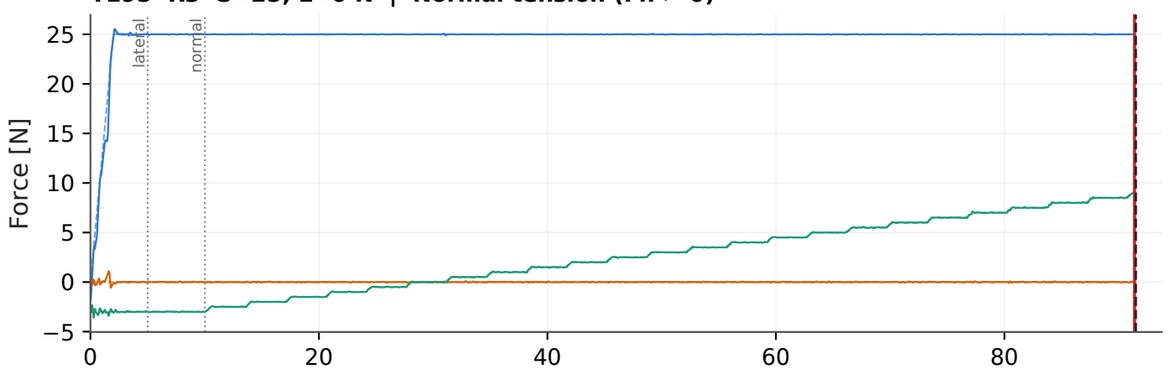
T194 R3 S=-15, L=20 N | Reverse shear / drag



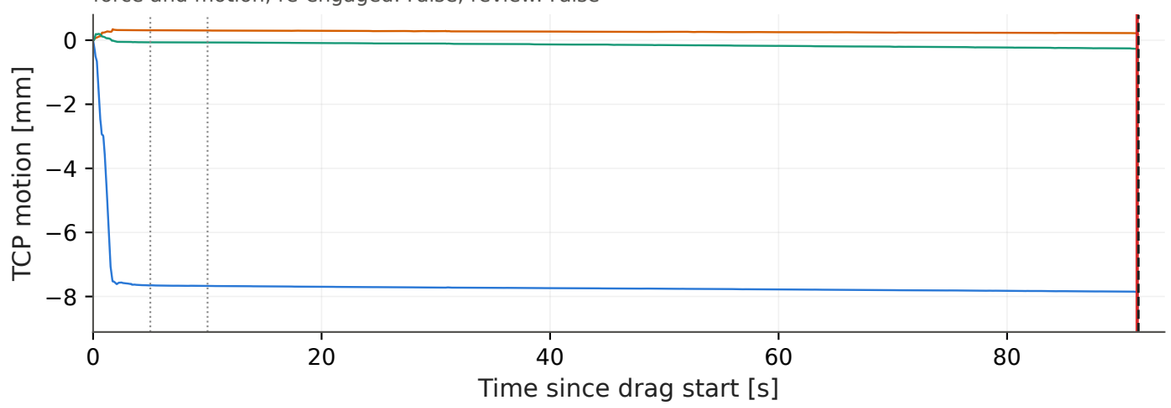
force and motion; re-engaged: True; review: True



T195 R3 S=25, L=0 N | Normal tension (Fn > 0)



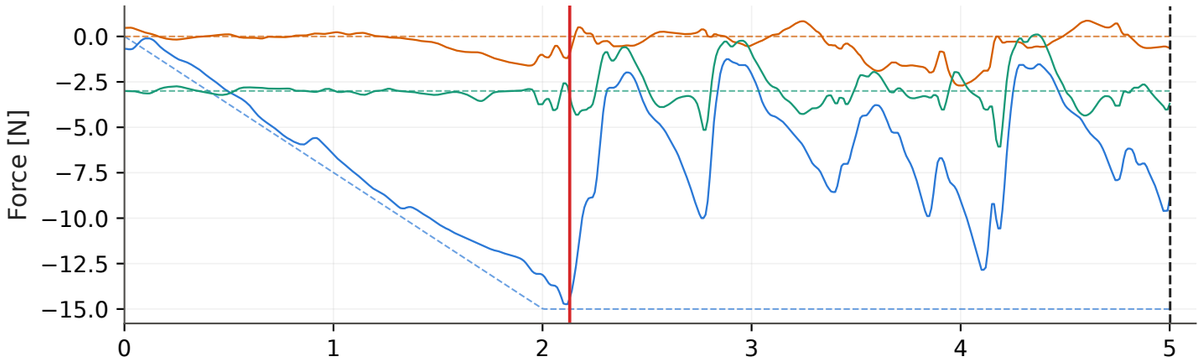
force and motion; re-engaged: False; review: False



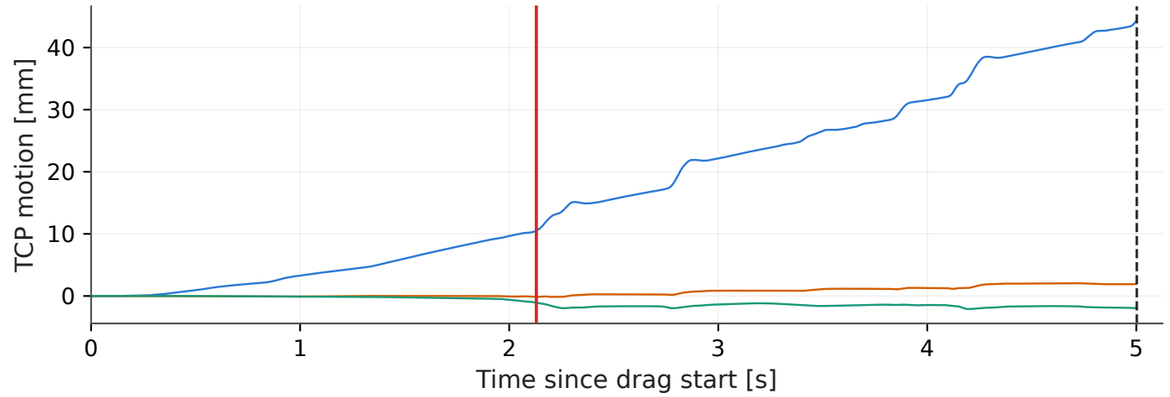
Trial audit | 196–198 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

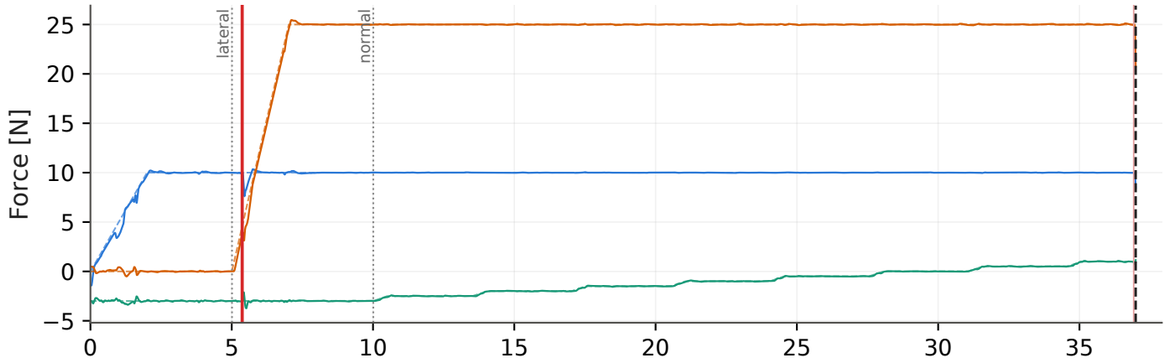
T196 R3 S=-15, L=25 N | Reverse shear / drag



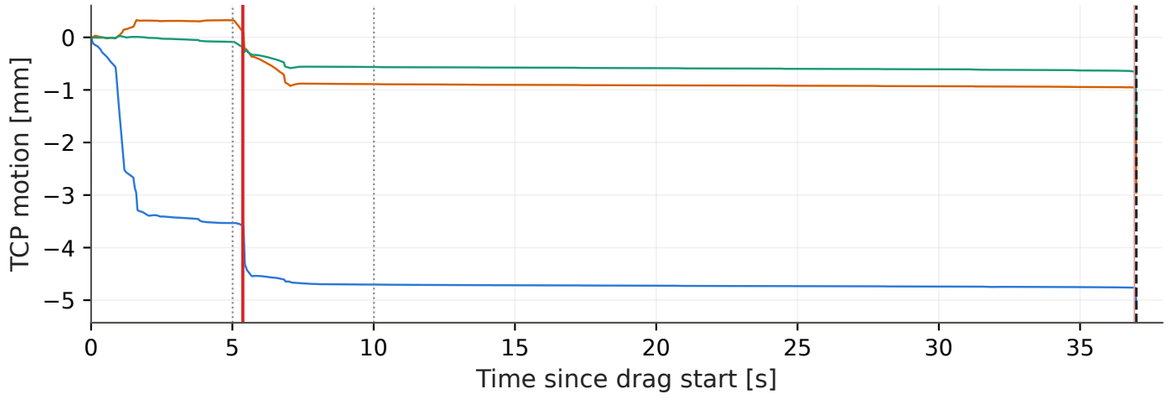
force and motion; re-engaged: False; review: False



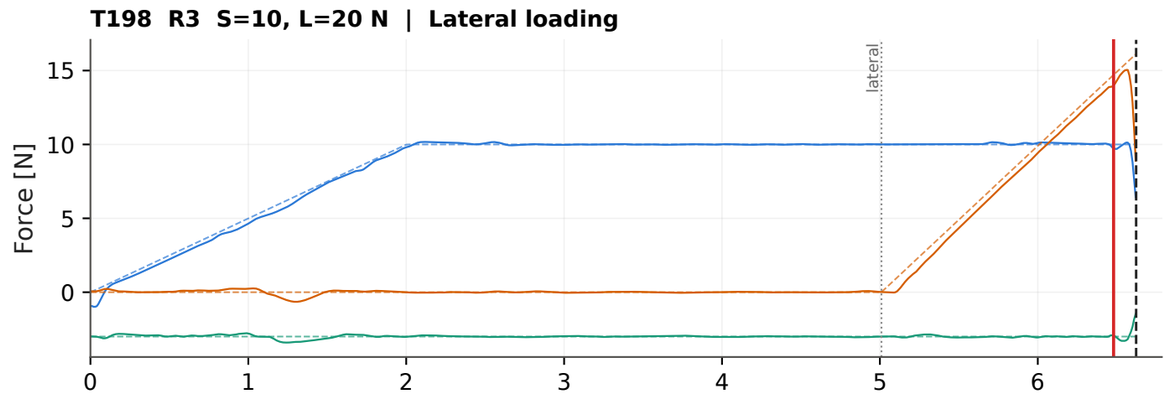
T197 R3 S=10, L=25 N | Lateral loading



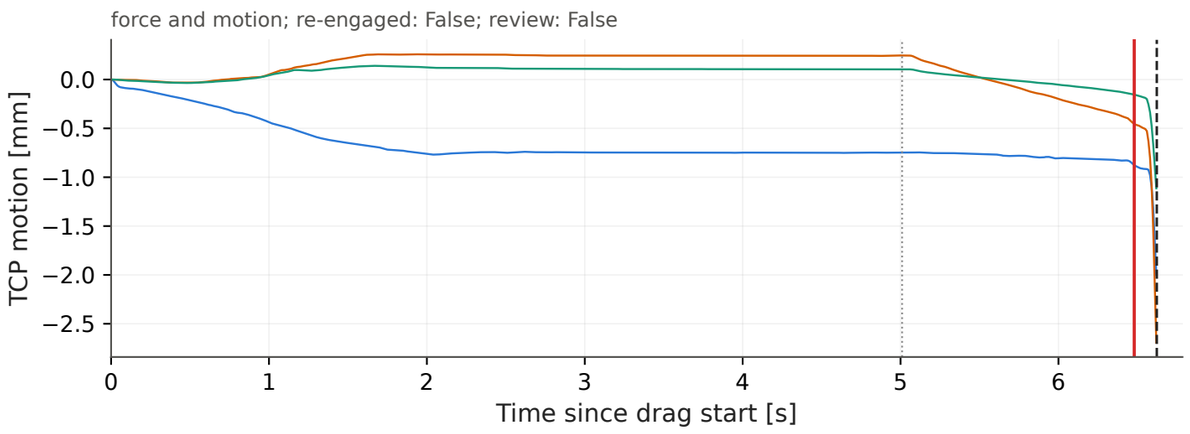
force and motion; re-engaged: True; review: True



T198 R3 S=10, L=20 N | Lateral loading



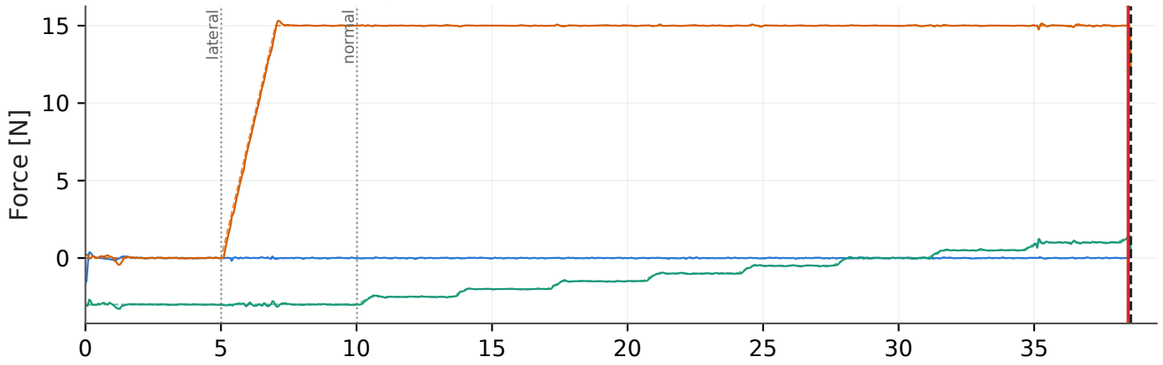
force and motion; re-engaged: False; review: False



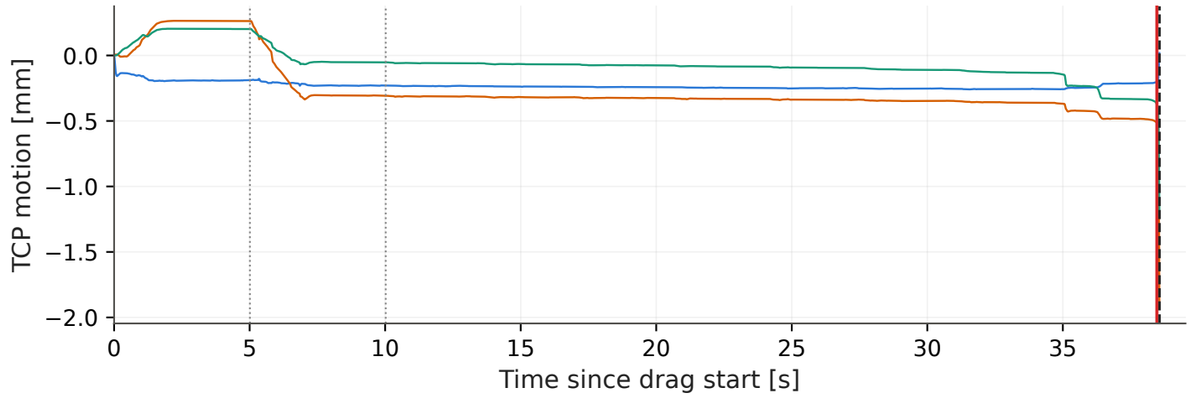
Trial audit | 199–201 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

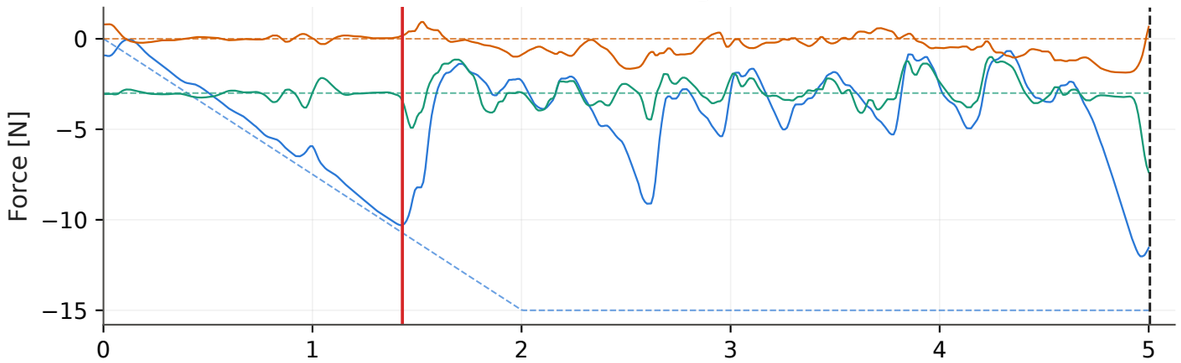
T199 R3 S=0, L=15 N | Normal tension (Fn > 0)



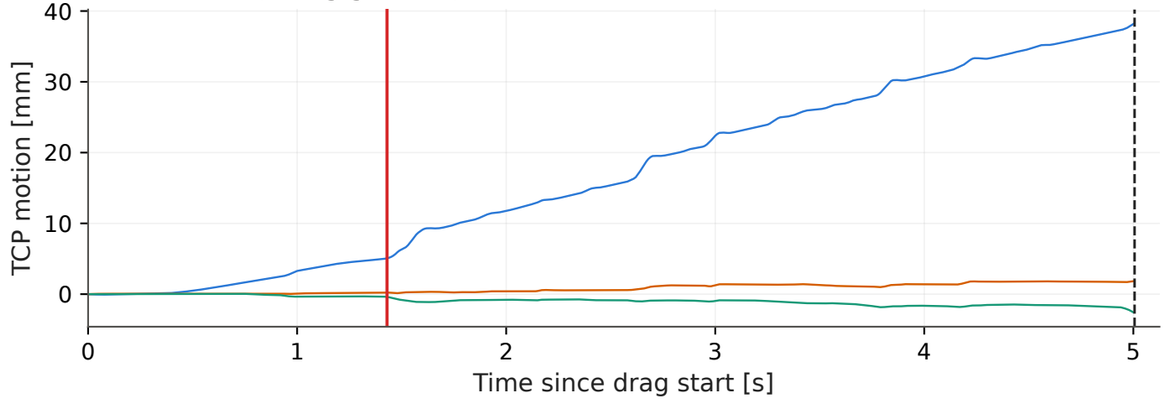
abrupt motion; re-engaged: False; review: False



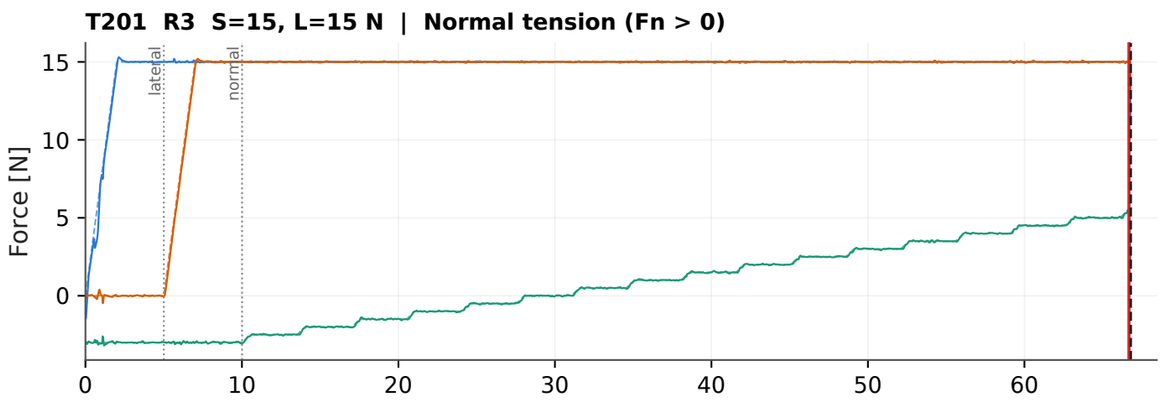
T200 R3 S=-15, L=30 N | Reverse shear / drag



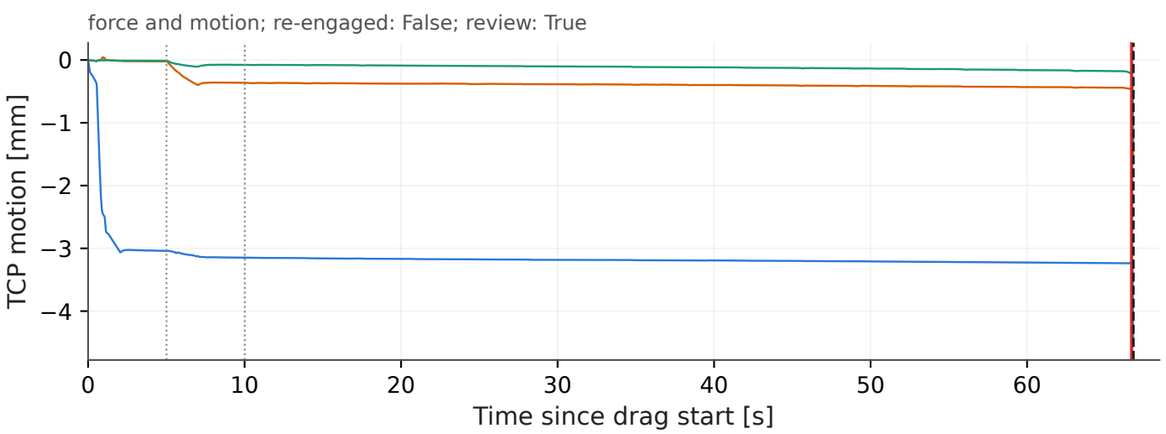
force and motion; re-engaged: False; review: False



T201 R3 S=15, L=15 N | Normal tension (Fn > 0)



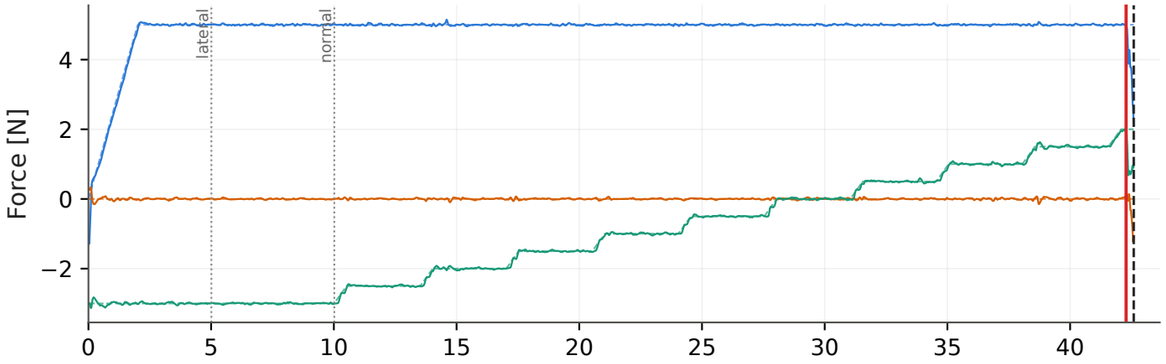
force and motion; re-engaged: False; review: True



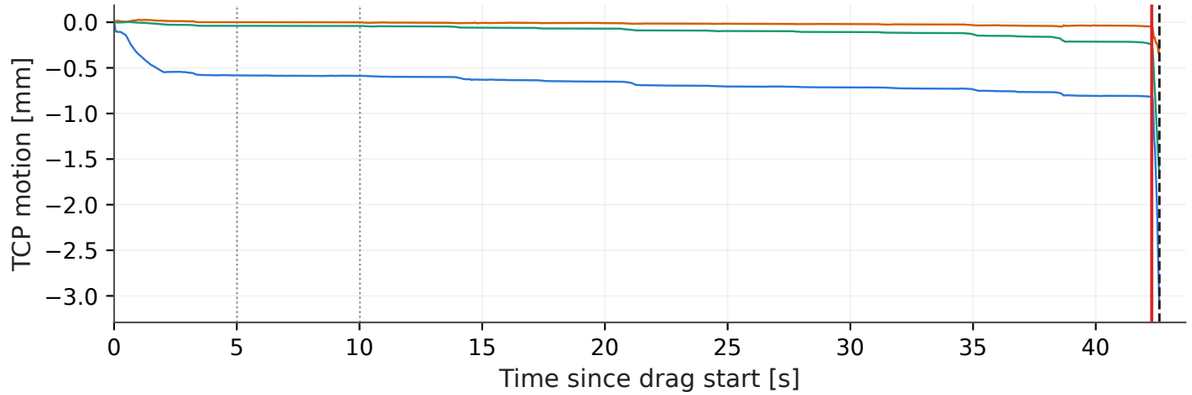
Trial audit | 202–204 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

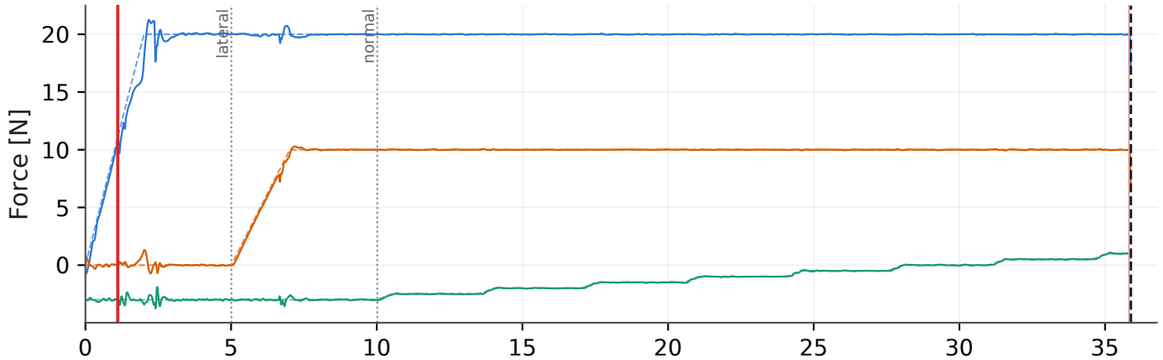
T202 R3 S=5, L=0 N | Normal tension (Fn > 0)



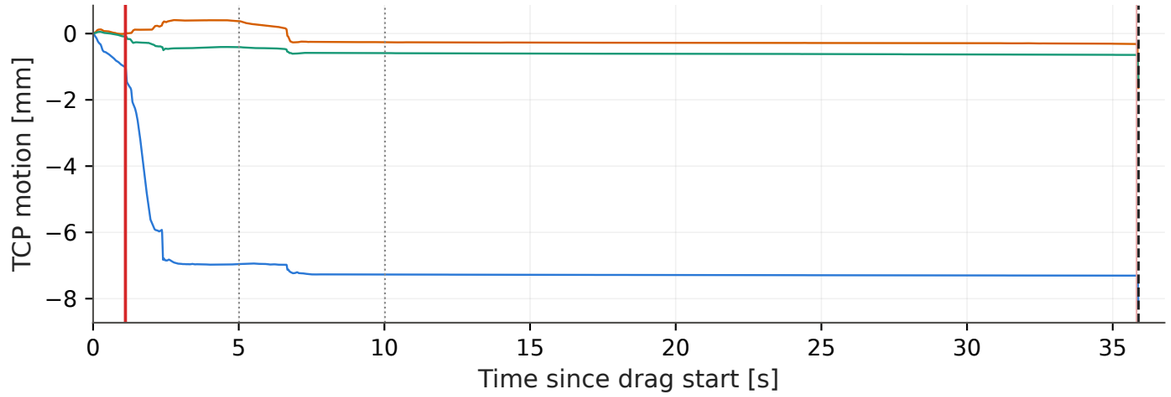
abrupt motion; re-engaged: False; review: False



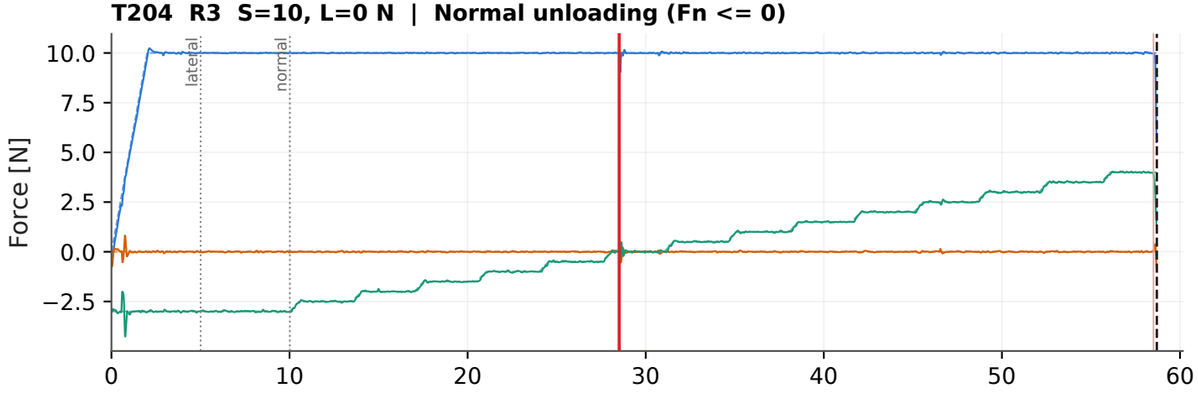
T203 R3 S=20, L=10 N | Forward shear / drag



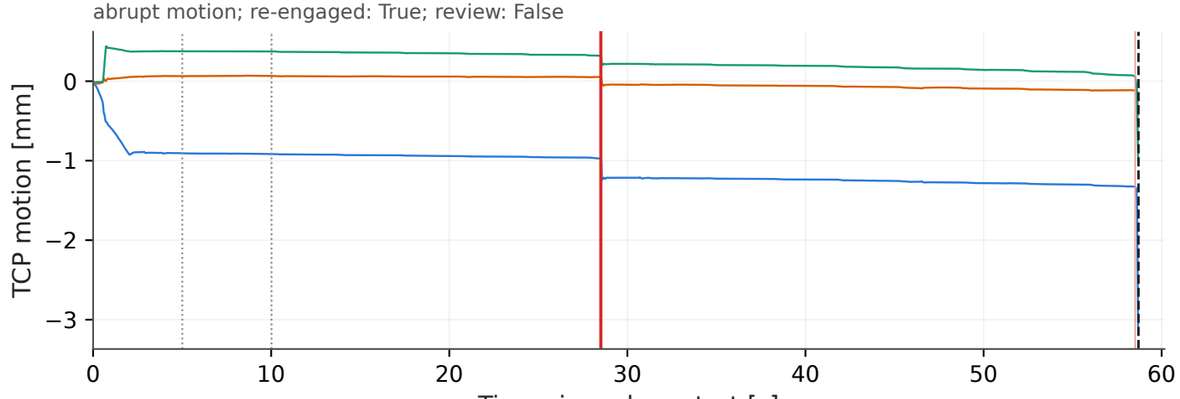
force and motion; re-engaged: True; review: False



T204 R3 S=10, L=0 N | Normal unloading (Fn <= 0)



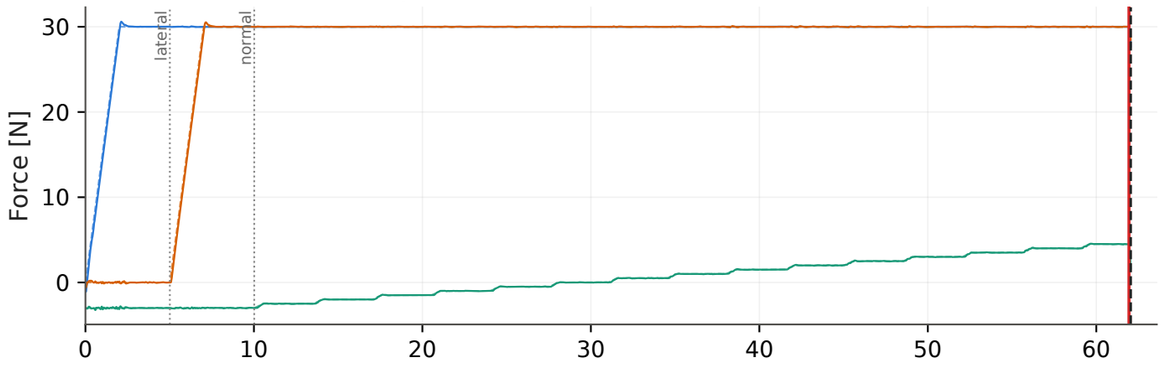
abrupt motion; re-engaged: True; review: False



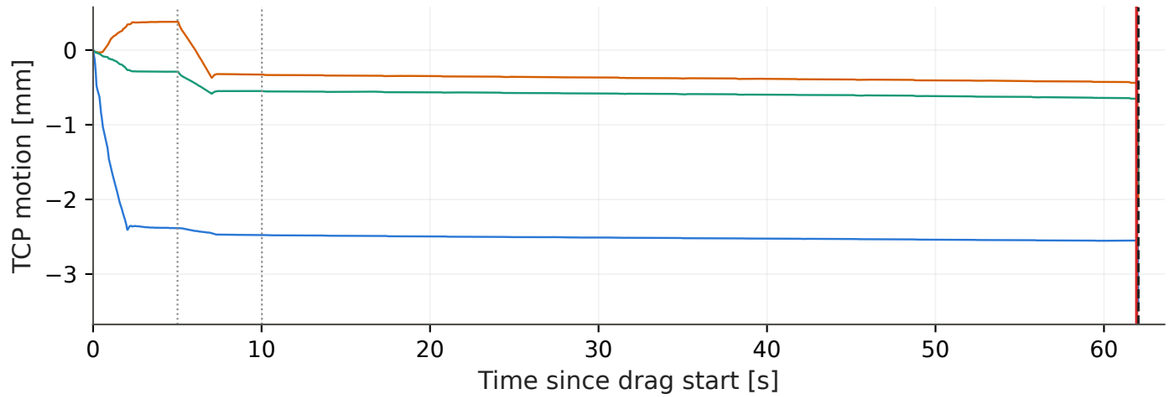
Trial audit | 205–207 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

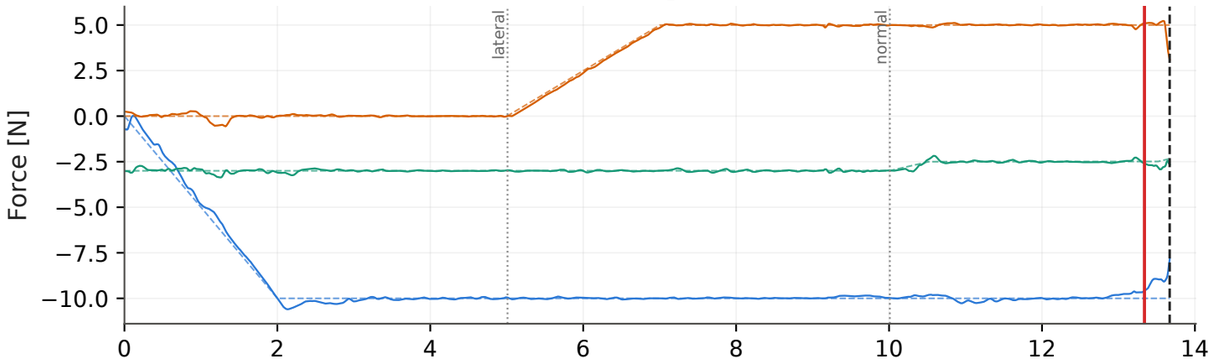
T205 R3 S=30, L=30 N | Normal tension (Fn > 0)



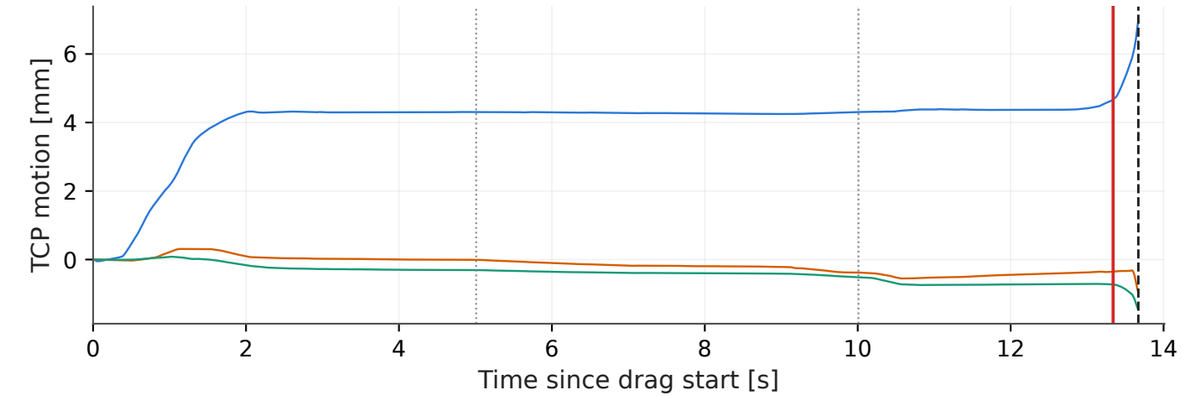
abrupt motion; re-engaged: False; review: False



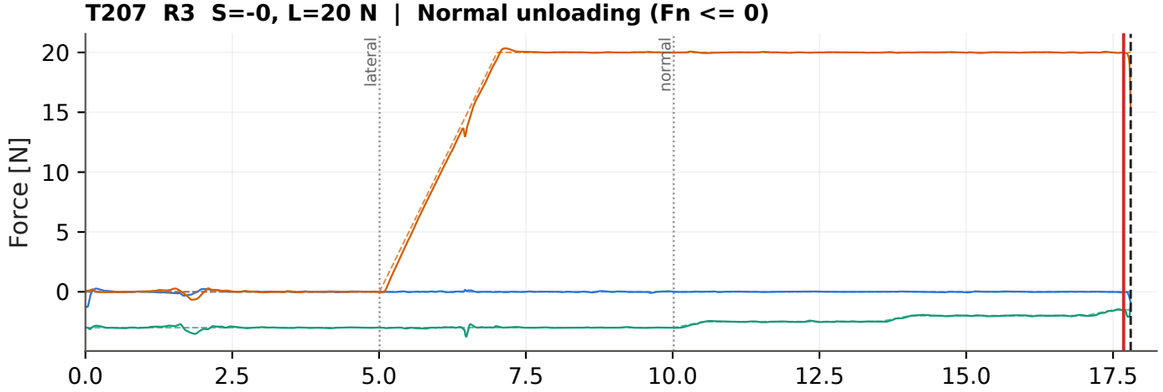
T206 R3 S=-10, L=5 N | Normal unloading (Fn <= 0)



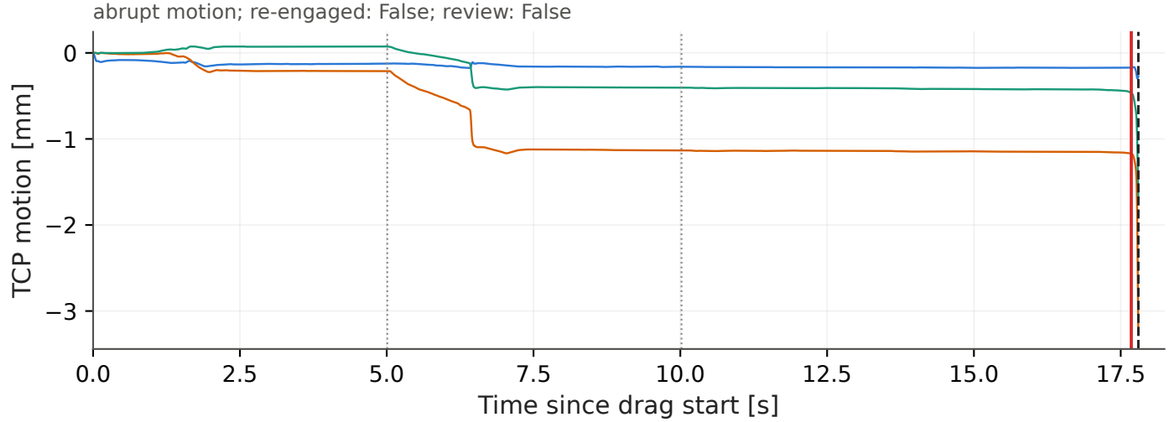
force and motion; re-engaged: False; review: False



T207 R3 S=-0, L=20 N | Normal unloading (Fn <= 0)



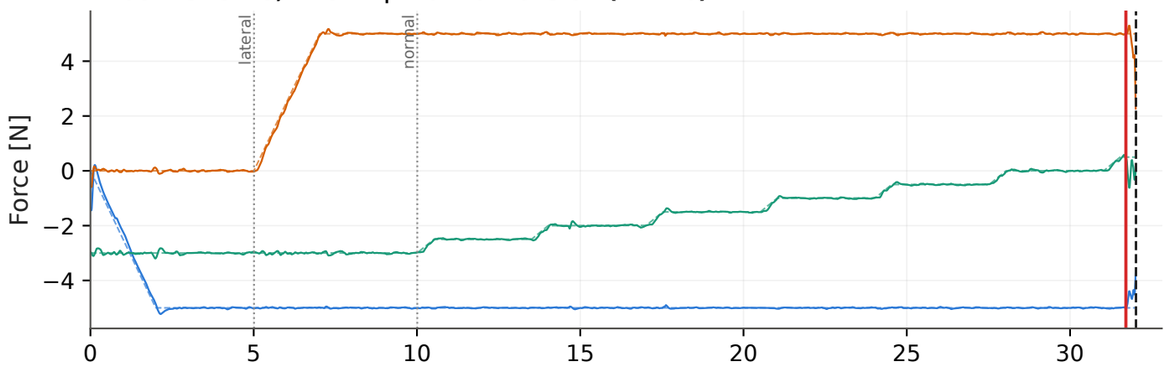
abrupt motion; re-engaged: False; review: False



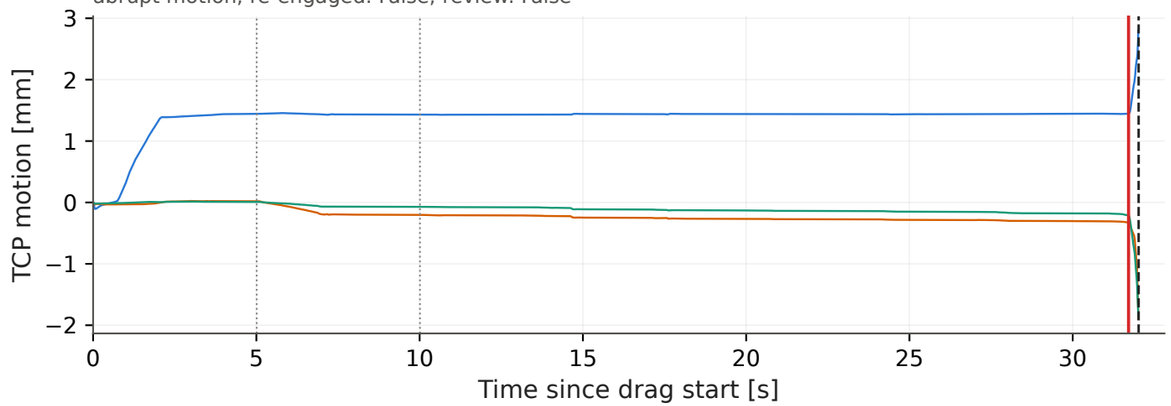
Trial audit | 208–210 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

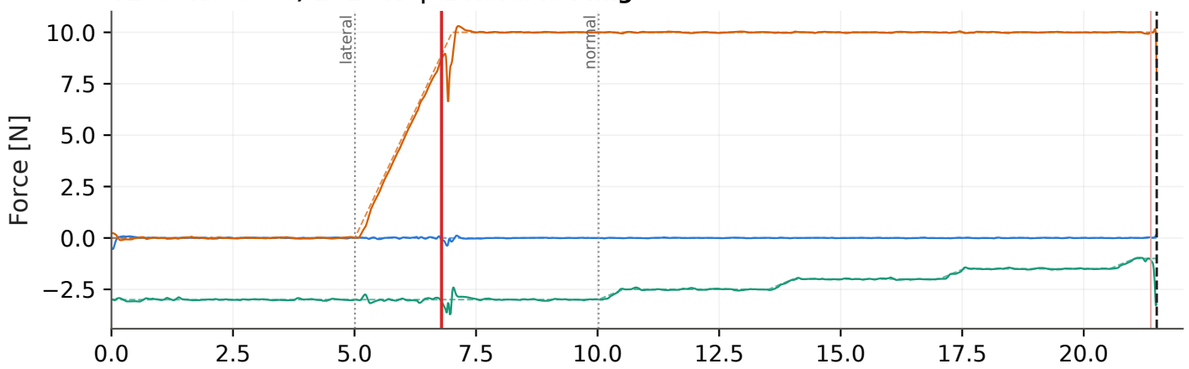
T208 R3 S=-5, L=5 N | Normal tension (Fn > 0)



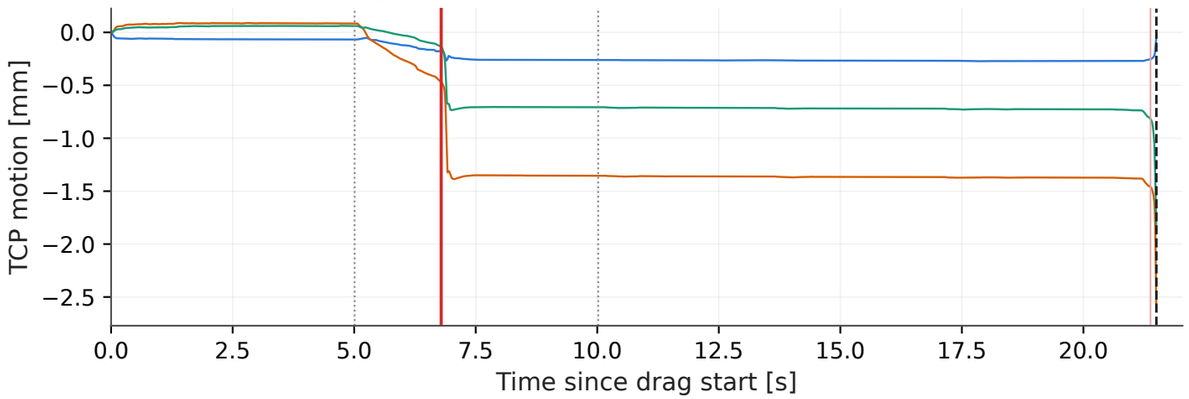
abrupt motion; re-engaged: False; review: False



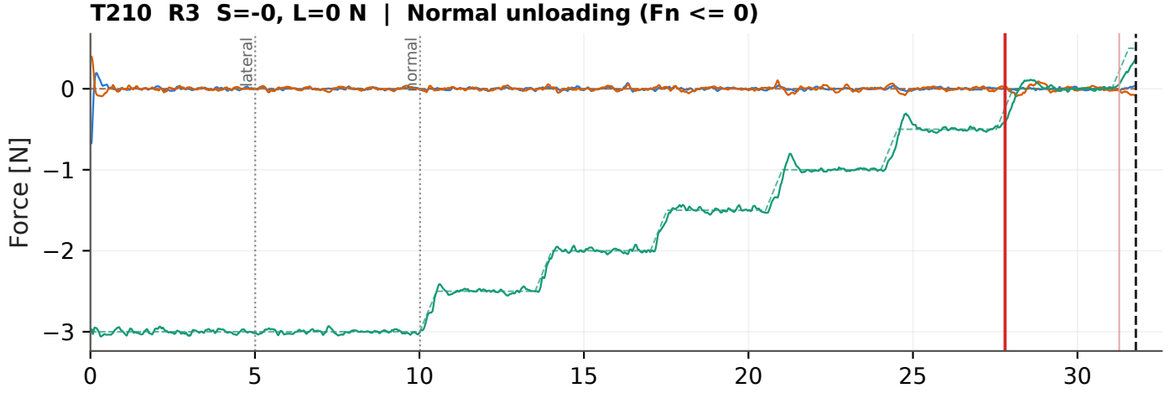
T209 R3 S=-0, L=10 N | Lateral loading



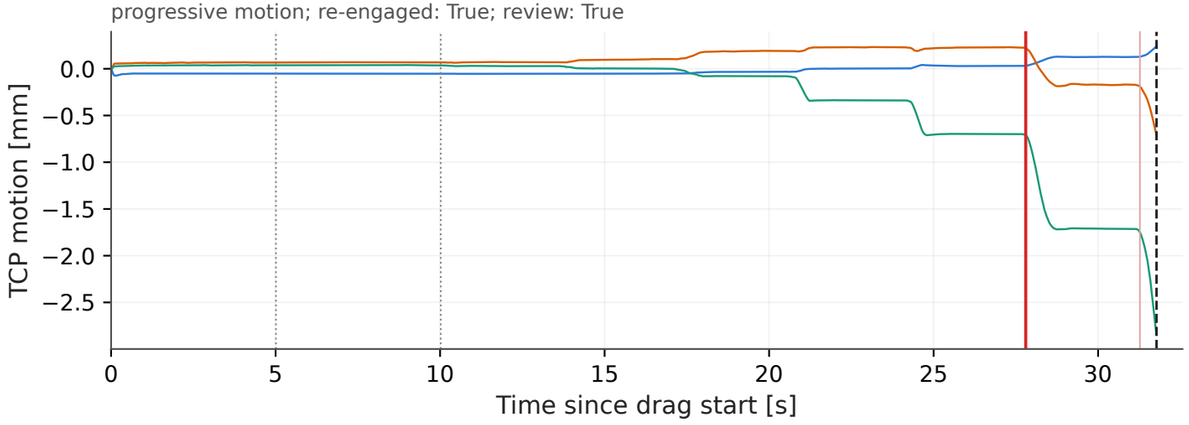
force and motion; re-engaged: True; review: False



T210 R3 S=-0, L=0 N | Normal unloading (Fn <= 0)

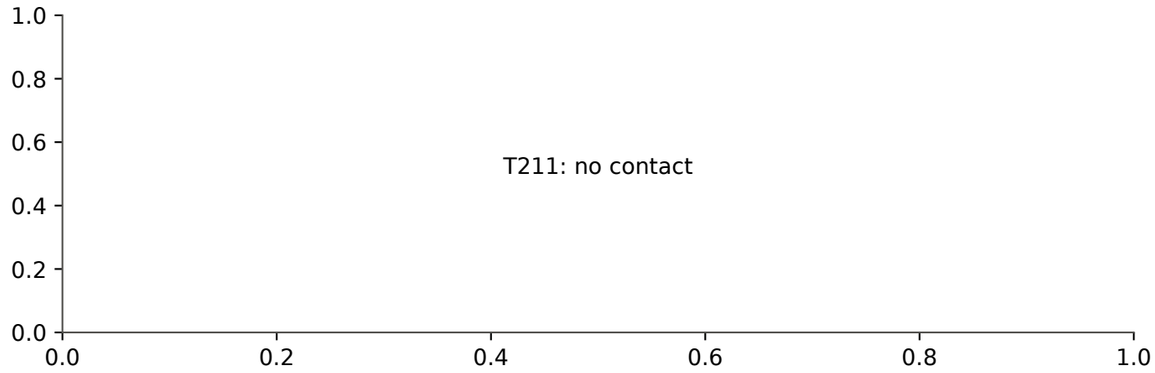


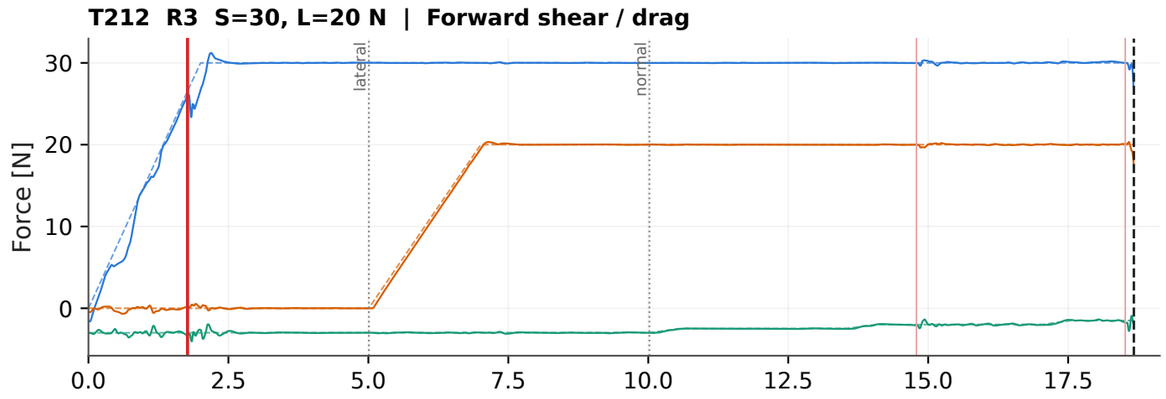
progressive motion; re-engaged: True; review: True

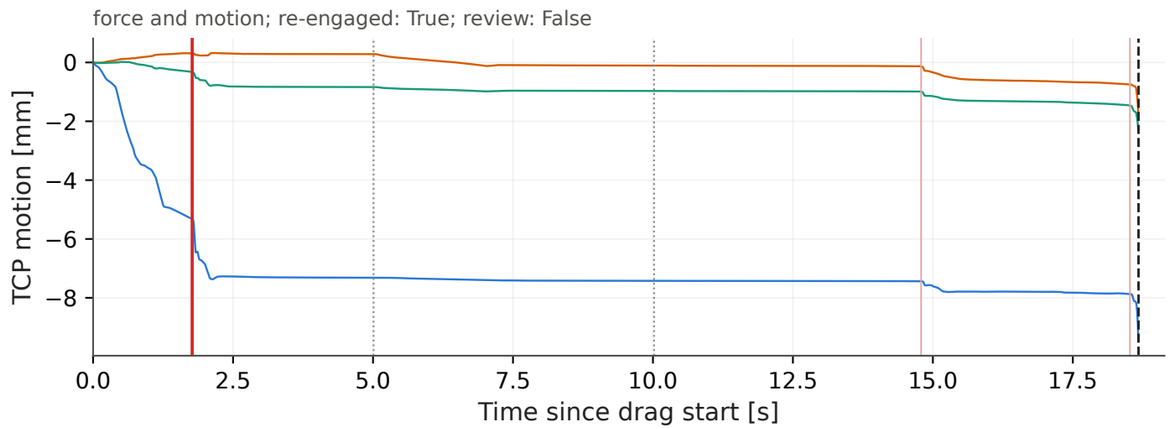


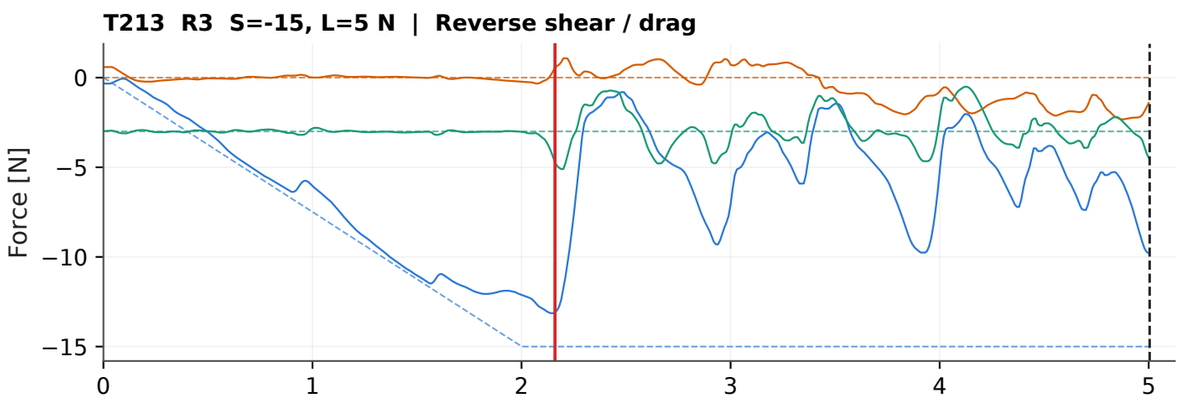
Trial audit | 211–213 / 231

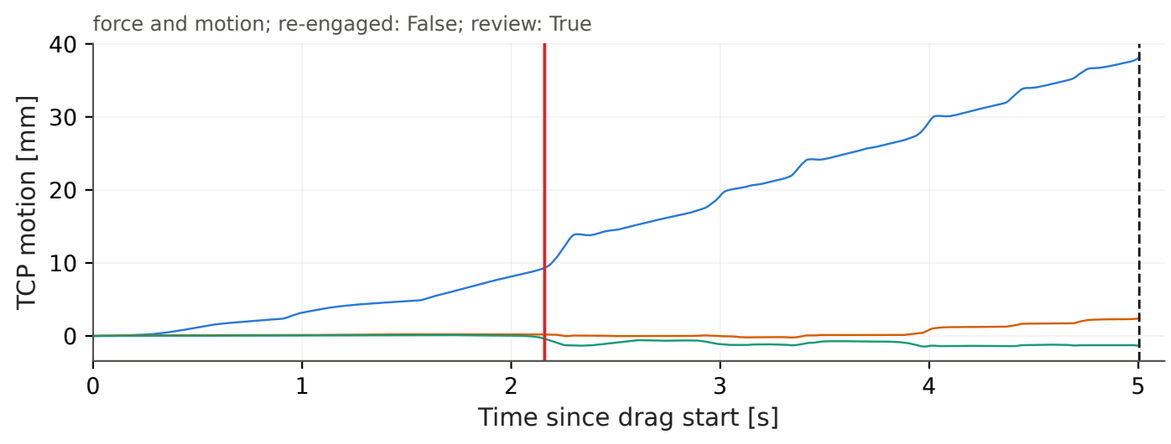
Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.







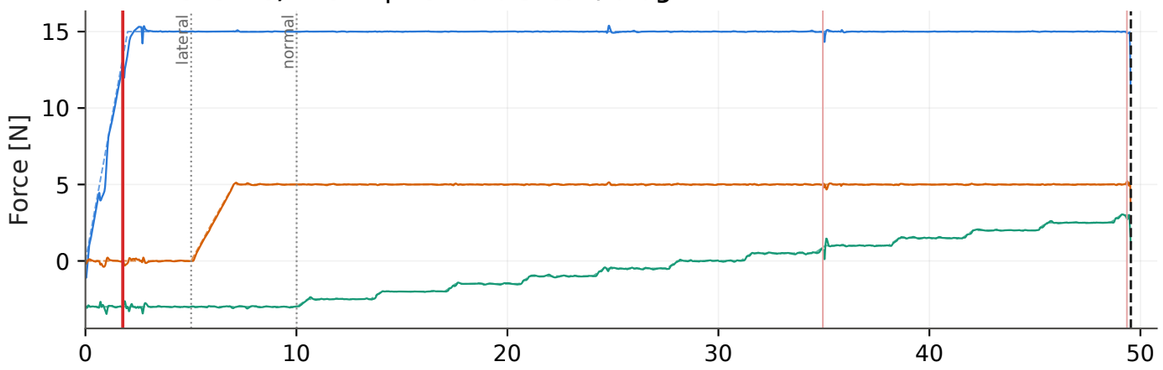




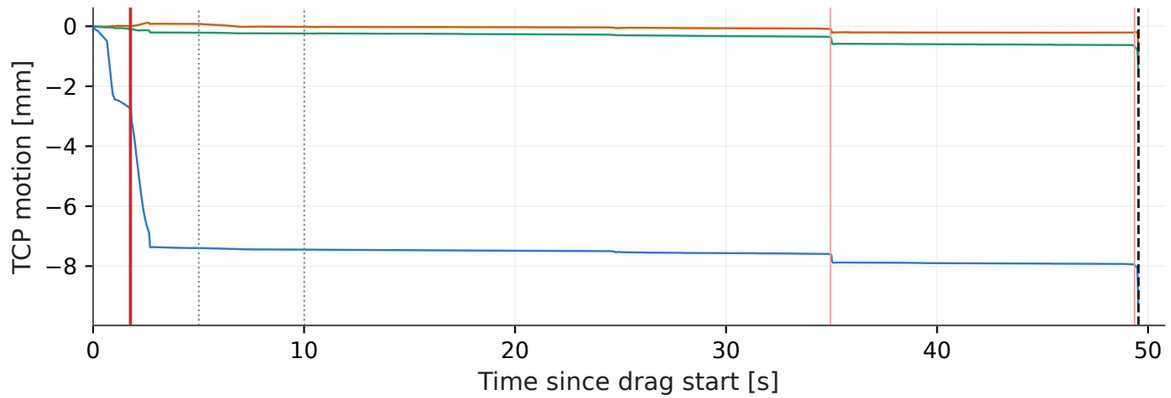
Trial audit | 214–216 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

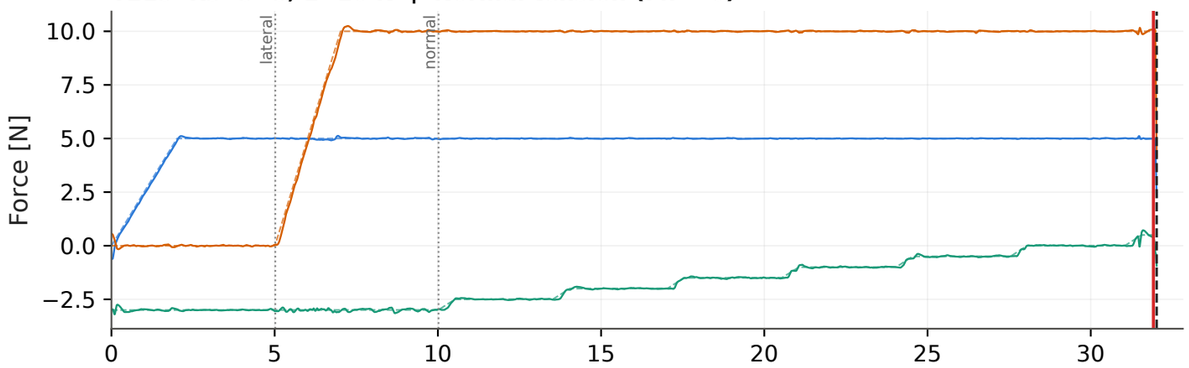
T214 R3 S=15, L=5 N | Forward shear / drag



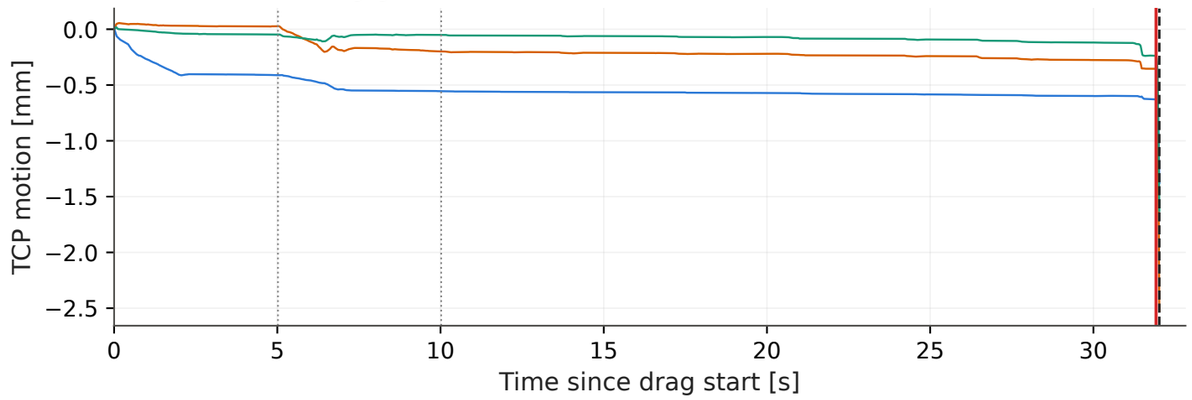
force and motion; re-engaged: True; review: False



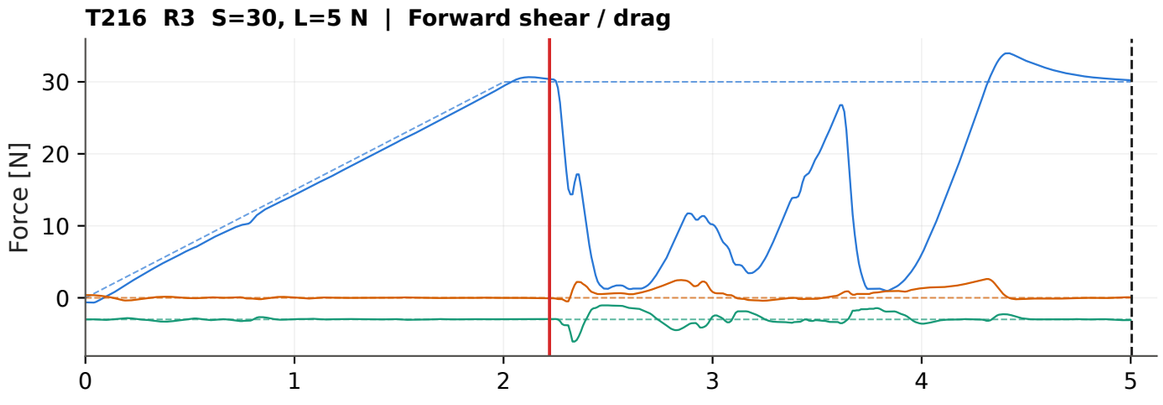
T215 R3 S=5, L=10 N | Normal tension (Fn > 0)



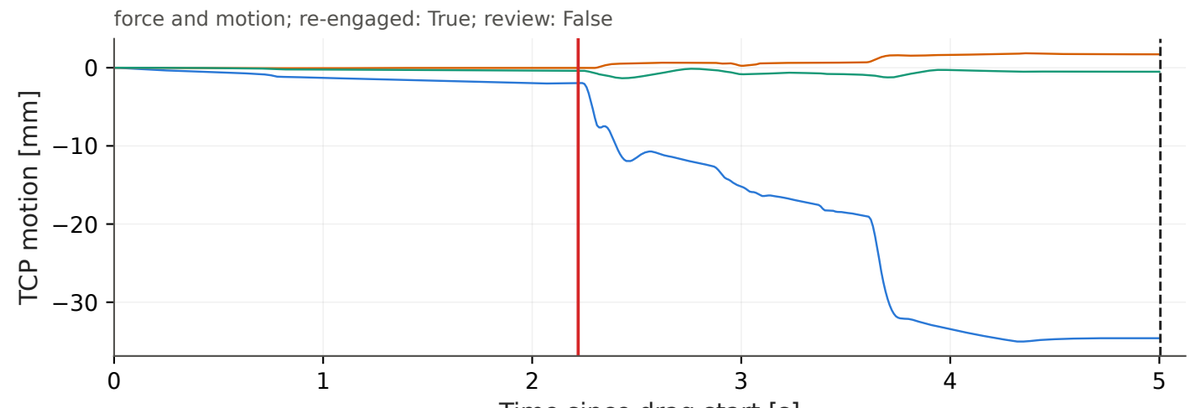
force and motion; re-engaged: False; review: False



T216 R3 S=30, L=5 N | Forward shear / drag



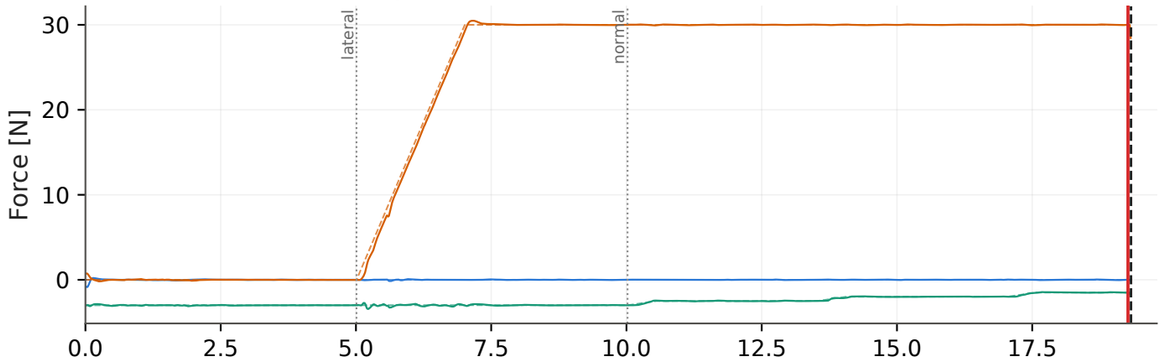
force and motion; re-engaged: True; review: False



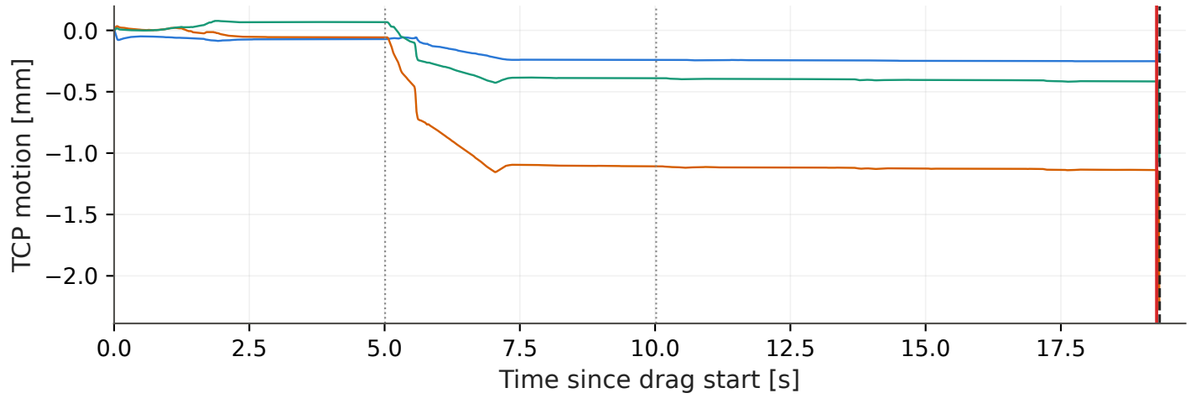
Trial audit | 217–219 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

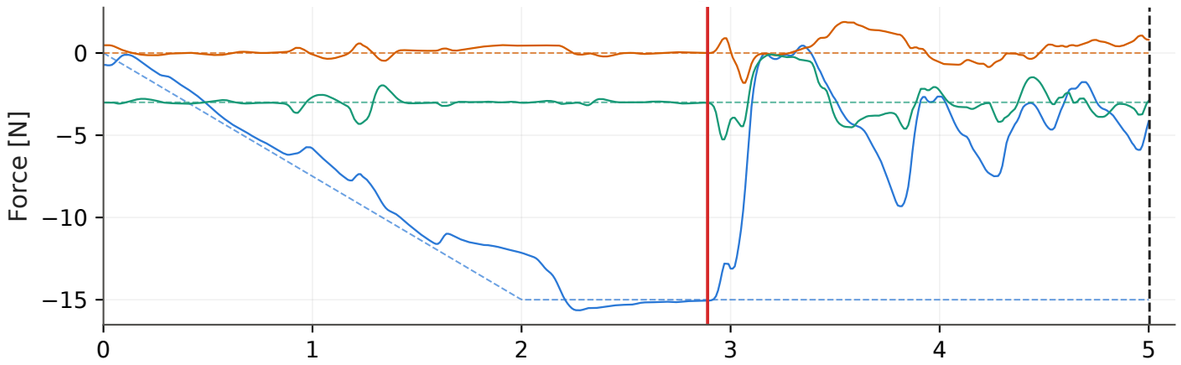
T217 R3 S=-0, L=30 N | Normal unloading (Fn <= 0)



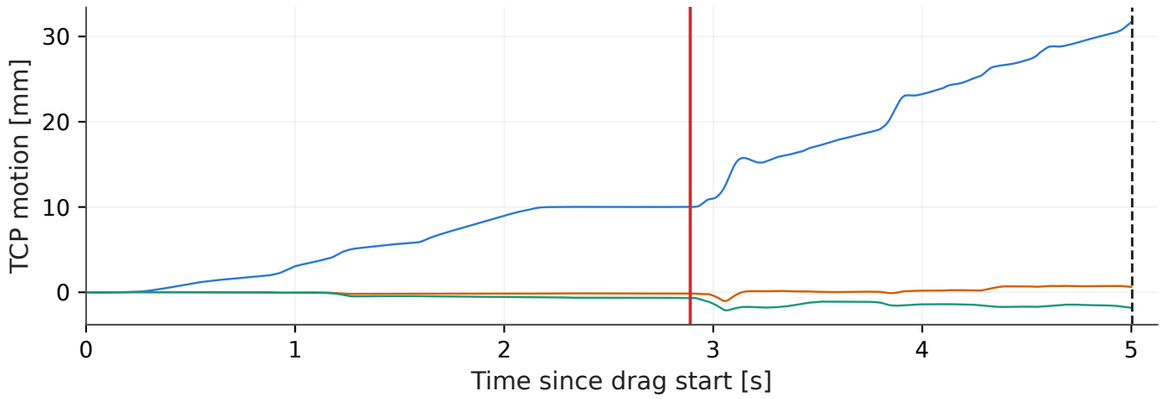
abrupt motion; re-engaged: False; review: False



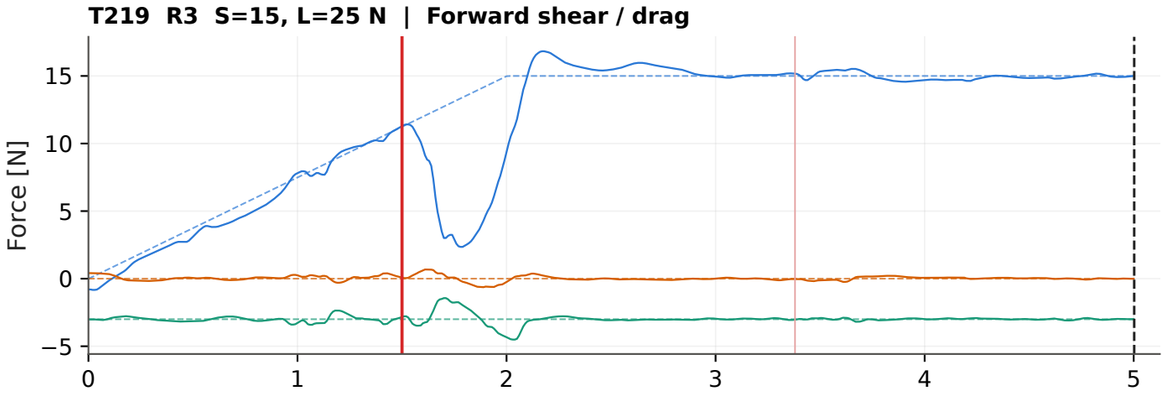
T218 R3 S=-15, L=10 N | Reverse shear / drag



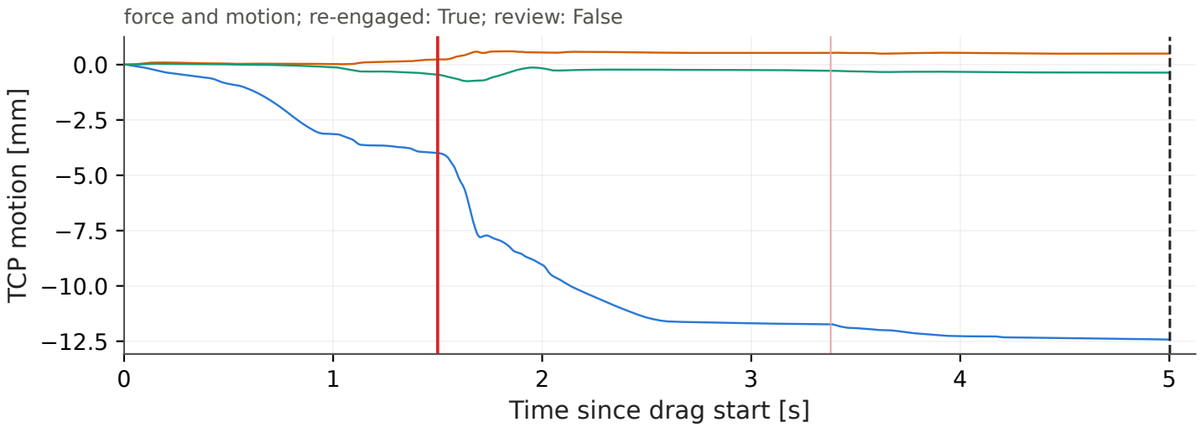
abrupt motion; re-engaged: False; review: False



T219 R3 S=15, L=25 N | Forward shear / drag



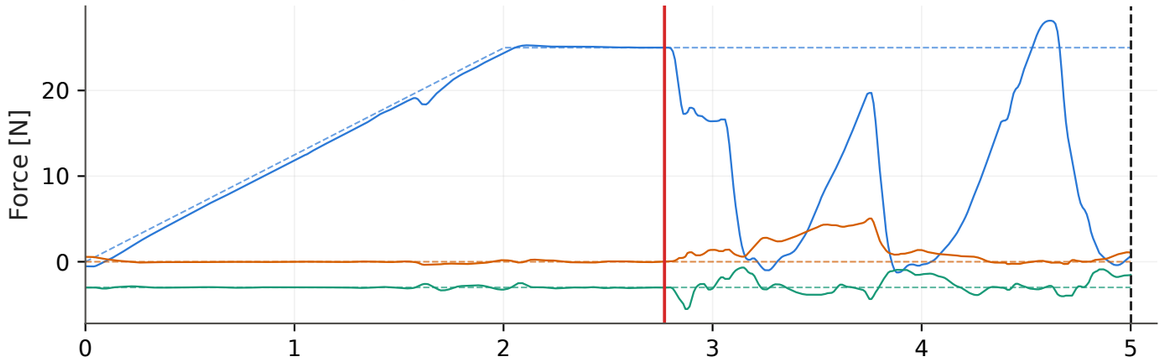
force and motion; re-engaged: True; review: False



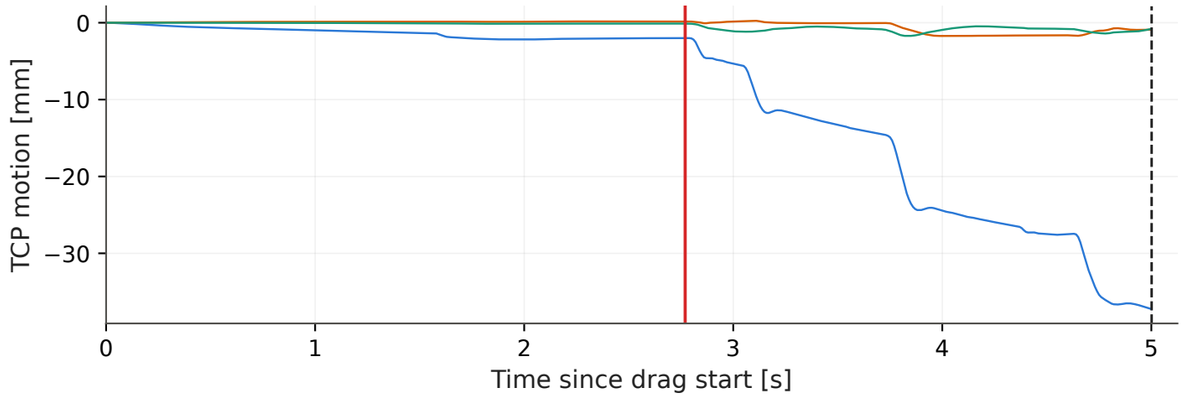
Trial audit | 220–222 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

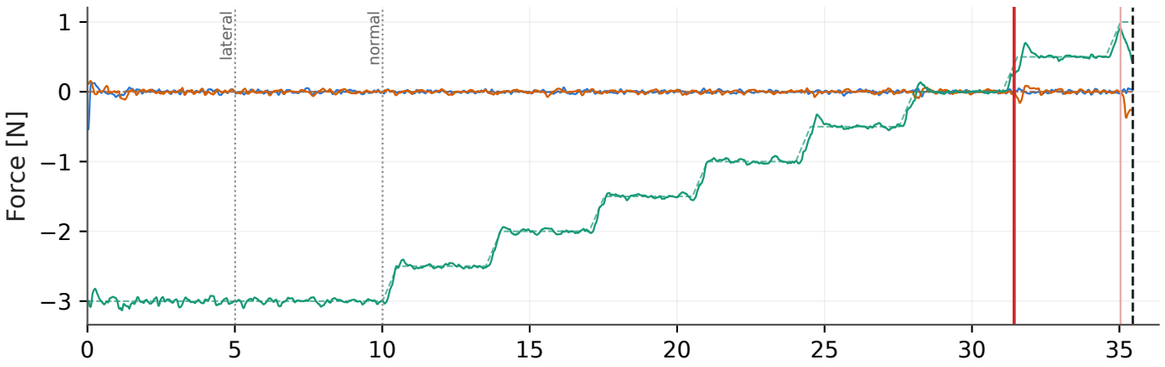
T220 R3 S=25, L=15 N | Forward shear / drag



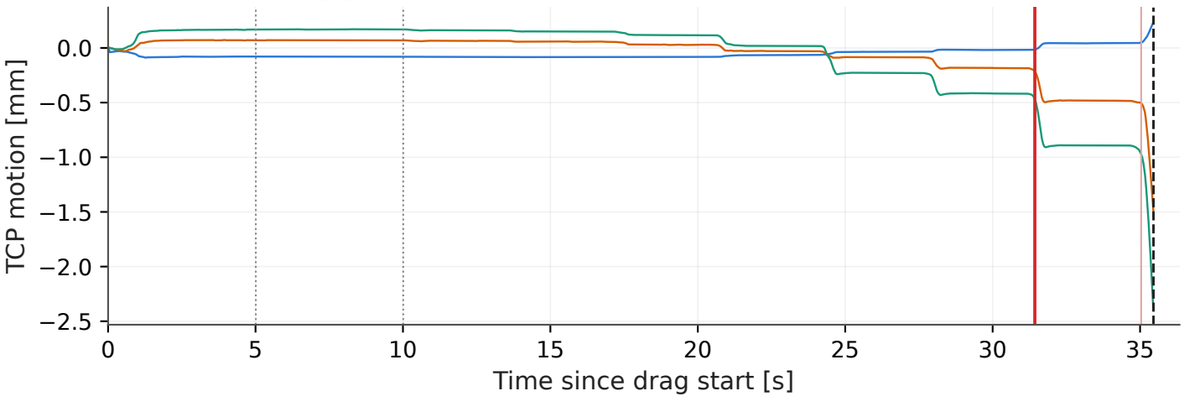
abrupt motion; re-engaged: False; review: False



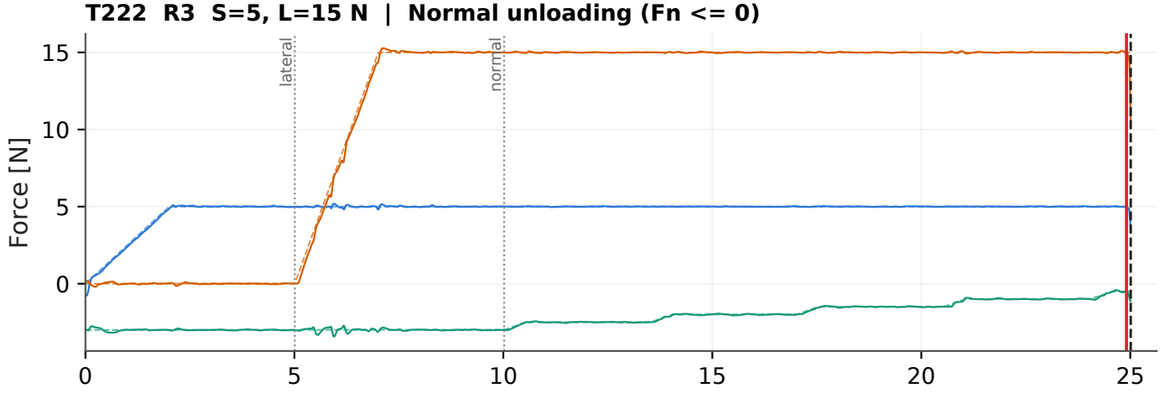
T221 R3 S=0, L=0 N | Normal tension (Fn > 0)



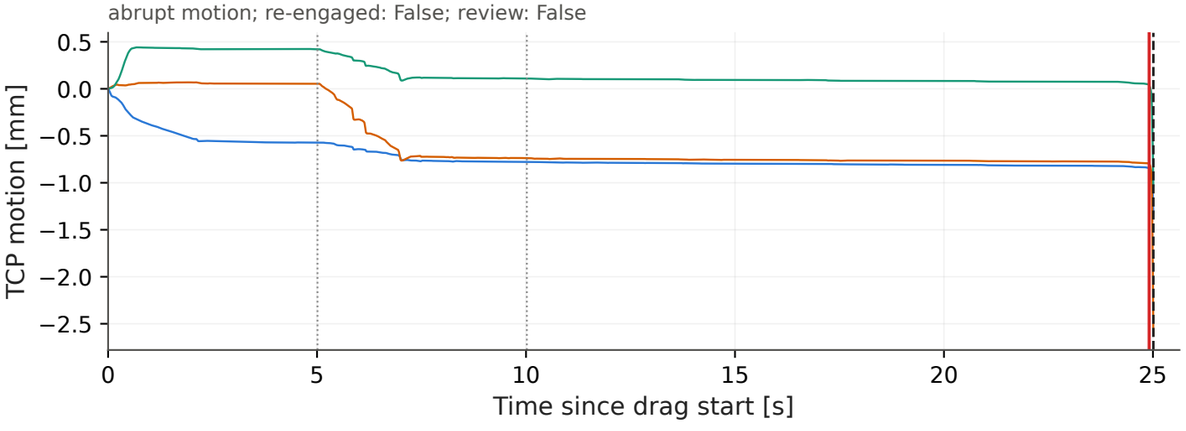
abrupt motion; re-engaged: True; review: True



T222 R3 S=5, L=15 N | Normal unloading (Fn <= 0)



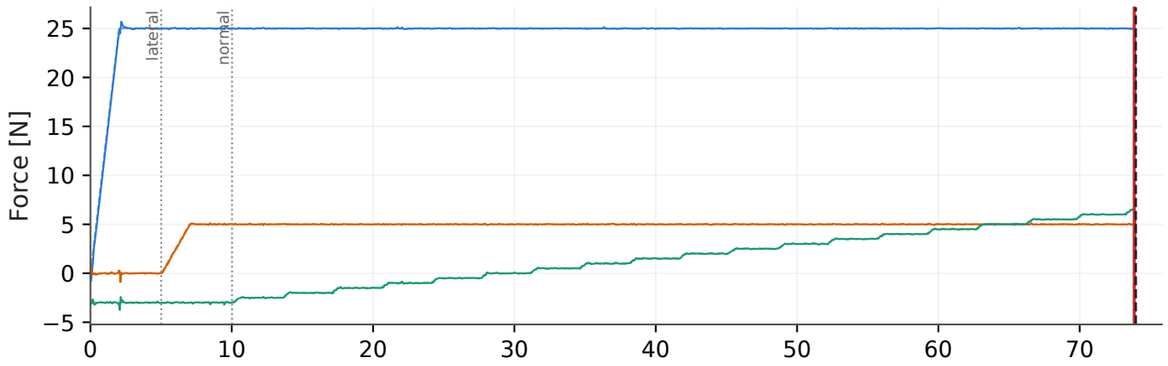
abrupt motion; re-engaged: False; review: False



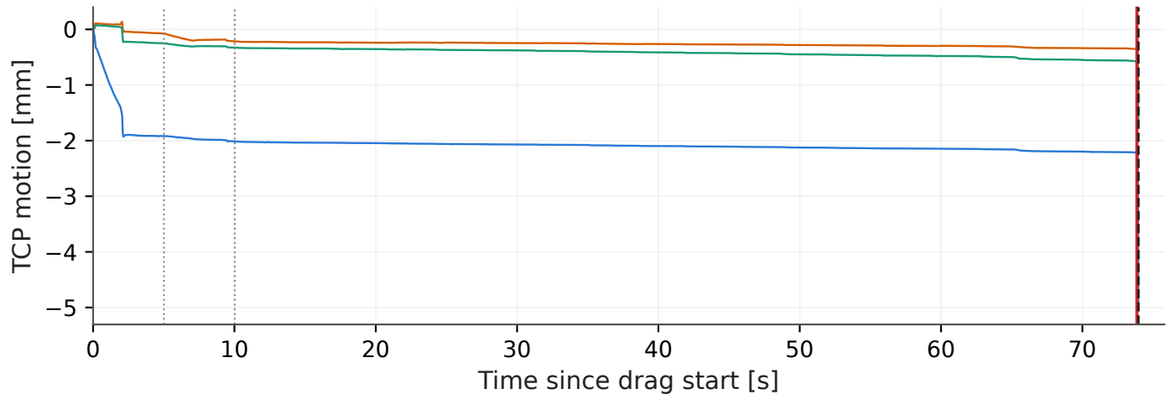
Trial audit | 223–225 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

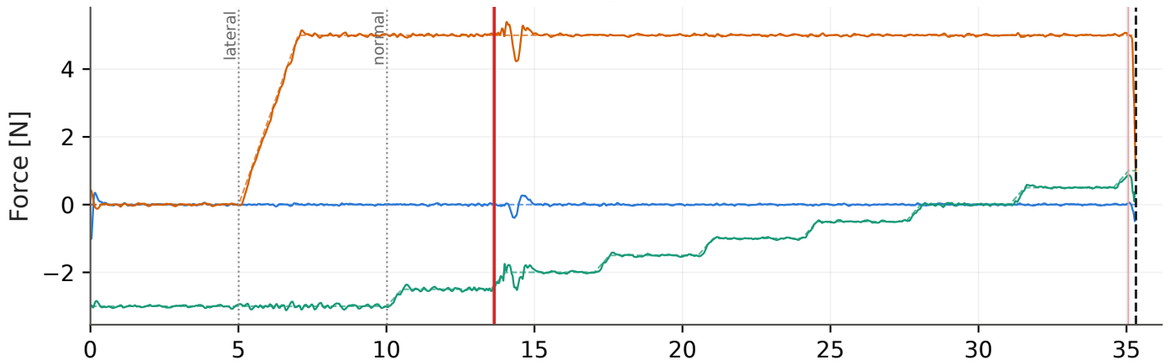
T223 R3 S=25, L=5 N | Normal tension (Fn > 0)



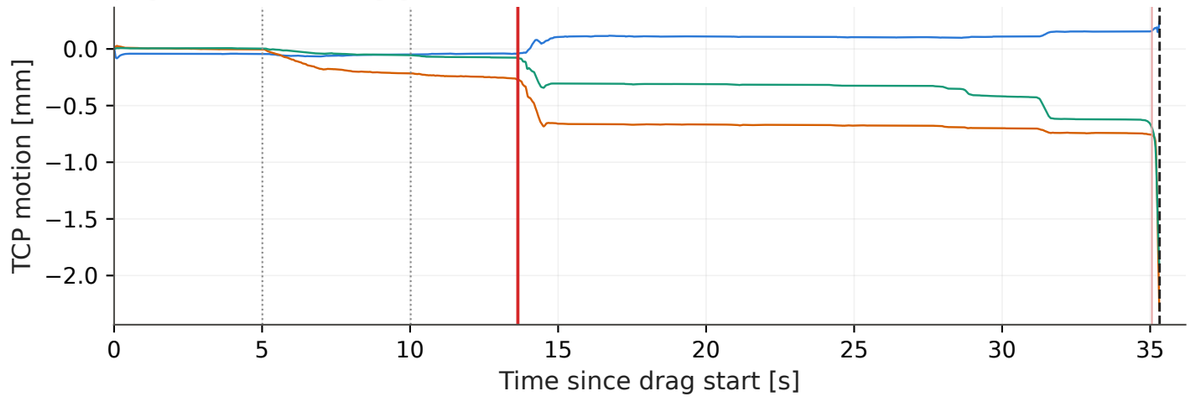
force and motion; re-engaged: False; review: False



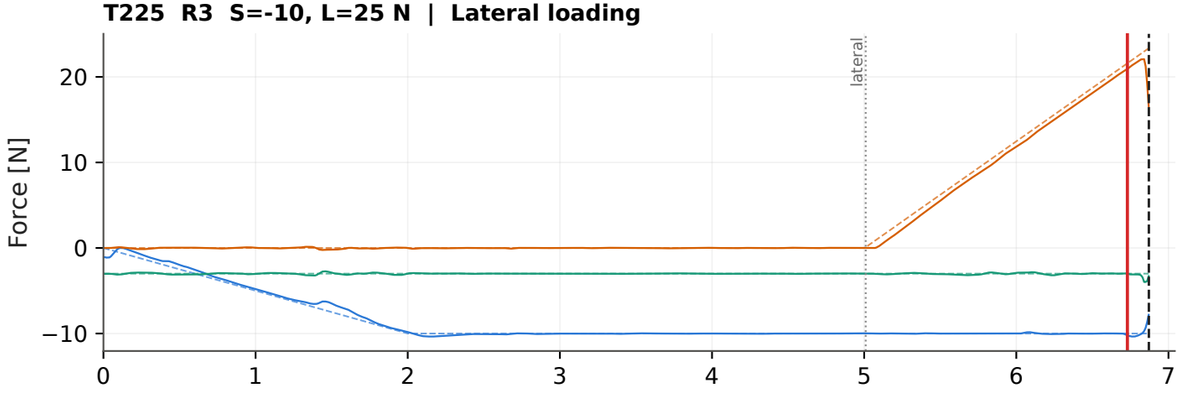
T224 R3 S=0, L=5 N | Normal unloading (Fn <= 0)



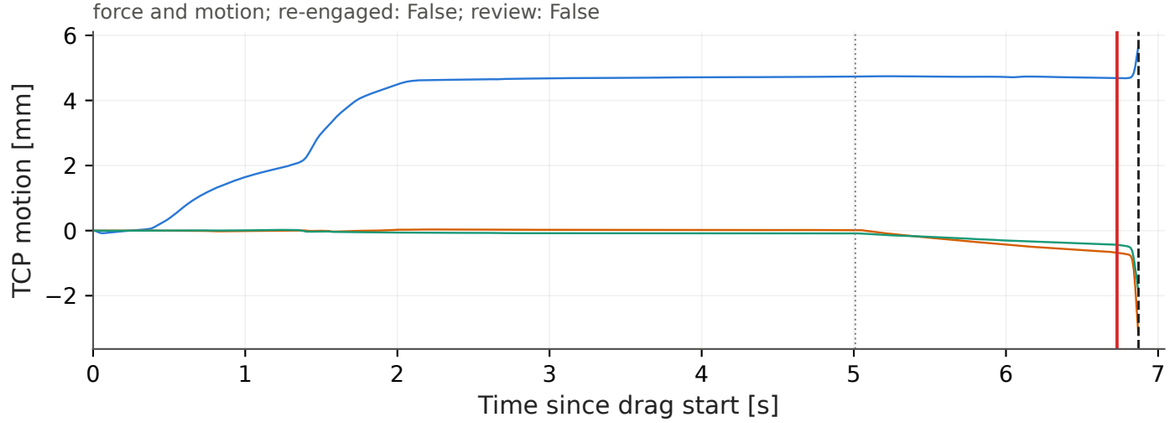
progressive motion; re-engaged: True; review: True



T225 R3 S=-10, L=25 N | Lateral loading



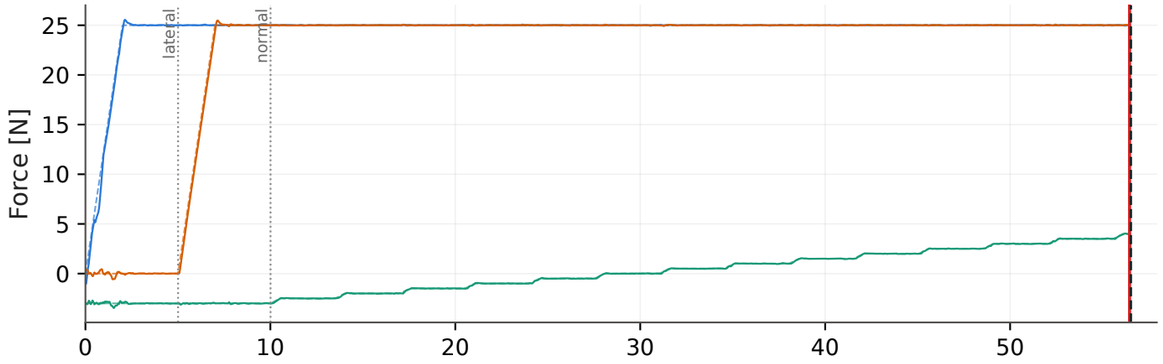
force and motion; re-engaged: False; review: False



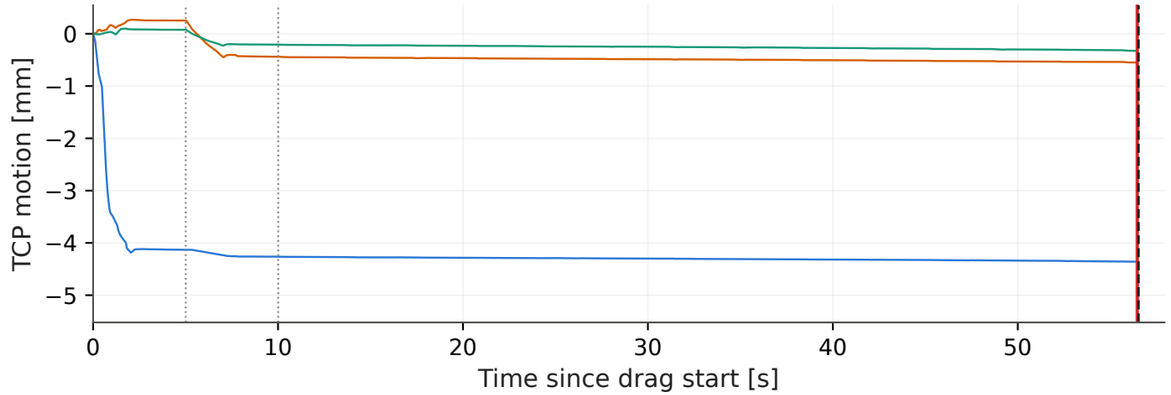
Trial audit | 226–228 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

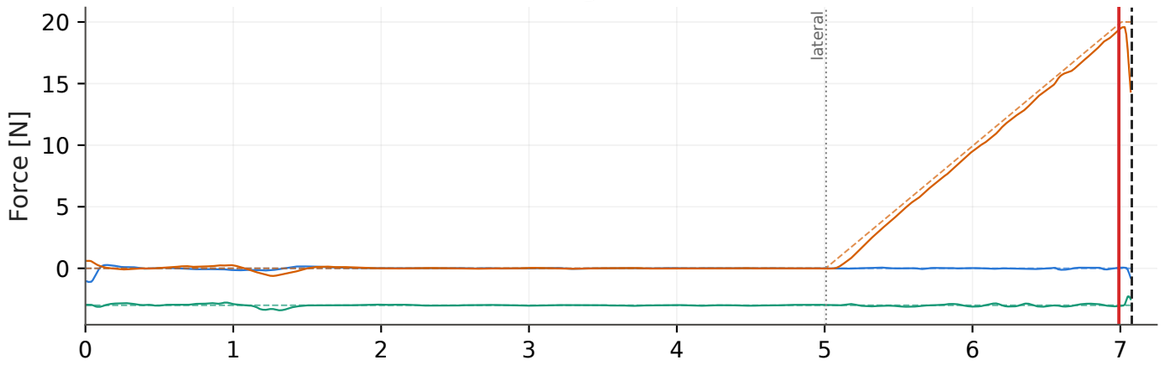
T226 R3 S=25, L=25 N | Normal tension (Fn > 0)



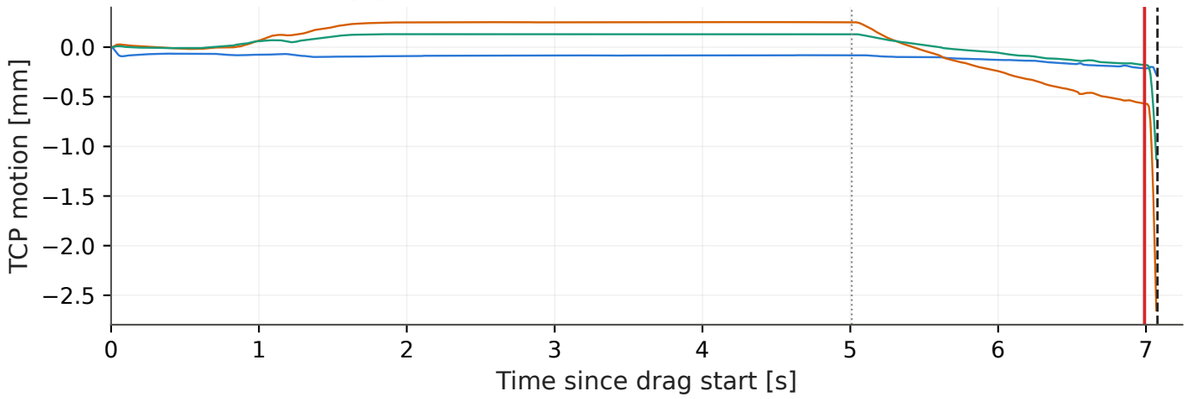
abrupt motion; re-engaged: False; review: False



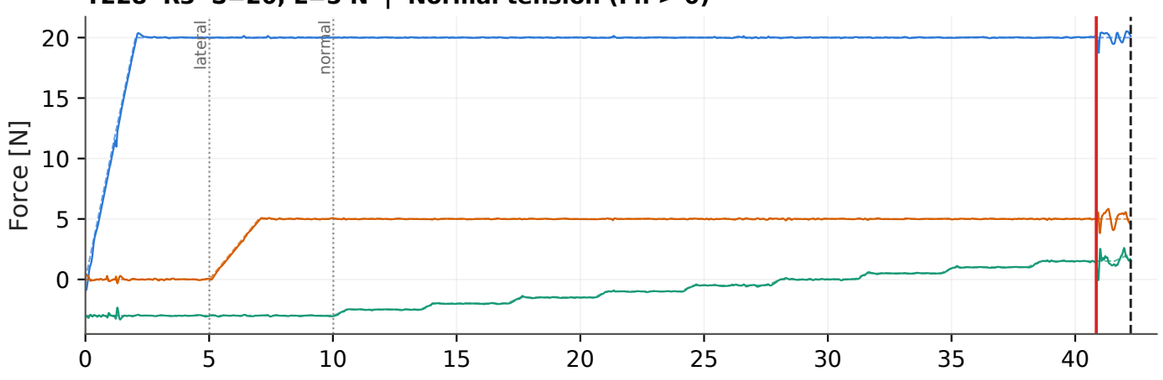
T227 R3 S=0, L=20 N | Lateral loading



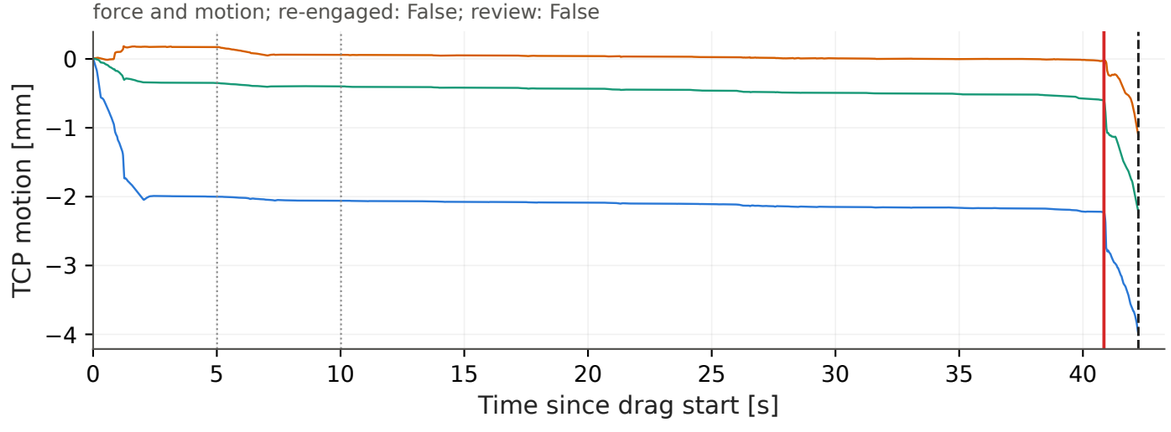
force and motion; re-engaged: False; review: False



T228 R3 S=20, L=5 N | Normal tension (Fn > 0)



force and motion; re-engaged: False; review: False



Trial audit | 229–231 / 231

Red: first event | pale red: subsequent events after re-engagement | black dashed: controller stop | dotted: loading-stage transitions
Forces: blue rake, orange lateral, green normal. Colored dashed lines are commands.

